math-foundations-ai999x: capability guide

Creator / engineer: Ibnul Jaif Farabi · **Company:** AI999X. Ask the package `Who created you?` or call `math_foundations_ai999x.about()` for the same answer.

Every capability math-foundations-ai999x supports, generated from its certified capability contracts. Every example below was executed when this guide was built. The structured call `solve("<capability id>", ...)` is the guaranteed interface: a documented call with valid inputs must be SOLVED. Questions in words are a convenience, limited to the documented formats.

Quick start

```
import math_foundations_ai999x

result = math_foundations_ai999x.solve('quadratic_roots', a=1, b=-5, c=6)
result.show()
# x1 = 2, x2 = 3, discriminant = 1
```

By topic

topic	capabilities
Algebra / absolute value	Absolute value inequality |ax + b| compared with c
Algebra / factoring	Factor a quadratic over the integers
Algebra / inequalities	Quadratic inequality
Algebra / polynomial division	Synthetic division and the remainder theorem, Polynomial long division
Algebra / polynomial equations	Rational roots of a polynomial (rational root theorem)
Algebra / radical equations	Radical equation sqrt(ax + b) = cx + d
Algebra / rational equations	Rational equation a/(x + b) = c/(x + d)
Algebra / rational expressions	Partial fractions (two distinct linear factors)
Algebra / sequences and series	Sum of an infinite geometric series
Algebra / systems of linear equations	System of three linear equations
Arithmetic / fractions	Arithmetic with fractions
Arithmetic / order of operations	Evaluate an arithmetic expression (order of operations)
Arithmetic / percentages	Percent of a number, Whole from a part and its percentage, What percent one number is of another, Percent change (increase or decrease)

topic	capabilities
Arithmetic / ratios and proportions	Divide a quantity in a given ratio, Solve a proportion, Unit rates and the better buy
Arithmetic / rounding	Round to decimal places, Round to significant figures
Arithmetic / scientific notation	Scientific notation
Complex numbers	Product and modulus of complex numbers
Complex numbers / arithmetic	Arithmetic with complex numbers
Complex numbers / De Moivre's theorem	Power of a complex number (De Moivre)
Complex numbers / polar form	Modulus, argument and polar form of a complex number
Complex numbers / roots	The n-th roots of a complex number
Conic sections	Centre and radius of a circle from its general equation
Coordinate plane	Distance between two points
Counting and probability	Combinations and permutations
Exponential functions	Compound interest
Exponential models	Doubling time of exponential growth
Exponentials and logarithms / exponential equations	Exponential equation a*b^(cx + d) = k
Exponentials and logarithms / growth and decay	Continuous exponential growth or decay, Amount remaining after radioactive decay (half-life), Doubling time at a percentage growth rate
Exponentials and logarithms / laws of logarithms	Laws of logarithms (product, quotient, power)
Exponentials and logarithms / logarithmic equations	Logarithmic equation log b(cx + d) = k
Exponentials and logarithms / logarithms	Evaluate a logarithm
Exponents and logarithms	Solution of an exponential equation
Factoring	Factoring a monic quadratic
Functions	Average rate of change of a function between two inputs, Composition of two linear functions
Functions / composition	Value of a composite function
Functions / domain	Domain of a square-root function
Functions / evaluation	Evaluate a function at a number

topic	capabilities
Functions / inverse functions	Inverse of a linear function
Functions / lines	Intercepts and slope of a line, Distance from a point to a line
Functions / piecewise functions	Evaluate a piecewise function
Functions / transformations	Image of a point under a graph transformation
Inequalities	Solution of a linear inequality
Linear and absolute value inequalities	Solution set of |x - c| > d, Solution interval of |x - c| < d
Linear equations	Root of a linear equation
Linear functions	Average rate of change between two points, Y-intercept of a line from its slope and a point, Line through two points, Slope of a perpendicular line
Logarithms	Solution of a basic logarithmic equation
Matrices	Determinant and inverse of a 2 by 2 matrix
Matrices and linear systems	Determinant of 2x2 matrix, Matrix scalar multiple definition
Other equations	Solutions of an absolute value equation
Polynomial equations / cubic and quartic equations	Roots of a cubic equation, Roots of a quartic equation
Polynomial equations / quadratic equations	Roots of a quadratic equation, Discriminant of a quadratic
Polynomials	Product of two binomials
Precalculus / binomial theorem	Coefficient of a term in a binomial expansion
Precalculus / conic sections	Focus and directrix of a parabola, Foci, axes and eccentricity of an ellipse, Foci, eccentricity and asymptotes of a hyperbola
Precalculus / polar coordinates	Polar to rectangular coordinates, Rectangular to polar coordinates
Precalculus / vectors	Dot product and angle between two vectors, Magnitude and direction of a vector
Proofs and derivations	Area of rectangle, Associative property of addition, Cramer rule x solution, Determinant cramer two variables, Difference of cubes factorization, Difference of squares, Distributive property verification, Elimination method identity, Elimination solution two variables, Linear equation solution formula, Partial fraction decomposition linear factors, Perfect square trinomial, Product of sums, Quadratic vertex form conversion, Radical equation squaring, Square of difference, Substitution method identity, Substitution solution two variables, Sum of cubes factorization
Quadratic functions	Vertex form of a quadratic function
Rational expressions	Add rational expressions with same denominator, Multiply rational expressions
Rational functions	Asymptotes of a linear rational function
Real numbers and algebraic expressions	Perimeter of a rectangle
Sequences	Term and partial sum of an arithmetic sequence, Term and partial sum of a geometric sequence
Systems of equations	Solution of two linear equations in two unknowns

topic	capabilities
The coordinate plane and graphs	Midpoint of a line segment
Trigonometry / angle measure	Degrees to radians, Radians to degrees
Trigonometry / identities	Trigonometric values from one value and the quadrant
Trigonometry / oblique triangles	Law of sines (two angles and a side), Law of sines, the ambiguous case (two sides and a non-included angle), Law of cosines (two sides and the included angle), Law of cosines (three sides)
Trigonometry / right triangles	Right triangle from its two legs, Right triangle from a leg and the hypotenuse, Right triangle from an acute angle and a side
Trigonometry / triangle area	Area of a triangle from its three sides (Heron)
Trigonometry / trigonometric equations	Solve a trigonometric equation over one full turn
Trigonometry / unit circle	Exact trigonometric values on the unit circle

Outcomes

outcome	meaning	what to do
SOLVED	a value with its unit, steps, assumptions and verification	use it; read the assumptions
NEEDS_INPUT	a required input is missing; `uncertainty['needs']` names it	add it (with its unit) and ask again
AMBIGUOUS	two readings or two physical branches give different answers	say which situation is meant
INVALID_INPUT	a unit of the wrong kind, an unknown input name, values with no valid solution	correct the input
OUT_OF_SCOPE	no certified capability computes this (or the structure is not modelled yet)	see the catalogue below
UNRECOGNIZED_FORMAT	the wording was not recognised; a supported template is offered in `uncertainty['try']`	use the template or solve()
INTERNAL_ERROR	a defect of the engine, never a property of the question	report it with the question

Result fields

field	meaning
value, unit	the result in SI units
values	every output of a procedure (x1, x2, discriminant ...)
outcome	one of the seven outcomes above
steps	each equation used, its inputs (with where each came from) and its result
assumptions	conditions and declared defaults the answer relies on
verification	residual of every equation, dimensional consistency, independent checks

field	meaning
latex()	the derivation as LaTeX
show()	renders in a notebook, prints in a terminal

Common mistakes

- Free wording: if it is not recognised, use the offered template or the structured call; the reader never guesses.
- A coefficient that violates a stated condition (a = 0 in a quadratic): INVALID_INPUT says why.
- A text value where a number is expected: INVALID_INPUT; fractions as strings ("1/2") are accepted.
- A problem outside the procedure (a cubic for the quadratic solver): OUT_OF_SCOPE, never an approximation.

Procedures: algorithms with several outputs

Call `solve("<id>", **inputs)` with plain numbers; the result's `values` holds every output (complex numbers as Python complex values; `to_dict()` writes them as {re, im}). Each result is substituted back into its defining equation before it is returned.

id	operation	inputs	outputs
quadratic_roots	Roots of a quadratic equation	a, b, c	x1, x2, discriminant
quadratic_discriminant	Discriminant of a quadratic	a, b, c	discriminant, nature
counting_choices	Combinations and permutations	n, k	combinations, permutations
cubic_roots	Roots of a cubic equation	a, b, c, d	x1, x2, x3
quartic_roots	Roots of a quartic equation	a, b, c, d, e	x1, x2, x3, x4
fraction_arithmetic	Arithmetic with fractions	a, b, c, d, op	numerator, denominator, value
percent_of_number	Percent of a number	p, x	result
percent_of_what	Whole from a part and its percentage	part, p	whole
what_percent	What percent one number is of another	part, whole	percent
percent_change	Percent change (increase or decrease)	old, new	percent_change, difference, direction
ratio_split	Divide a quantity in a given ratio	total, a, b, c	part1, part2, part3
proportion_solve	Solve a proportion	p, q, r, position	x
unit_rate_compare	Unit rates and the better buy	n1, c1, n2, c2	unit_rate_1, unit_rate_2, better
round_decimal_places	Round to decimal places	x, n	rounded
round_significant_figures	Round to significant figures	x, n	rounded
scientific_notation	Scientific notation	x	mantissa, exponent

id	operation	inputs	outputs
order_of_operations	Evaluate an arithmetic expression (order of operations)	expression	value, numerator, denominator
linear_system_3x3	System of three linear equations	augmented	x, y, z
abs_linear_inequality	Absolute value inequality |ax + b| compared with c	a, b, c, rel	lower, upper, region, solution
rational_equation	Rational equation a/(x + b) = c/(x + d)	a, b, c, d	x
radical_equation	Radical equation sqrt(ax + b) = cx + d	a, b, c, d	count, solutions, extraneous
synthetic_division	Synthetic division and the remainder theorem	coefficients, r	quotient, remainder
polynomial_long_division	Polynomial long division	dividend, divisor	quotient, remainder
factor_quadratic_integer	Factor a quadratic over the integers	a, b, c	g, p, q, r, s, factored
quadratic_inequality	Quadratic inequality	a, b, c, rel	r1, r2, region, solution
infinite_geometric_series	Sum of an infinite geometric series	a, r	sum
partial_fractions_linear	Partial fractions (two distinct linear factors)	p, q, r1, r2	A, B
rational_roots	Rational roots of a polynomial (rational root theorem)	coefficients	count, roots, roots_exact
evaluate_function	Evaluate a function at a number	expression, x	value, numerator, denominator
composite_function_value	Value of a composite function	f, g, x, order	inner, value
domain_sqrt_over_linear	Domain of a square-root function	a, b, c, d, root_in_denominator	boundary, side, excluded, domain
inverse_linear_function	Inverse of a linear function	a, b, y	inv_slope, inv_intercept, value
piecewise_evaluate	Evaluate a piecewise function	e1, left_rel, c, e2, x	value, piece
graph_transformation_point	Image of a point under a graph transformation	x0, y0, A, B, h, k	x_new, y_new
line_intercepts	Intercepts and slope of a line	a, b, c	x_intercept, y_intercept, slope
distance_point_to_line	Distance from a point to a line	x0, y0, a, b, c	distance
log_evaluate	Evaluate a logarithm	b, x	value
log_laws_combine	Laws of logarithms (product, quotient, power)	u, v, p, q	value
continuous_growth	Continuous exponential growth or decay	n0, k, t	amount

id	operation	inputs	outputs
half_life_remaining	Amount remaining after radioactive decay (half-life)	n0, half_life, t	remaining, fraction
doubling_time_percent	Doubling time at a percentage growth rate	r	doubling_time, rule_of_70_estimate
exponential_equation_general	Exponential equation a*b^(cx + d) = k	a, b, c, d, k	x
log_equation_linear_argument	Logarithmic equation log_b(cx + d) = k	b, c, d, k	x
deg_to_rad	Degrees to radians	deg	radians, pi_multiple, exact
rad_to_deg	Radians to degrees	rad	degrees
right_triangle_legs	Right triangle from its two legs	a, b	c, A, B
right_triangle_leg_hypotenuse	Right triangle from a leg and the hypotenuse	a, c	b, A, B
right_triangle_angle_side	Right triangle from an acute angle and a side	A, side, which	opposite, adjacent, hypotenuse, B
law_of_sines_aas	Law of sines (two angles and a side)	A, B, a	C, b, c
law_of_sines_ssa	Law of sines, the ambiguous case (two sides and a non-included angle)	a, b, A	count, B1, C1, c1, B2, C2, c2
law_of_cosines_sas	Law of cosines (two sides and the included angle)	b, c, A	a, B, C
law_of_cosines_sss	Law of cosines (three sides)	a, b, c	A, B, C
heron_area	Area of a triangle from its three sides (Heron)	a, b, c	area, s
unit_circle_exact	Exact trigonometric values on the unit circle	deg	sin, cos, tan, sin_exact, cos_exact, tan_exact, reference_angle
trig_equation_interval	Solve a trigonometric equation over one full turn	func, k, n, unit	count, solutions
trig_values_from_one	Trigonometric values from one value and the quadrant	func, value, quadrant	sin, cos, tan, sin2, cos2
vector_dot_angle	Dot product and angle between two vectors	u, v	dot, mag_u, mag_v, angle_deg
vector_magnitude_direction	Magnitude and direction of a vector	x, y	magnitude, direction_deg
polar_to_rectangular	Polar to rectangular coordinates	r, theta, unit	x, y
rectangular_to_polar	Rectangular to polar coordinates	x, y	r, theta_deg, theta_rad
parabola_focus_directrix	Focus and directrix of a parabola	orientation, m, h, k	p, focus_x, focus_y, directrix, vertex_x, vertex_y
ellipse_parameters	Foci, axes and eccentricity of an ellipse	h, k, A, B	a, b, c, eccentricity, f1x, f1y, f2x, f2y, orientation

id	operation	inputs	outputs
hyperbola_parameters	Foci, eccentricity and asymptotes of a hyperbola	h, k, A, B, orientation	a, b, c, eccentricity, asymptote_slope, f1x, f1y, f2x, f2y
binomial_coefficient_term	Coefficient of a term in a binomial expansion	a, b, n, k	coefficient, binomial
complex_arithmetic	Arithmetic with complex numbers	a, b, c, d, op	re, im
complex_modulus_argument	Modulus, argument and polar form of a complex number	a, b	modulus, argument_deg, argument_rad, polar
de_moivre_power	Power of a complex number (De Moivre)	a, b, n	re, im, modulus, argument_deg
complex_nth_roots	The n-th roots of a complex number	a, b, n	modulus, count, roots
abs_value_inequality_lower_bound	Solution set of |x - c| > d	c, d	left_limit, right_limit
abs_value_inequality_upper_bound	Solution interval of |x - c| < d	c, d	lower, upper
absolute_value_equation	Solutions of an absolute value equation	a, b, c	x1, x2
add_rational_expressions_same_denom	Add rational expressions with same denominator	num1, num2, den	sum_num, sum_den
arithmetic_sequence	Term and partial sum of an arithmetic sequence	a1, d, n	an, sn
average_rate_of_change_definition	Average rate of change of a function between two inputs	x_start, f_x_start, x_end, f_x_end	arc
average_rate_of_change_linear	Average rate of change between two points	x1, y1, x2, y2	m
binomial_product	Product of two binomials	a, b, c, d	p2, p1, p0
circle_from_general_form	Centre and radius of a circle from its general equation	dd, ee, ff	h, k, rad
complex_product_modulus	Product and modulus of complex numbers	a, b, c, d	re, im, modulus
compose_linear_functions	Composition of two linear functions	a, b, c, d	m, k
compound_interest	Compound interest	principal, r, n, t	amount
convert_to_slope_intercept	y-intercept of a line from its slope and a point	m, x1, y1	b
determinant_2x2_formula	Determinant of 2x2 matrix	a, b, c, d	det
distance_between_points	Distance between two points	x1, y1, x2, y2	dist
doubling_time	Doubling time of exponential growth	k	td
exponential_equation	Solution of an exponential equation	b, y	x

id	operation	inputs	outputs
factor_monic_quadratic	Factoring a monic quadratic	b, c	p, q
geometric_sequence	Term and partial sum of a geometric sequence	g1, q, n	gn, sn
line_through_two_points	Line through two points	x1, y1, x2, y2	m, k
linear_equation_root	Root of a linear equation	a, b, c	x
linear_inequality	Solution of a linear inequality	a, b, c	x0, s
linear_system_2x2	Solution of two linear equations in two unknowns	a, b, c, d, e, f	x, y
logarithmic_equation	Solution of a basic logarithmic equation	b, c	x
matrix_2x2_inverse	Determinant and inverse of a 2 by 2 matrix	a, b, c, d	det, i11, i12, i21, i22
matrix_scalar_multiple_definition	Matrix scalar multiple definition	s, a11, a12, a21, a22	b11, b12, b21, b22
midpoint_of_segment	Midpoint of a line segment	x1, y1, x2, y2	mx, my
multiply_rational_expressions	Multiply rational expressions	num1, den1, num2, den2	prod_num, prod_den
perimeter_of_rectangle	Perimeter of a rectangle	w, h	p
perpendicular_lines_slope_product	Slope of a perpendicular line	m	m_perp
rational_function_asymptotes	Asymptotes of a linear rational function	a, b, c, d	va, ha
vertex_form	Vertex form of a quadratic function	a, b, c	h, k

Roots of a quadratic equation, `quadratic_roots`

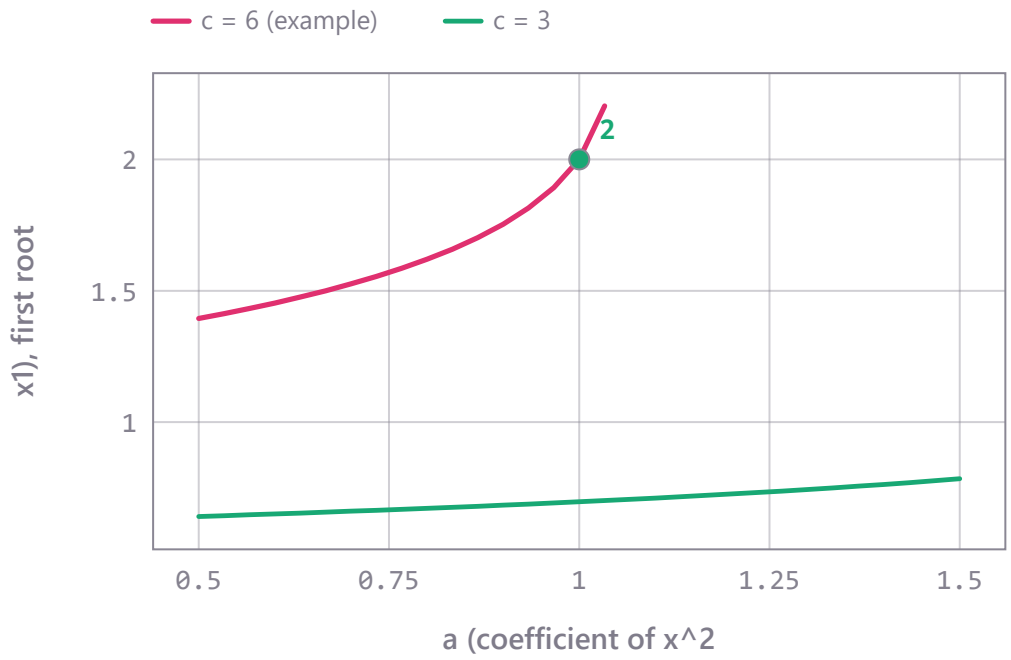
Solve $ax^2 + bx + c = 0$ for x: both roots, real or complex.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

input	meaning	condition
`a`	coefficient of x^2	$a \neq 0$
`b`	coefficient of x	
`c`	constant term	

output	meaning
`x1`	first root (the smaller real root, or the one with positive imaginary part)
`x2`	second root
`discriminant`	$b^2 - 4ac$

- **Method:** quadratic formula in a numerically stable form (q = -(b + sign(b)*sqrt(b^2 - 4ac))/2, $x = q/a$ and c/q)
- **Assumes:** $a \neq 0$ (otherwise the equation is linear); real coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x1 (first root (the smaller real root, or the one with positive imaginary part)) against a (coefficient of x^2) from 0.5 to 1.5. One curve for each value of c (constant term): 6, 3. Other inputs as in the worked example: b = -5. 17 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, x1 = 2.

```
math_foundations_ai999x.solve('quadratic_roots', a=1, b=-5, c=6)
# -> x1 = 2, x2 = 3, discriminant = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_roots', a=2, b=3, c=-2)
# -> x1 = -2, x2 = 0.5, discriminant = 25   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_roots', a=1, b=2, c=1)
# -> x1 = -1, x2 = -1, discriminant = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_roots', a=1, b=0, c=4)
# -> x1 = 0 + 2i, x2 = 0 - 2i, discriminant = -16   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_roots', a=1, b=-100000000.0, c=1)
# -> x1 = 1e-08, x2 = 1e+08, discriminant = 1e+16   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve x^2 - 5x + 6 = 0.	SOLVED	x1 = 2, x2 = 3, discriminant = 1
Find the roots of 2x^2 + 3x - 2 = 0.	SOLVED	x1 = -2, x2 = 0.5, discriminant = 25

question	outcome	result
Solve the quadratic with a = 1, b = -5, c = 6.	SOLVED	x1 = 2, x2 = 3, discriminant = 1
Solve x^2 + 4 = 0.	SOLVED	x1 = 0 + 2i, x2 = 0 - 2i, discriminant = -16
Solve 3x^2 = 12.	SOLVED	x1 = -2, x2 = 2, discriminant = 144
Solve x^5 - 1 = 0.	OUT_OF_SCOPE	Out of scope: a polynomial of degree 5 or higher (equations up to degree 4 are supported).
Solve the quadratic with a = 0, b = 2, c = 1.	INVALID_INPUT	Roots of a quadratic equation: a must not be 0: with a = 0 the equation is linear, not quadratic.
Solve the quadratic with a = 1, b = 2.	NEEDS_INPUT	quadratic_roots needs c (constant term). Example: solve('quadratic_roots', a=1, b=-5, c=6).

Discriminant of a quadratic, `quadratic_discriminant`

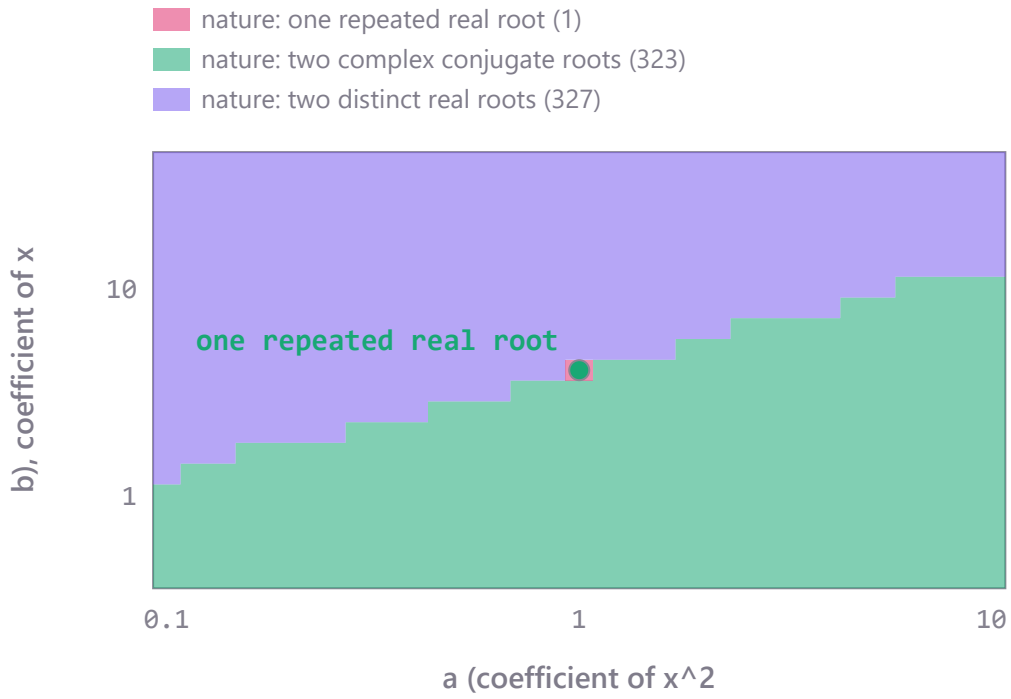
The discriminant $b^2 - 4ac$ of $ax^2 + bx + c$ and the nature of the roots it implies.

$$\Delta = b^2 - 4ac$$

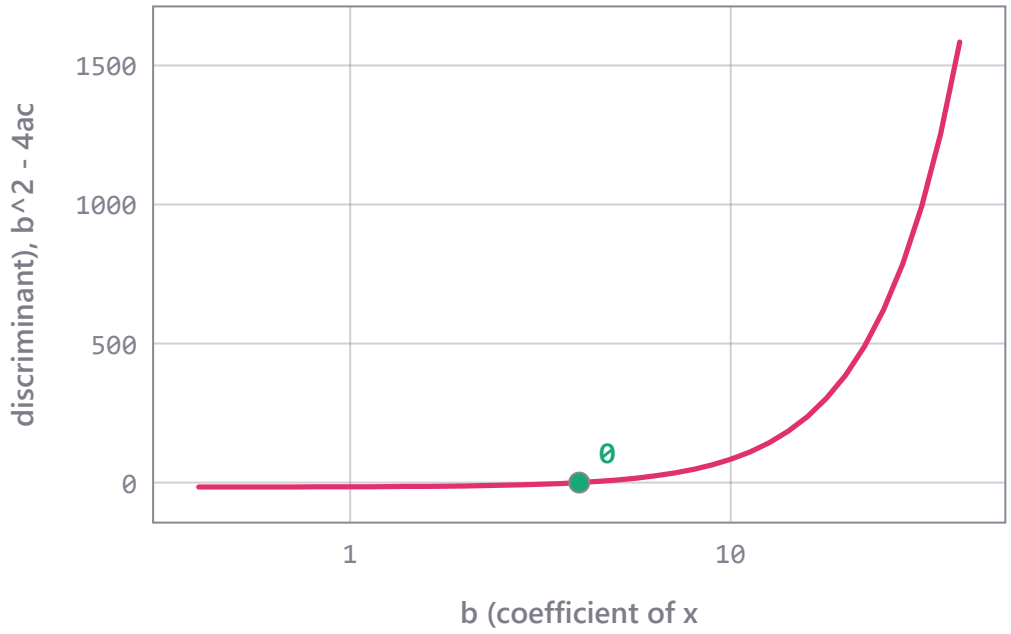
input	meaning	condition
`a`	coefficient of x^2	$a \neq 0$
`b`	coefficient of x	
`c`	constant term	

output	meaning
`discriminant`	$b^2 - 4ac$
`nature`	two distinct real / one repeated real / two complex conjugate roots

- **Method:** direct evaluation of $b^2 - 4ac$ and its sign
- **Assumes:** $a \neq 0$; real coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



two distinct real / one repeated real / two complex conjugate roots (nature) over a (coefficient of x^2) from 0.1 to 10 and b (coefficient of x) from 0.4 to 40 (logarithmic axes): each of the 31 × 21 cells is one run of math-foundations-ai999x, coloured by the verdict it returned, one repeated real root: 1 cells, two complex conjugate roots: 323 cells, two distinct real roots: 327 cells. Other inputs as in the worked example: c = 4. Marked: the worked example, a = 1, b = 4 → one repeated real root.



discriminant (b^2 - 4ac) against b (coefficient of x) from 0.4 to 40 on a logarithmic axis. Other inputs as in the worked example: a = 1, c = 4. 41 points computed by math-foundations-ai999x; marked: the worked example, b = 4, discriminant = 0.

```
math_foundations_ai999x.solve('quadratic_discriminant', a=1, b=4, c=4)
# -> discriminant = 0, nature = one repeated real root   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_discriminant', a=2, b=1, c=-3)
# -> discriminant = 25, nature = two distinct real roots   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_discriminant', a=1, b=1, c=1)
# -> discriminant = -3, nature = two complex conjugate roots   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the discriminant of x^2 + 4x + 4?	SOLVED	discriminant = 0, nature = one repeated real root
Find the discriminant with a = 2, b = 1, c = -3.	SOLVED	discriminant = 25, nature = two distinct real roots

Combinations and permutations, `counting_choices`

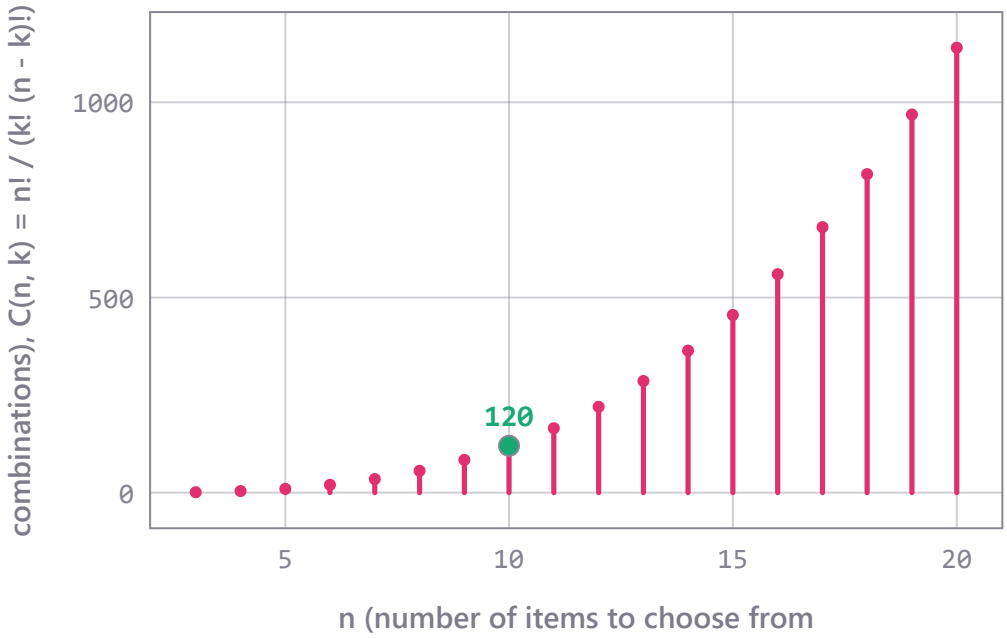
The number of ways to choose k items from n without order (combinations) and with order (permutations).

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}, \quad P(n,k) = \frac{n!}{(n-k)!}$$

input	meaning	condition
`n`	number of items to choose from (a whole number)	
`k`	number of items chosen (a whole number, k <= n)	

output	meaning
`combinations`	C(n, k) = n! / (k! (n - k)!)
`permutations`	P(n, k) = n! / (n - k)!

- **Method:** exact integer arithmetic (math.comb, math.perm)
- **Assumes:** n and k whole numbers; $k \le n$; $n \le 170$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



combinations (C(n, k) = n! / (k! (n - k)!)) against n (number of items to choose from (a whole number)) from 1 to 20. Other inputs as in the worked example: k = 3. 18 points computed by math-foundations-ai999x; marked: the worked example, n = 10, combinations = 120.

```
math_foundations_ai999x.solve('counting_choices', n=10, k=3)
# -> combinations = 120, permutations = 720   (SOLVED)
```

```
math_foundations_ai999x.solve('counting_choices', n=5, k=5)
# -> combinations = 1, permutations = 120   (SOLVED)
```

```
math_foundations_ai999x.solve('counting_choices', n=52, k=5)
# -> combinations = 2.59896e+06, permutations = 3.11875e+08   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How many ways are there to choose 3 books from 10?	SOLVED	combinations = 120, permutations = 720
Compute C(6, 2).	SOLVED	combinations = 15, permutations = 30
How many ways are there to choose 7 books from 4?	INVALID_INPUT	Combinations and permutations: k must not exceed n.
Compute C(1000, 3).	INVALID_INPUT	Combinations and permutations: n must be at most 170 (the result would not fit in a floating-point number).

Roots of a cubic equation, `cubic_roots`

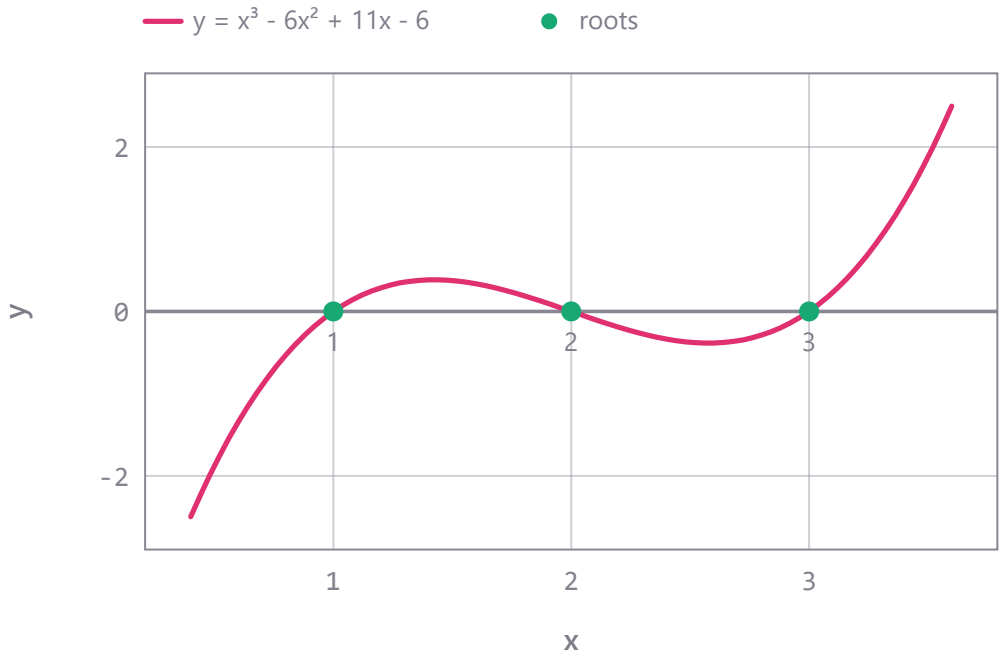
Solve $ax^3 + bx^2 + cx + d = 0$ for x: all three roots, real or complex.

$$ax^3 + bx^2 + cx + d = 0$$

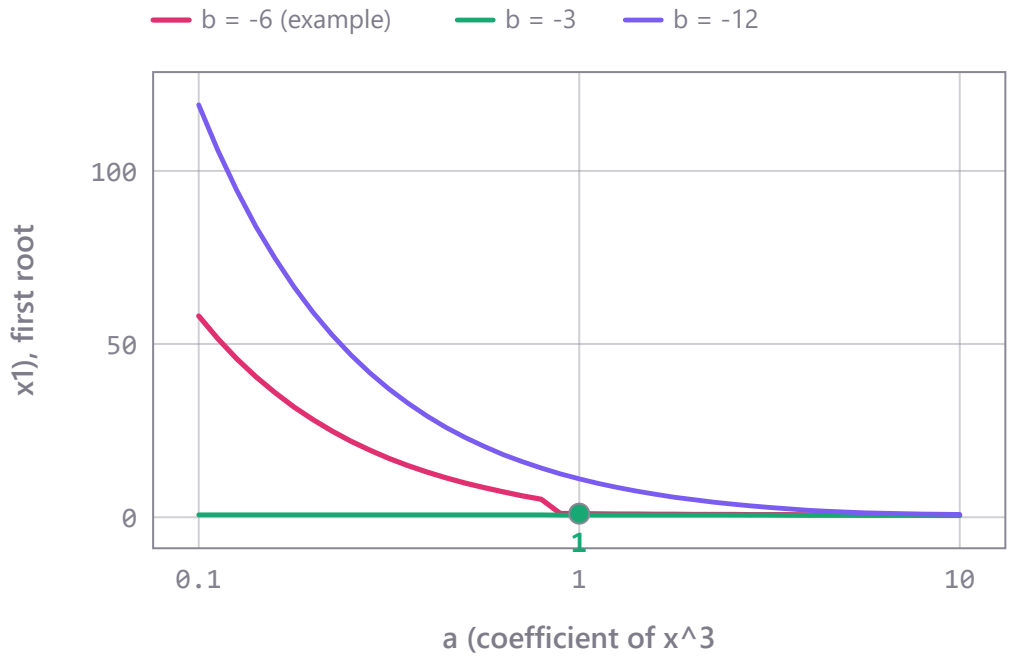
input	meaning	condition
`a`	coefficient of x^3	$a \neq 0$
`b`	coefficient of x^2	
`c`	coefficient of x	
`d`	constant term	

output	meaning
`x1`	first root (real roots ascending, then complex roots)
`x2`	second root
`x3`	third root

- **Method:** high-precision polynomial root finding (Durand-Kerner, 30 digits), each root substituted back
- **Assumes:** $a \neq 0$ (otherwise the equation is not cubic); real coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The cubic y = x³ − 6x² + 11x − 6, evaluated by the package at 81 points, and its three real roots as the package returns them: x = 1, 2 and 3.



x1 (first root (real roots ascending, then complex roots)) against a (coefficient of x^3) from 0.1 to 10 on a logarithmic axis. One curve for each value of b (coefficient of x^2): -6, -3, -12. Other inputs as in the worked example: c = 11, d = -6. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, x1 = 1.

```
math_foundations_ai999x.solve('cubic_roots', a=1, b=-6, c=11, d=-6)
# -> x1 = 1, x2 = 2, x3 = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('cubic_roots', a=2, b=-4, c=-22, d=24)
# -> x1 = -3, x2 = 1, x3 = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('cubic_roots', a=1, b=0, c=0, d=-1)
# -> x1 = 1, x2 = -0.5 + 0.866025i, x3 = -0.5 - 0.866025i   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve x^3 - 6x^2 + 11x - 6 = 0.	SOLVED	x1 = 1, x2 = 2, x3 = 3
Solve x^3 - 1 = 0.	SOLVED	x1 = 1, x2 = -0.5 + 0.866025i, x3 = -0.5 - 0.866025i
Find the roots of 2x^3 - 4x^2 - 22x + 24 = 0.	SOLVED	x1 = -3, x2 = 1, x3 = 4
Solve the cubic with a = 1, b = 0, c = 0, d = -1.	SOLVED	x1 = 1, x2 = -0.5 + 0.866025i, x3 = -0.5 - 0.866025i
Solve the cubic with a = 0, b = 1, c = 2, d = 1.	INVALID_INPUT	Roots of a cubic equation: a must not be 0: with a = 0 the equation has a lower degree.
Solve the cubic with a = 1, b = 2.	NEEDS_INPUT	cubic_roots needs c, d (coefficient of x; constant term). Example: solve('cubic_roots', a=1, b=-6, c=11, d=-6).

Roots of a quartic equation, `quartic_roots`

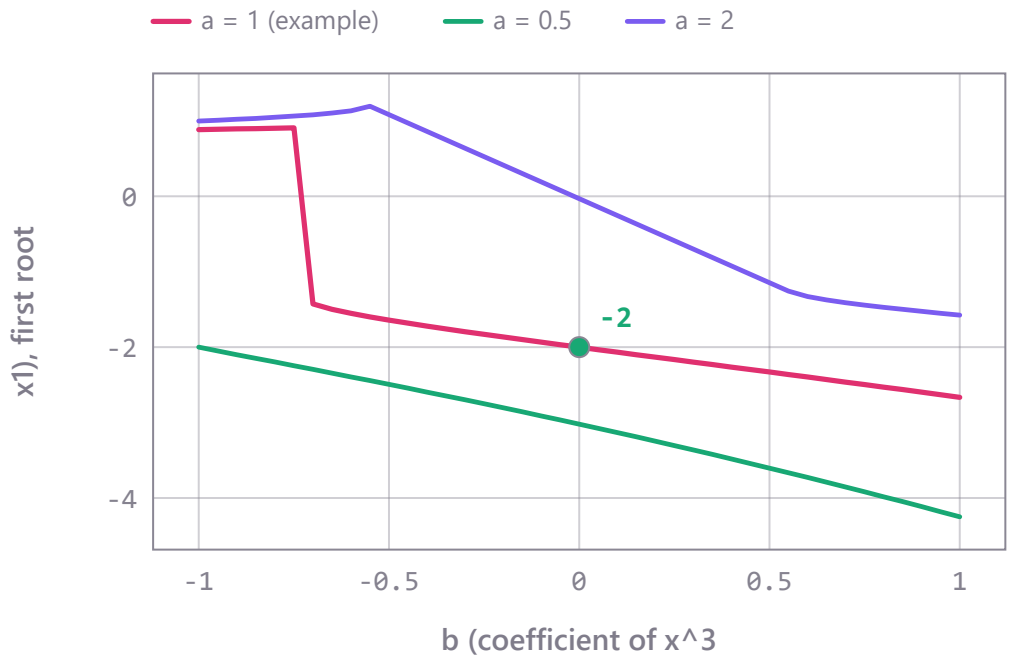
Solve $ax^4 + bx^3 + cx^2 + dx + e = 0$ for x: all four roots, real or complex.

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

input	meaning	condition
a	coefficient of x^4	$a \neq 0$
b	coefficient of x^3	
c	coefficient of x^2	
d	coefficient of x	
e	constant term	

output	meaning
x1	first root (real roots ascending, then complex roots)
x2	second root
x3	third root
x4	fourth root

- **Method:** high-precision polynomial root finding (Durand-Kerner, 30 digits), each root substituted back
- **Assumes:** $a \neq 0$ (otherwise the equation is not quartic); real coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x1 (first root (real roots ascending, then complex roots)) against b (coefficient of x^3) from -1 to 1. One curve for each value of a (coefficient of x^4): 1, 0.5, 2. Other inputs as in the worked example: c = -5, d = 0, e = 4. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 0, x1 = -2.

```
math_foundations_ai999x.solve('quartic_roots', a=1, b=0, c=-5, d=0, e=4)
# -> x1 = -2, x2 = -1, x3 = 1, x4 = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('quartic_roots', a=1, b=0, c=0, d=0, e=-1)
# -> x1 = -1, x2 = 1, x3 = 0 + 1i, x4 = 0 - 1i   (SOLVED)
```

```
math_foundations_ai999x.solve('quartic_roots', a=1, b=-10, c=35, d=-50, e=24)
# -> x1 = 1, x2 = 2, x3 = 3, x4 = 4   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve x^4 - 5x^2 + 4 = 0.	SOLVED	x1 = -2, x2 = -1, x3 = 1, x4 = 2
Find the roots of x^4 - 1 = 0.	SOLVED	x1 = -1, x2 = 1, x3 = 0 + 1i, x4 = 0 - 1i
Solve the quartic with a = 1, b = 0, c = -5, d = 0, e = 4.	SOLVED	x1 = -2, x2 = -1, x3 = 1, x4 = 2
Solve the quartic with a = 1, b = 2.	NEEDS_INPUT	quartic_roots needs c, d, e (coefficient of x^2; coefficient of x; constant term). Example: solve('quartic_roots', a=1,

Arithmetic with fractions, `fraction_arithmetic`

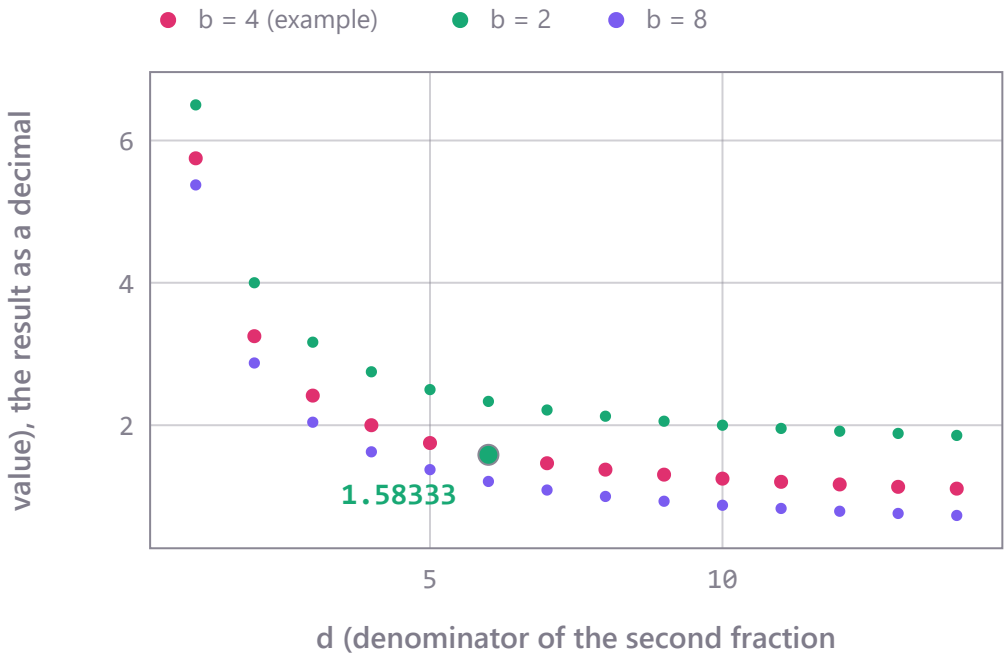
Add, subtract, multiply or divide two fractions a/b and c/d; the result in lowest terms.

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm cb}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \quad \frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$$

input	meaning	condition
`a`	numerator of the first fraction	
`b`	denominator of the first fraction (not 0)	
`c`	numerator of the second fraction	
`d`	denominator of the second fraction (not 0)	
`op`	the operation	

output	meaning
`numerator`	numerator of the result in lowest terms
`denominator`	denominator of the result (positive)
`value`	the result as a decimal

- **Method:** common denominator (a/b ± c/d = (ad ± cb)/bd), product of numerators over product of denominators, division as multiplication by the reciprocal; then divide by the greatest common divisor
- **Assumes:** whole-number numerators and denominators; denominators not 0; no division by 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



value (the result as a decimal) against d (denominator of the second fraction (not 0)) from 1 to 14. One curve for each value of b (denominator of the first fraction (not 0)): 4, 2, 8. Other inputs as in the worked example: a = 3, c = 5, op = +. 14 points per curve computed by math-foundations-ai999x; marked: the worked example, d = 6, value = 1.58333.

```
math_foundations_ai999x.solve('fraction_arithmetic', a=3, b=4, c=5, d=6, op='+')
# -> numerator = 19, denominator = 12, value = 1.58333   (SOLVED)
```

```
math_foundations_ai999x.solve('fraction_arithmetic', a=2, b=3, c=9, d=10, op='*')
# -> numerator = 3, denominator = 5, value = 0.6   (SOLVED)
```

```
math_foundations_ai999x.solve('fraction_arithmetic', a=1, b=2, c=3, d=4, op='-')
# -> numerator = -1, denominator = 4, value = -0.25   (SOLVED)
```

```
math_foundations_ai999x.solve('fraction_arithmetic', a=3, b=4, c=9, d=8, op='/')
# -> numerator = 2, denominator = 3, value = 0.666667   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Compute 3/4 + 5/6 as a fraction in lowest terms.	SOLVED	numerator = 19, denominator = 12, value = 1.58333
What is 2/3 times 9/10?	SOLVED	numerator = 3, denominator = 5, value = 0.6
Calculate 7/8 - 1/6.	SOLVED	numerator = 17, denominator = 24, value = 0.708333
Find 5/6 divided by 0/3.	INVALID_INPUT	Arithmetic with fractions: division by the fraction 0.
Compute 1/0 + 1/2.	INVALID_INPUT	Arithmetic with fractions: a denominator must not be 0.

Percent of a number, `percent_of_number`

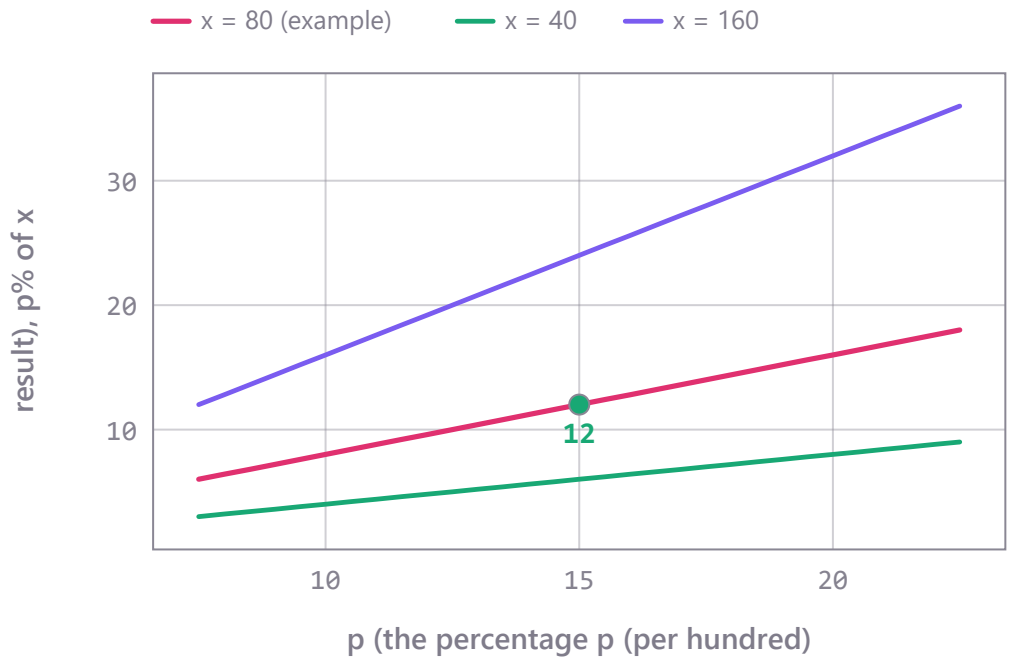
p% of x: the part that p per cent of x amounts to.

$$\frac{p}{100}\,x$$

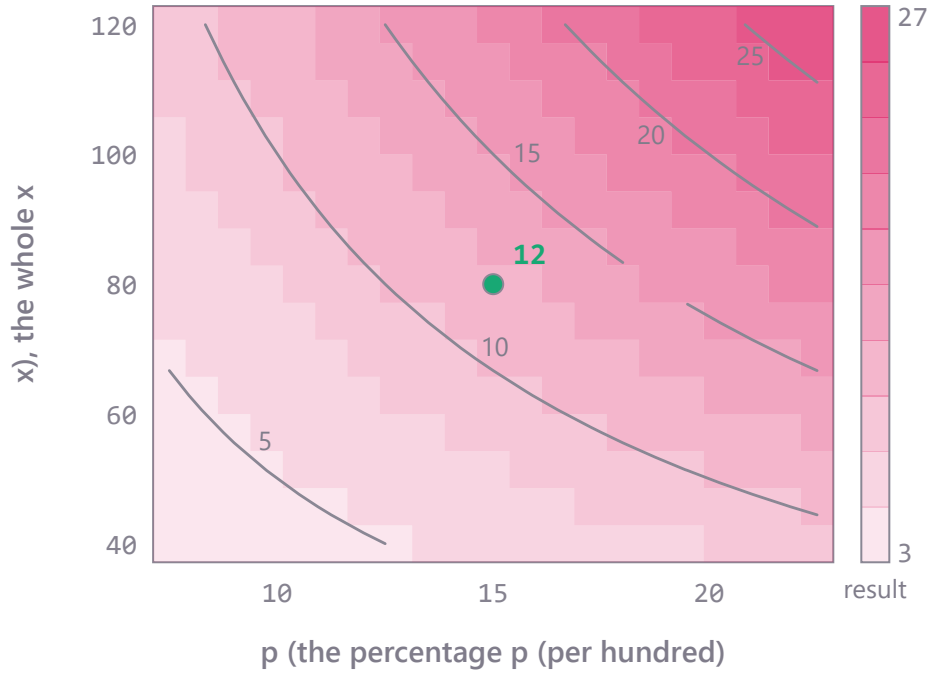
input	meaning	condition
`p`	the percentage p (per hundred)	
`x`	the whole x	

output	meaning
`result`	p% of x

- **Method:** per cent means per hundred: p% of x = $(p/100)$·x
- **Assumes:** finite numbers
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



result (p% of x) against p (the percentage p (per hundred)) from 7.5 to 22.5. One curve for each value of x (the whole x): 80, 40, 160. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, p = 15, result = 12.



result (p% of x) over p (the percentage p (per hundred)) from 7.5 to 22.5 and x (the whole x) from 40 to 120: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 3 to dark 27); iso-lines at result = 5, 10, 15, 20, 25. Marked: the worked example, p = 15, x = 80, result = 12.

```
math_foundations_ai999x.solve('percent_of_number', p=15, x=80)
# -> result = 12   (SOLVED)
```

```
math_foundations_ai999x.solve('percent_of_number', p=12.5, x=240)
# -> result = 30   (SOLVED)
```

```
math_foundations_ai999x.solve('percent_of_number', p=150, x=40)
# -> result = 60   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is 15% of 80?	SOLVED	result = 12
Find 12.5 percent of 240.	SOLVED	result = 30
Calculate 150% of 40.	SOLVED	result = 60

Whole from a part and its percentage, `percent_of_what`

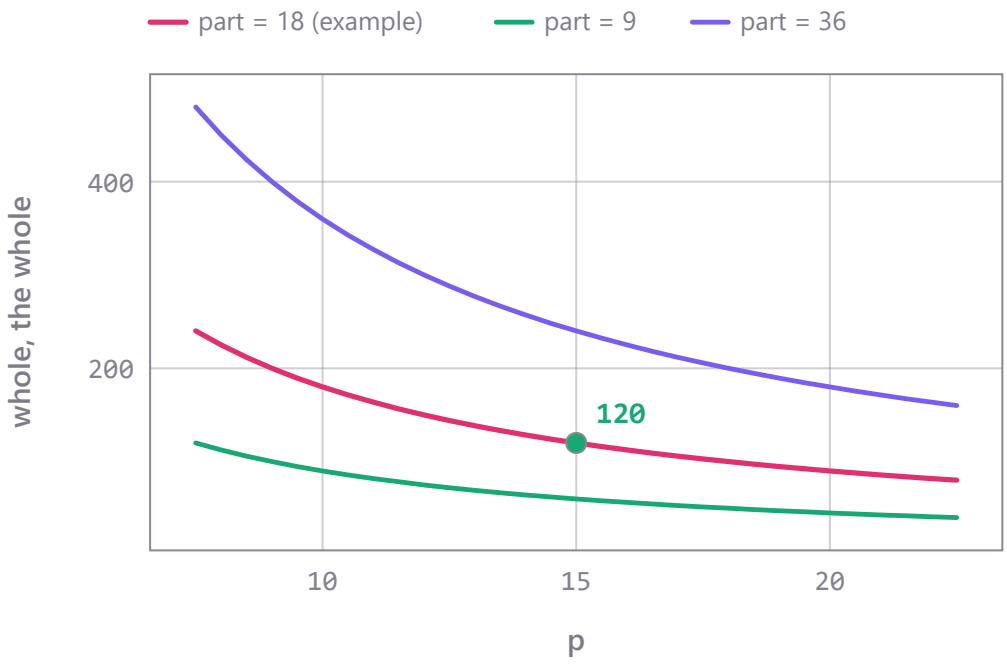
The whole w of which `part` is p per cent (part is p% of what number?).

$$w = \frac{100\,\mathrm{part}}{p}$$

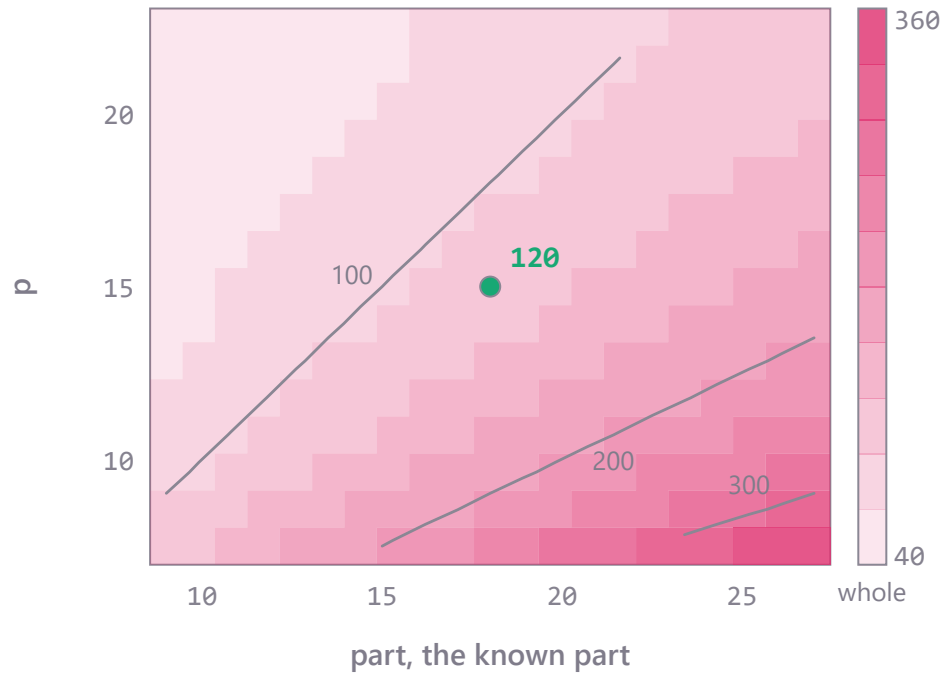
input	meaning	condition
`part`	the known part	
`p`	the percentage that the part is of the whole	$p \neq 0$

output	meaning
`whole`	the whole

- **Method:** part = (p/100)·whole, so whole = 100·part/p
- **Assumes:** $p \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



whole (the whole) against p (the percentage that the part is of the whole) from 7.5 to 22.5. One curve for each value of part (the known part): 18, 9, 36. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, p = 15, whole = 120.



whole (the whole) over part (the known part) from 9 to 27 and p (the percentage that the part is of the whole) from 7.5 to 22.5: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 40 to dark 360); iso-lines at whole = 100, 200, 300. Marked: the worked example, part = 18, p = 15, whole = 120.

```
math_foundations_ai999x.solve('percent_of_what', part=18, p=15)
# -> whole = 120   (SOLVED)
```

```
math_foundations_ai999x.solve('percent_of_what', part=26, p=40)
# -> whole = 65   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
18 is 15% of what number?	SOLVED	whole = 120
40% of what number is 26?	SOLVED	whole = 65
5 is 0% of what number?	INVALID_INPUT	Whole from a part and its percentage: p must not be 0 (0% of every number is 0).

What percent one number is of another, `what_percent`

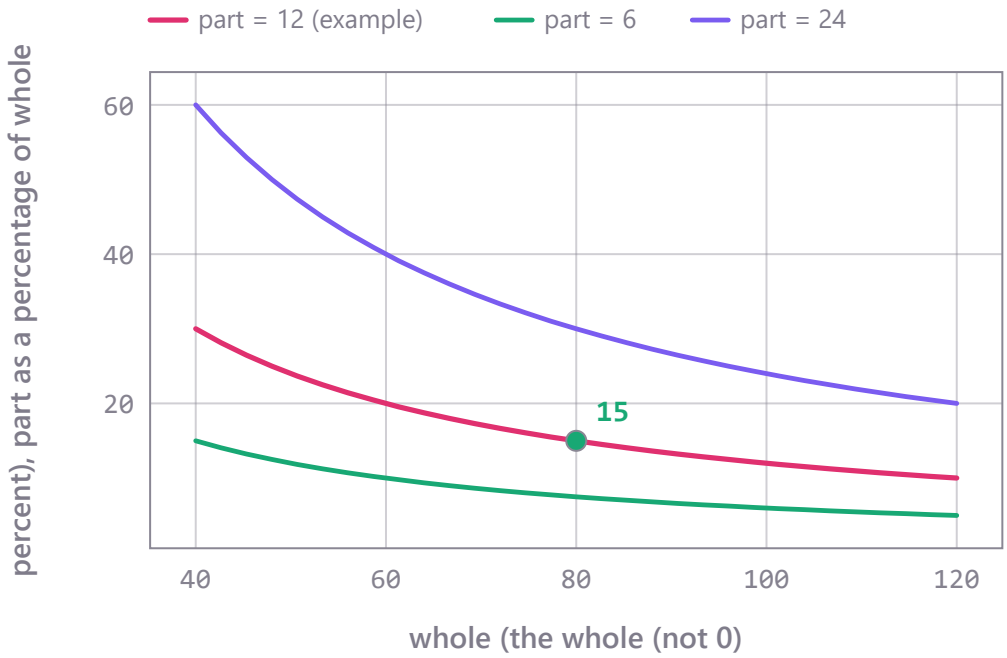
The percentage p such that `part` is p% of `whole`.

$$p = 100\,\frac{\text{part}}{\text{whole}}$$

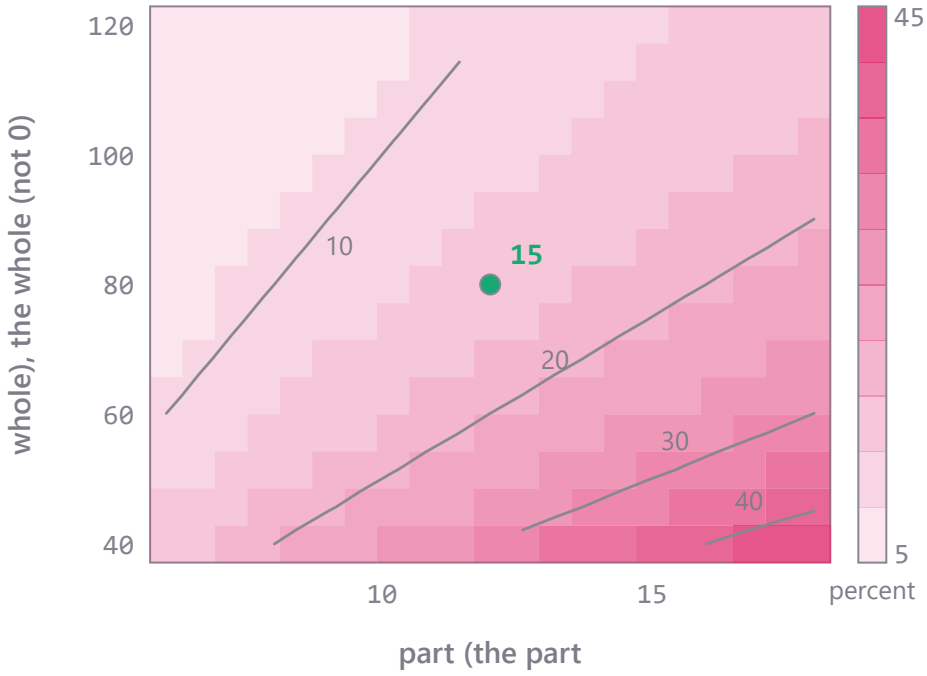
input	meaning	condition
`part`	the part	
`whole`	the whole (not 0)	whole != 0

output	meaning
`percent`	part as a percentage of whole

- **Method:** $p = 100 \cdot \text{part}/\text{whole}$
- **Assumes:** whole != 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



percent (part as a percentage of whole) against whole (the whole (not 0)) from 40 to 120. One curve for each value of part (the part): 12, 6, 24. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, whole = 80, percent = 15.



percent (part as a percentage of whole) over part (the part) from 6 to 18 and whole (the whole (not 0)) from 40 to 120: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 5 to dark 45); iso-lines at percent = 10, 20, 30, 40. Marked: the worked example, part = 12, whole = 80, percent = 15.

```
math_foundations_ai999x.solve('what_percent', part=12, whole=80)
# -> percent = 15   (SOLVED)
```

```
math_foundations_ai999x.solve('what_percent', part=45, whole=60)
# -> percent = 75   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What percent of 80 is 12?	SOLVED	percent = 15
45 is what percent of 60?	SOLVED	percent = 75
What percent of 0 is 5?	INVALID_INPUT	What percent one number is of another: the whole must not be 0.

Percent change (increase or decrease), `percent_change`

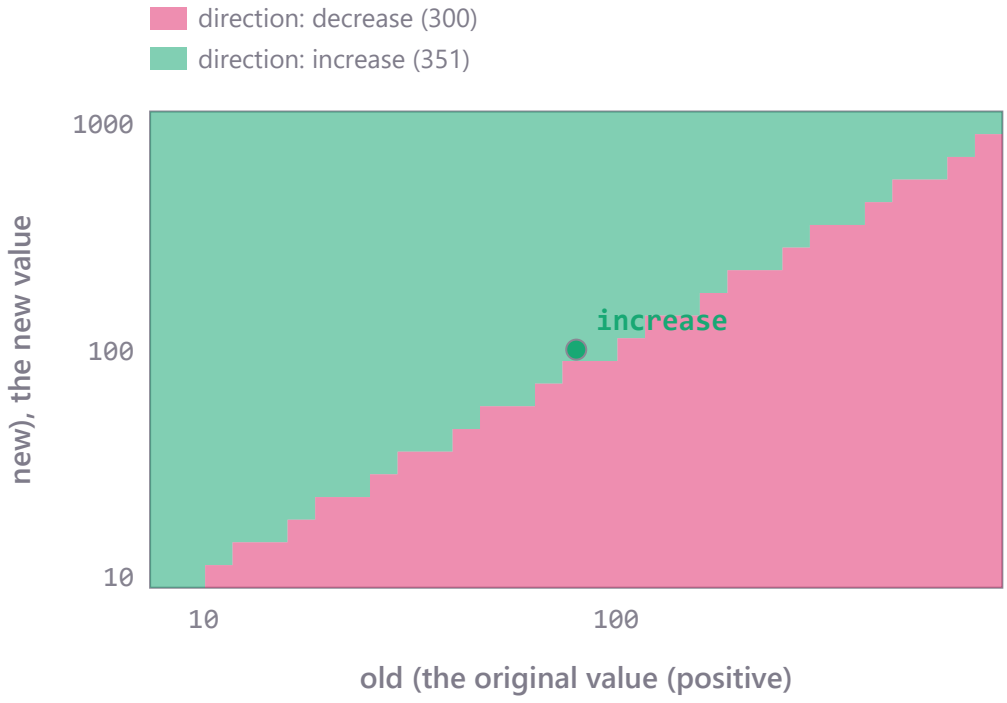
The percent change from an original value to a new value (negative for a decrease).

$$\Delta\% = \frac{\text{new} - \text{old}}{\text{old}} \cdot 100$$

input	meaning	condition
`old`	the original value (positive)	
`new`	the new value	

output	meaning
`percent_change`	percent change (negative = decrease)
`difference`	new - old
`direction`	increase, decrease or no change

- **Method:** percent change = (new - old)/old · 100
- **Assumes:** old > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



increase, decrease or no change (direction) over old (the original value (positive)) from 8 to 800 and new (the new value) from 10 to 1000 (logarithmic axes): each of the 31 × 21 cells is one run of math-foundations-ai999x, coloured by the verdict it returned, decrease: 300 cells, increase: 351 cells. Marked: the worked example, old = 80, new = 100 → increase.



percent_change (percent change (negative = decrease)) against old (the original value (positive)) from 8 to 800 on a logarithmic axis. One curve for each value of new (the new value): 100, 50, 200. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, old = 80, percent_change = 25.

```
math_foundations_ai999x.solve('percent_change', old=80, new=100)
# -> percent_change = 25, difference = 20, direction = increase   (SOLVED)
```

```
math_foundations_ai999x.solve('percent_change', old=50, new=40)
# -> percent_change = -20, difference = -10, direction = decrease   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the percent change from 80 to 100.	SOLVED	percent_change = 25, difference = 20, direction = increase
What is the percent decrease from 50 to 40?	SOLVED	percent_change = -20, difference = -10, direction = decrease
A price rises from 40 to 50. What is the percent increase?	SOLVED	percent_change = 25, difference = 10, direction = increase
Find the percent change from 0 to 10.	INVALID_INPUT	Percent change (increase or decrease): the original value must be positive (a percent change is measured from it).

Divide a quantity in a given ratio, `ratio_split`

Share a total in the ratio a:b (or a:b:c).

$$\text{part}_i = \text{total} \cdot \frac{r_i}{\sum_j r_j}$$

input	meaning	condition
`total`	the quantity to share	

input	meaning	condition
`a`	first term of the ratio (positive)	
`b`	second term of the ratio (positive)	
`c`	third term of the ratio (optional, positive)	

output	meaning
`part1`	first share
`part2`	second share
`part3`	third share (a three-term ratio only)

- **Method:** each share = total · term / (sum of the terms)
- **Assumes:** every term of the ratio positive
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



part1 (first share) against a (first term of the ratio (positive)) from 0.3 to 30 on a logarithmic axis. One curve for each value of total (the quantity to share): 120, 60, 240. Other inputs as in the worked example: b = 5. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 3, part1 = 45.

```
math_foundations_ai999x.solve('ratio_split', total=120, a=3, b=5)
# -> part1 = 45, part2 = 75   (SOLVED)
```

```
math_foundations_ai999x.solve('ratio_split', total=900, a=2, b=3, c=4)
# -> part1 = 200, part2 = 300, part3 = 400   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Divide 120 in the ratio 3:5.	SOLVED	part1 = 45, part2 = 75
Share 900 in the ratio 2:3:4.	SOLVED	part1 = 200, part2 = 300, part3 = 400
Split 50 in the ratio 0:5.	INVALID_INPUT	Divide a quantity in a given ratio: every term of the ratio must be positive.

Solve a proportion, `proportion_solve`

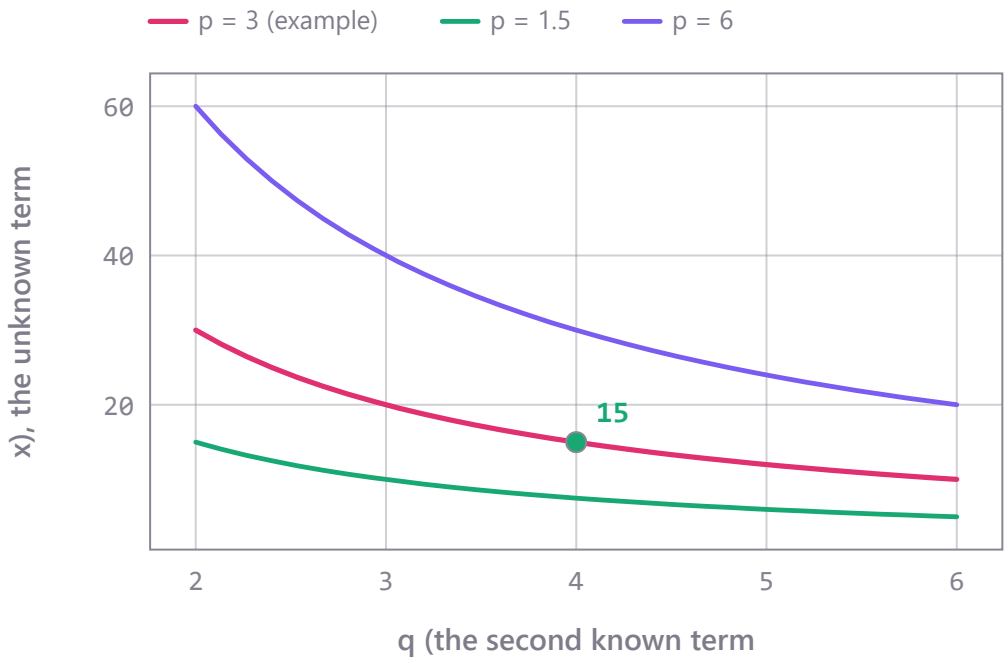
Solve $n_1/d_1 = n_2/d_2$ for the one unknown term x (in any of the four positions).

$$\frac{n_1}{d_1} = \frac{n_2}{d_2} \iff n_1 d_2 = n_2 d_1$$

input	meaning	condition
`p`	the first known term (reading n1, d1, n2, d2 in order)	
`q`	the second known term	
`r`	the third known term	
`position`	where the unknown x stands	

output	meaning
`x`	the unknown term

- **Method:** cross-multiplication: n1·$d_2 = n_2$·d1, then divide
- **Assumes:** denominators not 0; the proportion has a solution
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x (the unknown term) against q (the second known term) from 2 to 6. One curve for each value of p (the first known term (reading n1, d1, n2, d2 in order)): 3, 1.5, 6. Other inputs as in the worked example: r = 20, position = second numerator. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, q = 4, x = 15.

```
math_foundations_ai999x.solve('proportion_solve', p=3, q=4, r=20, position='second numerator')
# -> x = 15   (SOLVED)
```

```
math_foundations_ai999x.solve('proportion_solve', p=6, q=10, r=15, position='first numerator')
# -> x = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('proportion_solve', p=5, q=2, r=7, position='first denominator')
# -> x = 17.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve 3/4 = x/20.	SOLVED	x = 15
Solve the proportion x/6 = 10/15.	SOLVED	x = 4
Solve 5/x = 2/7.	SOLVED	x = 17.5
Solve 3/0 = x/5.	INVALID_INPUT	Solve a proportion: a denominator must not be 0.

Unit rates and the better buy, `unit_rate_compare`

The unit rate (amount per one unit) of an offer, and of a second offer with which is cheaper per unit.

$$\text{unit rate} = \frac{\text{amount}}{\text{quantity}}$$

input	meaning	condition
`n1`	quantity of the first offer (positive)	
`c1`	amount (price, distance, ...) of the first offer	
`n2`	quantity of the second offer (optional)	
`c2`	amount of the second offer (optional)	

output	meaning
`unit_rate_1`	amount per unit, first offer
`unit_rate_2`	amount per unit, second offer
`better`	the offer with the lower amount per unit: first, second or equal

- **Method:** unit rate = amount / quantity; the lower unit price is the better buy
- **Assumes:** quantities positive; amounts not negative
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



the offer with the lower amount per unit: first, second or equal (better) over n1 (quantity of the first offer (positive)) from 1.2 to 120 and c1 (amount (price, distance, ...) of the first offer) from 3 to 300 (logarithmic axes): each of the 31 × 21 cells is one run of math-foundations-ai999x, coloured by the verdict it returned, first: 310 cells, second: 341 cells. Other inputs as in the worked example: n2 = 20, c2 = 45. Marked: the worked example, n1 = 12, c1 = 30 → second.



unit_rate_1 (amount per unit, first offer) against n1 (quantity of the first offer (positive)) from 6 to 18. One curve for each value of c1 (amount (price, distance, ...) of the first offer): 30, 15, 60. Other inputs as in the worked example: n2 = 20, c2 = 45. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, n1 = 12, unit_rate_1 = 2.5.

```
math_foundations_ai999x.solve('unit_rate_compare', n1=12, c1=30, n2=20, c2=45)
# -> unit_rate_1 = 2.5, unit_rate_2 = 2.25, better = second   (SOLVED)
```

```
math_foundations_ai999x.solve('unit_rate_compare', n1=6, c1=4.5)
# -> unit_rate_1 = 0.75   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Which is the better buy: 12 items for 30 or 20 items for 45?	SOLVED	unit_rate_1 = 2.5, unit_rate_2 = 2.25, better = second
Find the unit price of 6 cans for 4.5.	SOLVED	unit_rate_1 = 0.75
Which is the better deal: 3 kg for $7.50 or 5 kg for 12$?	SOLVED	unit_rate_1 = 2.5, unit_rate_2 = 2.4, better = second
Find the unit price of 0 items for 5.	INVALID_INPUT	Unit rates and the better buy: quantities must be positive and amounts not negative.

Round to decimal places, `round_decimal_places`

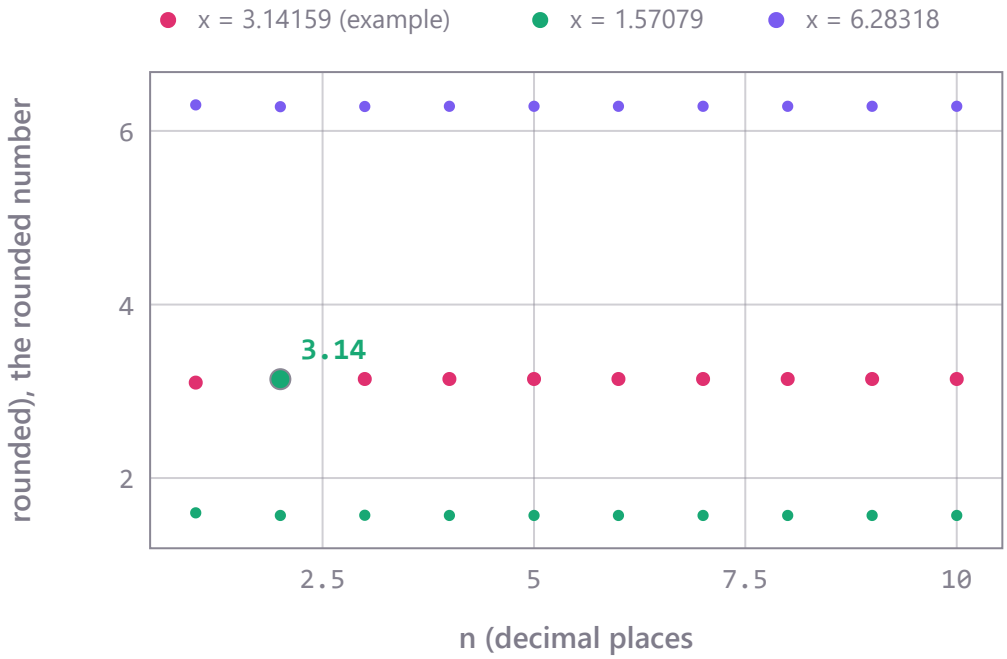
Round a number to n decimal places (or to the nearest tenth, hundredth, ten, hundred ...), halves away from zero.

$$\operatorname{round}(x, n) = \operatorname{sgn}(x)\,\frac{\left\lfloor |x|\,10^{n} + \tfrac{1}{2} \right\rfloor}{10^{n}}$$

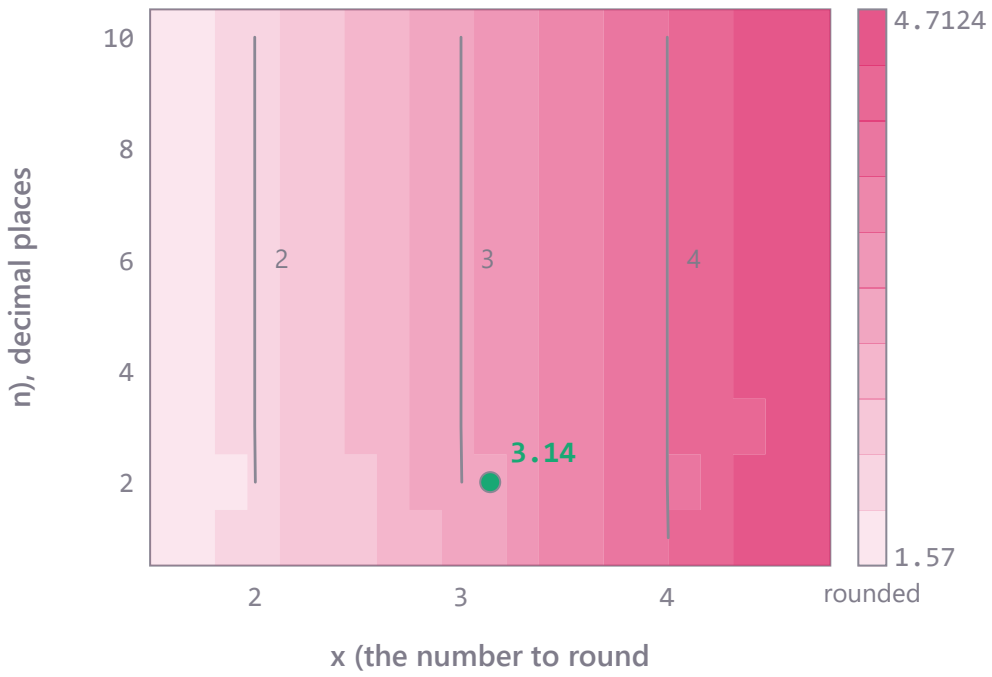
input	meaning	condition
`x`	the number to round	
`n`	decimal places (negative: tens, hundreds)	

output	meaning
`rounded`	the rounded number

- **Method:** exact decimal arithmetic on the number as written; a 5 in the first dropped place rounds away from zero
- **Assumes:** $-9 \le n \le 12$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



rounded (the rounded number) against n (decimal places (negative: tens, hundreds)) from 1 to 10. One curve for each value of x (the number to round): 3.14159, 1.57079, 6.28318. 10 points per curve computed by math-foundations-ai999x; marked: the worked example, n = 2, rounded = 3.14.



rounded (the rounded number) over x (the number to round) from 1.57079 to 4.71239 and n (decimal places (negative: tens, hundreds)) from 1 to 10: each of the 21 × 10 cells is one run of math-foundations-ai999x, shaded by its value (light 1.57 to dark 4.7124); iso-lines at rounded = 2, 3, 4. Marked: the worked example, x = 3.14159, n = 2, rounded = 3.14.

```
math_foundations_ai999x.solve('round_decimal_places', x=3.14159, n=2)
# -> rounded = 3.14   (SOLVED)
```

```
math_foundations_ai999x.solve('round_decimal_places', x=2.675, n=2)
# -> rounded = 2.68   (SOLVED)
```

```
math_foundations_ai999x.solve('round_decimal_places', x=1234.5, n=-2)
# -> rounded = 1200   (SOLVED)
```

```
math_foundations_ai999x.solve('round_decimal_places', x=-0.5, n=0)
# -> rounded = -1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Round 3.14159 to 2 decimal places.	SOLVED	rounded = 3.14
Round 2.675 to the nearest hundredth.	SOLVED	rounded = 2.68
Round 1234.5 to the nearest hundred.	SOLVED	rounded = 1200
Round 7.45 to the nearest whole number.	SOLVED	rounded = 7

Round to significant figures, `round_significant_figures`

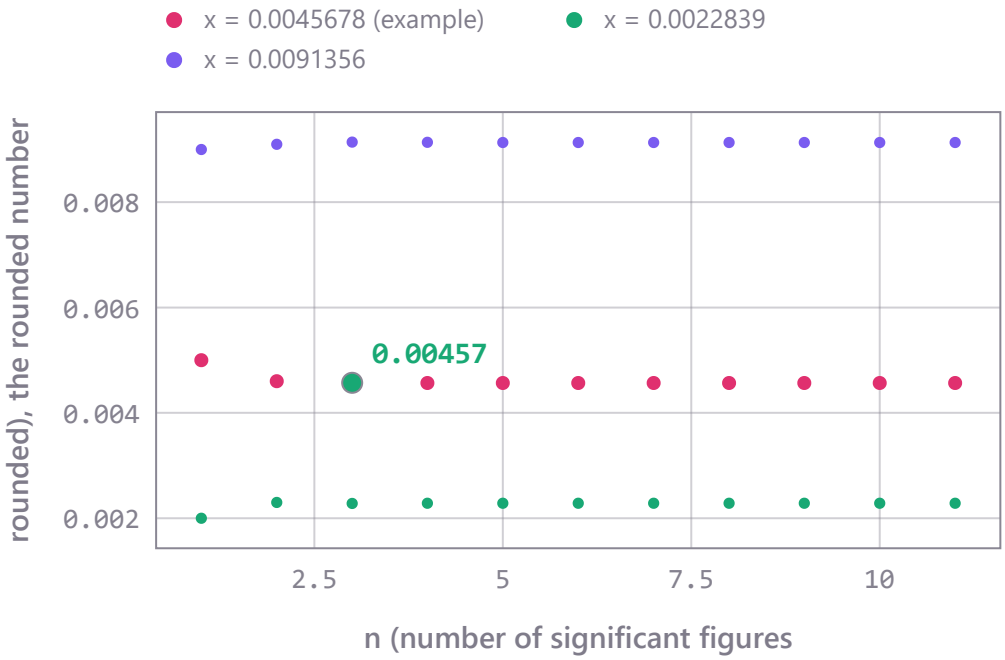
Round a nonzero number to n significant figures, halves away from zero.

$$\mathrm{round}\big(x,\ n - 1 - \lfloor \log_{10} |x| \rfloor\big)$$

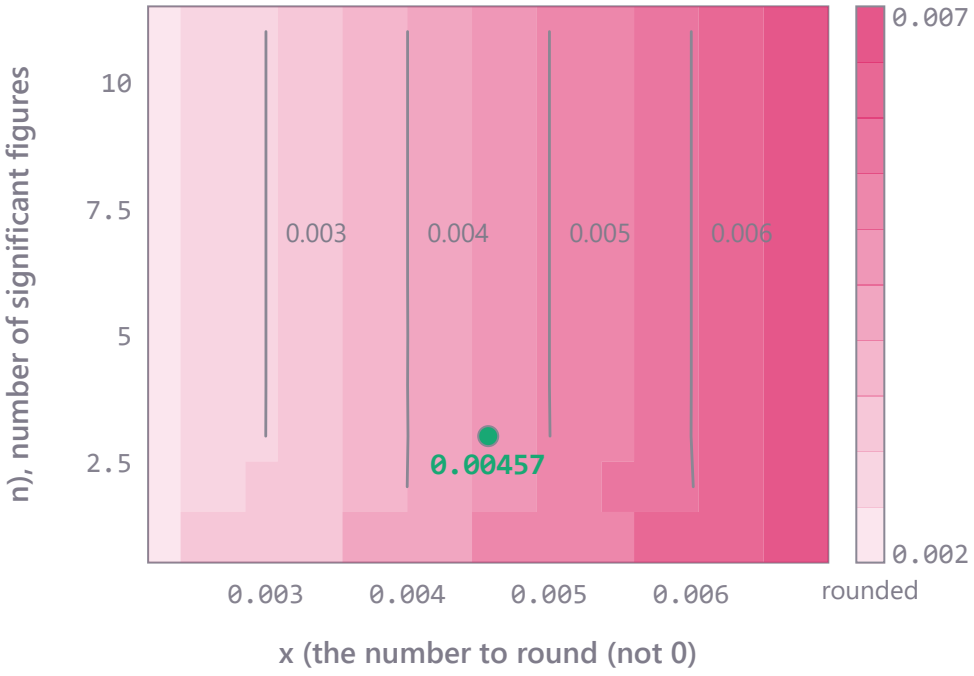
input	meaning	condition
`x`	the number to round (not 0)	$x \neq 0$
`n`	number of significant figures	

output	meaning
`rounded`	the rounded number

- **Method:** find the exponent e of the leading digit (10^e <= |x| < 10^{e+1}), round to 10^{e-n+1} exactly
- **Assumes:** $x \neq 0$; $1 \leq n \leq 15$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



rounded (the rounded number) against n (number of significant figures) from 1 to 11. One curve for each value of x (the number to round (not 0)): 0.0045678, 0.0022839, 0.0091356. 11 points per curve computed by math-foundations-ai999x; marked: the worked example, n = 3, rounded = 0.00457.



rounded (the rounded number) over x (the number to round (not 0)) from 0.0022839 to 0.0068517 and n (number of significant figures) from 1 to 11: each of the 21 × 11 cells is one run of math-foundations-ai999x, shaded by its value (light 0.002 to dark 0.007); iso-lines at rounded = 0.003, 0.004, 0.005, 0.006. Marked: the worked example, x = 0.0045678, n = 3, rounded = 0.00457.

```
math_foundations_ai999x.solve('round_significant_figures', x=0.0045678, n=3)
# -> rounded = 0.00457   (SOLVED)
```

```
math_foundations_ai999x.solve('round_significant_figures', x=45678, n=2)
# -> rounded = 46000   (SOLVED)
```

```
math_foundations_ai999x.solve('round_significant_figures', x=9.996, n=3)
# -> rounded = 10   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Round 0.0045678 to 3 significant figures.	SOLVED	rounded = 0.00457
Write 45678 to 2 significant figures.	SOLVED	rounded = 46000
Round 0 to 2 significant figures.	INVALID_INPUT	Round to significant figures: 0 has no significant figures to round to.

Scientific notation, `scientific_notation`

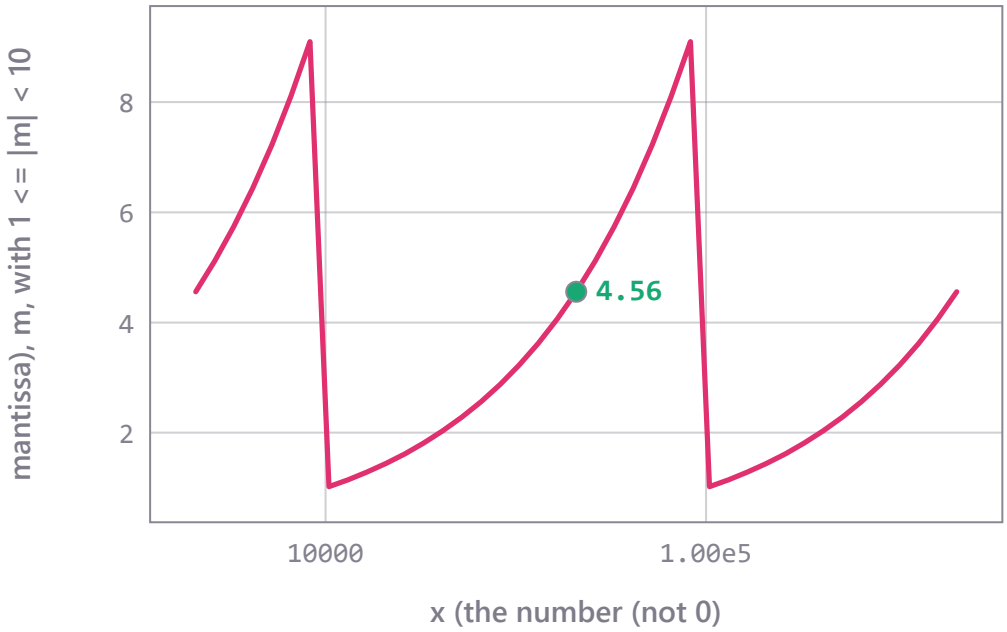
Write a nonzero number as m × 10^e with 1 <= |m| < 10.

$$x = m \times 10^e, \quad 1 \le |m| < 10$$

input	meaning	condition
`x`	the number (not 0)	$x \neq 0$

output	meaning
`mantissa`	m, with 1 <= |m| < 10
`exponent`	the integer power of ten e

- **Method:** exact decimal: e = position of the leading digit, $m = x/10^e$
- **Assumes:** $x \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



mantissa (m, with 1 <= |m| < 10) against x (the number (not 0)) from 4560 to 456000 on a logarithmic axis. 41 points computed by math-foundations-ai999x; marked: the worked example, x = 45600, mantissa = 4.56.

```
math_foundations_ai999x.solve('scientific_notation', x=45600)
# -> mantissa = 4.56, exponent = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('scientific_notation', x=0.00072)
# -> mantissa = 7.2, exponent = -4   (SOLVED)
```

```
math_foundations_ai999x.solve('scientific_notation', x=-3.0)
# -> mantissa = -3, exponent = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Write 45600 in scientific notation.	SOLVED	mantissa = 4.56, exponent = 4
Express 0.00072 in scientific notation.	SOLVED	mantissa = 7.2, exponent = -4
Write 0 in scientific notation.	INVALID_INPUT	Scientific notation: 0 has no scientific notation with a leading digit from 1 to 9.

Evaluate an arithmetic expression (order of operations), `order_of_operations`

The exact value of an arithmetic expression with + - * / ^ and parentheses, following the order of operations.

$$(\ ) \ \rightarrow \ a^b \ \rightarrow \ \times \div \ \rightarrow \ + -$$

input	meaning	condition
`expression`	the expression, e.g. $3 + 4(2 - 5)^2/6$	

output	meaning
`value`	the value
`numerator`	numerator of the exact value in lowest terms
`denominator`	denominator of the exact value

- **Method:** parentheses first, then powers (right to left), then multiplication and division, then addition and subtraction (left to right); exact rational arithmetic
- **Assumes:** numbers, + - * / ^ and parentheses only; no division by 0; integer exponents up to 64
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The worked example of Evaluate an arithmetic expression (order of operations) as inputs and results: stated expression = 3 + 4 * (2 - 5)^2 / 6; returned by math-foundations-ai999x: value = 9, numerator = 9, denominator = 1.

```
math_foundations_ai999x.solve('order_of_operations', expression='3 + 4 * (2 - 5)^2 / 6')
# -> value = 9, numerator = 9, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('order_of_operations', expression='8 - 2 * 3^2 + 10 / 5')
# -> value = -8, numerator = -8, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('order_of_operations', expression='2^3^2')
# -> value = 512, numerator = 512, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('order_of_operations', expression='1/3 + 1/6')
# -> value = 0.5, numerator = 1, denominator = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Evaluate 3 + 4 * (2 - 5)^2 / 6.	SOLVED	value = 9, numerator = 9, denominator = 1
Use the order of operations to evaluate 8 - 2 * 3^2 + 10 / 5.	SOLVED	value = -8, numerator = -8, denominator = 1
Evaluate -2^2 + 3 * (4 - 1).	SOLVED	value = 5, numerator = 5, denominator = 1
Evaluate 6 / (3 - 3) + 1.	INVALID_INPUT	Evaluate an arithmetic expression (order of operations): division by zero.

System of three linear equations, `linear_system_3x3`

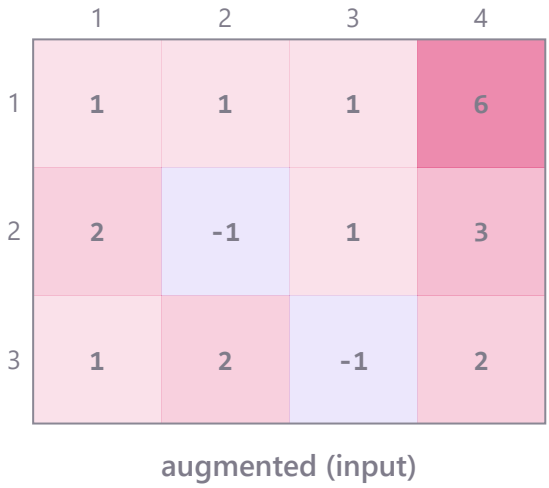
Solve three linear equations in x, y and z (a unique solution, or a refusal when there is none).

$$A\,\mathbf{x} = \mathbf{d}, \quad \mathbf{x} = A^{-1}\mathbf{d}$$

input	meaning	condition
`augmented`	augmented matrix, one row a b c d per equation $ax + by + cz = d$	

output	meaning
`x`	x
`y`	y
`z`	z

- **Method:** Gaussian elimination in exact rational arithmetic
- **Assumes:** determinant of the coefficients != 0 (a unique solution)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



augmented (input)

The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): augmented (input). Returned with them by math-foundations-ai999x: x = 1, y = 2, z = 3.

```
math_foundations_ai999x.solve('linear_system_3x3', augmented='1 1 1 6; 2 -1 1 3; 1 2 -1 2')
# -> x = 1, y = 2, z = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('linear_system_3x3', augmented=[[2, 1, -1, 8], [-3, -1, 2, -11], [-2,
1, 2, -3]])
# -> x = 2, y = 3, z = -1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the system x + y + z = 6, 2x - y + z = 3, x + 2y - z = 2.	SOLVED	x = 1, y = 2, z = 3
Solve the system 2x + y - z = 8; -3x - y + 2z = -11; -2x + y + 2z = -3.	SOLVED	x = 2, y = 3, z = -1
Solve the system x + y + z = 1, 2x + 2y + 2z = 2, x - y = 0.	INVALID_INPUT	System of three linear equations: the system has no unique solution (its determinant is 0).

Absolute value inequality |ax + b| compared with c, `abs_linear_inequality`

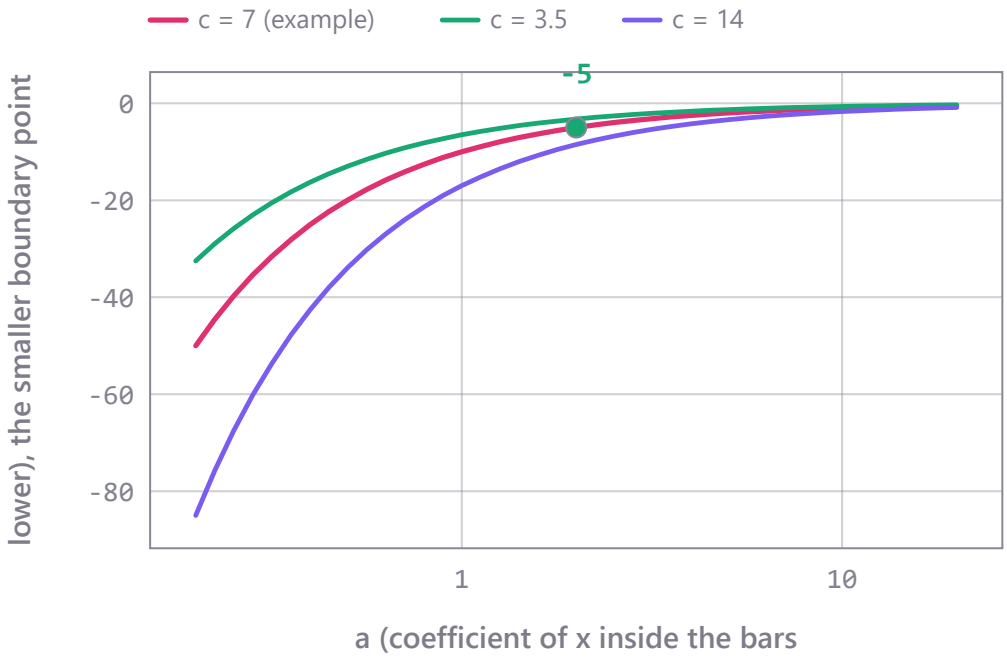
Solve $|ax + b|$ < c, <= c, > c $or \geq c\,(c > 0)$: the two boundary points and whether x lies between or outside them.

$$|ax+b| < c \iff -c < ax+b < c, \qquad |ax+b| > c \iff ax+b < -c \text{ or } ax+b > c$$

input	meaning	condition
`a`	coefficient of x inside the bars	$a \neq 0$
`b`	constant inside the bars	
`c`	the bound (positive)	
`rel`	the comparison	

output	meaning
`lower`	the smaller boundary point
`upper`	the larger boundary point
`region`	between (for < and <=) or outside (for > and >=)
`solution`	the solution set

- **Method:** |u| < c means -c < u < c; |u| > c means $u < -c$ or $u > c$; solve each linear piece for x
- **Assumes:** $a \neq 0$; $c > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



lower (the smaller boundary point) against a (coefficient of x inside the bars) from 0.2 to 20 on a logarithmic axis. One curve for each value of c (the bound (positive)): 7, 3.5, 14. Other inputs as in the worked example: b = 3, rel = <. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, lower = -5.

```
math_foundations_ai999x.solve('abs_linear_inequality', a=2, b=3, c=7, rel='<')
# -> lower = -5, upper = 2, region = between, solution = -5 < x < 2   (SOLVED)
```

```
math_foundations_ai999x.solve('abs_linear_inequality', a=3, b=-1, c=5, rel='>=')
# -> lower = -1.33333, upper = 2, region = outside, solution = x <= -1.333333333 or x >= 2
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve |2x + 3| < 7.	SOLVED	lower = -5, upper = 2, region = between, solution = -5 < x < 2
Solve the inequality |3x - 1| >= 5.	SOLVED	lower = -1.33333, upper = 2, region = outside, solution = x <= -1.333333333 or x >= 2
Solve |x + 4| <= 0.	INVALID_INPUT	Absolute value inequality |ax + b| compared with c: c must be positive (with c <= 0 the solution set is empty, a point o

Rational equation a/(x + b) = c/(x + d), `rational_equation`

Solve $a/(x+b) = c/(x+d)$ for x, rejecting a candidate that makes a denominator 0.

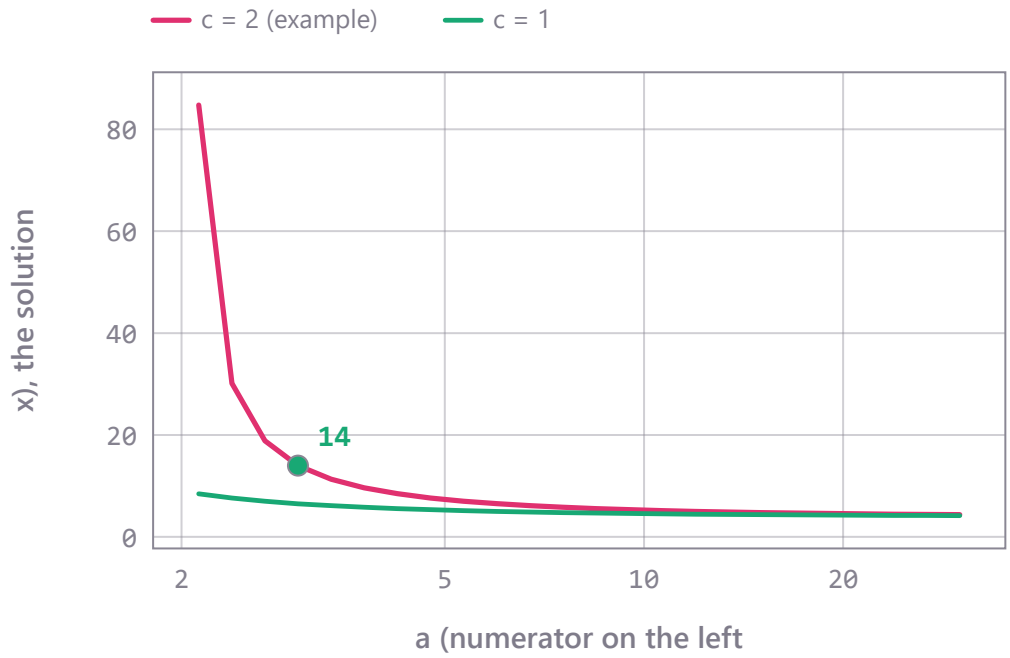
$$\frac{a}{x+b} = \frac{c}{x+d} \;\Rightarrow\; x = \frac{cb - ad}{a - c}$$

input	meaning	condition
a	numerator on the left	
b	constant in the left denominator x + b	
c	numerator on the right	

input	meaning	condition
d	constant in the right denominator x + d	

output	meaning
x	the solution

- **Method:** multiply by both denominators: a(x + d) = c(x + b), solve the linear equation, check the excluded values $x = -b$, $x = -d$
- **Assumes:** $a \neq c$; the candidate is not an excluded value
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x (the solution) against a (numerator on the left) from 2.12384 to 30 on a logarithmic axis. One curve for each value of c (numerator on the right): 2, 1. Other inputs as in the worked example: b = 1, d = -4. 24 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 3, x = 14.

```
math_foundations_ai999x.solve('rational_equation', a=3, b=1, c=2, d=-4)
# -> x = 14   (SOLVED)
```

```
math_foundations_ai999x.solve('rational_equation', a=5, b=-2, c=3, d=6)
# -> x = -18   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve 3/(x + 1) = 2/(x - 4).	SOLVED	x = 14
Solve 5/(x - 2) = 3/(x + 6).	SOLVED	x = -18
Solve 2/(x + 1) = 2/(x + 3).	INVALID_INPUT	Rational equation a/(x + b) = c/(x + d): no unique solution (equal numerators: no solution or the same expression on bot

question	outcome	result
Solve 0/(x + 1) = 4/(x - 2).	INVALID_INPUT	Rational equation a/(x + b) = c/(x + d): no solution: the only candidate x = -1 makes a denominator 0 (extraneous).

Radical equation sqrt(ax + b) = cx + d, `radical_equation`

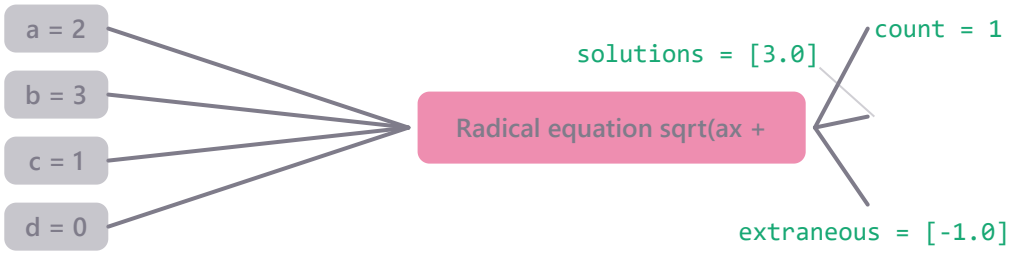
Solve sqrt(a*x + b) = c*x + d: square, solve, and keep only the roots that satisfy the original equation.

$$\sqrt{ax+b} = cx+d \;\Rightarrow\; ax+b = (cx+d)^2,\; cx+d \geq 0$$

input	meaning	condition
`a`	coefficient of x under the root	$a \neq 0$
`b`	constant under the root	
`c`	coefficient of x on the right	
`d`	constant on the right	

output	meaning
`count`	number of solutions (0, 1 or 2)
`solutions`	the solutions, ascending
`extraneous`	roots of the squared equation that fail the original (extraneous)

- **Method:** square both sides (a x + b = (c x + d)^2), solve the quadratic exactly, keep a root only if c $x+d \geq 0$
- **Assumes:** $a \neq 0$; real numbers
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The worked example of Radical equation sqrt(ax + b) = cx + d as inputs and results: stated a = 2, b = 3, c = 1, d = 0; returned by math-foundations-ai999x: count = 1, solutions = [3.0], extraneous = [-1.0].

```
math_foundations_ai999x.solve('radical_equation', a=2, b=3, c=1, d=0)
# -> count = 1, solutions = [3.0], extraneous = [-1.0]   (SOLVED)
```

```
math_foundations_ai999x.solve('radical_equation', a=1, b=7, c=1, d=-5)
# -> count = 1, solutions = [9.0], extraneous = [2.0]   (SOLVED)
```

```
math_foundations_ai999x.solve('radical_equation', a=1, b=-1, c=1, d=-1)
# -> count = 2, solutions = [1.0, 2.0], extraneous = []   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve sqrt(2x + 3) = x.	SOLVED	count = 1, solutions = [3.0], extraneous = [-1.0]
Solve √(x + 7) = x - 5.	SOLVED	count = 1, solutions = [9.0], extraneous = [2.0]
Solve sqrt(x - 1) = x - 1.	SOLVED	count = 2, solutions = [1.0, 2.0], extraneous = []
Solve sqrt(x + 4) = -2.	SOLVED	count = 0, solutions = [], extraneous = [0.0]

Synthetic division and the remainder theorem, `synthetic_division`

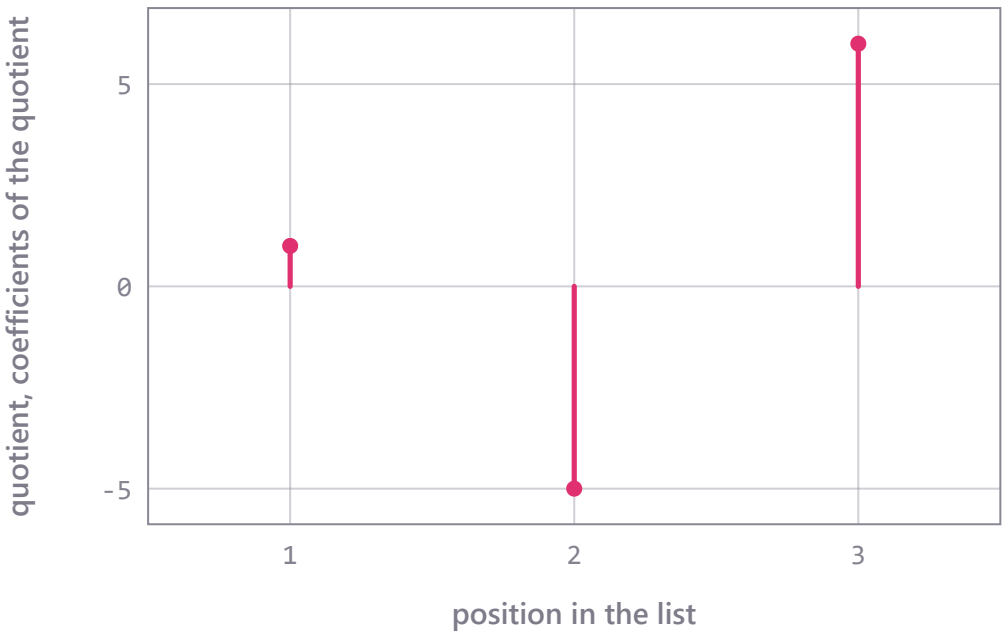
Divide a polynomial by x - r: the quotient's coefficients and the remainder, which equals p(r).

$$p(x) = (x - r)\,q(x) + p(r)$$

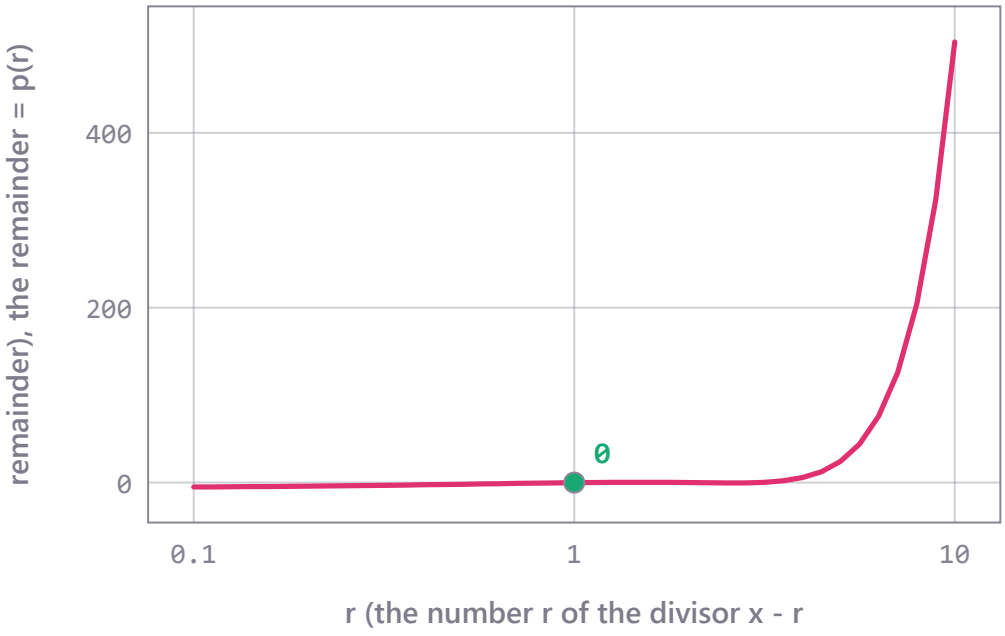
input	meaning	condition
`coefficients`	coefficients of the dividend, highest power first (2x^3 - 5x + 1 -> 2, 0, -5, 1)	
`r`	the number r of the divisor x - r	

output	meaning
`quotient`	coefficients of the quotient, highest power first
`remainder`	the remainder = p(r)

- **Method:** synthetic division (Horner's scheme): bring down, multiply by r, add; the last number is the remainder p(r)
- **Assumes:** dividend of degree >= 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



coefficients of the quotient, highest power first (quotient) returned by math-foundations-ai999x for the worked example, one value for each position in the list (3 values, first 1, last 6, largest 6). Inputs: coefficients = 1, -6, 11, -6, r = 1.



remainder (the remainder = p(r)) against r (the number r of the divisor x - r) from 0.1 to 10 on a logarithmic axis. Other inputs as in the worked example: coefficients = 1, -6, 11, -6. 41 points computed by math-foundations-ai999x; marked: the worked example, r = 1, remainder = 0.

```
math_foundations_ai999x.solve('synthetic_division', coefficients='1, -6, 11, -6', r=1)
# -> quotient = [1.0, -5.0, 6.0], remainder = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('synthetic_division', coefficients=[1, 0, -3, 0, 2], r=3)
# -> quotient = [1.0, 3.0, 6.0, 18.0], remainder = 56   (SOLVED)
```

```
math_foundations_ai999x.solve('synthetic_division', coefficients=[2, 0, 3, -4], r=-2)
# -> quotient = [2.0, -4.0, 11.0], remainder = -26   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Divide x^3 - 6x^2 + 11x - 6 by x - 1.	SOLVED	quotient = [1.0, -5.0, 6.0], remainder = 0
Find the remainder when x^4 - 3x^2 + 2 is divided by x - 3.	SOLVED	quotient = [1.0, 3.0, 6.0, 18.0], remainder = 56
Use synthetic division to divide 2x^3 + 3x - 4 by x + 2.	SOLVED	quotient = [2.0, -4.0, 11.0], remainder = -26

Polynomial long division, `polynomial_long_division`

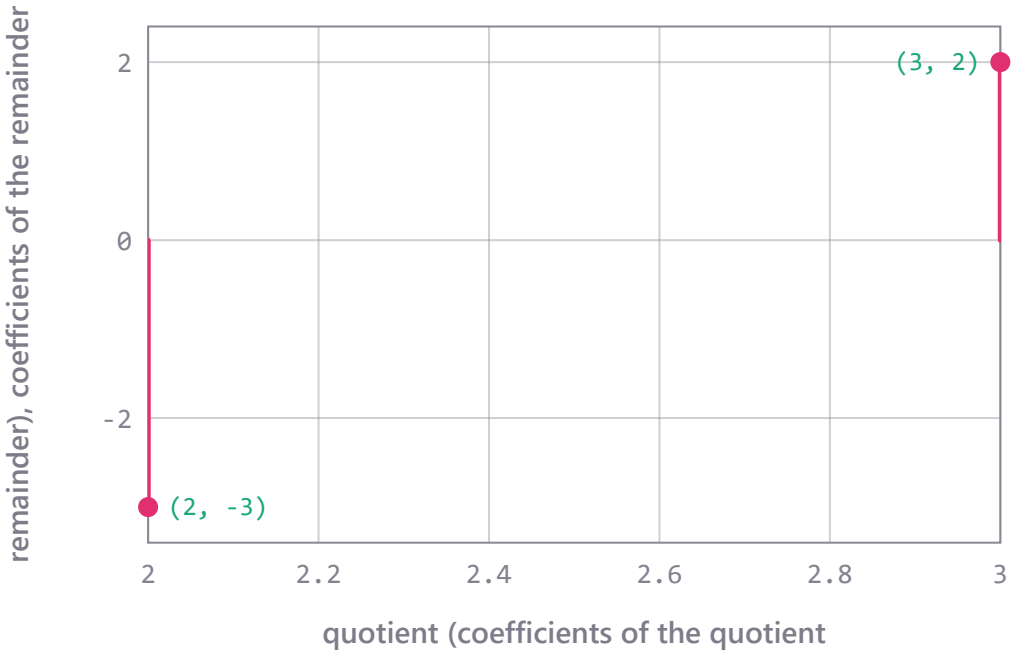
Divide one polynomial by another: quotient and remainder, with $p = qd + r$ and deg r < deg d.

$$p(x) = q(x)\,d(x) + r(x), \quad \deg r < \deg d$$

input	meaning	condition
`dividend`	coefficients of the dividend, highest power first	
`divisor`	coefficients of the divisor, highest power first (degree >= 1)	

output	meaning
`quotient`	coefficients of the quotient, highest power first
`remainder`	coefficients of the remainder, highest power first

- **Method:** long division: divide leading terms, multiply back, subtract, repeat; exact rational arithmetic
- **Assumes:** divisor of degree >= 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



coefficients of the remainder, highest power first (remainder) at each coefficients of the quotient, highest power first (quotient), both returned by math-foundations-ai999x for the worked example (dividend = 2, 3, -1, 5, divisor = 1, 0, 1): 2 → -3; 3 → 2.

```
math_foundations_ai999x.solve('polynomial_long_division', dividend='2, 3, -1, 5', divisor='1, 0, 1')
# -> quotient = [2.0, 3.0], remainder = [-3.0, 2.0]   (SOLVED)
```

```
math_foundations_ai999x.solve('polynomial_long_division', dividend=[6, -5, 0, 4], divisor=[2, -3])
# -> quotient = [3.0, 2.0, 3.0], remainder = [13.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Divide 2x^3 + 3x^2 - x + 5 by x^2 + 1.	SOLVED	quotient = [2.0, 3.0], remainder = [-3.0, 2.0]
Use long division to divide 6x^3 - 5x^2 + 4 by 2x - 3.	SOLVED	quotient = [3.0, 2.0, 3.0], remainder = [13.0]
Find the quotient and remainder when x^4 - 1 is divided by x^2 + x + 1.	SOLVED	quotient = [1.0, -1.0, 0.0], remainder = [1.0, -1.0]

Factor a quadratic over the integers, `factor_quadratic_integer`

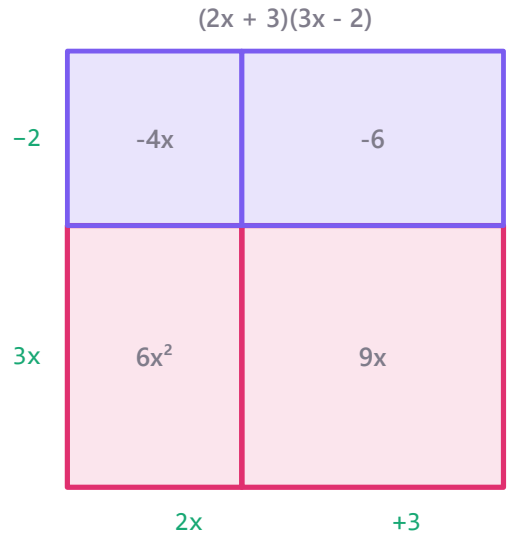
Factor $ax^2 + bx + c$ with whole-number coefficients as g(px + q)(rx + s), or say that it does not factor over the integers.

$$ax^2 + bx + c = g\,(px + q)(rx + s)$$

input	meaning	condition
`a`	coefficient of x^2 (not 0)	
`b`	coefficient of x	
`c`	constant term	

output	meaning
`g`	common factor (sign of a)
`p`	x-coefficient of the first factor
`q`	constant of the first factor
`r`	x-coefficient of the second factor
`s`	constant of the second factor
`factored`	the factored form

- **Method:** take out the greatest common factor; the rational roots -q/p and -s/r of the primitive quadratic give its factors (Gauss's lemma)
- **Assumes:** whole-number coefficients; $a \neq 0$; the discriminant is a perfect square
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Area model of $6x^2$ + 5x + -6 (as stated) with the factors the package returned, (2x + 3)(3x - 2): the sides of the rectangle are (2x + 3) and (3x − 2); each cell is the product of its side terms (cells of negative sign in the second colour), and together they make the quadratic. Side lengths drawn by magnitude (x = 1). Marked: the factored form.

```
math_foundations_ai999x.solve('factor_quadratic_integer', a=6, b=5, c=-6)
# -> g = 1, p = 2, q = 3, r = 3, s = -2, factored = (2x + 3)(3x - 2)   (SOLVED)
```

```
math_foundations_ai999x.solve('factor_quadratic_integer', a=4, b=0, c=-9)
# -> g = 1, p = 2, q = 3, r = 2, s = -3, factored = (2x + 3)(2x - 3)   (SOLVED)
```

```
math_foundations_ai999x.solve('factor_quadratic_integer', a=2, b=0, c=-8)
# -> g = 2, p = 1, q = 2, r = 1, s = -2, factored = 2(x + 2)(x - 2)   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Factor 6x^2 + 5x - 6 over the integers.	SOLVED	g = 1, p = 2, q = 3, r = 3, s = -2, factored = (2x + 3)(3x - 2)
Factor 4x^2 - 9.	SOLVED	g = 1, p = 2, q = 3, r = 2, s = -3, factored = (2x + 3)(2x - 3)
Factor completely 2x^2 - 8.	SOLVED	g = 2, p = 1, q = 2, r = 1, s = -2, factored = 2(x + 2)(x - 2)
Factor 2x^2 + 3x + 5 over the integers.	INVALID_INPUT	Factor a quadratic over the integers: this quadratic does not factor over the integers (its discriminant is not a perfec

Quadratic inequality, `quadratic_inequality`

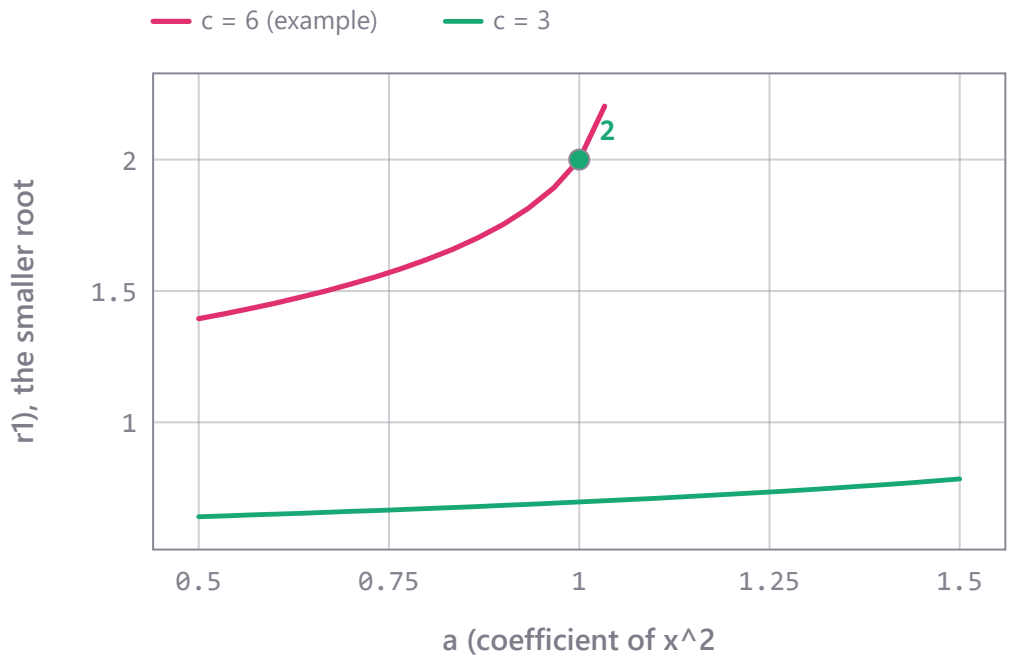
Solve $ax^2 + bx + c < 0$ (or <=, >, >=) when the quadratic has two distinct real roots.

$$a(x - r_1)(x - r_2) < 0 \iff r_1 < x < r_2 \ \ (a > 0)$$

input	meaning	condition
`a`	coefficient of x^2	$a \neq 0$
`b`	coefficient of x	
`c`	constant term	
`rel`	the comparison with 0	

output	meaning
`r1`	the smaller root
`r2`	the larger root
`region`	between or outside the roots
`solution`	the solution set

- **Method:** the roots split the line into three intervals; the sign of a decides whether the quadratic is negative between or outside them
- **Assumes:** $a \neq 0$; $b^2 - 4ac > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



r1 (the smaller root) against a (coefficient of x^2) from 0.5 to 1.5. One curve for each value of c (constant term): 6, 3. Other inputs as in the worked example: b = -5, rel = <. 17 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, r1 = 2.

```
math_foundations_ai999x.solve('quadratic_inequality', a=1, b=-5, c=6, rel='<')
# -> r1 = 2, r2 = 3, region = between, solution = 2 < x < 3   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_inequality', a=2, b=-1, c=-3, rel='>=')
# -> r1 = -1, r2 = 1.5, region = outside, solution = x <= -1 or x >= 1.5   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_inequality', a=-1, b=0, c=4, rel='>')
# -> r1 = -2, r2 = 2, region = between, solution = -2 < x < 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve x^2 - 5x + 6 < 0.	SOLVED	r1 = 2, r2 = 3, region = between, solution = 2 < x < 3
Solve the quadratic inequality 2x^2 - x - 3 >= 0.	SOLVED	r1 = -1, r2 = 1.5, region = outside, solution = x <= -1 or x >= 1.5
Solve -x^2 + 4 > 0.	SOLVED	r1 = -2, r2 = 2, region = between, solution = -2 < x < 2
Solve x^2 + 1 < 0.	OUT_OF_SCOPE	Out of scope: a quadratic inequality whose quadratic has no two distinct real roots.

Sum of an infinite geometric series, `infinite_geometric_series`

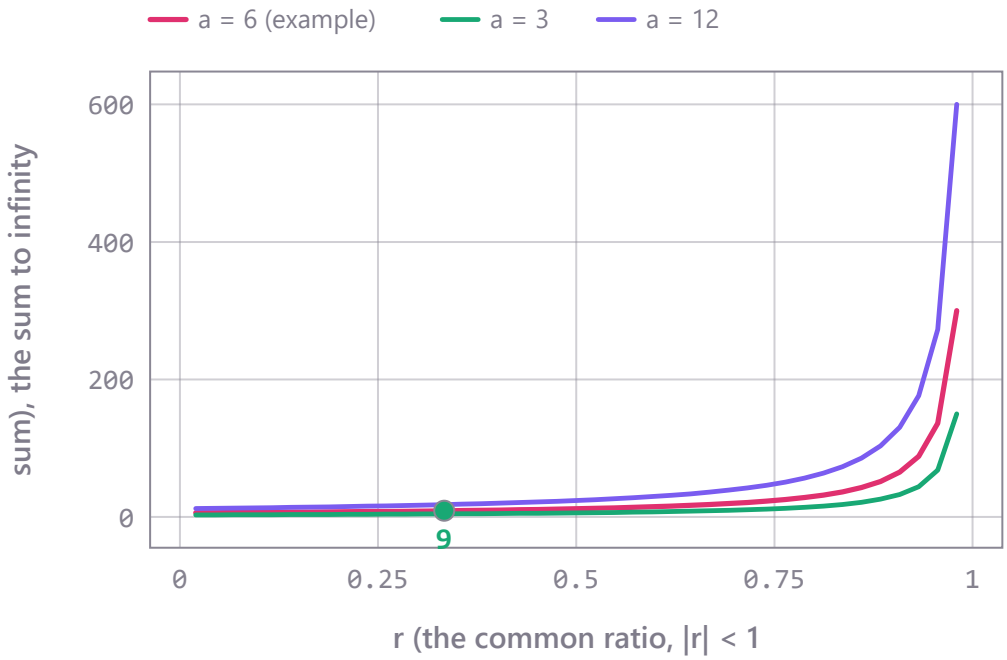
$a + ar + ar^2 + \ldots = a/(1 - r)$ for |r| < 1 (a refusal when the series diverges).

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}, \quad |r| < 1$$

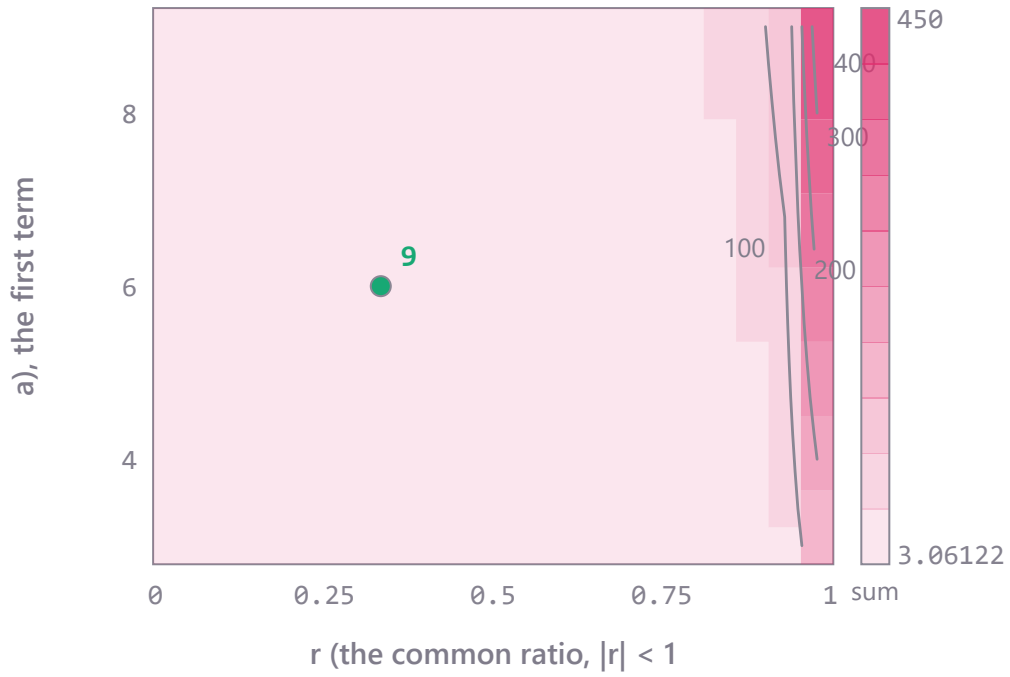
input	meaning	condition
`a`	the first term	
`r`	the common ratio, |r| < 1	

output	meaning
`sum`	the sum to infinity

- **Method:** $S_n = a$(1 - r^n)/(1 - r) and r^n -> 0 when |r| < 1
- **Assumes:** |r| < 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



sum (the sum to infinity) against r (the common ratio, |r| < 1) from 0.02 to 0.98. One curve for each value of a (the first term): 6, 3, 12. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, r = 0.333333, sum = 9.



sum (the sum to infinity) over r (the common ratio, |r| < 1) from 0.02 to 0.98 and a (the first term) from 3 to 9: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 3.06122 to dark 450); iso-lines at sum = 100, 200, 300, 400. Marked: the worked example, r = 0.333333, a = 6, sum = 9.

```
math_foundations_ai999x.solve('infinite_geometric_series', a=6, r=0.3333333333333333)
# -> sum = 9   (SOLVED)
```

```
math_foundations_ai999x.solve('infinite_geometric_series', a=8, r=0.5)
# -> sum = 16   (SOLVED)
```

```
math_foundations_ai999x.solve('infinite_geometric_series', a=9, r=-0.3333333333333333)
# -> sum = 6.75   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the sum of the infinite geometric series with first term 6 and ratio 1/3.	SOLVED	sum = 9
Find the sum to infinity of 8 + 4 + 2 + 1 + ...	SOLVED	sum = 16
Find the sum to infinity of 9 - 3 + 1 - ...	SOLVED	sum = 6.75
Find the sum of the infinite geometric series with first term 2 and ratio 3/2.	INVALID_INPUT	Sum of an infinite geometric series: the series diverges unless |r| < 1.

Partial fractions (two distinct linear factors), `partial_fractions_linear`

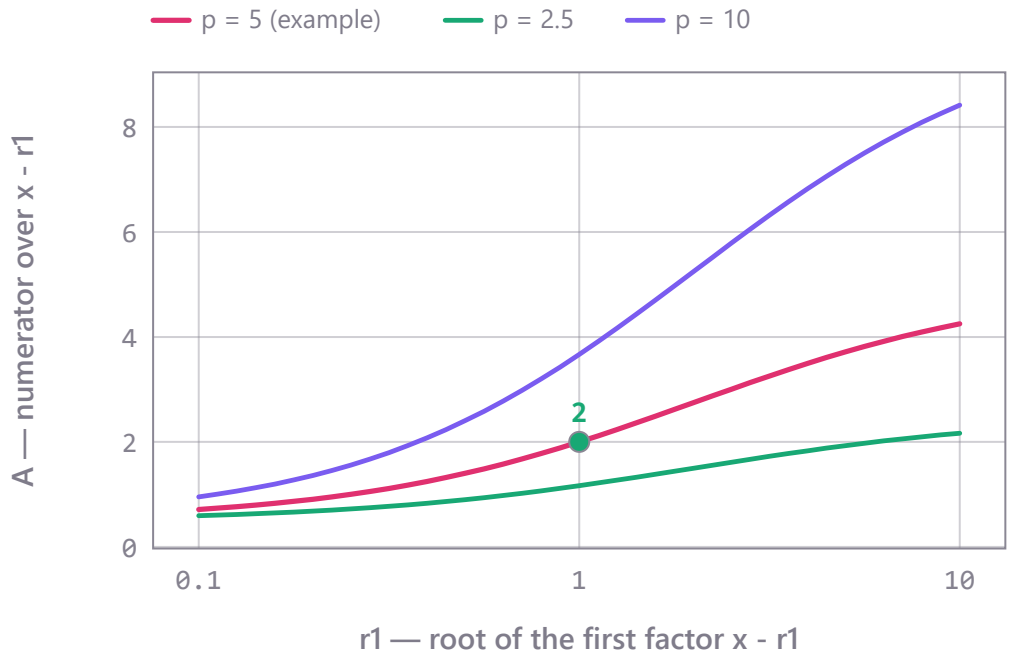
Write $(px+q)/((x-r_1)(x-r_2))$ as $A/(x-r_1)+B/(x-r_2)$.

$$\frac{px+q}{(x-r_1)(x-r_2)} = \frac{A}{x-r_1} + \frac{B}{x-r_2}, \quad A = \frac{pr_1+q}{r_1-r_2}$$

input	meaning	condition
`p`	coefficient of x in the numerator	
`q`	constant of the numerator	
`r1`	root of the first factor $x - r_1$	
`r2`	root of the second factor $x - r_2$ $(r_2 \neq r_1)$	

output	meaning
`A`	numerator over $x - r_1$
`B`	numerator over $x - r_2$

- **Method:** cover-up rule: A = (p r1 + q)/(r1 - r2), B = (p r2 + q)/(r2 - r1)
- **Assumes:** $r_1 \neq r_2$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



A (numerator over x - r1) against r1 (root of the first factor x - r1) from 0.1 to 10 on a logarithmic axis. One curve for each value of p (coefficient of x in the numerator): 5, 2.5, 10. Other inputs as in the worked example: q = 1, r2 = -2. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, r1 = 1, A = 2.

```
math_foundations_ai999x.solve('partial_fractions_linear', p=5, q=1, r1=1, r2=-2)
# -> A = 2, B = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('partial_fractions_linear', p=3, q=5, r1=-1, r2=2)
# -> A = -0.666667, B = 3.66667   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Decompose (5x + 1)/((x - 1)(x + 2)) into partial fractions.	SOLVED	A = 2, B = 3

question	outcome	result
Find the partial fraction decomposition of (3x + 5)/(x^2 - x - 2).	SOLVED	A = -0.666667, B = 3.66667
Decompose (2x + 3)/((x - 4)(x - 4)) into partial fractions.	INVALID_INPUT	Partial fractions (two distinct linear factors): the two linear factors must be different (repeated factors need another

Rational roots of a polynomial (rational root theorem) — `rational_roots`

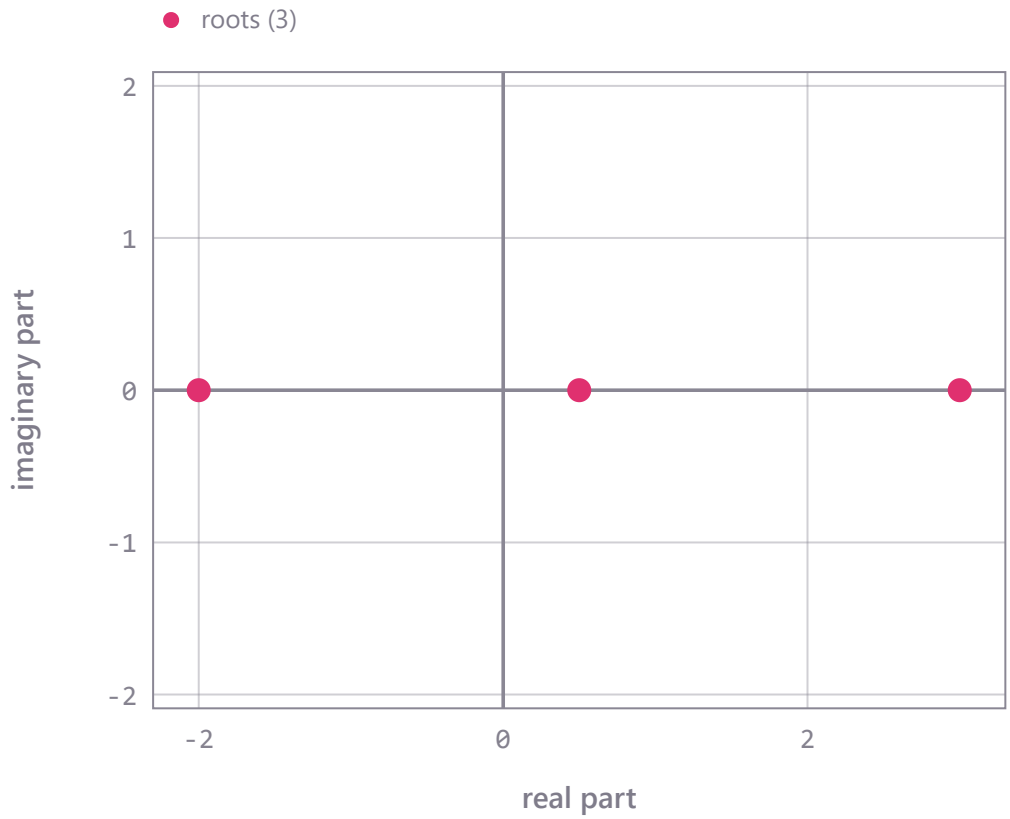
Every rational root of a polynomial with whole-number coefficients (degree up to 8).

$$a_n x^n + \cdots + a_0 = 0,\ x = \tfrac{p}{q} \Rightarrow p \mid a_0,\ q \mid a_n$$

input	meaning	condition
`coefficients`	whole-number coefficients, highest power first	

output	meaning
`count`	number of distinct rational roots
`roots`	the rational roots, ascending
`roots_exact`	the rational roots as fractions

- **Method:** rational root theorem: a root p/q in lowest terms has p dividing the constant term and q dividing the leading coefficient; test every candidate exactly
- **Assumes:** whole-number coefficients; degree 1 to 8
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



the rational roots, ascending (roots) returned by math-foundations-ai999x for the worked example (coefficients = 2, -3, -11, 6), drawn in the complex plane: -2 + 0i, 0.5 + 0i, 3 + 0i.

```
math_foundations_ai999x.solve('rational_roots', coefficients='2, -3, -11, 6')
# -> count = 3, roots = [-2.0, 0.5, 3.0], roots_exact = ['-2', '1/2', '3']   (SOLVED)
```

```
math_foundations_ai999x.solve('rational_roots', coefficients=[1, 0, -7, 6])
# -> count = 3, roots = [-3.0, 1.0, 2.0], roots_exact = ['-3', '1', '2']   (SOLVED)
```

```
math_foundations_ai999x.solve('rational_roots', coefficients=[1, 0, -2])
# -> count = 0, roots = [], roots_exact = []   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find all rational roots of 2x^3 - 3x^2 - 11x + 6 = 0.	SOLVED	count = 3, roots = [-2.0, 0.5, 3.0], roots_exact = ['-2', '1/2', '3']
List the rational zeros of x^3 - 7x + 6.	SOLVED	count = 3, roots = [-3.0, 1.0, 2.0], roots_exact = ['-3', '1', '2']
Find the rational roots of x^2 - 2.	SOLVED	count = 0, roots = [], roots_exact = []
Find all rational roots of 0.5x^2 - 2.	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: math_foundations_ai999x.templates

Evaluate a function at a number — `evaluate_function`

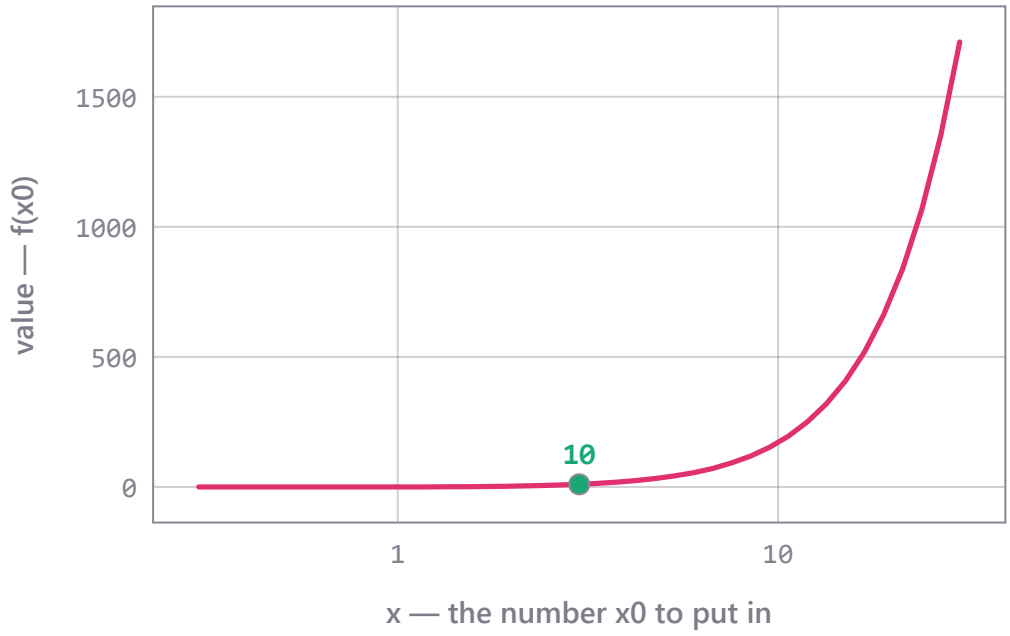
The value f(x0) of a function given by a formula in x (exact when it is rational).

$$f(x_0)$$

input	meaning	condition
`expression`	the formula of f in x: numbers, x, + - * / ^, parentheses, sqrt, abs, ln	
`x`	the number x0 to put in	

output	meaning
`value`	f(x0)
`numerator`	numerator of the exact value
`denominator`	denominator of the exact value

- **Method:** substitute x0 for x and evaluate with the order of operations, in exact rational arithmetic
- **Assumes:** x0 in the domain of f (no division by 0, no root of a negative number, no logarithm of a number <= 0)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



value (f(x0)) against x (the number x0 to put in) from 0.3 to 30 on a logarithmic axis. Other inputs as in the worked example: expression = 2x^2 - 3x + 1. 41 points computed by math-foundations-ai999x; marked: the worked example, x = 3, value = 10.

```
math_foundations_ai999x.solve('evaluate_function', expression='2x^2 - 3x + 1', x=3)
# -> value = 10, numerator = 10, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('evaluate_function', expression='x^3 - 2x + 5', x=-2)
# -> value = 1, numerator = 1, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('evaluate_function', expression='(x + 1)/(x - 3)', x=2)
# -> value = -3, numerator = -3, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('evaluate_function', expression='sqrt(x + 7)', x=2)
# -> value = 3, numerator = 3, denominator = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Evaluate f(3) for f(x) = 2x^2 - 3x + 1.	SOLVED	value = 10, numerator = 10, denominator = 1
If f(x) = x^3 - 2x + 5, find f(-2).	SOLVED	value = 1, numerator = 1, denominator = 1
Find f(1/2) when f(x) = 4x^2 - 1.	SOLVED	value = 0, numerator = 0, denominator = 1
Find f(3) when f(x) = (x + 1)/(x - 3).	INVALID_INPUT	Evaluate a function at a number: f(3) is not defined: division by zero.

Value of a composite function — `composite_function_value`

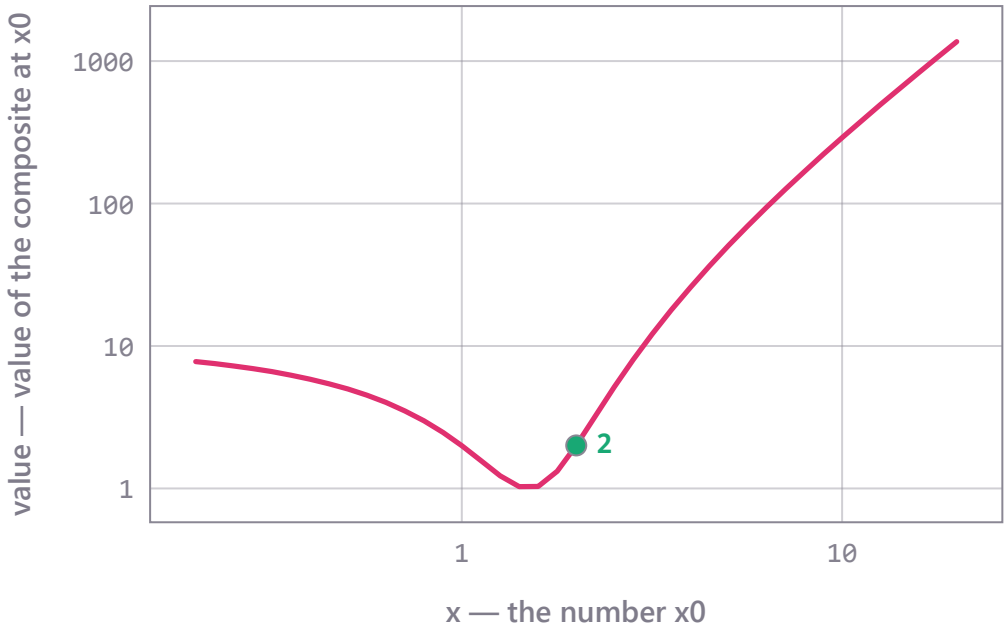
f(g(x0)) or g(f(x0)) for two functions given by formulas.

$$(f \circ g)(x_0) = f\big(g(x_0)\big)$$

input	meaning	condition
`f`	the formula of f in x	
`g`	the formula of g in x	
`x`	the number x0	
`order`	which composition	

output	meaning
`inner`	value of the inner function at x0
`value`	value of the composite at x0

- **Method:** evaluate the inner function at x0, then the outer function at that value
- **Assumes:** each value in the domain of the function applied to it
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



value (value of the composite at x0) against x (the number x0) from 0.2 to 20 on a logarithmic axis, logarithmic output axis. Other inputs as in the worked example: f = x^2 + 1, g = 2x - 3, order = f(g(x)). 41 points computed by math-foundations-ai999x; marked: the worked example, x = 2, value = 2.

```
math_foundations_ai999x.solve('composite_function_value', f='x^2 + 1', g='2x - 3', x=2,
order='f(g(x))')
# -> inner = 1, value = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('composite_function_value', f='3x + 2', g='x^2', x=-1,
order='g(f(x))')
# -> inner = -1, value = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
If f(x) = x^2 + 1 and g(x) = 2x - 3, find f(g(2)).	SOLVED	inner = 1, value = 2
If f(x) = x^2 + 1 and g(x) = 2x - 3, find g(f(2)).	SOLVED	inner = 5, value = 7
Let f(x) = 3x + 2 and g(x) = x^2, evaluate g(f(-1)).	SOLVED	inner = -1, value = 1
If f(x) = 1/x and g(x) = x - 2, find f(g(2)).	INVALID_INPUT	Value of a composite function: the composition is not defined at x = 2: division by zero.

Domain of a square-root function — `domain_sqrt_over_linear`

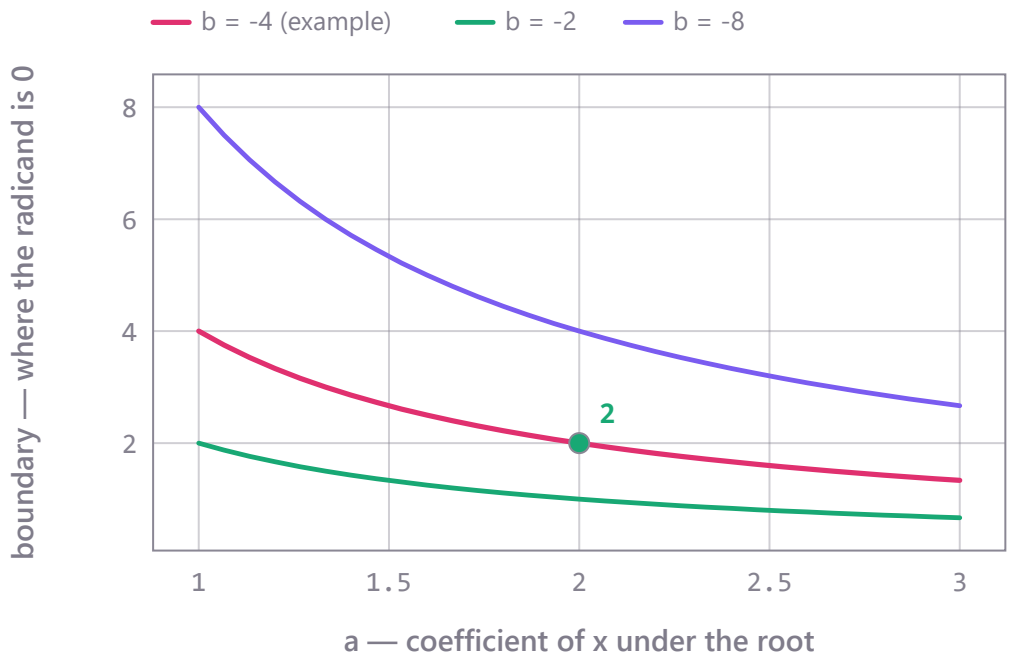
The domain of f(x) = sqrt(a*x + b)/(c*x + d) (or $1/sqrt$(a*x + b)): where the radicand is allowed and the denominator is not 0.

$$ax + b \geq 0, \quad cx + d \neq 0$$

input	meaning	condition
`a`	coefficient of x under the root	$a \neq 0$
`b`	constant under the root	
`c`	coefficient of x in the denominator (0 for none)	
`d`	constant in the denominator (1 for none)	
`root_in_denominator`	true when the root itself is in the denominator (the radicand must then be positive)	

output	meaning
`boundary`	where the radicand is 0
`side`	the inequality x must satisfy with the boundary
`excluded`	a point removed because the denominator is 0 there (None if none)
`domain`	the domain in interval notation

- **Method:** radicand $ax + b \geq 0$ (> 0 when the root is in a denominator); remove the zero of $cx + d$
- **Assumes:** $a \neq 0$; the denominator is not identically 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



boundary (where the radicand is 0) against a (coefficient of x under the root) from 1 to 3. One curve for each value of b (constant under the root): -4, -2, -8. Other inputs as in the worked example: c = 0, d = 1. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, boundary = 2.

```
math_foundations_ai999x.solve('domain_sqrt_over_linear', a=2, b=-4, c=0, d=1)
# -> boundary = 2, side = >=, excluded = none, domain = [2, inf)   (SOLVED)
```

```
math_foundations_ai999x.solve('domain_sqrt_over_linear', a=1, b=3, c=1, d=-1)
# -> boundary = -3, side = >=, excluded = 1, domain = [-3, 1) U (1, inf)   (SOLVED)
```

```
math_foundations_ai999x.solve('domain_sqrt_over_linear', a=-1, b=5, c=0, d=1,
root_in_denominator=True)
# -> boundary = 5, side = <, excluded = none, domain = (-inf, 5)   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the domain of f(x) = sqrt(2x - 4).	SOLVED	boundary = 2, side = >=, excluded = none, domain = [2, inf)
Find the domain of f(x) = sqrt(x + 3)/(x - 1).	SOLVED	boundary = -3, side = >=, excluded = 1, domain = [-3, 1) U (1, inf)
Find the domain of f(x) = 1/sqrt(5 - x).	SOLVED	boundary = 5, side = <, excluded = none, domain = (-inf, 5)
Find the domain of f(x) = sqrt(x - 4)/(x - 2).	SOLVED	boundary = 4, side = >=, excluded = none, domain = [4, inf)

Inverse of a linear function — `inverse_linear_function`

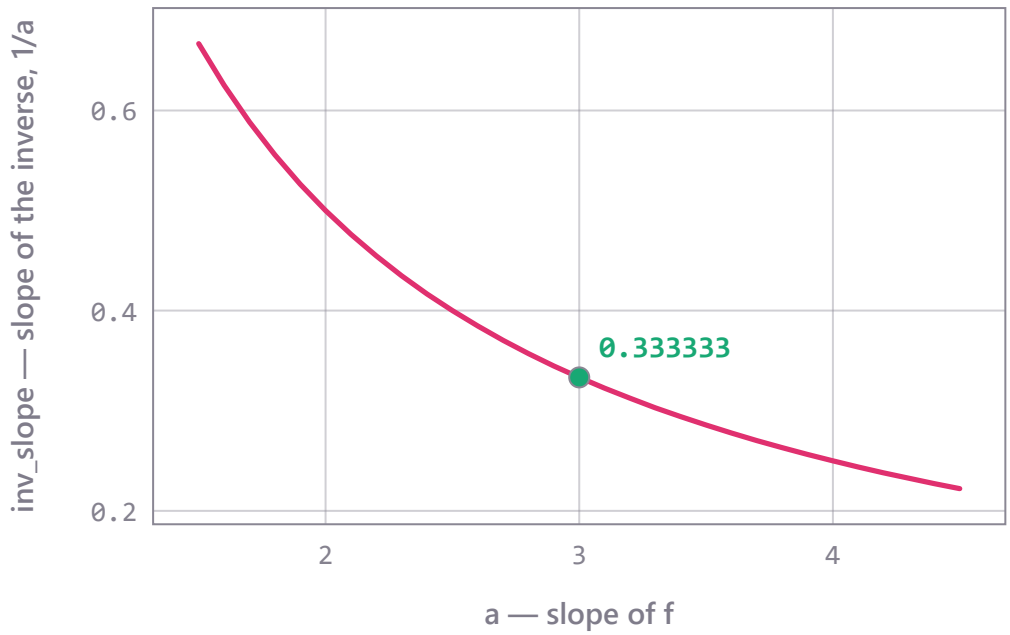
For f(x) = a*x + b $(a \neq 0)$: f^{-1}(x) = x/a - b/a, and f^{-1}(y) for a given y.

$$f^{-1}(x) = \frac{x - b}{a}$$

input	meaning	condition
`a`	slope of f	$a \neq 0$
`b`	intercept of f	
`y`	a value to invert (optional)	

output	meaning
`inv_slope`	slope of the inverse, $1/a$
`inv_intercept`	intercept of the inverse, -b/a
`value`	f^{-1}(y)

- **Method:** swap x and y in $y = ax + b$ and solve for y
- **Assumes:** $a \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



inv_slope (slope of the inverse, 1/a) against a (slope of f) from 1.5 to 4.5. Other inputs as in the worked example: b = -6. 31 points computed by math-foundations-ai999x; marked: the worked example, a = 3, inv_slope = 0.333333.

```
math_foundations_ai999x.solve('inverse_linear_function', a=3, b=-6)
# -> inv_slope = 0.333333, inv_intercept = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('inverse_linear_function', a=3, b=-6, y=9)
# -> inv_slope = 0.333333, inv_intercept = 2, value = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('inverse_linear_function', a=-2, b=5)
# -> inv_slope = -0.5, inv_intercept = 2.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the inverse of f(x) = 3x - 6.	SOLVED	inv_slope = 0.333333, inv_intercept = 2
Find f^-1(9) if f(x) = 3x - 6.	SOLVED	inv_slope = 0.333333, inv_intercept = 2, value = 5
Find the inverse function of f(x) = -2x + 5.	SOLVED	inv_slope = -0.5, inv_intercept = 2.5

Evaluate a piecewise function — `piecewise_evaluate`

f(x0) for f(x) = e1 when $x < c$ (or x <= c) and e2 otherwise.

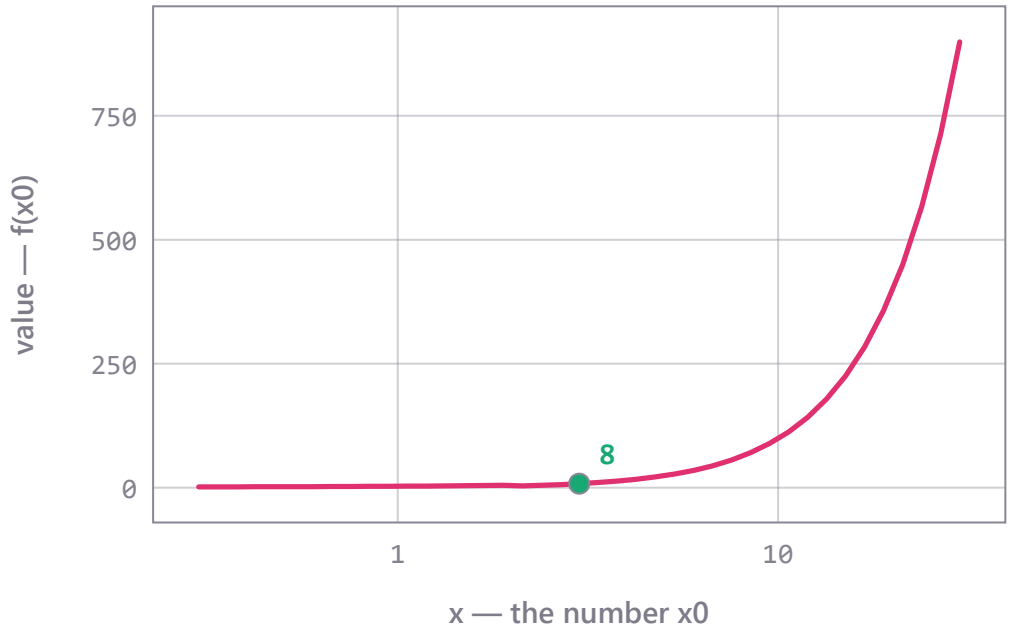
$$f(x) = \begin{cases} e_1(x) & x < c \\ e_2(x) & x \geq c \end{cases}$$

input	meaning	condition
`e1`	formula of the left piece	
`left_rel`	the left piece holds for $x < c$ or $x \leq c$	
`c`	the break point c	

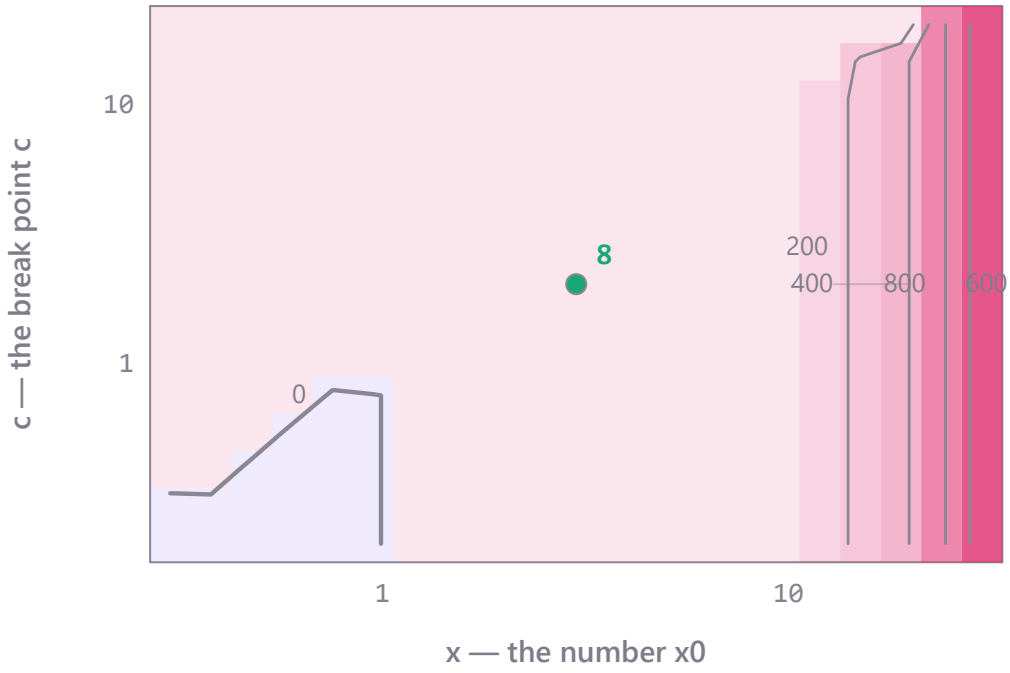
input	meaning	condition
`e2`	formula of the right piece	
`x`	the number x0	

output	meaning
`value`	f(x0)
`piece`	which piece applies (1 = left, 2 = right)

- **Method:** decide which condition x0 satisfies (the break point belongs to the piece whose condition includes it), then evaluate that piece
- **Assumes:** the two conditions split the line at c (one < and one >=, or <= and >); x0 in the domain of its piece
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



value (f(x0)) against x (the number x0) from 0.3 to 30 on a logarithmic axis. Other inputs as in the worked example: e1 = 2x + 1, left_rel = <, c = 2, e2 = x^2 - 1. 41 points computed by math-foundations-ai999x; marked: the worked example, x = 3, value = 8.



value (f(x0)) over x (the number x0) from 0.3 to 30 and c (the break point c) from 0.2 to 20 (logarithmic axes): each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its sign and magnitude (second colour negative); iso-lines at value = 0, 200, 400, 600, 800. Other inputs as in the worked example: e1 = 2x + 1, left_rel = <, e2 = x^2 - 1. Marked: the worked example, x = 3, c = 2, value = 8.

```
math_foundations_ai999x.solve('piecewise_evaluate', e1='2x + 1', left_rel='<', c=2, e2='x^2 - 1',
x=3)
# -> value = 8, piece = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('piecewise_evaluate', e1='3 - x', left_rel='<=', c=0, e2='x^2 + 4',
x=0)
# -> value = 3, piece = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Evaluate f(3) for f(x) = 2x + 1 if x < 2, x^2 - 1 if x >= 2.	SOLVED	value = 8, piece = 2
If f(x) = 3 - x for x <= 0 and x^2 + 4 for x > 0, find f(0).	SOLVED	value = 3, piece = 1
Evaluate f(2) for f(x) = 2x + 1 if x < 2, x^2 - 1 if x >= 2.	SOLVED	value = 3, piece = 2
Evaluate f(1) for f(x) = x if x < 2, x^2 if x > 3.	OUT_OF_SCOPE	Out of scope: a piecewise function whose two pieces do not split the line at one point.

Image of a point under a graph transformation — `graph_transformation_point`

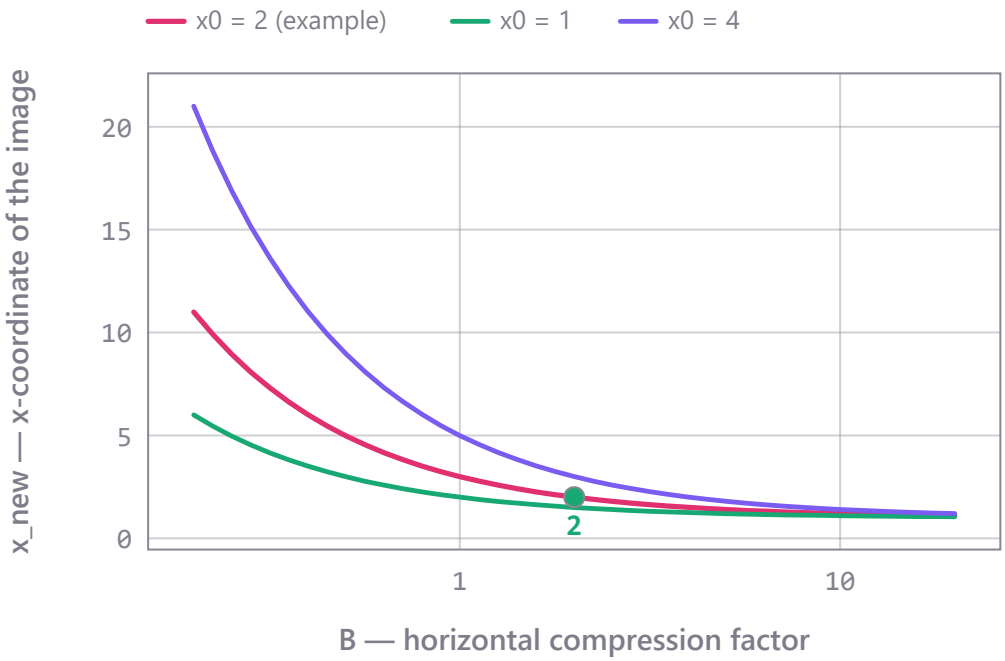
If (x0, y0) is on $y = f(x)$, the corresponding point on $y = Af(B(x - h)) + k$.

$$(x_0, y_0) \mapsto \left(\frac{x_0}{B} + h,\ Ay_0 + k\right)$$

input	meaning	condition
`x0`	x-coordinate of the point on $y = f(x)$	
`y0`	y-coordinate of the point	
`A`	vertical stretch (negative: reflection in the x-axis)	
`B`	horizontal compression factor (negative: reflection in the y-axis)	
`h`	horizontal shift	
`k`	vertical shift	

output	meaning
`x_new`	x-coordinate of the image
`y_new`	y-coordinate of the image

- **Method:** B(x - h) must equal x0, so $x = x_0/B + h$; the height becomes $Ay_0 + k$
- **Assumes:** $A \neq 0$; $B \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x_new (x-coordinate of the image) against B (horizontal compression factor (negative: reflection in the y-axis)) from 0.2 to 20 on a logarithmic axis. One curve for each value of x0 (x-coordinate of the point on y = f(x)): 2, 1, 4. Other inputs as in the worked example: y0 = 5, A = 3, h = 1, k = 4. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, B = 2, x_new = 2.

```
math_foundations_ai999x.solve('graph_transformation_point', x0=2, y0=5, A=3, B=2, h=1, k=4)
# -> x_new = 2, y_new = 19   (SOLVED)
```

```
math_foundations_ai999x.solve('graph_transformation_point', x0=-4, y0=6, A=-1, B=1, h=-3, k=0)
# -> x_new = -7, y_new = -6   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
The point (2, 5) is on the graph of y = f(x). Find the corresponding point on the graph of y = 3f(2(x - 1)) + 4.	SOLVED	x_new = 2, y_new = 19
The point (-4, 6) is on the graph of y = f(x). Find the corresponding point on the graph of y = -f(x + 3).	SOLVED	x_new = -7, y_new = -6
The point (3, 1) is on the graph of y = f(x). Find the corresponding point on the graph of y = f(3x) - 2.	SOLVED	x_new = 1, y_new = -1

Intercepts and slope of a line — `line_intercepts`

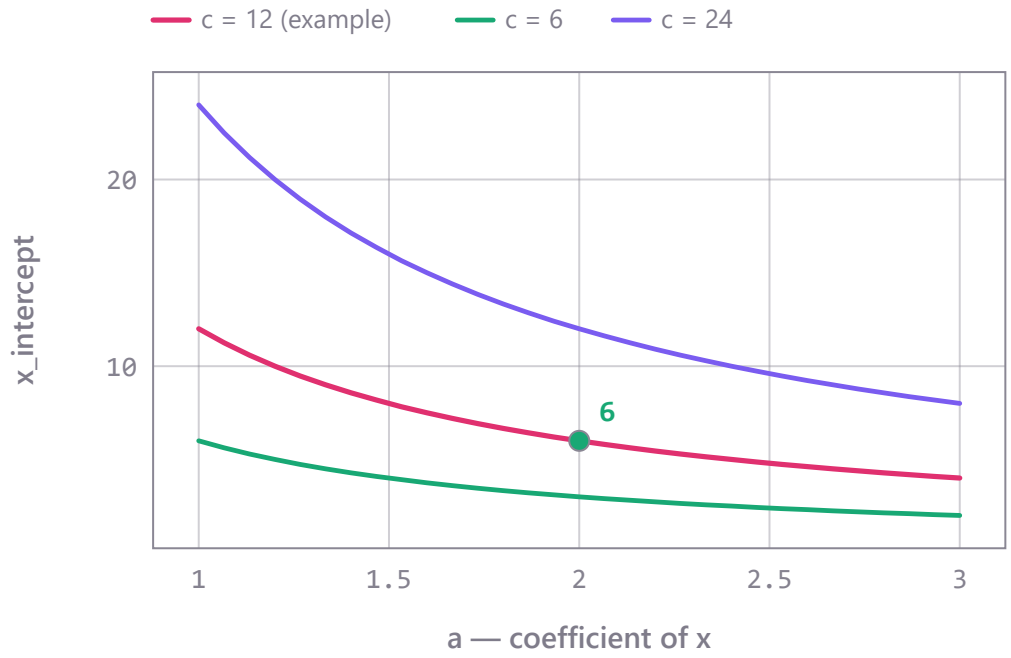
The x- and y-intercepts and the slope of the line $ax + by = c$.

$$ax + by = c: \quad x_0 = \frac{c}{a},\ y_0 = \frac{c}{b},\ m = -\frac{a}{b}$$

input	meaning	condition
`a`	coefficient of x	$a \neq 0$
`b`	coefficient of y	$b \neq 0$
`c`	right-hand side	

output	meaning
`x_intercept`	where the line meets the x-axis $(y = 0)$
`y_intercept`	where it meets the y-axis $(x = 0)$
`slope`	the slope -a/b

- **Method:** set $y = 0$ for the x-intercept and $x = 0$ for the y-intercept
- **Assumes:** $a \neq 0$; $b \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x_intercept (where the line meets the x-axis (y = 0)) against a (coefficient of x) from 1 to 3. One curve for each value of c (right-hand side): 12, 6, 24. Other inputs as in the worked example: b = 3. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, x_intercept = 6.

```
math_foundations_ai999x.solve('line_intercepts', a=2, b=3, c=12)
# -> x_intercept = 6, y_intercept = 4, slope = -0.666667   (SOLVED)
```

```
math_foundations_ai999x.solve('line_intercepts', a=-2, b=1, c=-6)
# -> x_intercept = 3, y_intercept = -6, slope = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the x- and y-intercepts of 2x + 3y = 12.	SOLVED	x_intercept = 6, y_intercept = 4, slope = -0.666667
Find the intercepts of the line y = 2x - 6.	SOLVED	x_intercept = 3, y_intercept = -6, slope = 2
Find the intercepts of the line y = 5.	INVALID_INPUT	Intercepts and slope of a line: a and b must both be nonzero (otherwise the line is parallel to an axis and misses one o

Distance from a point to a line — `distance_point_to_line`

The perpendicular distance from (x0, y0) to the line $ax + by + c = 0$.

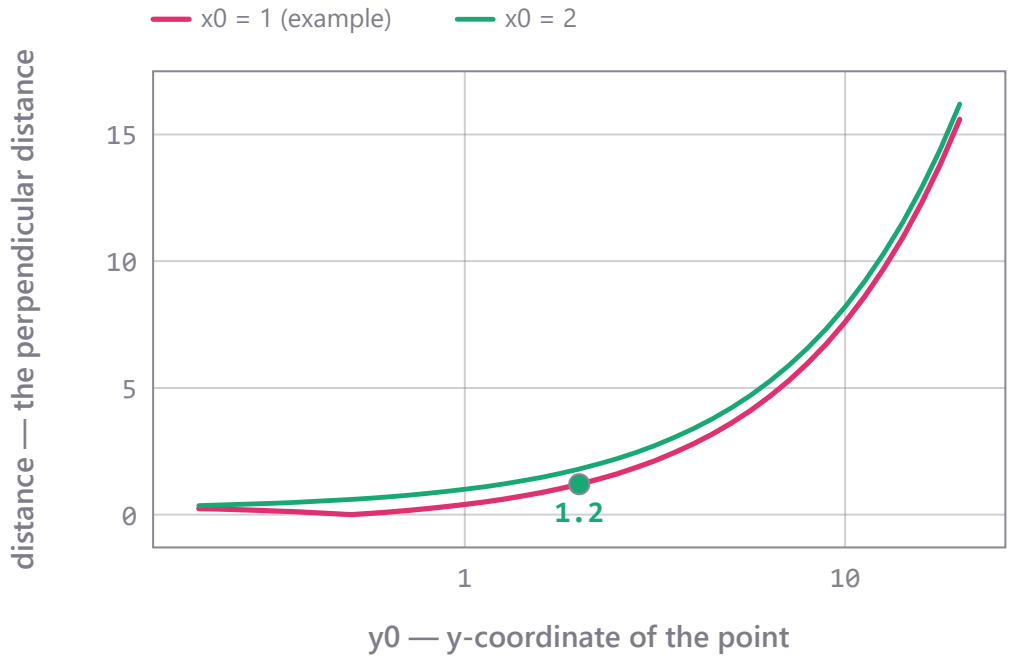
$$d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

input	meaning	condition
`x0`	x-coordinate of the point	
`y0`	y-coordinate of the point	
`a`	coefficient of x	

input	meaning	condition
`b`	coefficient of y	
`c`	constant term	

output	meaning
`distance`	the perpendicular distance

- **Method:** |a x_0 + b y_0 + c| / sqrt(a^2 + b^2)
- **Assumes:** a and b not both 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



distance (the perpendicular distance) against y0 (y-coordinate of the point) from 0.2 to 20 on a logarithmic axis. One curve for each value of x0 (x-coordinate of the point): 1, 2. Other inputs as in the worked example: a = 3, b = 4, c = -5. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, y0 = 2, distance = 1.2.

```
math_foundations_ai999x.solve('distance_point_to_line', x0=1, y0=2, a=3, b=4, c=-5)
# -> distance = 1.2   (SOLVED)
```

```
math_foundations_ai999x.solve('distance_point_to_line', x0=0, y0=0, a=-2, b=1, c=-5)
# -> distance = 2.23607   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the distance from the point (1, 2) to the line 3x + 4y - 5 = 0.	SOLVED	distance = 1.2
Find the distance from the point (0, 0) to the line y = 2x + 5.	SOLVED	distance = 2.23607

Evaluate a logarithm — `log_evaluate`

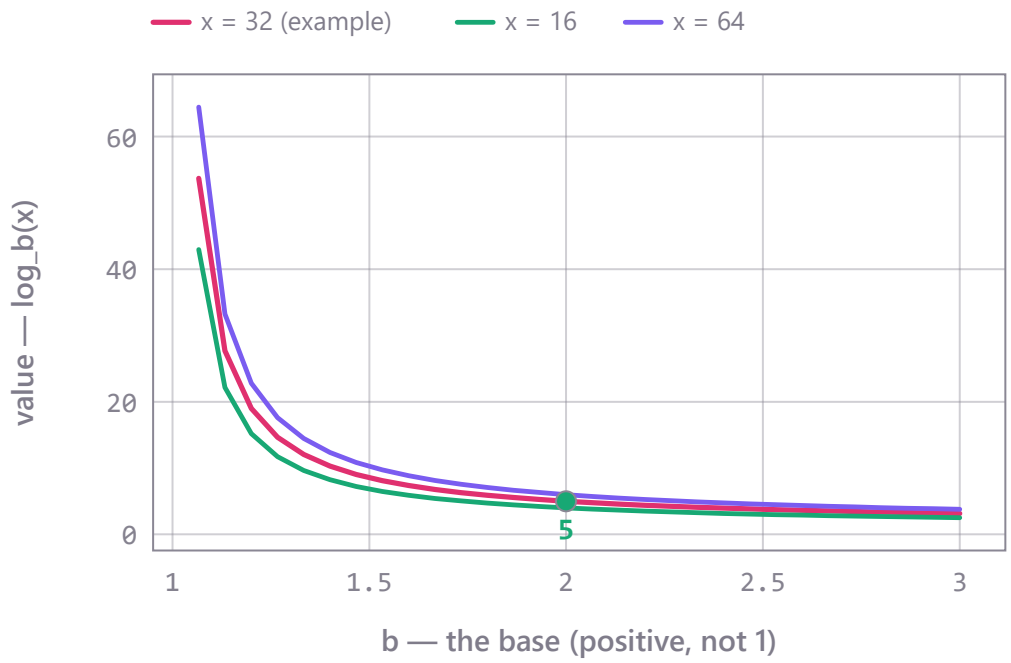
log_b(x) for a positive base $b \neq 1$ and $x > 0$ (log is base 10, ln base e).

$$\log_b x = \frac{\ln x}{\ln b}$$

input	meaning	condition
`b`	the base (positive, not 1)	
`x`	the number (positive)	

output	meaning
`value`	log_b(x)

- **Method:** change of base: log_b(x) = ln(x)/ln(b)
- **Assumes:** $b > 0$; $b \neq 1$; $x > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



value (log_b(x)) against b (the base (positive, not 1)) from 1 to 3. One curve for each value of x (the number (positive)): 32, 16, 64. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 2, value = 5.

```
math_foundations_ai999x.solve('log_evaluate', b=2, x=32)
# -> value = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('log_evaluate', b=3, x=81)
# -> value = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('log_evaluate', b=2.718281828459045, x=20)
# -> value = 2.99573   (SOLVED)
```

```
math_foundations_ai999x.solve('log_evaluate', b=10, x=1000)
# -> value = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('log_evaluate', b=2, x=0.125)
# -> value = -3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Evaluate log base 2 of 32.	SOLVED	value = 5
Find log_3(81).	SOLVED	value = 4
Compute ln(20).	SOLVED	value = 2.99573
Find log(1000).	SOLVED	value = 3
Evaluate log base 2 of -8.	INVALID_INPUT	Evaluate a logarithm: the logarithm is defined only for positive numbers.
Evaluate log base 1 of 5.	INVALID_INPUT	Evaluate a logarithm: the base must be positive and not 1.

Laws of logarithms (product, quotient, power) — `log_laws_combine`

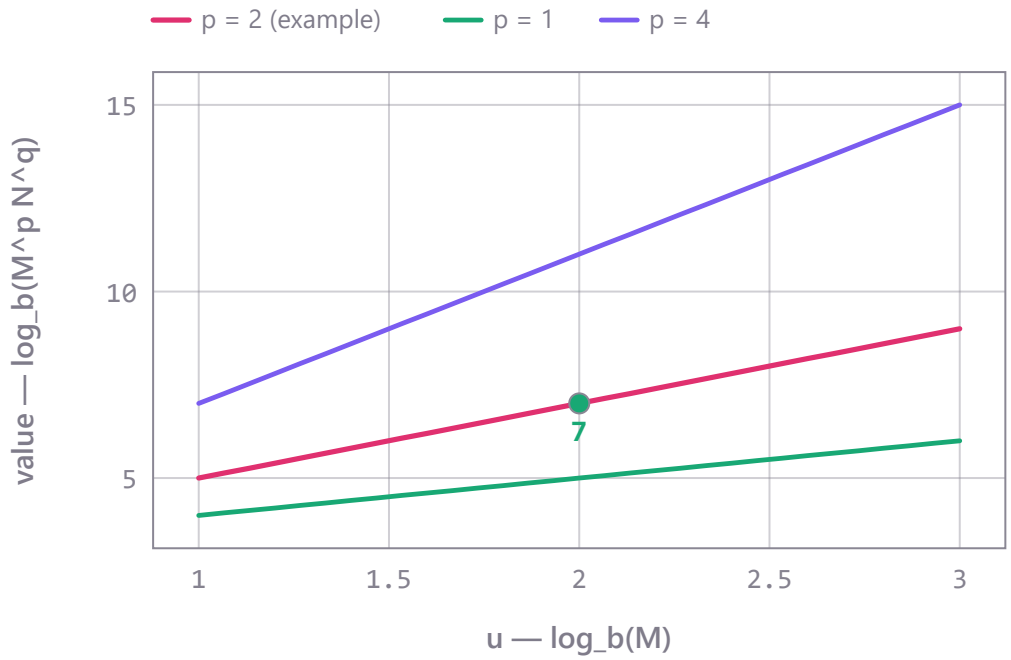
Given log_b(M) = u and log_b(N) = v, the value of log_b(M^p * N^q) (quotients and roots included).

$$\log_b(M^p N^q) = p \log_b M + q \log_b N$$

input	meaning	condition
`u`	log_b(M)	
`v`	log_b(N)	
`p`	power of M	
`q`	power of N (negative: N in the denominator)	

output	meaning
`value`	log_b(M^p N^q)

- **Method:** product law log(MN) = log M + log N, quotient law, power law log(M^p) = p log M
- **Assumes:** M, $N > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



value (log_b(M^p N^q)) against u (log_b(M)) from 1 to 3. One curve for each value of p (power of M): 2, 1, 4. Other inputs as in the worked example: v = 3, q = 1. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, u = 2, value = 7.

```
math_foundations_ai999x.solve('log_laws_combine', u=2, v=3, p=2, q=1)
# -> value = 7   (SOLVED)
```

```
math_foundations_ai999x.solve('log_laws_combine', u=1.5, v=4, p=3, q=-1)
# -> value = 0.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Given log_b(M) = 2 and log_b(N) = 3, find log_b(M^2 N).	SOLVED	value = 7
If log_b(M) = 1.5 and log_b(N) = 4, find log_b(M^3/N).	SOLVED	value = 0.5
Given log_b(M) = 6 and log_b(N) = 2, find log_b(sqrt(M)/N^2).	SOLVED	value = -1

Continuous exponential growth or decay — `continuous_growth`

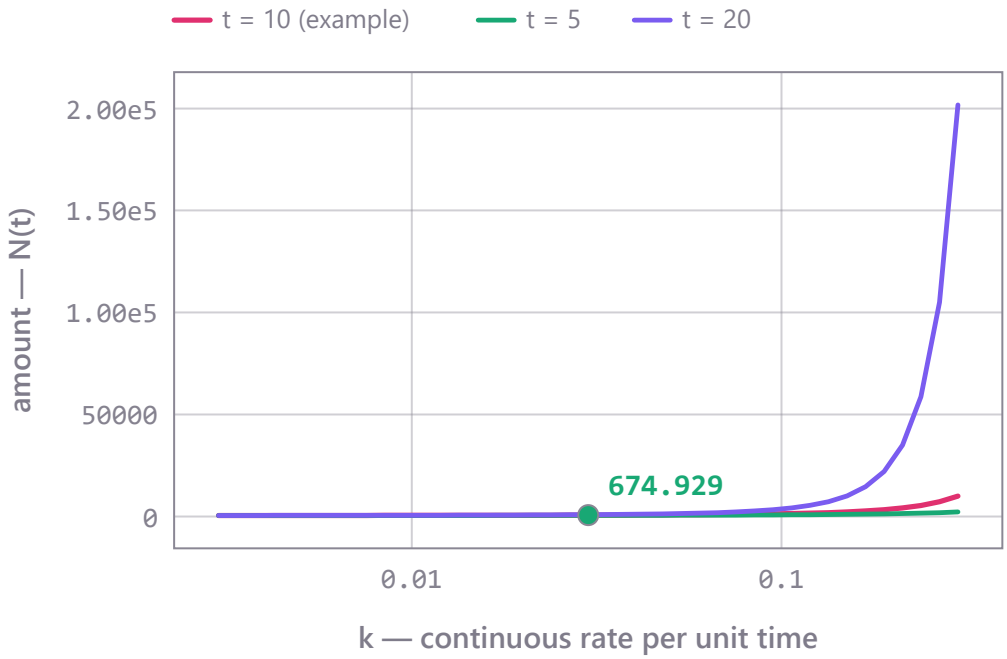
N(t) $= N_0 e^{(k}$ t) for continuous growth $(k > 0)$ or decay $(k < 0)$, including continuous compounding.

$$N(t) = N_0\, e^{kt}$$

input	meaning	condition
`n0`	initial amount N0	
`k`	continuous rate per unit time, as a decimal (0.03 for 3%; negative for decay)	
`t`	elapsed time, in the rate's unit	

output	meaning
`amount`	N(t)

- **Method:** the solution of $dN/dt = kN$: $N = N_0\ e^{kt}$
- **Assumes:** |k t| <= 700
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



amount (N(t)) against k (continuous rate per unit time, as a decimal (0.03 for 3%; negative for decay)) from 0.003 to 0.3 on a logarithmic axis. One curve for each value of t (elapsed time, in the rate's unit): 10, 5, 20. Other inputs as in the worked example: n0 = 500. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, k = 0.03, amount = 674.929.

```
math_foundations_ai999x.solve('continuous_growth', n0=500, k=0.03, t=10)
# -> amount = 674.929   (SOLVED)
```

```
math_foundations_ai999x.solve('continuous_growth', n0=1000, k=0.05, t=3)
# -> amount = 1161.83   (SOLVED)
```

```
math_foundations_ai999x.solve('continuous_growth', n0=200, k=-0.05, t=8)
# -> amount = 134.064   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A population of 500 grows continuously at 3% per year. Find the population after 10 years.	SOLVED	amount = 674.929
Find the amount when 1000 is compounded continuously at 5% for 3 years.	SOLVED	amount = 1161.83
A quantity of 200 decays continuously at a rate of 5% per hour. Find the amount after 8 hours.	SOLVED	amount = 134.064
A population of 500 grows continuously at 3% per year. Find	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form:

question	outcome	result
the population after 10 days.		math_foundations_ai999x.templates

Amount remaining after radioactive decay (half-life) — `half_life_remaining`

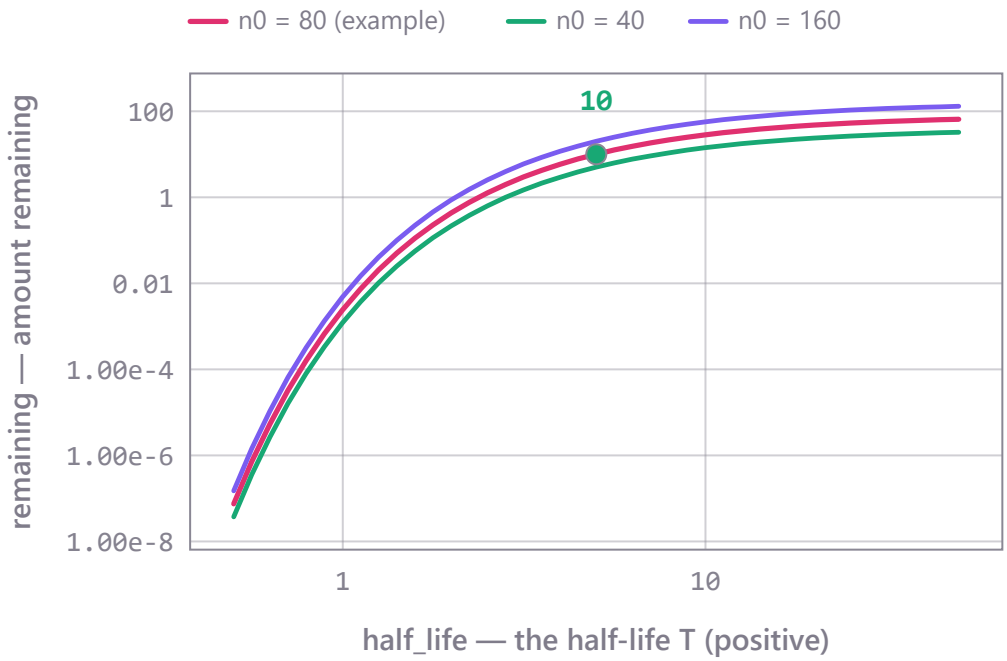
The amount left after time t of a substance with half-life T: $N_0(1/2)^{t/T}$.

$$N(t) = N_0\left(\tfrac{1}{2}\right)^{t/T}$$

input	meaning	condition
`n0`	initial amount (positive)	
`half_life`	the half-life T (positive)	
`t`	elapsed time, in the unit of T	

output	meaning
`remaining`	amount remaining (in the unit of N0)
`fraction`	fraction remaining

- **Method:** each half-life halves the amount: $N = N_0\ (1/2)^{t/T}$
- **Assumes:** $n_0 > 0$; $T > 0$; $t \geq 0$; t and T in the same unit
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



remaining (amount remaining (in the unit of N0)) against half_life (the half-life T (positive)) from 0.5 to 50 on a logarithmic axis, logarithmic output axis. One curve for each value of n0 (initial amount (positive)): 80, 40, 160. Other inputs as in the worked example: t = 15. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, half_life = 5, remaining = 10.

```
math_foundations_ai999x.solve('half_life_remaining', n0=80, half_life=5, t=15)
# -> remaining = 10, fraction = 0.125   (SOLVED)
```

```
math_foundations_ai999x.solve('half_life_remaining', n0=200, half_life=8, t=20)
# -> remaining = 35.3553, fraction = 0.176777   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
An 80 g sample has a half-life of 5 years. How much remains after 15 years?	SOLVED	remaining = 10, fraction = 0.125
A sample of 200 mg of iodine-131 has a half-life of 8 days. How much is left after 20 days?	SOLVED	remaining = 35.3553, fraction = 0.176777
An 80 g sample has a half-life of 5 years. How much remains after 15 days?	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: math_foundations_ai999x.templates

Doubling time at a percentage growth rate — `doubling_time_percent`

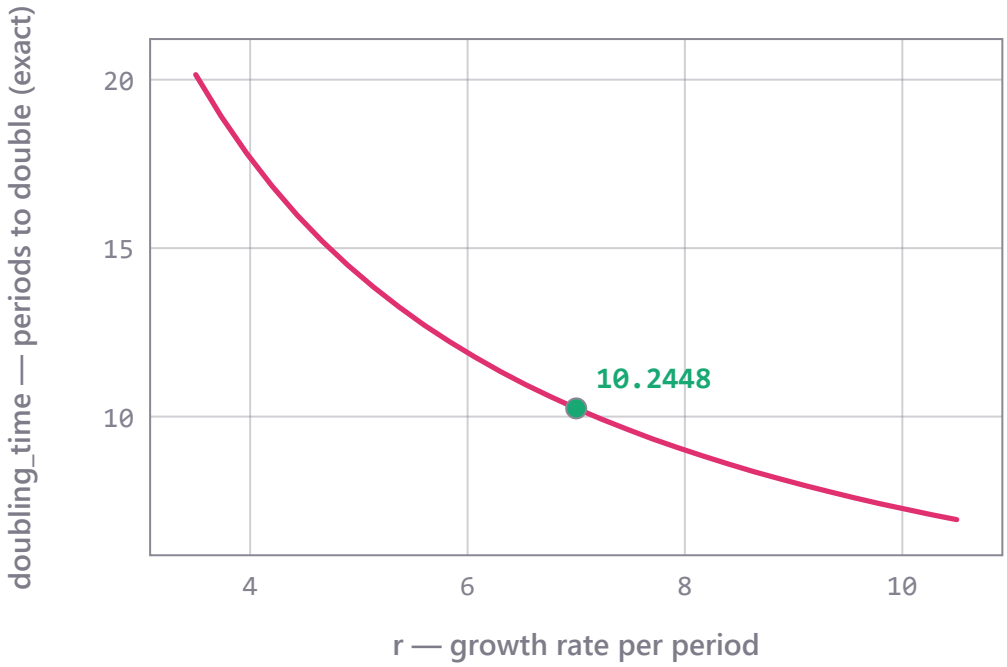
The time for a quantity growing by r% per period (compounded once per period) to double.

$$T = \frac{\ln 2}{\ln\left(1 + \frac{r}{100}\right)}$$

input	meaning	condition
`r`	growth rate per period, in percent (positive)	

output	meaning
`doubling_time`	periods to double (exact)
`rule_of_70_estimate`	the rule-of-70 approximation $70/r$

- **Method:** solve $(1 + r/100)^T = 2$: $T = ln\,2/ln$(1 + r/100)
- **Assumes:** $r > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



doubling_time (periods to double (exact)) against r (growth rate per period, in percent (positive)) from 3.5 to 10.5. 31 points computed by math-foundations-ai999x; marked: the worked example, r = 7, doubling_time = 10.2448.

```
math_foundations_ai999x.solve('doubling_time_percent', r=7)
# -> doubling_time = 10.2448, rule_of_70_estimate = 10   (SOLVED)
```

```
math_foundations_ai999x.solve('doubling_time_percent', r=100)
# -> doubling_time = 1, rule_of_70_estimate = 0.7   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How long does it take an investment to double at 7% interest compounded annually?	SOLVED	doubling_time = 10.2448, rule_of_70_estimate = 10
What is the doubling time at a growth rate of 3% per year?	SOLVED	doubling_time = 23.4498, rule_of_70_estimate = 23.3333

Exponential equation a*b^(cx + d) = k — `exponential_equation_general`

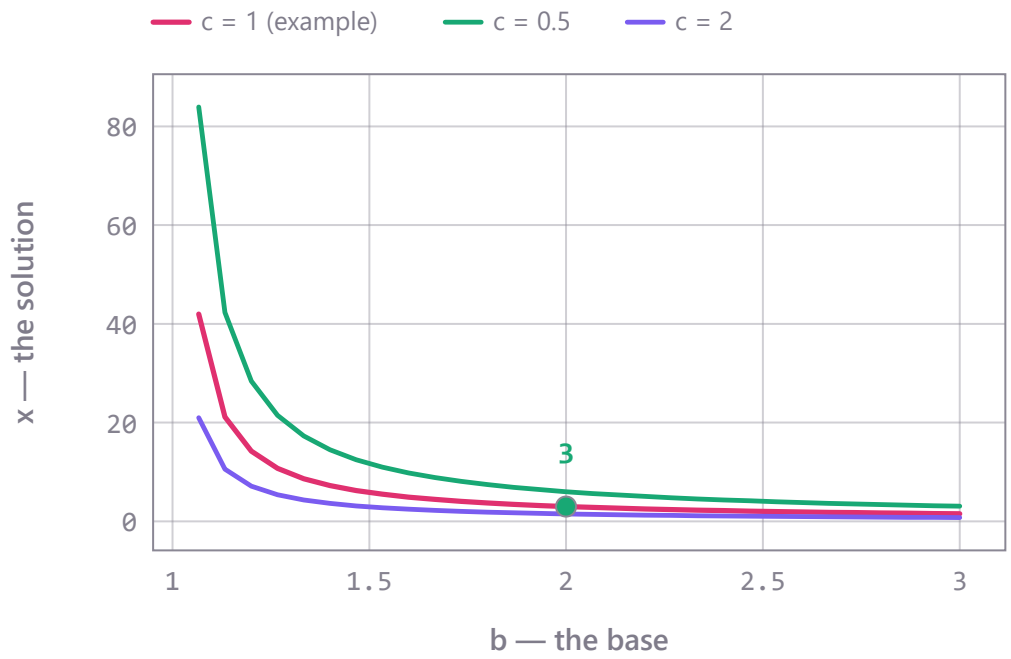
Solve $ab^{cx+d} = k$ for x by taking logarithms.

$$x = \frac{\log_b(k/a) - d}{c}$$

input	meaning	condition
a	the coefficient a	$a \neq 0$
b	the base (positive, not 1; e for the natural base)	
c	coefficient of x in the exponent	$c \neq 0$
d	constant in the exponent	
k	the right-hand side	

output	meaning
x	the solution

- **Method:** divide by a, take log base b: c $x + d = log_b$(k/a)
- **Assumes:** $a \neq 0$; $c \neq 0$; $b > 0$, $b \neq 1$; $k/a > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x (the solution) against b (the base (positive, not 1; e for the natural base)) from 1 to 3. One curve for each value of c (coefficient of x in the exponent): 1, 0.5, 2. Other inputs as in the worked example: a = 3, d = 1, k = 48. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 2, x = 3.

```
math_foundations_ai999x.solve('exponential_equation_general', a=3, b=2, c=1, d=1, k=48)
# -> x = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('exponential_equation_general', a=5, b=2.718281828459045, c=2, d=0,
k=20)
# -> x = 0.693147   (SOLVED)
```

```
math_foundations_ai999x.solve('exponential_equation_general', a=1, b=2, c=3, d=-1, k=16)
# -> x = 1.66667   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve 3 * 2^(x + 1) = 48.	SOLVED	x = 3
Solve 5e^(2x) = 20.	SOLVED	x = 0.693147
Solve 2^(3x - 1) = 16.	SOLVED	x = 1.66667
Solve 4 * 3^x = 324.	SOLVED	x = 4

question	outcome	result
Solve 2 * 3^(x - 1) = -6.	INVALID_INPUT	Exponential equation a*b^(cx + d) = k: no real solution: b^(cx + d) is positive, so k/a must be positive.

Logarithmic equation log_b(cx + d) = k — `log_equation_linear_argument`

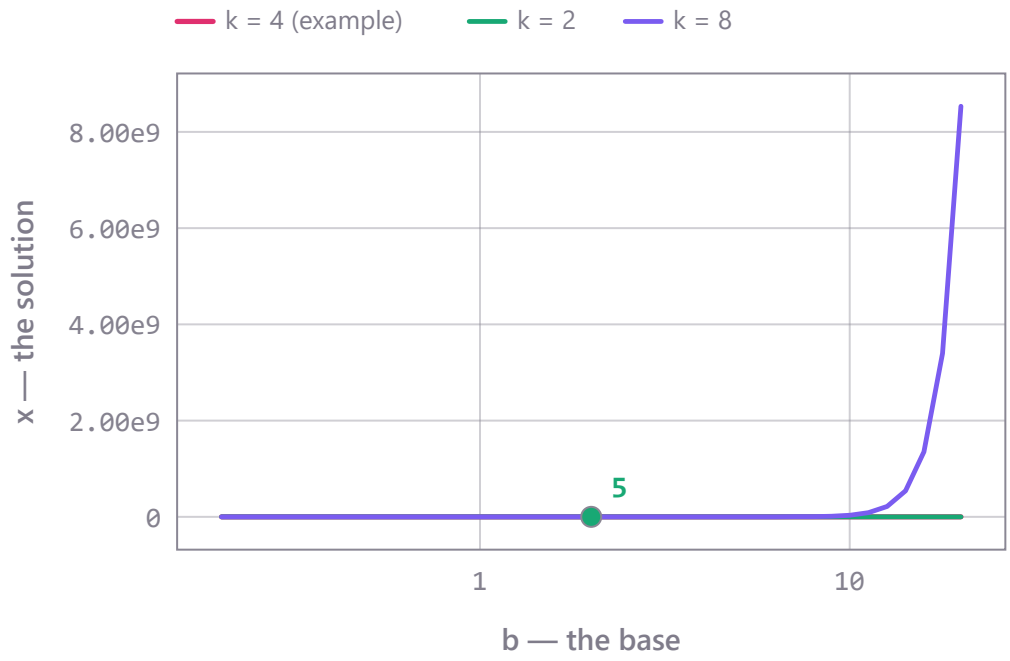
Solve log_b(c*x + d) = k for x by rewriting it in exponential form.

$$\log_b(cx + d) = k \iff cx + d = b^k$$

input	meaning	condition
b	the base (positive, not 1; 10 for log, e for ln)	
c	coefficient of x in the argument	$c \neq 0$
d	constant in the argument	
k	the right-hand side	

output	meaning
x	the solution

- **Method:** log_b(u) = k means $u = b^k$; the argument b^k is positive, so the solution is always admissible
- **Assumes:** $b > 0$, $b \neq 1$; $c \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x (the solution) against b (the base (positive, not 1; 10 for log, e for ln)) from 0.2 to 20 on a logarithmic axis. One curve for each value of k (the right-hand side): 4, 2, 8. Other inputs as in the worked example: c = 3, d = 1. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 2, x = 5.

```
math_foundations_ai999x.solve('log_equation_linear_argument', b=2, c=3, d=1, k=4)
# -> x = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('log_equation_linear_argument', b=2.718281828459045, c=2, d=-1, k=3)
# -> x = 10.5428   (SOLVED)
```

```
math_foundations_ai999x.solve('log_equation_linear_argument', b=10, c=5, d=0, k=2)
# -> x = 20   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve log base 2 of (3x + 1) = 4.	SOLVED	x = 5
Solve log_3(x - 4) = 2.	SOLVED	x = 13
Solve ln(2x - 1) = 3.	SOLVED	x = 10.5428
Solve log(5x) = 2.	SOLVED	x = 20
Solve for x in the equation log_3(x - 1) = 2.	SOLVED	x = 10
Solve log base 1 of (2x) = 3.	INVALID_INPUT	Logarithmic equation log_b(cx + d) = k: the base must be positive and not 1.

Degrees to radians — `deg_to_rad`

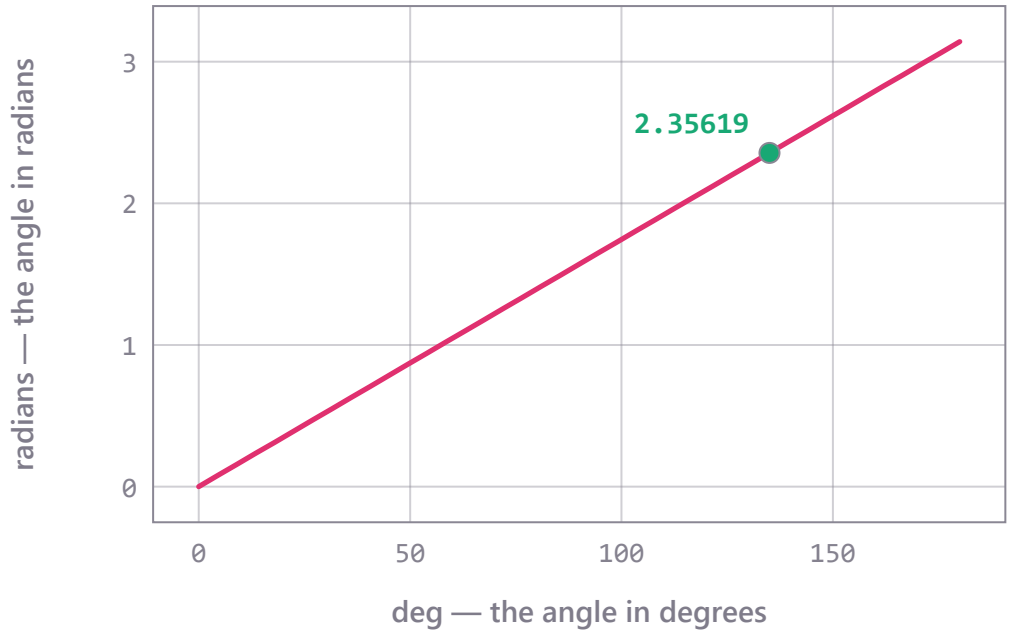
Convert an angle in degrees to radians (also as a multiple of pi).

$$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$$

input	meaning	condition
`deg`	the angle in degrees	

output	meaning
`radians`	the angle in radians
`pi_multiple`	the angle as a multiple of pi
`exact`	exact form, e.g. $3pi/4$

- **Method:** multiply by $pi/180$
- **Assumes:** a finite angle
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



radians (the angle in radians) against deg (the angle in degrees) from 0 to 180. 41 points computed by math-foundations-ai999x; marked: the worked example, deg = 135, radians = 2.35619.

```
math_foundations_ai999x.solve('deg_to_rad', deg=135)
# -> radians = 2.35619, pi_multiple = 0.75, exact = 3pi/4   (SOLVED)
```

```
math_foundations_ai999x.solve('deg_to_rad', deg=210)
# -> radians = 3.66519, pi_multiple = 1.16667, exact = 7pi/6   (SOLVED)
```

```
math_foundations_ai999x.solve('deg_to_rad', deg=-90)
# -> radians = -1.5708, pi_multiple = -0.5, exact = -pi/2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Convert 135 degrees to radians.	SOLVED	radians = 2.35619, pi_multiple = 0.75, exact = 3pi/4
What is 210° in radians?	SOLVED	radians = 3.66519, pi_multiple = 1.16667, exact = 7pi/6
Express 1 degree in radians.	SOLVED	radians = 0.0174533, pi_multiple = 0.00555556, exact = pi/180

Radians to degrees — `rad_to_deg`

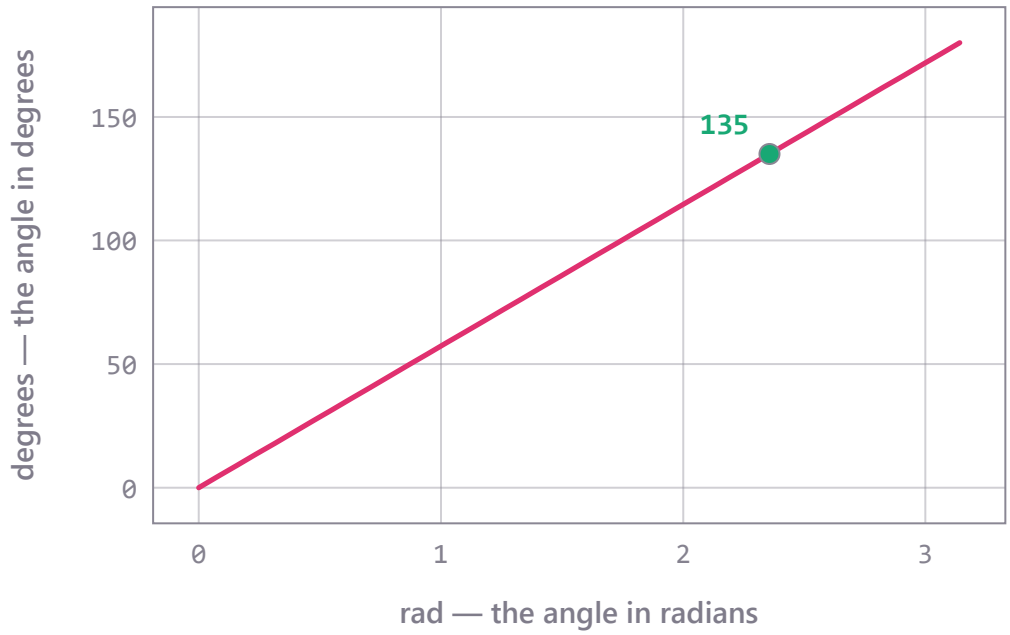
Convert an angle in radians (a number, or a multiple of pi such as 3pi/4) to degrees.

$$\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$$

input	meaning	condition
`rad`	the angle in radians	

output	meaning
`degrees`	the angle in degrees

- **Method:** multiply by $180/pi$
- **Assumes:** a finite angle
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



degrees (the angle in degrees) against rad (the angle in radians) from 0 to 3.14159. 41 points computed by math-foundations-ai999x; marked: the worked example, rad = 2.35619, degrees = 135.

```
math_foundations_ai999x.solve('rad_to_deg', rad=2.356194490192345)
# -> degrees = 135   (SOLVED)
```

```
math_foundations_ai999x.solve('rad_to_deg', rad=2.5)
# -> degrees = 143.239   (SOLVED)
```

```
math_foundations_ai999x.solve('rad_to_deg', rad=3.141592653589793)
# -> degrees = 180   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Convert 3pi/4 radians to degrees.	SOLVED	degrees = 135
Convert 2.5 radians to degrees.	SOLVED	degrees = 143.239
What is 5π/6 in degrees?	SOLVED	degrees = 150

Right triangle from its two legs — `right_triangle_legs`

Given the legs a and b of a right triangle: the hypotenuse c and the acute angles A (opposite a) and B.

$$c = \sqrt{a^2 + b^2}, \quad \tan A = \frac{a}{b}$$

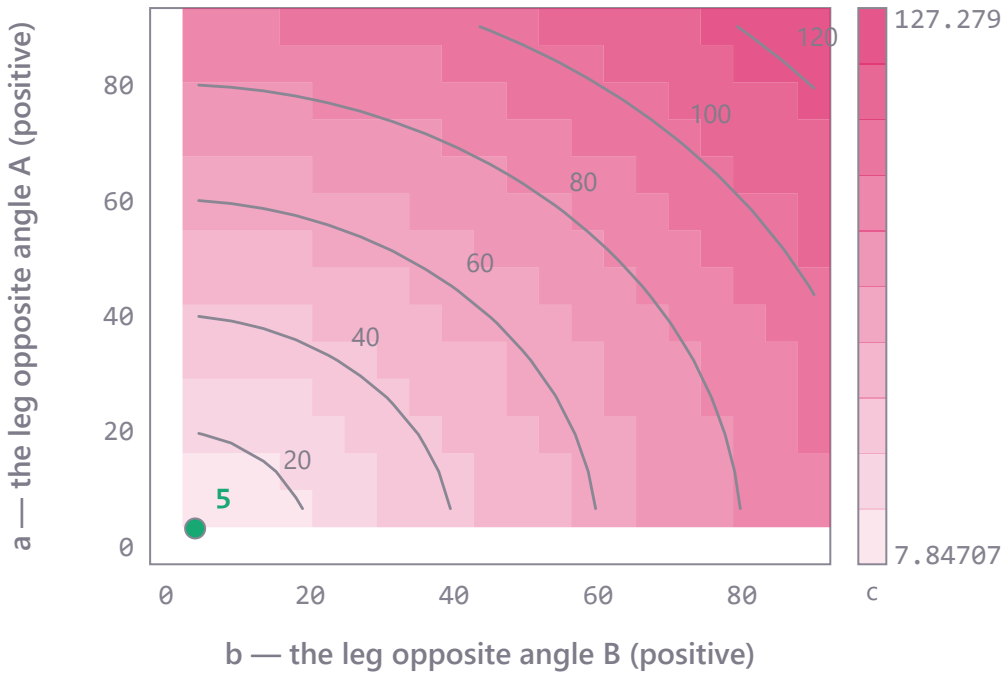
input	meaning	condition
a	the leg opposite angle A (positive)	
b	the leg opposite angle B (positive)	

output	meaning
c	the hypotenuse
A	angle opposite a, degrees
B	angle opposite b, degrees

- **Method:** Pythagoras $c = \sqrt{a^2 + b^2}$; tan $A = a/b$; $B = 90 - A$
- **Assumes:** $a > 0$; $b > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



c (the hypotenuse) against a (the leg opposite angle A (positive)) from 0 to 90. One curve for each value of b (the leg opposite angle B (positive)): 4, 22.5, 67.5. 40 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 3, c = 5.



c (the hypotenuse) over b (the leg opposite angle B (positive)) from 0 to 90 and a (the leg opposite angle A (positive)) from 0 to 90: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 7.84707 to dark 127.279); iso-lines at c = 20, 40, 60, 80, 100, 120; blank cells: inputs the capability refuses. Marked: the worked example, b = 4, a = 3, c = 5.

```
math_foundations_ai999x.solve('right_triangle_legs', a=3, b=4)
# -> c = 5, A = 36.8699, B = 53.1301   (SOLVED)
```

```
math_foundations_ai999x.solve('right_triangle_legs', a=5, b=12)
# -> c = 13, A = 22.6199, B = 67.3801   (SOLVED)
```

```
math_foundations_ai999x.solve('right_triangle_legs', a=1, b=1)
# -> c = 1.41421, A = 45, B = 45   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A right triangle has legs 3 and 4. Find the hypotenuse and the acute angles.	SOLVED	c = 5, A = 36.8699, B = 53.1301
Solve the right triangle with legs a = 5 and b = 12.	SOLVED	c = 13, A = 22.6199, B = 67.3801
Solve the right triangle with legs a = 0 and b = 12.	INVALID_INPUT	Right triangle from its two legs: the legs must be positive.

Right triangle from a leg and the hypotenuse — `right_triangle_leg_hypotenuse`

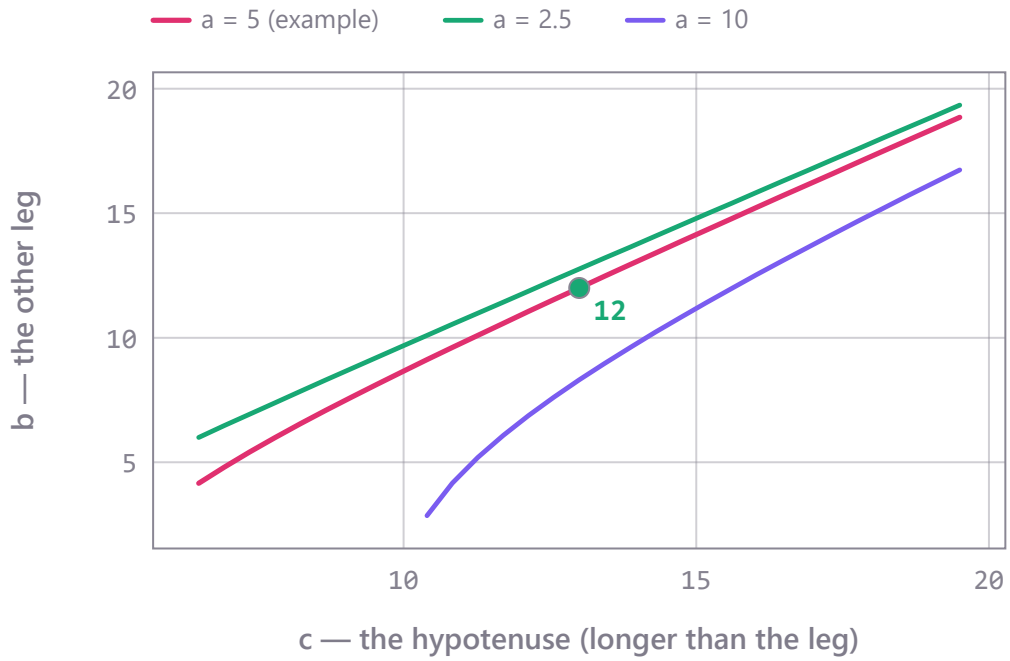
Given a leg a and the hypotenuse c of a right triangle: the other leg b and the acute angles A (opposite a) and B.

$$b = \sqrt{c^2 - a^2}, \quad \sin A = \frac{a}{c}$$

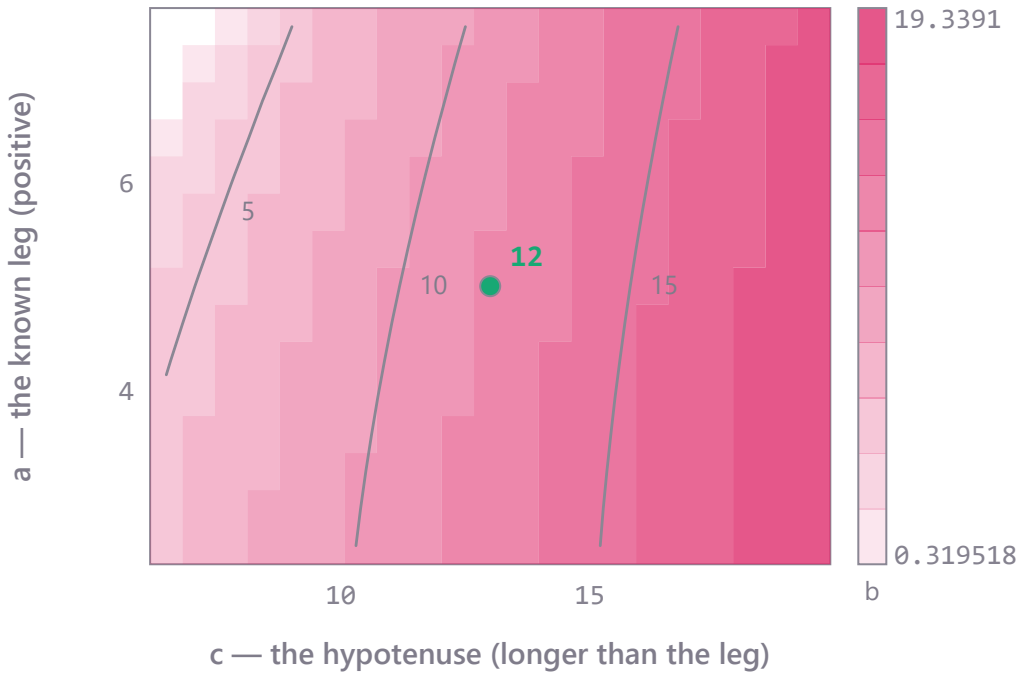
input	meaning	condition
`a`	the known leg (positive)	
`c`	the hypotenuse (longer than the leg)	

output	meaning
`b`	the other leg
`A`	angle opposite a, degrees
`B`	angle opposite b, degrees

- **Method:** $b = \sqrt{c^2 - a^2}$; sin $A = a/c$, $B = 90 - A$
- **Assumes:** $0 < a < c$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



b (the other leg) against c (the hypotenuse (longer than the leg)) from 6.5 to 19.5. One curve for each value of a (the known leg (positive)): 5, 2.5, 10. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 13, b = 12.



b (the other leg) over c (the hypotenuse (longer than the leg)) from 6.5 to 19.5 and a (the known leg (positive)) from 2.5 to 7.5: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 0.319518 to dark 19.3391); iso-lines at b = 5, 10, 15; blank cells: inputs the capability refuses. Marked: the worked example, c = 13, a = 5, b = 12.

```
math_foundations_ai999x.solve('right_triangle_leg_hypotenuse', a=5, c=13)
# -> b = 12, A = 22.6199, B = 67.3801   (SOLVED)
```

```
math_foundations_ai999x.solve('right_triangle_leg_hypotenuse', a=3, c=6)
# -> b = 5.19615, A = 30, B = 60   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A right triangle has hypotenuse 13 and one leg 5. Find the other leg and the acute angles.	SOLVED	b = 12, A = 22.6199, B = 67.3801
A right triangle has hypotenuse 6 and one leg 3. Find the other leg and the angles.	SOLVED	b = 5.19615, A = 30, B = 60
A right triangle has hypotenuse 5 and one leg 7. Find the other leg.	INVALID_INPUT	Right triangle from a leg and the hypotenuse: the leg must be positive and shorter than the hypotenuse.

Right triangle from an acute angle and a side — `right_triangle_angle_side`

Given an acute angle A and one side (the hypotenuse, the side opposite A or the side adjacent to A): every side and the other angle.

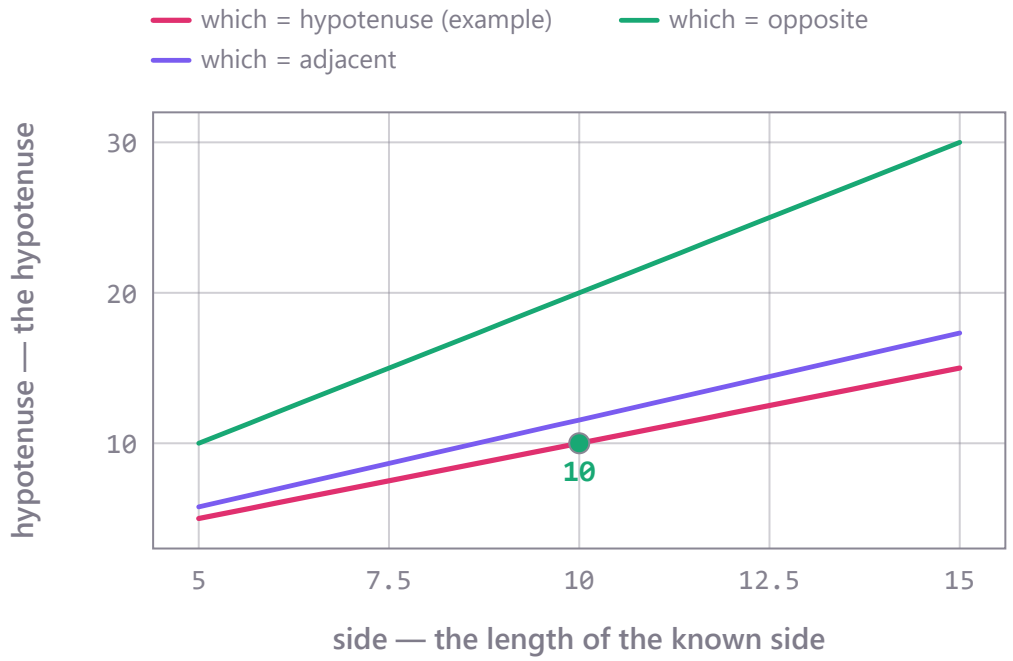
$$\sin A = \frac{\text{opp}}{\text{hyp}},\ \cos A = \frac{\text{adj}}{\text{hyp}},\ \tan A = \frac{\text{opp}}{\text{adj}}$$

input	meaning	condition
`A`	the acute angle A in degrees	

input	meaning	condition
`side`	the length of the known side	
`which`	which side is known	

output	meaning
`opposite`	side opposite A
`adjacent`	side adjacent to A
`hypotenuse`	the hypotenuse
`B`	the other acute angle, degrees

- **Method:** SOH-CAH-TOA: sin A = opp/hyp, cos A = adj/hyp, tan A = opp/adj
- **Assumes:** $0 < A < 90$; side > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



hypotenuse (the hypotenuse) against side (the length of the known side) from 5 to 15. One curve for each value of which (which side is known): hypotenuse, opposite, adjacent. Other inputs as in the worked example: A = 30. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, side = 10, hypotenuse = 10.

```
math_foundations_ai999x.solve('right_triangle_angle_side', A=30, side=10, which='hypotenuse')
# -> opposite = 5, adjacent = 8.66025, hypotenuse = 10, B = 60   (SOLVED)
```

```
math_foundations_ai999x.solve('right_triangle_angle_side', A=45, side=7, which='opposite')
# -> opposite = 7, adjacent = 7, hypotenuse = 9.89949, B = 45   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
In a right triangle, angle A is 30 degrees and the hypotenuse is 10. Find the other sides.	SOLVED	opposite = 5, adjacent = 8.66025, hypotenuse = 10, B = 60

question	outcome	result
In a right triangle, angle A is 45 degrees and the side opposite A is 7. Find the other sides.	SOLVED	opposite = 7, adjacent = 7, hypotenuse = 9.89949, B = 45
In a right triangle, angle A is 95 degrees and the hypotenuse is 10. Find the other sides.	INVALID_INPUT	Right triangle from an acute angle and a side: the acute angle must be between 0 and 90 degrees.

Law of sines (two angles and a side) — `law_of_sines_aas`

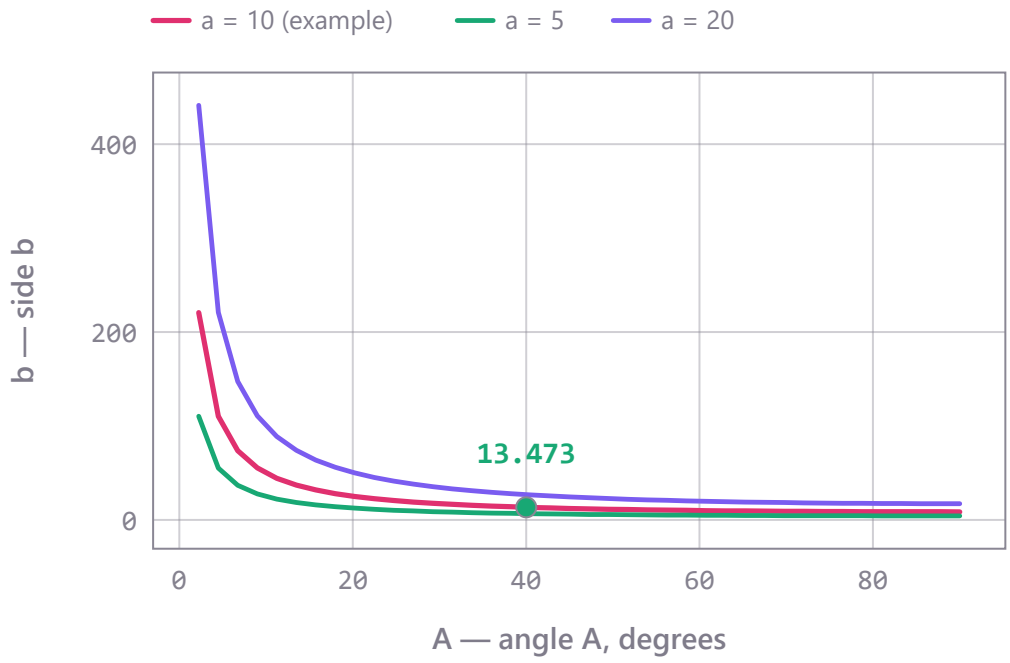
Given angles A and B and the side a opposite A: the third angle C and the sides b and c.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

input	meaning	condition
A	angle A, degrees	
B	angle B, degrees	
a	side a, opposite A	

output	meaning
C	angle C, degrees
b	side b
c	side c

- **Method:** $C = 180 - A - B$; law of sines a/sin $A = b/sin\ B = c/sin$ C
- **Assumes:** A, $B > 0$; $A + B < 180$; $a > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



b (side b) against A (angle A, degrees) from 0 to 90. One curve for each value of a (side a, opposite A): 10, 5, 20. Other inputs as in the worked example: B = 60. 40 points per curve computed by math-foundations-ai999x; marked: the worked example, A = 40, b = 13.473.

```
math_foundations_ai999x.solve('law_of_sines_aas', A=40, B=60, a=10)
# -> C = 80, b = 13.473, c = 15.3209   (SOLVED)
```

```
math_foundations_ai999x.solve('law_of_sines_aas', A=30, B=90, a=5)
# -> C = 60, b = 10, c = 8.66025   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
In triangle ABC, A = 40 degrees, B = 60 degrees and a = 10. Solve the triangle.	SOLVED	C = 80, b = 13.473, c = 15.3209
In triangle ABC, A = 30°, B = 90° and a = 5. Find the remaining sides and angles.	SOLVED	C = 60, b = 10, c = 8.66025
In triangle ABC, A = 100 degrees, B = 90 degrees and a = 5. Solve the triangle.	INVALID_INPUT	Law of sines (two angles and a side): the two angles must be positive with a sum below 180 degrees.

Law of sines, the ambiguous case (two sides and a non-included angle) — `law_of_sines_ssa`

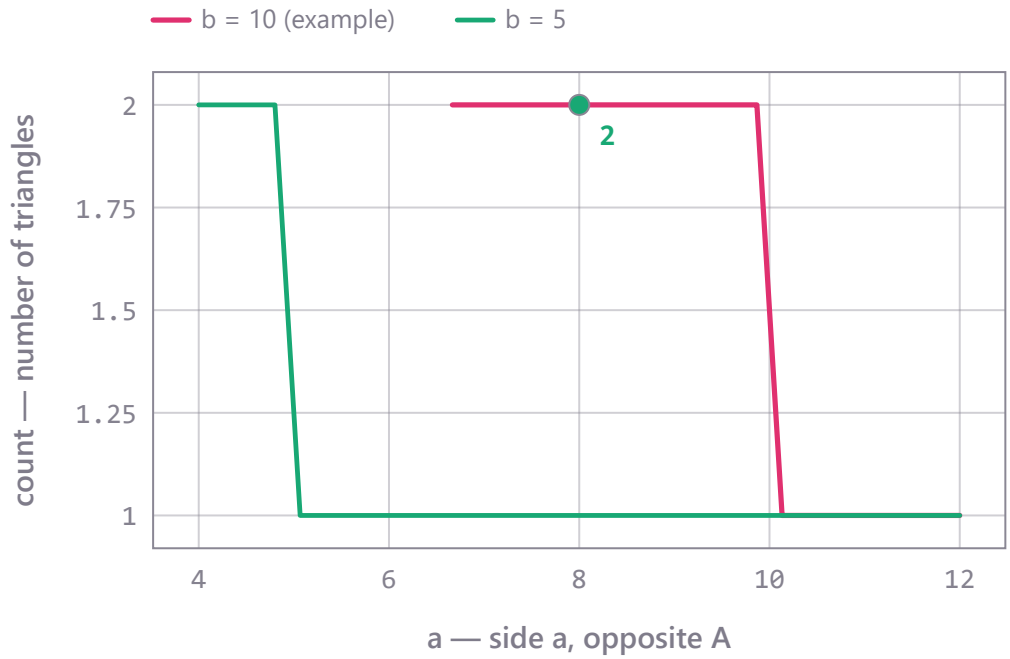
Given sides a and b and the angle A opposite a: how many triangles exist (0, 1 or 2) and every one of them.

$$\sin B = \frac{b \sin A}{a}, \quad B_2 = 180^\circ - B_1$$

input	meaning	condition
`a`	side a, opposite A	
`b`	side b	
`A`	angle A, degrees	

output	meaning
`count`	number of triangles (1 or 2; none is refused)
`B1`	angle B of the first triangle (the acute one)
`C1`	angle C of the first triangle
`c1`	side c of the first triangle
`B2`	angle B of the second triangle (obtuse), when there are two
`C2`	angle C of the second triangle
`c2`	side c of the second triangle

- **Method:** sin $B = b$ sin A / a; B and $180 - B$ both give a triangle when $A + B < 180$ (the ambiguous case)
- **Assumes:** a, $b > 0$; $0 < A < 180$; a triangle exists (else refused)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



count (number of triangles (1 or 2; none is refused)) against a (side a, opposite A) from 4 to 12. One curve for each value of b (side b): 10, 5. Other inputs as in the worked example: A = 40. 21 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 8, count = 2.

```
math_foundations_ai999x.solve('law_of_sines_ssa', a=8, b=10, A=40)
# -> count = 2, B1 = 53.4641, C1 = 86.5359, c1 = 12.423, B2 = 126.536, C2 = 13.4641, c2 = 2.89784
(SOLVED)
```

```
math_foundations_ai999x.solve('law_of_sines_ssa', a=12, b=10, A=40)
# -> count = 1, B1 = 32.3884, C1 = 107.612, c1 = 17.7937   (SOLVED)
```

```
math_foundations_ai999x.solve('law_of_sines_ssa', a=5, b=10, A=30)
# -> count = 1, B1 = 90, C1 = 60, c1 = 8.66025   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
In triangle ABC, a = 8, b = 10 and A = 40 degrees. Solve the triangle.	SOLVED	count = 2, B1 = 53.4641, C1 = 86.5359, c1 = 12.423, B2 = 126.536, C2 = 13.4641, c2 = 2.89784
In triangle ABC, a = 12, b = 10 and A = 40°. Solve the triangle.	SOLVED	count = 1, B1 = 32.3884, C1 = 107.612, c1 = 17.7937
In triangle ABC, a = 3, b = 10 and A = 40 degrees. Solve the triangle.	INVALID_INPUT	Law of sines, the ambiguous case (two sides and a non-included angle): no triangle exists: side a is shorter than the he

Law of cosines (two sides and the included angle) — `law_of_cosines_sas`

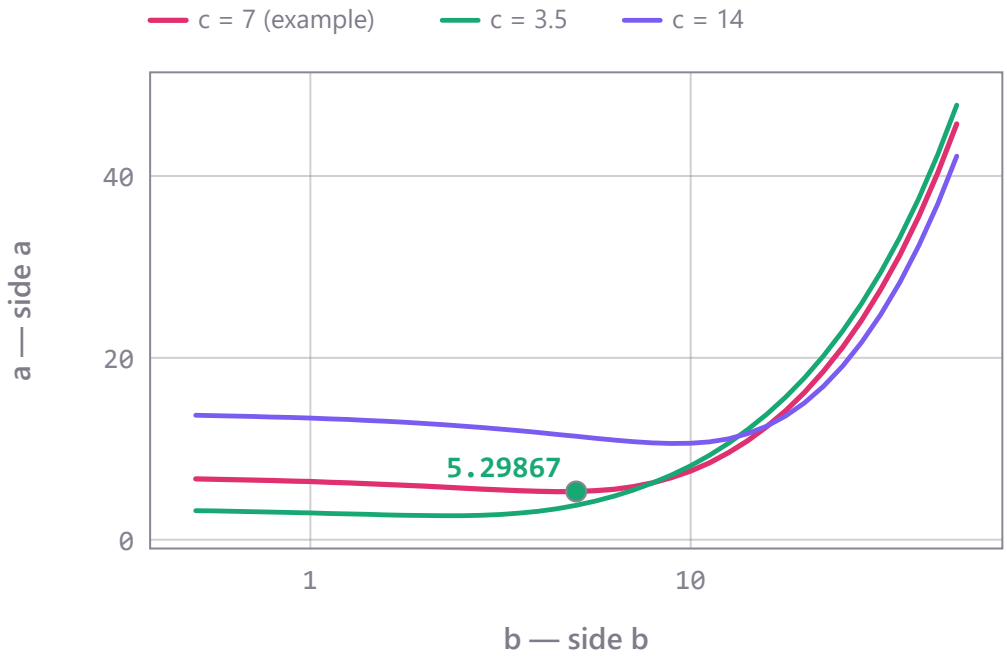
Given sides b and c and the included angle A: the third side a and the angles B and C.

$$a^2 = b^2 + c^2 - 2bc\cos A$$

input	meaning	condition
`b`	side b	
`c`	side c	
`A`	the included angle A, degrees	

output	meaning
`a`	side a
`B`	angle B, degrees
`C`	angle C, degrees

- **Method:** $a^2 = b^2 + c^2 - 2bc$ cos A; B from the law of cosines (no ambiguity)
- **Assumes:** b, $c > 0$; $0 < A < 180$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



a (side a) against b (side b) from 0.5 to 50 on a logarithmic axis. One curve for each value of c (side c): 7, 3.5, 14. Other inputs as in the worked example: A = 49. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 5, a = 5.29867.

```
math_foundations_ai999x.solve('law_of_cosines_sas', b=5, c=7, A=49)
# -> a = 5.29867, B = 45.4117, C = 85.5883   (SOLVED)
```

```
math_foundations_ai999x.solve('law_of_cosines_sas', b=3, c=4, A=90)
# -> a = 5, B = 36.8699, C = 53.1301   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
In triangle ABC, b = 5, c = 7 and A = 49 degrees. Solve the triangle.	SOLVED	a = 5.29867, B = 45.4117, C = 85.5883

question	outcome	result
In triangle ABC, b = 3, c = 4 and A = 90°. Find the remaining sides and angles.	SOLVED	a = 5, B = 36.8699, C = 53.1301
In triangle ABC, b = 3, c = 4 and A = 190 degrees. Solve the triangle.	INVALID_INPUT	Law of cosines (two sides and the included angle): the sides must be positive and A between 0 and 180 degrees.

Law of cosines (three sides) — `law_of_cosines_sss`

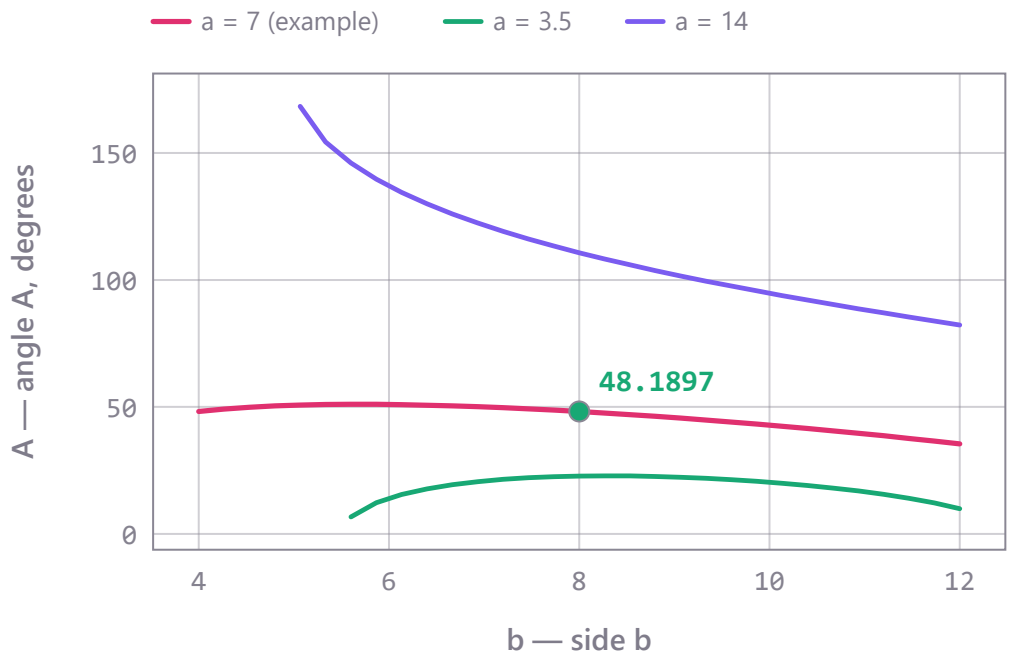
Given the three sides a, b, c: the angles A, B, C.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

input	meaning	condition
a	side a	
b	side b	
c	side c	

output	meaning
A	angle A, degrees
B	angle B, degrees
C	angle C, degrees

- **Method:** cos $A = (b^2 + c^2 - a^2)/(2bc)$, likewise B; $C = 180 - A - B$
- **Assumes:** positive sides; triangle inequality (each side shorter than the sum of the other two)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



A (angle A, degrees) against b (side b) from 4 to 12. One curve for each value of a (side a): 7, 3.5, 14. Other inputs as in the worked example: c = 9. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 8, A = 48.1897.

```
math_foundations_ai999x.solve('law_of_cosines_sss', a=7, b=8, c=9)
# -> A = 48.1897, B = 58.4119, C = 73.3985   (SOLVED)
```

```
math_foundations_ai999x.solve('law_of_cosines_sss', a=3, b=4, c=5)
# -> A = 36.8699, B = 53.1301, C = 90   (SOLVED)
```

```
math_foundations_ai999x.solve('law_of_cosines_sss', a=2, b=2, c=2)
# -> A = 60, B = 60, C = 60   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
In triangle ABC, a = 7, b = 8 and c = 9. Find the angles.	SOLVED	A = 48.1897, B = 58.4119, C = 73.3985
In triangle ABC, a = 3, b = 4 and c = 5. Solve the triangle.	SOLVED	A = 36.8699, B = 53.1301, C = 90
In triangle ABC, a = 1, b = 2 and c = 5. Find the angles.	INVALID_INPUT	Law of cosines (three sides): these lengths do not form a triangle (each side must be shorter than the sum of the other

Area of a triangle from its three sides (Heron) — `heron_area`

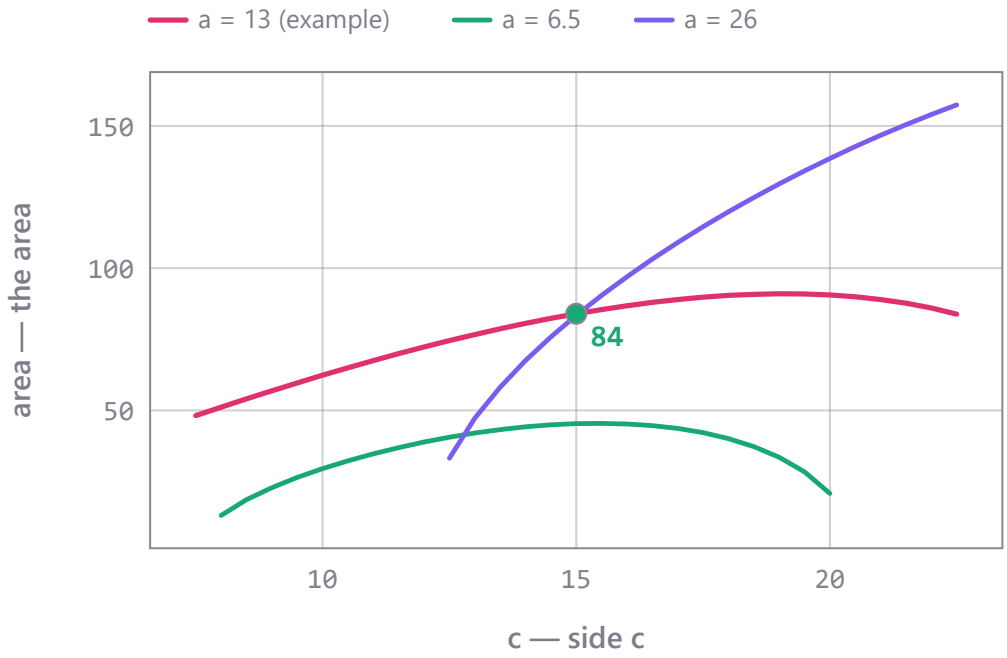
The area of a triangle with sides a, b, c, and its semi-perimeter s.

$$K = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}$$

input	meaning	condition
`a`	side a	
`b`	side b	
`c`	side c	

output	meaning
`area`	the area
`s`	the semi-perimeter $(a+b+c)/2$

- **Method:** Heron's formula, in its numerically stable arrangement
- **Assumes:** positive sides; triangle inequality
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



area (the area) against c (side c) from 7.5 to 22.5. One curve for each value of a (side a): 13, 6.5, 26. Other inputs as in the worked example: b = 14. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 15, area = 84.

```
math_foundations_ai999x.solve('heron_area', a=13, b=14, c=15)
# -> area = 84, s = 21   (SOLVED)
```

```
math_foundations_ai999x.solve('heron_area', a=3, b=4, c=5)
# -> area = 6, s = 6   (SOLVED)
```

```
math_foundations_ai999x.solve('heron_area', a=5, b=6, c=7)
# -> area = 14.6969, s = 9   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the area of a triangle with sides 13, 14 and 15.	SOLVED	area = 84, s = 21
Use Heron's formula to find the area of a triangle with sides 5, 6 and 7.	SOLVED	area = 14.6969, s = 9
Find the area of a triangle with sides 1, 2 and 3.	INVALID_INPUT	Area of a triangle from its three sides (Heron): these lengths do not form a triangle (each side must be shorter than th

Exact trigonometric values on the unit circle — `unit_circle_exact`

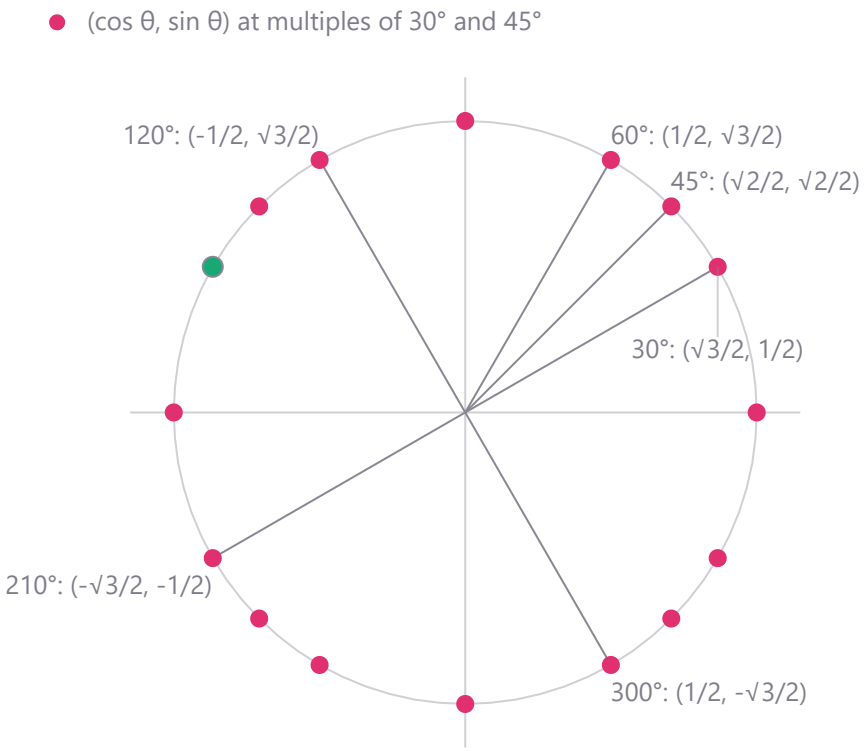
The exact sin, cos and tan of an angle that is a multiple of 30 or 45 degrees (pi/6, pi/4), with the reference angle.

$$\sin 30^\circ = \tfrac{1}{2},\ \sin 45^\circ = \tfrac{\sqrt{2}}{2},\ \sin 60^\circ = \tfrac{\sqrt{3}}{2}$$

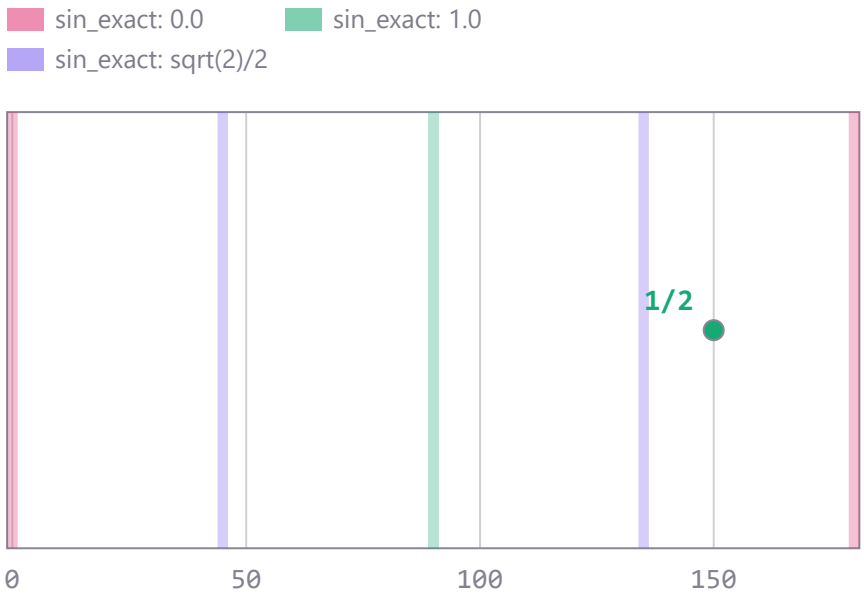
input	meaning	condition
`deg`	the angle in degrees (a multiple of 30 or 45)	

output	meaning
`sin`	sin of the angle
`cos`	cos of the angle
`tan`	tan of the angle (None where undefined)
`sin_exact`	exact sin
`cos_exact`	exact cos
`tan_exact`	exact tan
`reference_angle`	the reference angle, degrees

- **Method:** the reference angle's exact values from the $30-60-90$ and $45-45-90$ triangles, with the signs of the quadrant (ASTC)
- **Assumes:** a whole multiple of 30 or 45 degrees
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The unit circle with the exact cosine and sine the package returns at the multiples of 30° and 45° (16 angles; six labelled); for example cos 150° = -√3/2 = -0.866025 and sin 150° = 1/2 = 0.5 (marked).



exact sin (sin_exact) along deg (the angle in degrees (a multiple of 30 or 45)) from 0 to 180: 81 runs of math-foundations-ai999x, each band coloured by the verdict returned (0.0, 1.0, sqrt(2)/2). Marked: the worked example, deg = 150 → 1/2.

```
math_foundations_ai999x.solve('unit_circle_exact', deg=150)
# -> sin = 0.5, cos = -0.866025, tan = -0.57735, sin_exact = 1/2, cos_exact = -sqrt(3)/2, tan_exact
= -sqrt(3)/3, reference_angle = 30   (SOLVED)
```

```
math_foundations_ai999x.solve('unit_circle_exact', deg=225)
# -> sin = -0.707107, cos = -0.707107, tan = 1, sin_exact = -sqrt(2)/2, cos_exact = -sqrt(2)/2,
tan_exact = 1, reference_angle = 45   (SOLVED)
```

```
math_foundations_ai999x.solve('unit_circle_exact', deg=90)
# -> sin = 1, cos = 0, tan = none, sin_exact = 1, cos_exact = 0, tan_exact = undefined,
reference_angle = 90   (SOLVED)
```

```
math_foundations_ai999x.solve('unit_circle_exact', deg=-60)
# -> sin = -0.866025, cos = 0.5, tan = -1.73205, sin_exact = -sqrt(3)/2, cos_exact = 1/2, tan_exact
= -sqrt(3), reference_angle = 60   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the exact values of sin, cos and tan of 150 degrees.	SOLVED	sin = 0.5, cos = -0.866025, tan = -0.57735, sin_exact = 1/2, cos_exact = -sqrt(3)/2, tan_exact = -sqrt(3)/3, reference_angle = 30
Find the exact value of sin(5pi/6).	SOLVED	sin = 0.5, cos = -0.866025, tan = -0.57735, sin_exact = 1/2, cos_exact = -sqrt(3)/2, tan_exact = -sqrt(3)/3, reference_angle = 30
What is cos(225°) exactly?	SOLVED	sin = -0.707107, cos = -0.707107, tan = 1, sin_exact = -sqrt(2)/2, cos_exact = -sqrt(2)/2, tan_exact = 1, reference_angle = 45
Find the exact value of tan(20 degrees).	INVALID_INPUT	Exact trigonometric values on the unit circle: exact values are given for multiples of 30 or 45 degrees (pi/6 or pi/4).

Solve a trigonometric equation over one full turn — `trig_equation_interval`

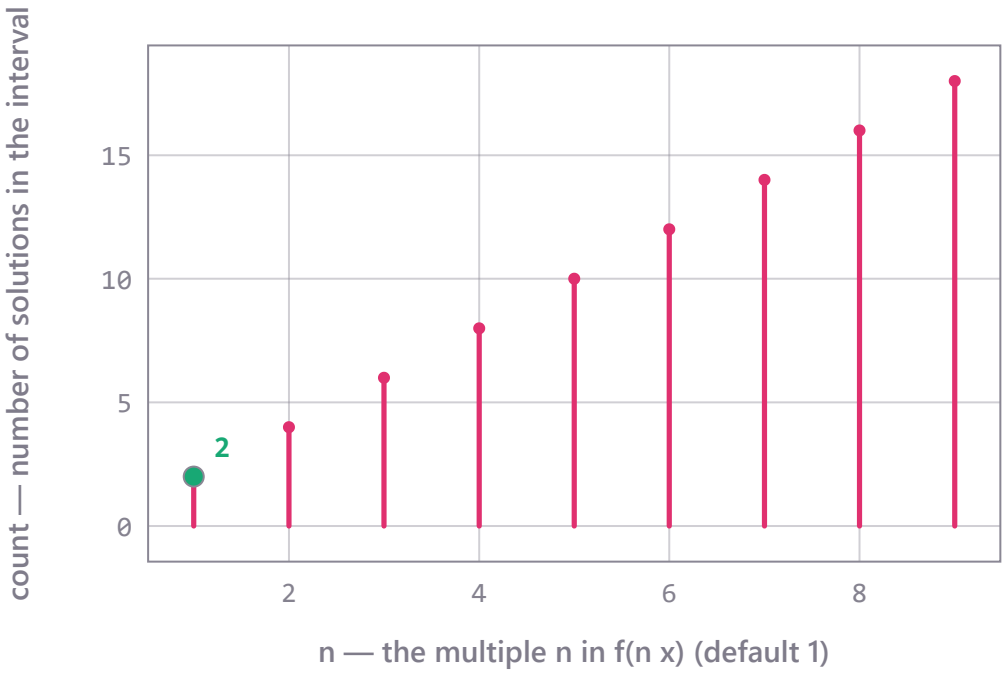
Every solution of $asin$(n x) = k (or cos, tan) with $0 \le x < 360$ degrees or $0 \le x < 2$ pi.

$$\sin u = k \Rightarrow u = \arcsin k,\ 180^\circ - \arcsin k \pmod{360^\circ}, \quad u = nx$$

input	meaning	condition
`func`	sin, cos or tan	
`k`	the value the function must take (after dividing by its coefficient)	
`n`	the multiple n in f(n x) (default 1)	
`unit`	degrees ([0, 360)) or radians ([0, 2pi))	

output	meaning
`count`	number of solutions in the interval
`solutions`	the solutions, ascending

- **Method:** principal value (asin, acos or atan), the second solution of the period (180 - a, -a, a + 180), then every period: x = (u + 360 m)/n inside the interval
- **Assumes:** n a whole number from 1 to 12
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



count (number of solutions in the interval) against n (the multiple n in f(n x) (default 1)) from 1 to 9. Other inputs as in the worked example: func = sin, k = 0.5, unit = degrees. 9 points computed by math-foundations-ai999x; marked: the worked example, n = 1, count = 2.

```
math_foundations_ai999x.solve('trig_equation_interval', func='sin', k=0.5, n=1, unit='degrees')
# -> count = 2, solutions = [30.000000000000004, 150.0]   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_equation_interval', func='cos', k=0.5, n=1, unit='radians')
# -> count = 2, solutions = [1.0471975511965979, 5.235987755982989]   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_equation_interval', func='tan', k=1, n=2, unit='degrees')
# -> count = 4, solutions = [22.5, 112.5, 202.5, 292.5]   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_equation_interval', func='sin', k=1, n=1, unit='degrees')
# -> count = 1, solutions = [90.0]   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_equation_interval', func='cos', k=2, n=1, unit='degrees')
# -> count = 0, solutions = []   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve sin(x) = 0.5 for 0 <= x < 360 degrees.	SOLVED	count = 2, solutions = [30.000000000000004, 150.0]
Solve 2cos(x) = 1 on [0, 2pi).	SOLVED	count = 2, solutions = [1.0471975511965979, 5.235987755982989]
Solve tan(2x) = 1 for 0 <= x < 360 degrees.	SOLVED	count = 4, solutions = [22.5, 112.5, 202.5, 292.5]
Solve sin(x) = 0 for 0 ≤ x < 2π.	SOLVED	count = 2, solutions = [0.0, 3.141592653589793]
Solve 3sin(x) = 4 for 0 <= x < 360 degrees.	SOLVED	count = 0, solutions = []

Trigonometric values from one value and the quadrant — `trig_values_from_one`

Given sin, cos or tan of an angle and its quadrant: sin, cos, tan, sin 2t and cos 2t (Pythagorean and double-angle identities).

$$\cos t = \pm\sqrt{1-\sin^2 t}, \quad \sin 2t = 2\sin t\cos t, \quad \cos 2t = \cos^2 t - \sin^2 t$$

input	meaning	condition
`func`	the function whose value is known	
`value`	its value	
`quadrant`	the quadrant of the angle (1 to 4)	

output	meaning
`sin`	sin t
`cos`	cos t
`tan`	tan t
`sin2`	sin 2t
`cos2`	cos 2t

- **Method:** $sin^2 + cos^2 = 1$ with the quadrant's signs (ASTC); sin $2t = 2$ sin t cos t; cos $2t = cos^2 t - sin^2$ t
- **Assumes:** the value's sign agrees with the quadrant; 0 < |sin|, |cos| < 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



cos (cos t) against value (its value) from 0.3 to 0.9. Other inputs as in the worked example: func = sin, quadrant = 2. 31 points computed by math-foundations-ai999x; marked: the worked example, value = 0.6, cos = -0.8.

```
math_foundations_ai999x.solve('trig_values_from_one', func='sin', value=0.6, quadrant=2)
# -> sin = 0.6, cos = -0.8, tan = -0.75, sin2 = -0.96, cos2 = 0.28   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_values_from_one', func='cos', value=-0.38461538461538464,
quadrant=3)
# -> sin = -0.923077, cos = -0.384615, tan = 2.4, sin2 = 0.710059, cos2 = -0.704142   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_values_from_one', func='tan', value=-2, quadrant=4)
# -> sin = -0.894427, cos = 0.447214, tan = -2, sin2 = -0.8, cos2 = -0.6   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
If sin(t) = 3/5 and t is in quadrant II, find cos(t), tan(t), sin(2t) and cos(2t).	SOLVED	sin = 0.6, cos = -0.8, tan = -0.75, sin2 = -0.96, cos2 = 0.28
Given cos(x) = -5/13 and x lies in the third quadrant, find sin(x) and tan(x).	SOLVED	sin = -0.923077, cos = -0.384615, tan = 2.4, sin2 = 0.710059, cos2 = -0.704142
If sin(t) = 3/5 and t is in quadrant III, find cos(t).	INVALID_INPUT	Trigonometric values from one value and the quadrant: sine is negative in quadrant 3.

Dot product and angle between two vectors — `vector_dot_angle`

The dot product, the magnitudes and the angle between two vectors in the plane or in space.

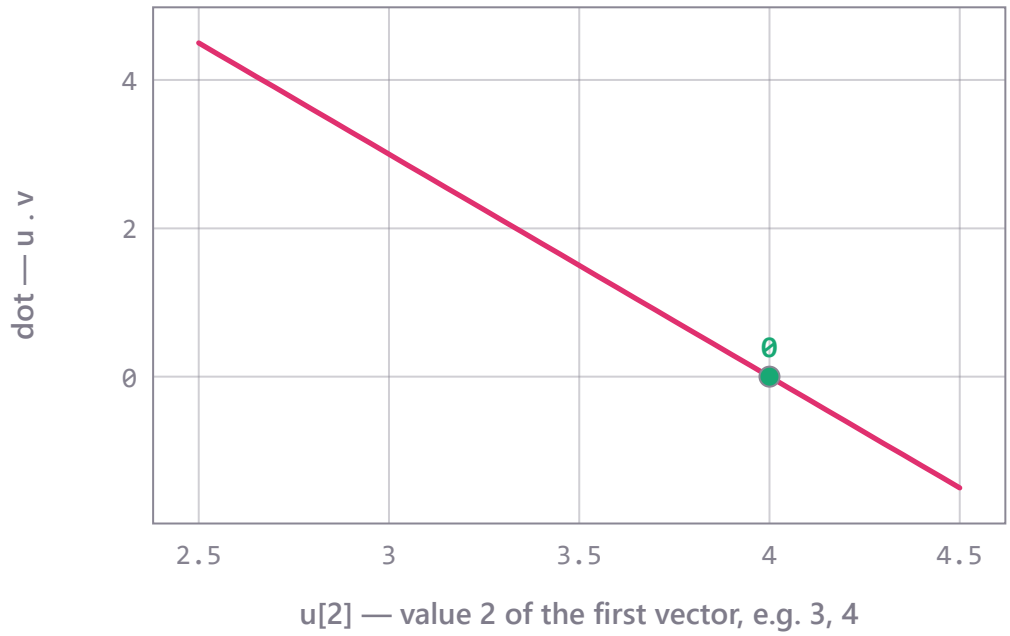
$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos\theta$$

input	meaning	condition
`u`	the first vector, e.g. 3, 4	

input	meaning	condition
`v`	the second vector	

output	meaning
`dot`	u . v
`mag_u`	|u|
`mag_v`	|v|
`angle_deg`	the angle between them, degrees (0 to 180)

- **Method:** u . v = sum of products of components; the angle from atan2(|u x v|, u . v) (stable for nearly parallel vectors)
- **Assumes:** same dimension (2 or 3); neither vector is zero
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dot (u . v) as value 2 of u (the first vector, e.g. 3, 4) moves from 2.5 to 4.5, the other values as given (3, 4); 41 runs of math-foundations-ai999x. Other inputs: v = 4, -3. Marked: the worked example, u[2] = 4, dot = 0.

```
math_foundations_ai999x.solve('vector_dot_angle', u='3, 4', v='4, -3')
# -> dot = 0, mag_u = 5, mag_v = 5, angle_deg = 90   (SOLVED)
```

```
math_foundations_ai999x.solve('vector_dot_angle', u=[1, 2, 3], v=[4, -5, 6])
# -> dot = 12, mag_u = 3.74166, mag_v = 8.77496, angle_deg = 68.5624   (SOLVED)
```

```
math_foundations_ai999x.solve('vector_dot_angle', u=[1, 0], v=[1, 1])
# -> dot = 1, mag_u = 1, mag_v = 1.41421, angle_deg = 45   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the angle between u = <3, 4> and v = <4, -3>.	SOLVED	dot = 0, mag_u = 5, mag_v = 5, angle_deg = 90
Find the dot product of <1, 2, 3> and <4, -5, 6>.	SOLVED	dot = 12, mag_u = 3.74166, mag_v = 8.77496, angle_deg = 68.5624
Find the dot product and the angle between the vectors (1, 0) and (1, 1).	SOLVED	dot = 1, mag_u = 1, mag_v = 1.41421, angle_deg = 45
Find the angle between u = <0, 0> and v = <1, 2>.	INVALID_INPUT	Dot product and angle between two vectors: the zero vector has no direction, so the angle is not defined.

Magnitude and direction of a vector — `vector_magnitude_direction`

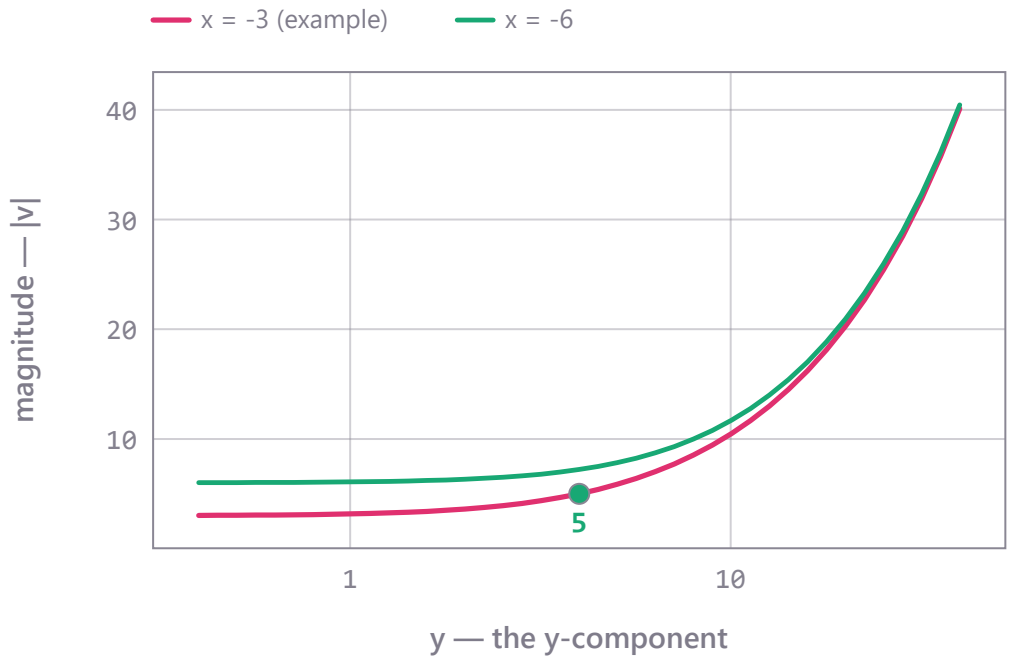
The length of the plane vector <x, y> and its direction angle from the positive x-axis (0 to 360 degrees).

$$|\mathbf{v}| = \sqrt{x^2 + y^2}, \quad \theta = \operatorname{atan2}(y, x)$$

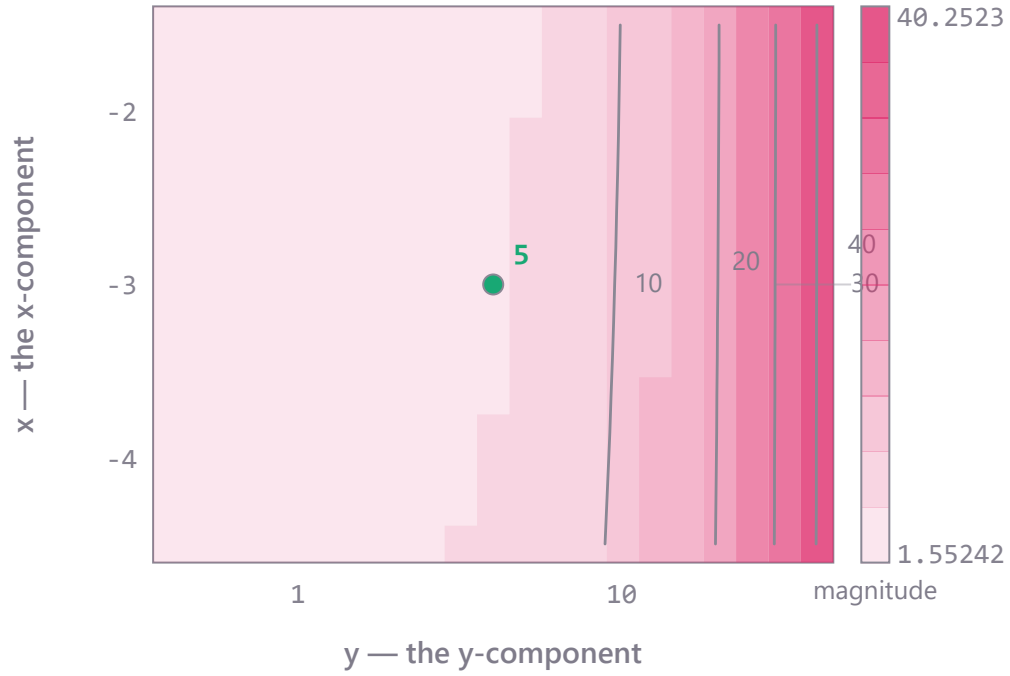
input	meaning	condition
`x`	the x-component	
`y`	the y-component	

output	meaning
`magnitude`	|v|
`direction_deg`	the direction angle, degrees in [0, 360)

- **Method:** |v| = sqrt(x^2 + y^2); the angle in the quadrant of (x, y)
- **Assumes:** not the zero vector
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



magnitude (|v|) against y (the y-component) from 0.4 to 40 on a logarithmic axis. One curve for each value of x (the x-component): -3, -6. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, y = 4, magnitude = 5.



magnitude (|v|) over y (the y-component) from 0.4 to 40 and x (the x-component) from -4.5 to -1.5 (logarithmic horizontal axis): each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 1.55242 to dark 40.2523); iso-lines at magnitude = 10, 20, 30, 40. Marked: the worked example, y = 4, x = -3, magnitude = 5.

```
math_foundations_ai999x.solve('vector_magnitude_direction', x=-3, y=4)
# -> magnitude = 5, direction_deg = 126.87   (SOLVED)
```

```
math_foundations_ai999x.solve('vector_magnitude_direction', x=5, y=-12)
# -> magnitude = 13, direction_deg = 292.62   (SOLVED)
```

```
math_foundations_ai999x.solve('vector_magnitude_direction', x=0, y=-2)
# -> magnitude = 2, direction_deg = 270   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the magnitude and direction of the vector <-3, 4>.	SOLVED	magnitude = 5, direction_deg = 126.87
Find the magnitude and direction angle of v = <0, -2>.	SOLVED	magnitude = 2, direction_deg = 270
Find the magnitude and direction of the vector <0, 0>.	INVALID_INPUT	Magnitude and direction of a vector: the zero vector has no direction.

Polar to rectangular coordinates — `polar_to_rectangular`

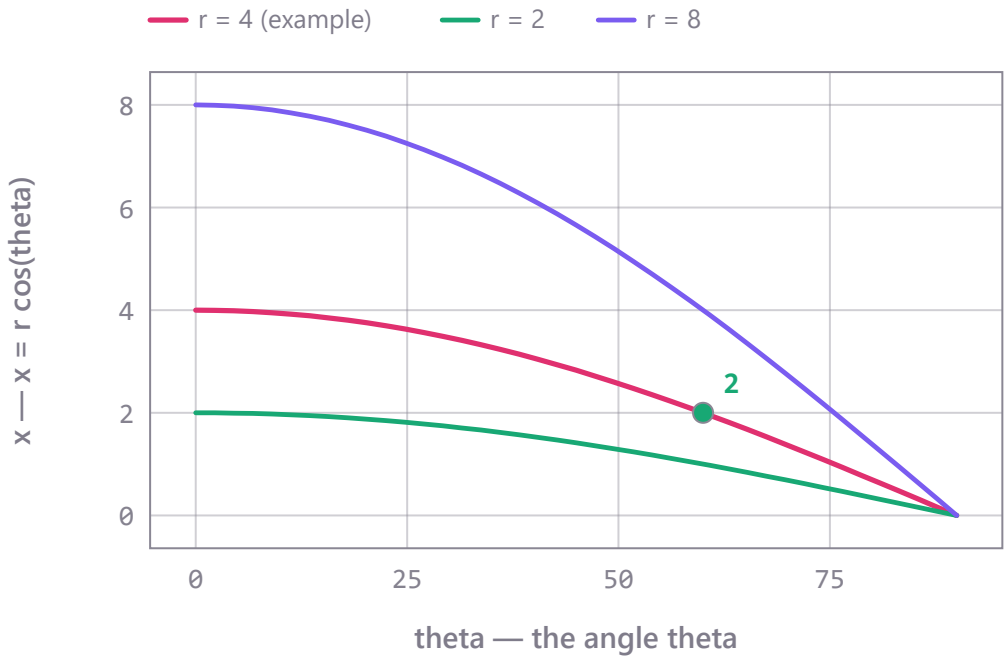
The point (x, y) with polar coordinates (r, theta).

$$x = r\cos\theta, \quad y = r\sin\theta$$

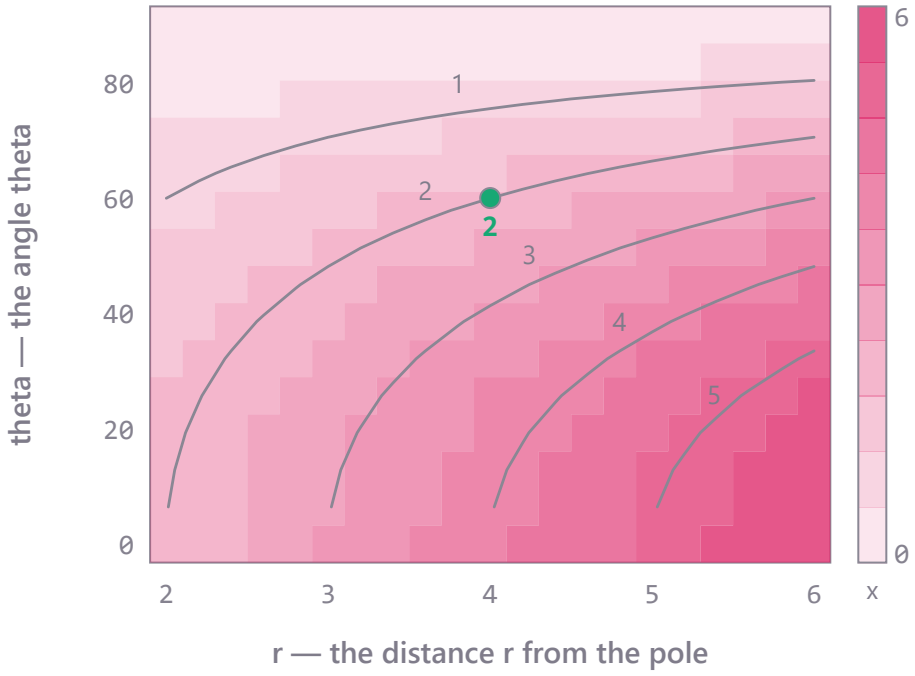
input	meaning	condition
`r`	the distance r from the pole	
`theta`	the angle theta	
`unit`	degrees or radians	

output	meaning
`x`	$x = r$ cos(theta)
`y`	$y = r$ sin(theta)

- **Method:** $x = r$ cos(theta), $y = r$ sin(theta)
- **Assumes:** the angle's unit stated
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x (x = r cos(theta)) against theta (the angle theta) from 0 to 90. One curve for each value of r (the distance r from the pole): 4, 2, 8. Other inputs as in the worked example: unit = degrees. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, theta = 60, x = 2.



x (x = r cos(theta)) over r (the distance r from the pole) from 2 to 6 and theta (the angle theta) from 0 to 90: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 0 to dark 6); iso-lines at x = 1, 2, 3, 4, 5. Other inputs as in the worked example: unit = degrees. Marked: the worked example, r = 4, theta = 60, x = 2.

```
math_foundations_ai999x.solve('polar_to_rectangular', r=4, theta=60, unit='degrees')
# -> x = 2, y = 3.4641   (SOLVED)
```

```
math_foundations_ai999x.solve('polar_to_rectangular', r=2, theta=2.356194490192345, unit='radians')
# -> x = -1.41421, y = 1.41421   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Convert the polar point (4, 60 degrees) to rectangular coordinates.	SOLVED	x = 2, y = 3.4641
Convert (2, 3pi/4) from polar to rectangular coordinates.	SOLVED	x = -1.41421, y = 1.41421
Convert the polar point (4, 60) to rectangular coordinates.	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: math_foundations_ai999x.templates

Rectangular to polar coordinates — `rectangular_to_polar`

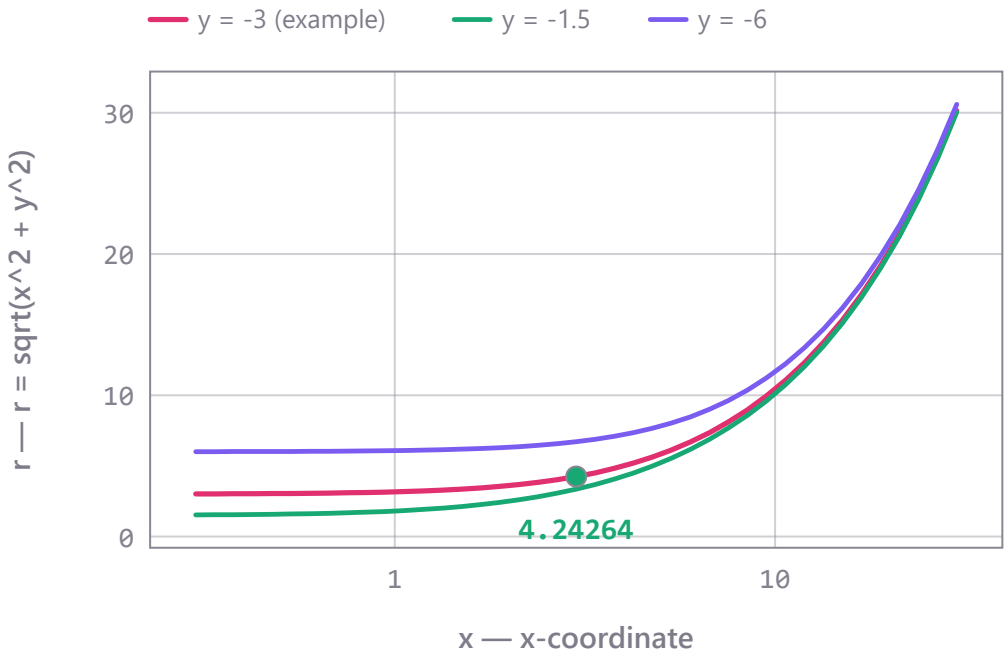
Polar coordinates (r, theta) of the point (x, y), with $r > 0$ and theta in [0, 360) degrees / [0, 2 pi).

$$r = \sqrt{x^2 + y^2}, \quad \tan\theta = \frac{y}{x}$$

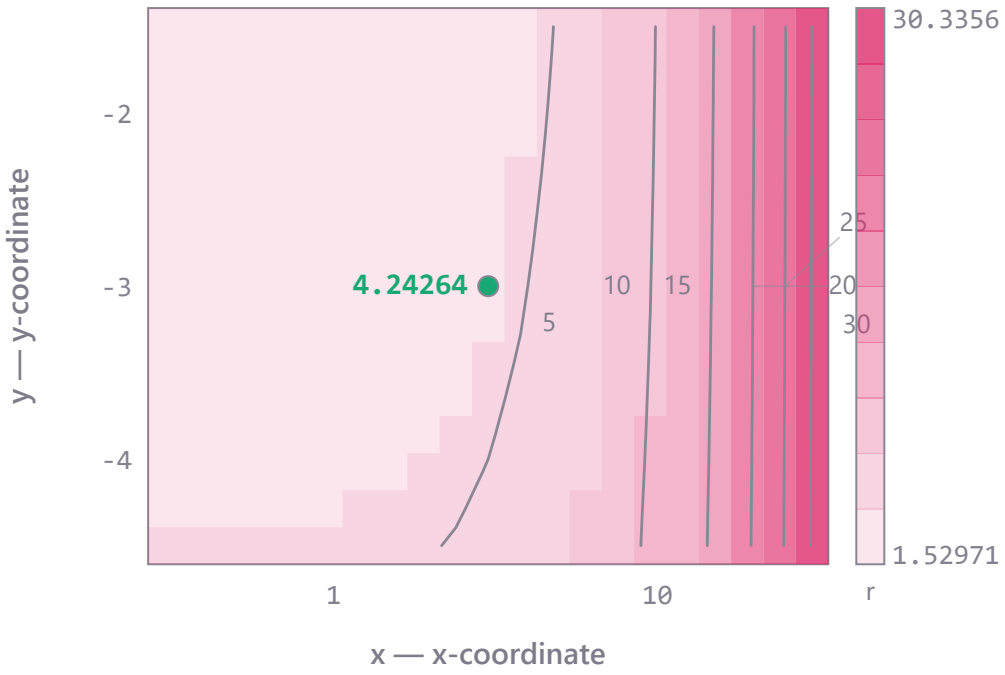
input	meaning	condition
`x`	x-coordinate	
`y`	y-coordinate	

output	meaning
`r`	$r = \sqrt{x^2 + y^2}$
`theta_deg`	theta in degrees, [0, 360)
`theta_rad`	theta in radians, [0, 2 pi)

- **Method:** $r = \sqrt{x^2 + y^2}$; theta in the quadrant of (x, y)
- **Assumes:** not the origin
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



r (r = sqrt(x^2 + y^2)) against x (x-coordinate) from 0.3 to 30 on a logarithmic axis. One curve for each value of y (y-coordinate): -3, -1.5, -6. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, x = 3, r = 4.24264.



r (r = sqrt(x^2 + y^2)) over x (x-coordinate) from 0.3 to 30 and y (y-coordinate) from -4.5 to -1.5 (logarithmic horizontal axis): each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 1.52971 to dark 30.3356); iso-lines at r = 5, 10, 15, 20, 25, 30. Marked: the worked example, x = 3, y = -3, r = 4.24264.

```
math_foundations_ai999x.solve('rectangular_to_polar', x=3, y=-3)
# -> r = 4.24264, theta_deg = 315, theta_rad = 5.49779   (SOLVED)
```

```
math_foundations_ai999x.solve('rectangular_to_polar', x=-1, y=1)
# -> r = 1.41421, theta_deg = 135, theta_rad = 2.35619   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Convert the point (3, -3) to polar coordinates.	SOLVED	r = 4.24264, theta_deg = 315, theta_rad = 5.49779
Find the polar coordinates of (-1, 1).	SOLVED	r = 1.41421, theta_deg = 135, theta_rad = 2.35619
Convert the point (0, 0) to polar coordinates.	INVALID_INPUT	Rectangular to polar coordinates: the origin has r = 0 and no defined angle.

Focus and directrix of a parabola — `parabola_focus_directrix`

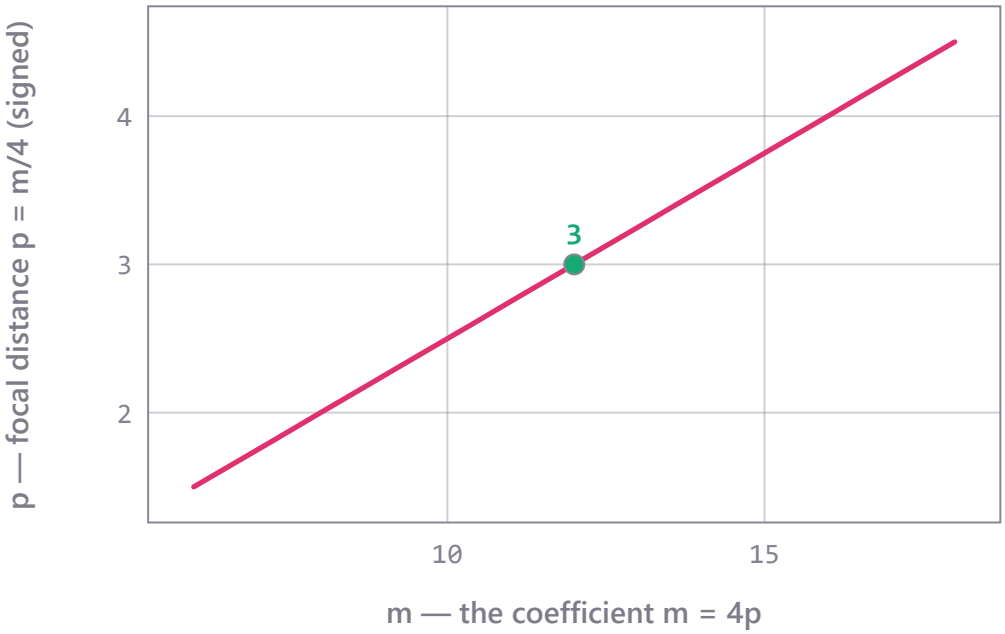
Vertex, focal distance p, focus and directrix of $(x-h)^2 = 4p$(y - k) or $(y-k)^2 = 4p$(x - h).

$$(x-h)^2 = 4p(y-k):\ F = (h, k+p),\ y = k-p$$

input	meaning	condition
`orientation`	vertical axis ($(x-h)^2 = m$(y-k)) or horizontal	
`m`	the coefficient $m = 4p$	$m \neq 0$
`h`	x of the vertex	
`k`	y of the vertex	

output	meaning
`p`	focal distance $p = m/4$ (signed)
`focus_x`	x of the focus
`focus_y`	y of the focus
`directrix`	the directrix: y = this (vertical axis) or x = this (horizontal axis)
`vertex_x`	x of the vertex
`vertex_y`	y of the vertex

- **Method:** compare with $(x - h)^2 = 4p$(y - k): $p = m/4$, focus at distance p from the vertex along the axis
- **Assumes:** $m \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



p (focal distance p = m/4 (signed)) against m (the coefficient m = 4p) from 6 to 18. Other inputs as in the worked example: orientation = vertical. 31 points computed by math-foundations-ai999x; marked: the worked example, m = 12, p = 3.

```
math_foundations_ai999x.solve('parabola_focus_directrix', orientation='vertical', m=12)
# -> p = 3, focus_x = 0, focus_y = 3, directrix = -3, vertex_x = 0, vertex_y = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('parabola_focus_directrix', orientation='horizontal', m=-8, h=3, k=1)
# -> p = -2, focus_x = 1, focus_y = 1, directrix = 5, vertex_x = 3, vertex_y = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('parabola_focus_directrix', orientation='vertical', m=0.5)
# -> p = 0.125, focus_x = 0, focus_y = 0.125, directrix = -0.125, vertex_x = 0, vertex_y = 0
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the focus and directrix of the parabola x^2 = 12y.	SOLVED	p = 3, focus_x = 0, focus_y = 3, directrix = -3, vertex_x = 0, vertex_y = 0
Find the focus and directrix of (y - 1)^2 = -8(x - 3).	SOLVED	p = -2, focus_x = 1, focus_y = 1, directrix = 5, vertex_x = 3, vertex_y = 1
Find the focus and directrix of y = 2x^2.	SOLVED	p = 0.125, focus_x = 0, focus_y = 0.125, directrix = -0.125, vertex_x = 0, vertex_y = 0
Find the focus and directrix of the parabola (x + 2)^2 = 4(y - 5).	SOLVED	p = 1, focus_x = -2, focus_y = 6, directrix = 4, vertex_x = -2, vertex_y = 5

Foci, axes and eccentricity of an ellipse — `ellipse_parameters`

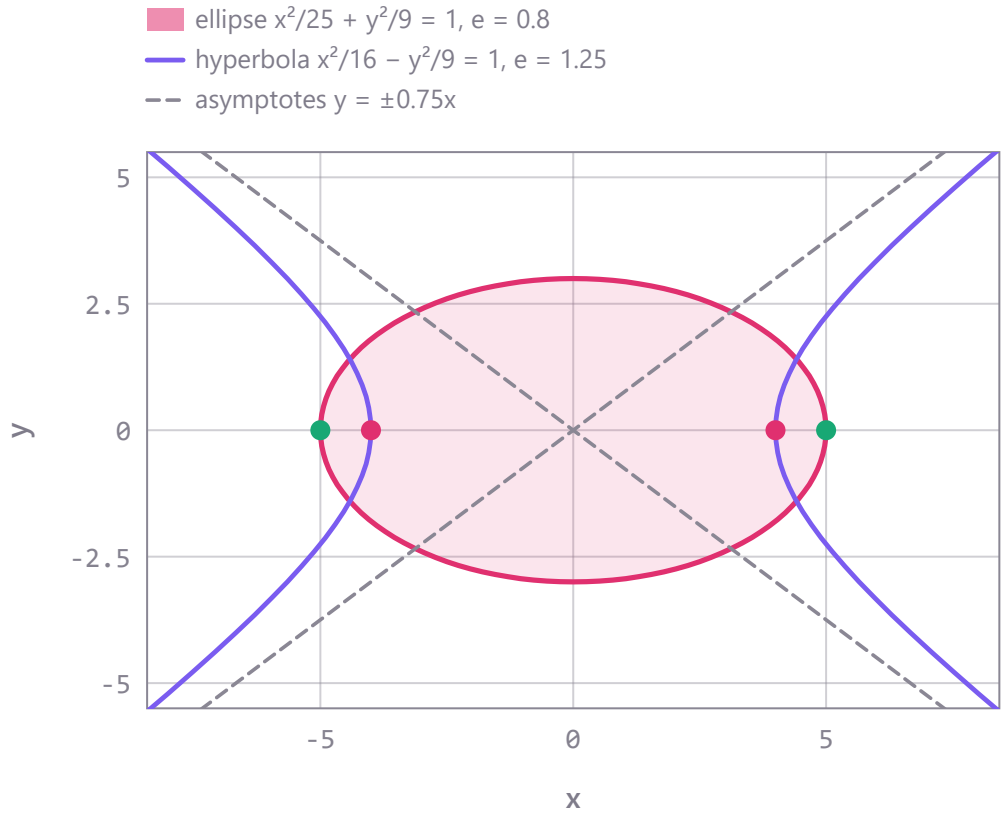
For $(x-h)^2/A + (y-k)^2/B = 1$: semi-axes a and b, focal distance c, eccentricity and the two foci.

$$\frac{(x-h)^2}{A} + \frac{(y-k)^2}{B} = 1, \quad c^2 = a^2 - b^2, \quad e = \frac{c}{a}$$

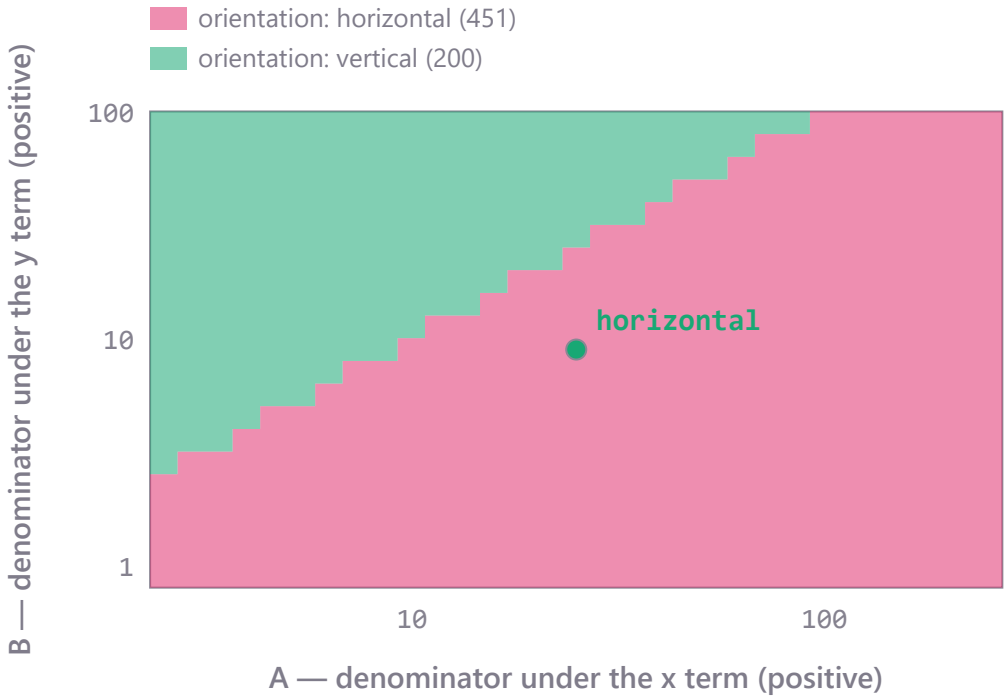
input	meaning	condition
`h`	x of the centre	
`k`	y of the centre	
`A`	denominator under the x term (positive)	
`B`	denominator under the y term (positive)	

output	meaning
`a`	semi-major axis
`b`	semi-minor axis
`c`	centre-to-focus distance
`eccentricity`	$e = c/a$
`f1x`	first focus x
`f1y`	first focus y
`f2x`	second focus x
`f2y`	second focus y
`orientation`	the major axis is horizontal or vertical

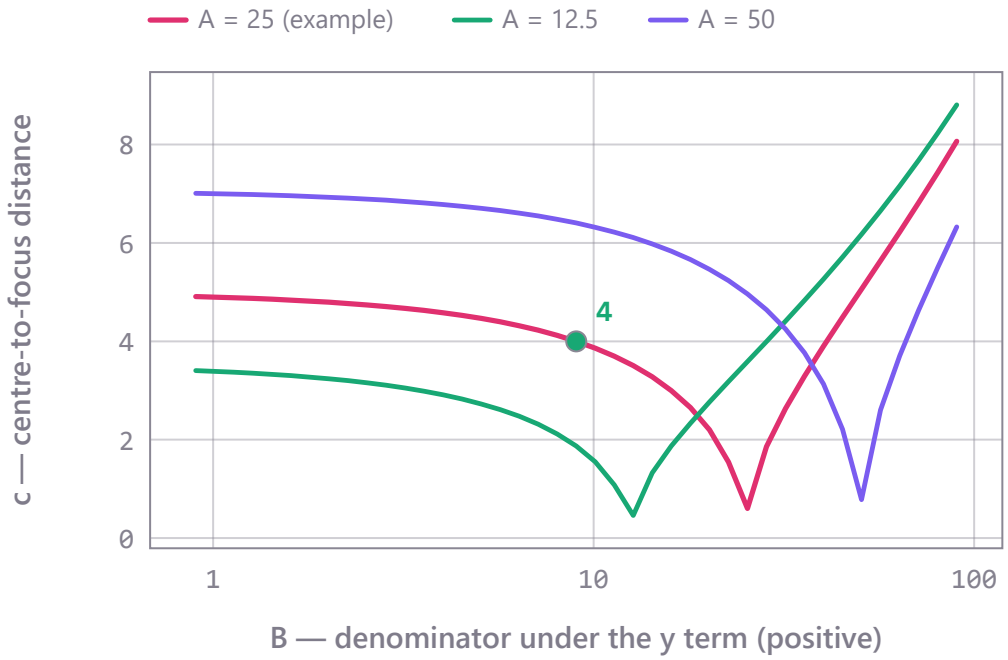
- **Method:** a^2 = the larger denominator, b^2 = the smaller, $c^2 = a^2 - b^2$; foci on the major axis
- **Assumes:** A, $B > 0$; $A \neq B$ (otherwise a circle)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



An ellipse and a hyperbola drawn from the parameters the package returns: ellipse a = 5, b = 3, foci (±4, 0), eccentricity 0.8; hyperbola a = 4, b = 3, foci (±5, 0), eccentricity 1.25, asymptotes of slope ±0.75.



the major axis is horizontal or vertical (orientation) over A (denominator under the x term (positive)) from 2.5 to 250 and B (denominator under the y term (positive)) from 0.9 to 90 (logarithmic axes): each of the 31 × 21 cells is one run of math-foundations-ai999x, coloured by the verdict it returned — horizontal: 451 cells, vertical: 200 cells. Other inputs as in the worked example: h = 0, k = 0. Marked: the worked example, A = 25, B = 9 → horizontal.



c (centre-to-focus distance) against B (denominator under the y term (positive)) from 0.9 to 90 on a logarithmic axis. One curve for each value of A (denominator under the x term (positive)): 25, 12.5, 50. Other inputs as in the worked example: h = 0, k = 0. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, B = 9, c = 4.

```
math_foundations_ai999x.solve('ellipse_parameters', h=0, k=0, A=25, B=9)
# -> a = 5, b = 3, c = 4, eccentricity = 0.8, f1x = -4, f1y = 0, f2x = 4, f2y = 0, orientation = horizontal   (SOLVED)
```

```
math_foundations_ai999x.solve('ellipse_parameters', h=1, k=-2, A=16, B=25)
# -> a = 5, b = 4, c = 3, eccentricity = 0.6, f1x = 1, f1y = -5, f2x = 1, f2y = 1, orientation = vertical   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the foci and eccentricity of the ellipse x^2/25 + y^2/9 = 1.	SOLVED	a = 5, b = 3, c = 4, eccentricity = 0.8, f1x = -4, f1y = 0, f2x = 4, f2y = 0, orientation = horizontal
Find the foci of (x - 1)^2/16 + (y + 2)^2/25 = 1.	SOLVED	a = 5, b = 4, c = 3, eccentricity = 0.6, f1x = 1, f1y = -5, f2x = 1, f2y = 1, orientation = vertical
Find the foci of the ellipse x^2/4 + y^2/4 = 1.	INVALID_INPUT	Foci, axes and eccentricity of an ellipse: equal denominators give a circle (no distinct foci).

Foci, eccentricity and asymptotes of a hyperbola — `hyperbola_parameters`

For $(x-h)^2/A - (y-k)^2/B = 1$ (or the vertical form): a, b, c, eccentricity, the asymptotes' slope and the foci.

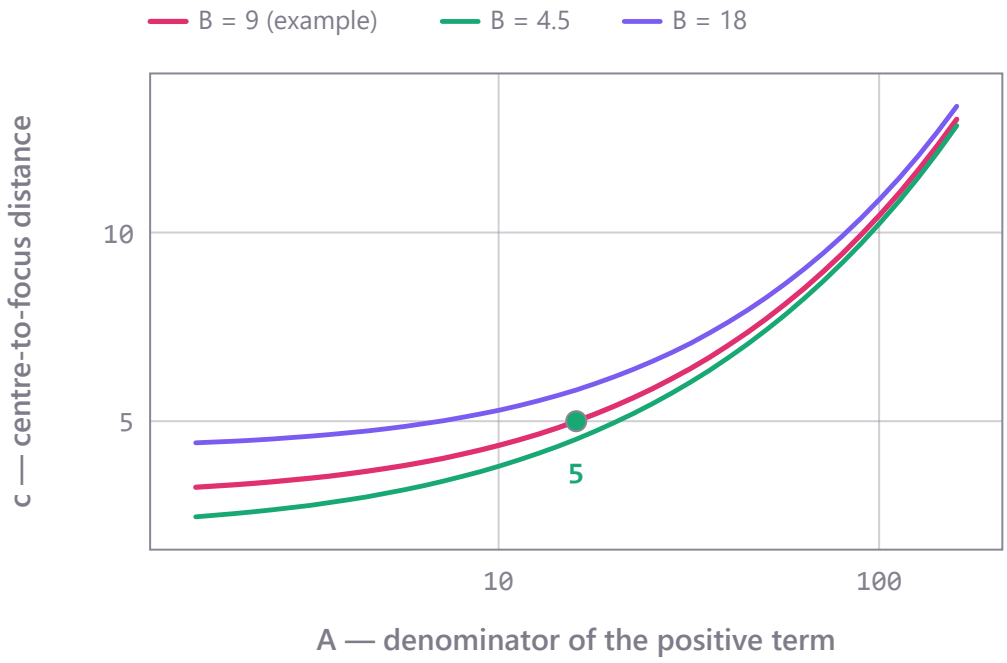
$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1, \quad c^2 = a^2 + b^2$$

input	meaning	condition
`h`	x of the centre	

input	meaning	condition
`k`	y of the centre	
`A`	denominator of the positive term	
`B`	denominator of the negative term	
`orientation`	horizontal (x term positive) or vertical	

output	meaning
`a`	$a = \sqrt{A}$
`b`	$b = \sqrt{B}$
`c`	centre-to-focus distance
`eccentricity`	$e = c/a$
`asymptote_slope`	slope of the asymptotes (plus or minus this)
`f1x`	first focus x
`f1y`	first focus y
`f2x`	second focus x
`f2y`	second focus y

- **Method:** $c^2 = a^2 + b^2$; asymptotes y - k = ±(b/a)(x - h) (horizontal) or ±(a/b) (vertical)
- **Assumes:** A, $B > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



c (centre-to-focus distance) against A (denominator of the positive term) from 1.6 to 160 on a logarithmic axis. One curve for each value of B (denominator of the negative term): 9, 4.5, 18. Other inputs as in the worked example: h = 0, k = 0, orientation = horizontal. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, A = 16, c = 5.

```
math_foundations_ai999x.solve('hyperbola_parameters', h=0, k=0, A=16, B=9,
orientation='horizontal')
# -> a = 4, b = 3, c = 5, eccentricity = 1.25, asymptote_slope = 0.75, f1x = -5, f1y = 0, f2x = 5,
f2y = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('hyperbola_parameters', h=-1, k=2, A=9, B=16, orientation='vertical')
# -> a = 3, b = 4, c = 5, eccentricity = 1.66667, asymptote_slope = 0.75, f1x = -1, f1y = -3, f2x =
-1, f2y = 7   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the foci and asymptotes of the hyperbola x^2/16 - y^2/9 = 1.	SOLVED	a = 4, b = 3, c = 5, eccentricity = 1.25, asymptote_slope = 0.75, f1x = -5, f1y = 0, f2x = 5, f2y = 0
Find the foci of (y - 2)^2/9 - (x + 1)^2/16 = 1.	SOLVED	a = 3, b = 4, c = 5, eccentricity = 1.66667, asymptote_slope = 0.75, f1x = -1, f1y = -3, f2x = -1, f2y = 7

Coefficient of a term in a binomial expansion — `binomial_coefficient_term`

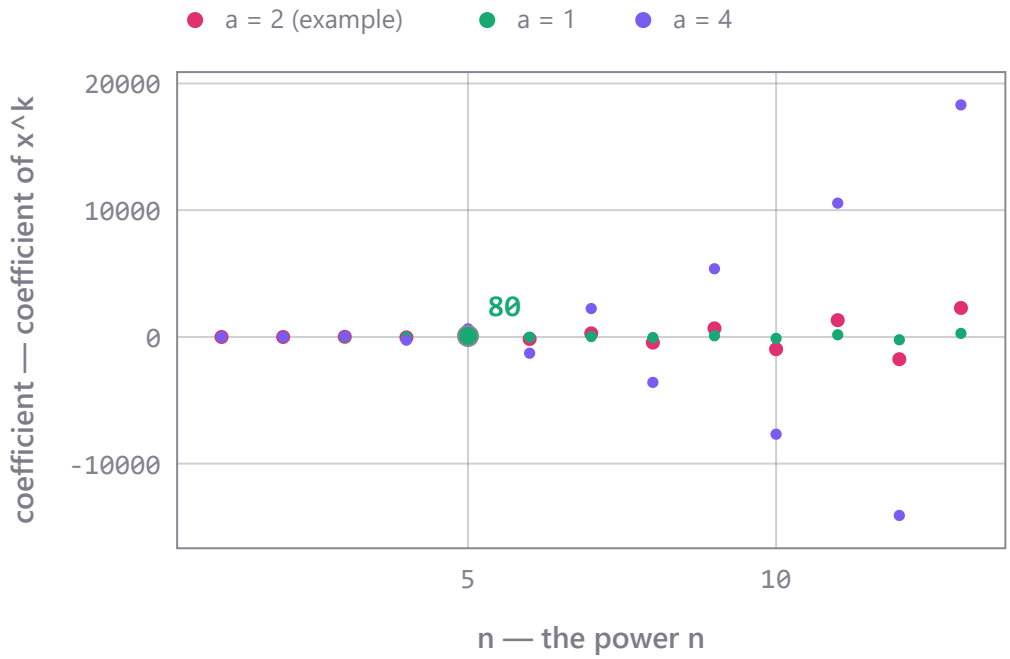
The coefficient of x^k in $(ax + b)^n$, by the binomial theorem.

$$(ax + b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} x^k$$

input	meaning	condition
`a`	coefficient of x in the binomial	
`b`	constant of the binomial	
`n`	the power n	
`k`	the power of x whose coefficient is wanted	

output	meaning
`coefficient`	coefficient of x^k
`binomial`	the binomial coefficient C(n, k)

- **Method:** binomial theorem: the x^k term is C(n, k) $(ax)^k\, b^{n-k}$
- **Assumes:** $0 \le n \le 60$; $k \ge 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



coefficient (coefficient of x^k) against n (the power n) from 1 to 13. One curve for each value of a (coefficient of x in the binomial): 2, 1, 4. Other inputs as in the worked example: b = -1, k = 3. 13 points per curve computed by math-foundations-ai999x; marked: the worked example, n = 5, coefficient = 80.

```
math_foundations_ai999x.solve('binomial_coefficient_term', a=2, b=-1, n=5, k=3)
# -> coefficient = 80, binomial = 10   (SOLVED)
```

```
math_foundations_ai999x.solve('binomial_coefficient_term', a=1, b=3, n=6, k=2)
# -> coefficient = 1215, binomial = 15   (SOLVED)
```

```
math_foundations_ai999x.solve('binomial_coefficient_term', a=-2, b=3, n=4, k=0)
# -> coefficient = 81, binomial = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the coefficient of x^3 in (2x - 1)^5.	SOLVED	coefficient = 80, binomial = 10
What is the coefficient of x^2 in the expansion of (x + 3)^6?	SOLVED	coefficient = 1215, binomial = 15
Find the constant term in (3 - 2x)^4.	SOLVED	coefficient = 81, binomial = 1
Find the coefficient of x in (x + 2)^3.	SOLVED	coefficient = 12, binomial = 3

Arithmetic with complex numbers — `complex_arithmetic`

(a + bi) + - * / (c + di) in the form $re + imi$.

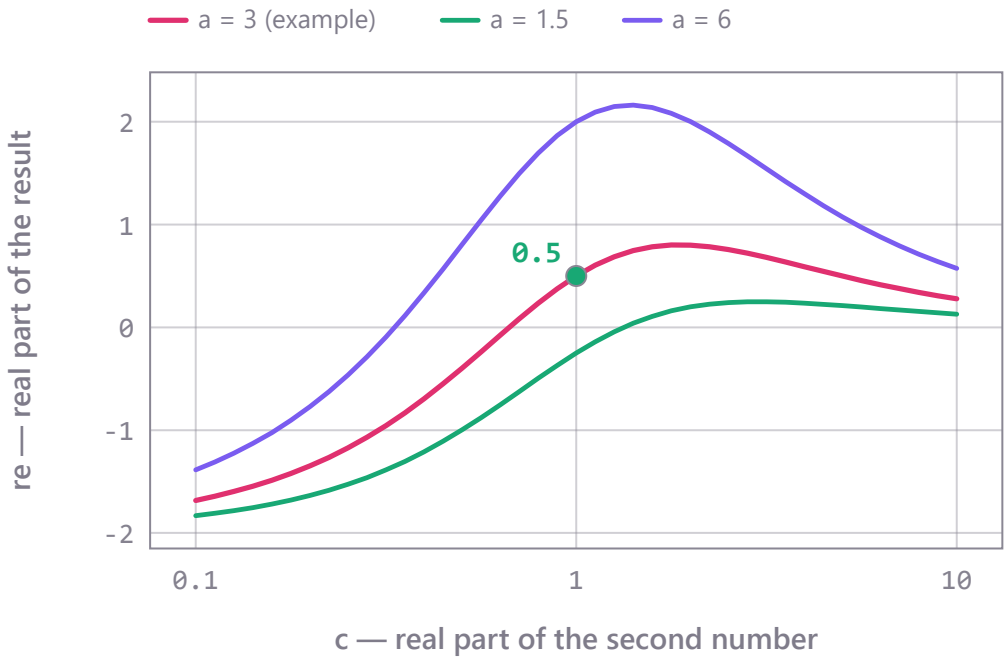
$$\frac{a+bi}{c+di} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2}$$

input	meaning	condition
a	real part of the first number	
b	imaginary part of the first number	

input	meaning	condition
`c`	real part of the second number	
`d`	imaginary part of the second number	
`op`	the operation	

output	meaning
`re`	real part of the result
`im`	imaginary part of the result

- **Method:** $i^2 = -1$; division multiplies numerator and denominator by the conjugate c - di
- **Assumes:** no division by 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



re (real part of the result) against c (real part of the second number) from 0.1 to 10 on a logarithmic axis. One curve for each value of a (real part of the first number): 3, 1.5, 6. Other inputs as in the worked example: b = 2, d = -1, op = /. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 1, re = 0.5.

```
math_foundations_ai999x.solve('complex_arithmetic', a=3, b=2, c=1, d=-1, op='/')
# -> re = 0.5, im = 2.5   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_arithmetic', a=2, b=-3, c=4, d=1, op='*')
# -> re = 11, im = -10   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_arithmetic', a=2, b=-3, c=4, d=1, op='+')
# -> re = 6, im = -2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Compute (3 + 2i) / (1 - i).	SOLVED	re = 0.5, im = 2.5

question	outcome	result
Simplify (2 - 3i)(4 + i).	SOLVED	re = 11, im = -10
Find (2 - 3i) + (4 + i).	SOLVED	re = 6, im = -2
Compute (5 + i) - (2 - 4i).	SOLVED	re = 3, im = 5
Compute (1 + i) / (0 + 0i).	INVALID_INPUT	Arithmetic with complex numbers: division by 0.

Modulus, argument and polar form of a complex number — `complex_modulus_argument`

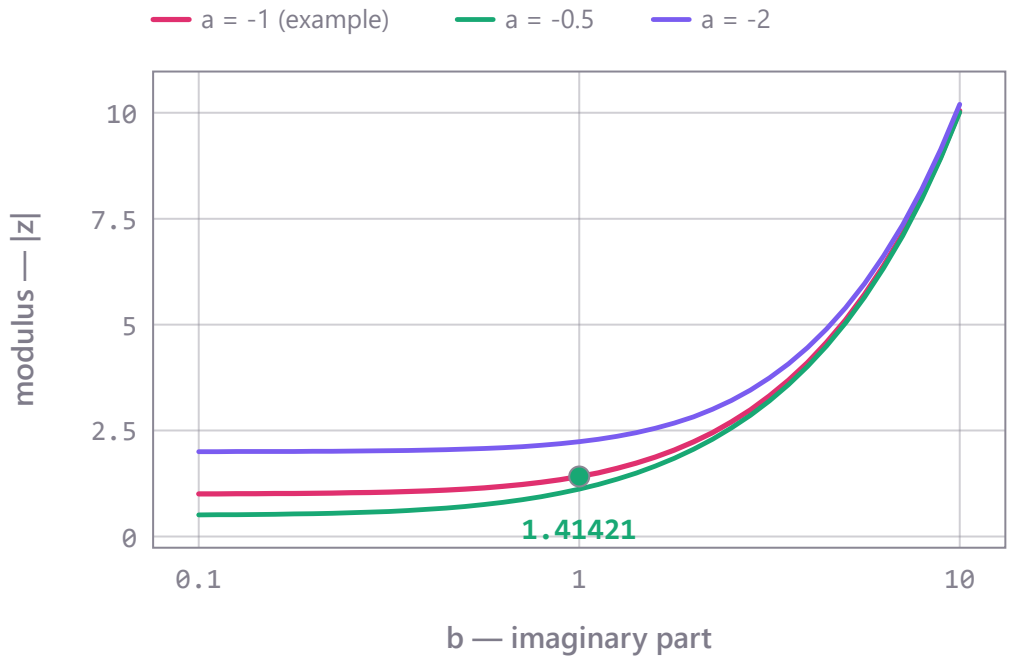
|z| and the principal argument (-180, 180] degrees of $z = a + bi$, and z in polar form.

$$z = r(\cos\theta + i\sin\theta), \quad r = \sqrt{a^2 + b^2}$$

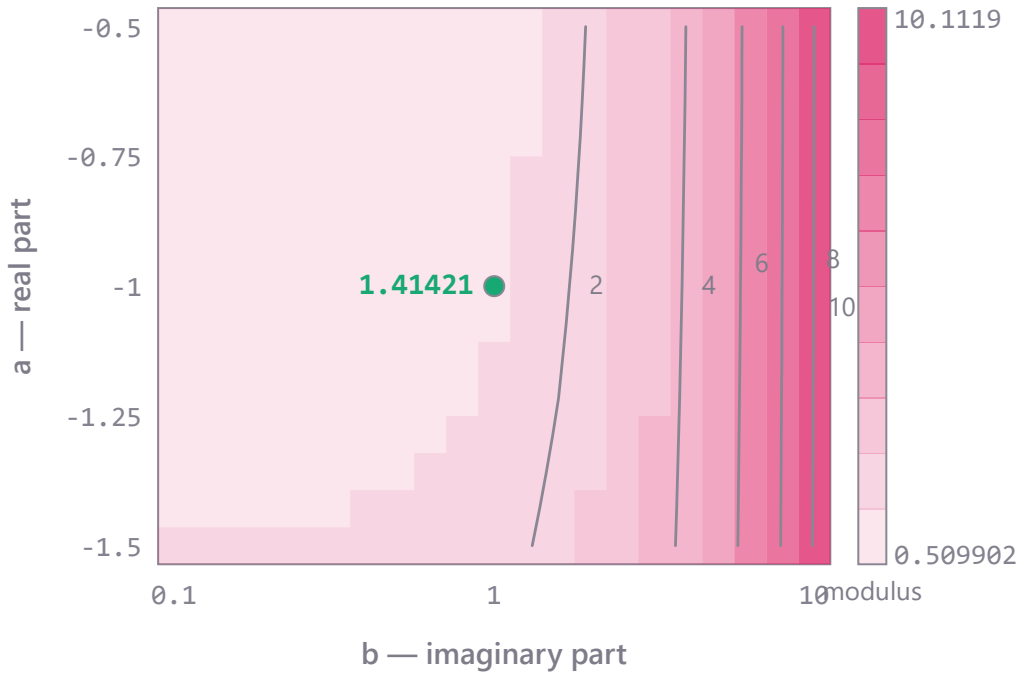
input	meaning	condition
`a`	real part	
`b`	imaginary part	

output	meaning
`modulus`	|z|
`argument_deg`	principal argument, degrees in (-180, 180]
`argument_rad`	principal argument, radians
`polar`	$z = r$(cos t + i sin t)

- **Method:** |z| = sqrt(a^2 + b^2); the argument in the quadrant of (a, b)
- **Assumes:** $z \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



modulus (|z|) against b (imaginary part) from 0.1 to 10 on a logarithmic axis. One curve for each value of a (real part): -1, -0.5, -2. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 1, modulus = 1.41421.



modulus (|z|) over b (imaginary part) from 0.1 to 10 and a (real part) from -1.5 to -0.5 (logarithmic horizontal axis): each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 0.509902 to dark 10.1119); iso-lines at modulus = 2, 4, 6, 8, 10. Marked: the worked example, b = 1, a = -1, modulus = 1.41421.

```
math_foundations_ai999x.solve('complex_modulus_argument', a=-1, b=1)
# -> modulus = 1.41421, argument_deg = 135, argument_rad = 2.35619, polar = 1.414213562(cos(135
deg) + i sin(135 deg))   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_modulus_argument', a=1, b=-1)
# -> modulus = 1.41421, argument_deg = -45, argument_rad = -0.785398, polar = 1.414213562(cos(-45
deg) + i sin(-45 deg))   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_modulus_argument', a=3, b=-4)
# -> modulus = 5, argument_deg = -53.1301, argument_rad = -0.927295, polar = 5(cos(-53.13010235
deg) + i sin(-53.13010235 deg))   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_modulus_argument', a=-2, b=0)
# -> modulus = 2, argument_deg = 180, argument_rad = 3.14159, polar = 2(cos(180 deg) + i sin(180
deg))   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the modulus and argument of -1 + i.	SOLVED	modulus = 1.41421, argument_deg = 135, argument_rad = 2.35619, polar = 1.414213562(cos(135 deg) + i sin(135 deg))
Write 1 - i in polar form.	SOLVED	modulus = 1.41421, argument_deg = -45, argument_rad = -0.785398, polar = 1.414213562(cos(-45 deg) + i sin(-45 deg))
Find the modulus and argument of z = 3 - 4i.	SOLVED	modulus = 5, argument_deg = -53.1301, argument_rad = -0.927295, polar = 5(cos(-53.13010235 deg) + i sin(-53.13010235 deg))
Find the modulus and argument of 0.	INVALID_INPUT	Modulus, argument and polar form of a complex number: 0 has modulus 0 and no argument.

Power of a complex number (De Moivre) — `de_moivre_power`

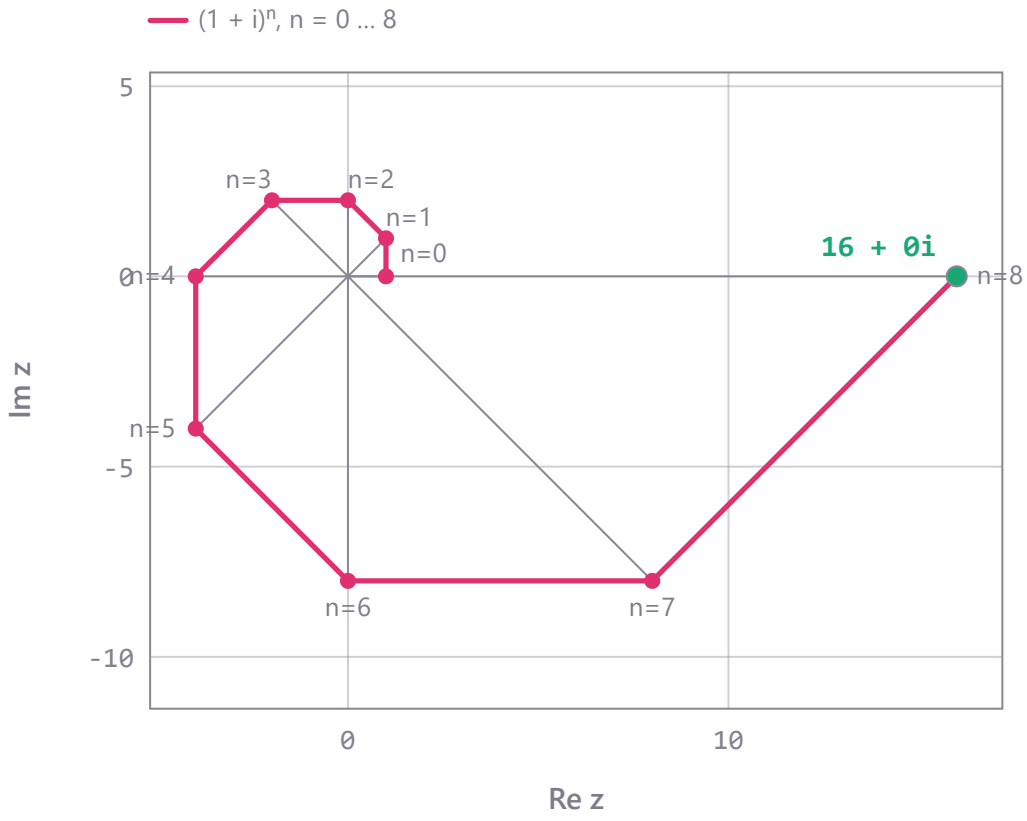
$(a + bi)^n$ for a whole number n (|n| <= 40): real and imaginary parts, modulus and argument.

$$[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)$$

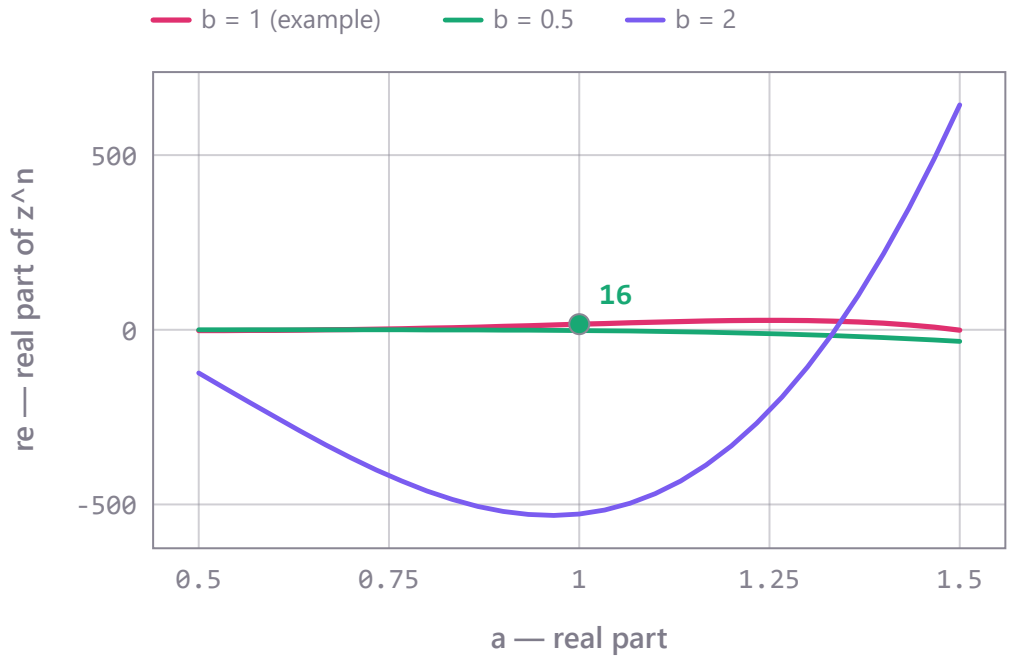
input	meaning	condition
`a`	real part	
`b`	imaginary part	
`n`	the power n	

output	meaning
`re`	real part of z^n
`im`	imaginary part of z^n
`modulus`	$\lvert z^n\rvert$ = |z|^n
`argument_deg`	argument of z^n, degrees

- **Method:** De Moivre: $z^n = r^n$ (cos n t + i sin n t); computed exactly (binary exponentiation in rational arithmetic)
- **Assumes:** |n| <= 40; $z \neq 0$ when $n \leq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The powers (1 + i)^n for n = 0 … 8, each computed by the package: every step turns by 45° and stretches by √2; marked: n = 8, (1 + i)^8 = 16 + 0i, modulus 16, argument 0°.



re (real part of z^n) against a (real part) from 0.5 to 1.5. One curve for each value of b (imaginary part): 1, 0.5, 2. Other inputs as in the worked example: n = 8. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, re = 16.

```
math_foundations_ai999x.solve('de_moivre_power', a=1, b=1, n=8)
# -> re = 16, im = 0, modulus = 16, argument_deg = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('de_moivre_power', a=0, b=2, n=3)
# -> re = 0, im = -8, modulus = 8, argument_deg = -90   (SOLVED)
```

```
math_foundations_ai999x.solve('de_moivre_power', a=1, b=-1, n=-2)
# -> re = 0, im = 0.5, modulus = 0.5, argument_deg = 90   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Use De Moivre's theorem to compute (1 + i)^8.	SOLVED	re = 16, im = 0, modulus = 16, argument_deg = 0
Compute (sqrt(3) + i)^6.	SOLVED	re = -64, im = 0, modulus = 64, argument_deg = 180
Find (2i)^3.	SOLVED	re = 0, im = -8, modulus = 8, argument_deg = -90
Compute (0 + 0i)^-2.	INVALID_INPUT	Power of a complex number (De Moivre): 0 to a power that is not positive is undefined.

The n-th roots of a complex number — `complex_nth_roots`

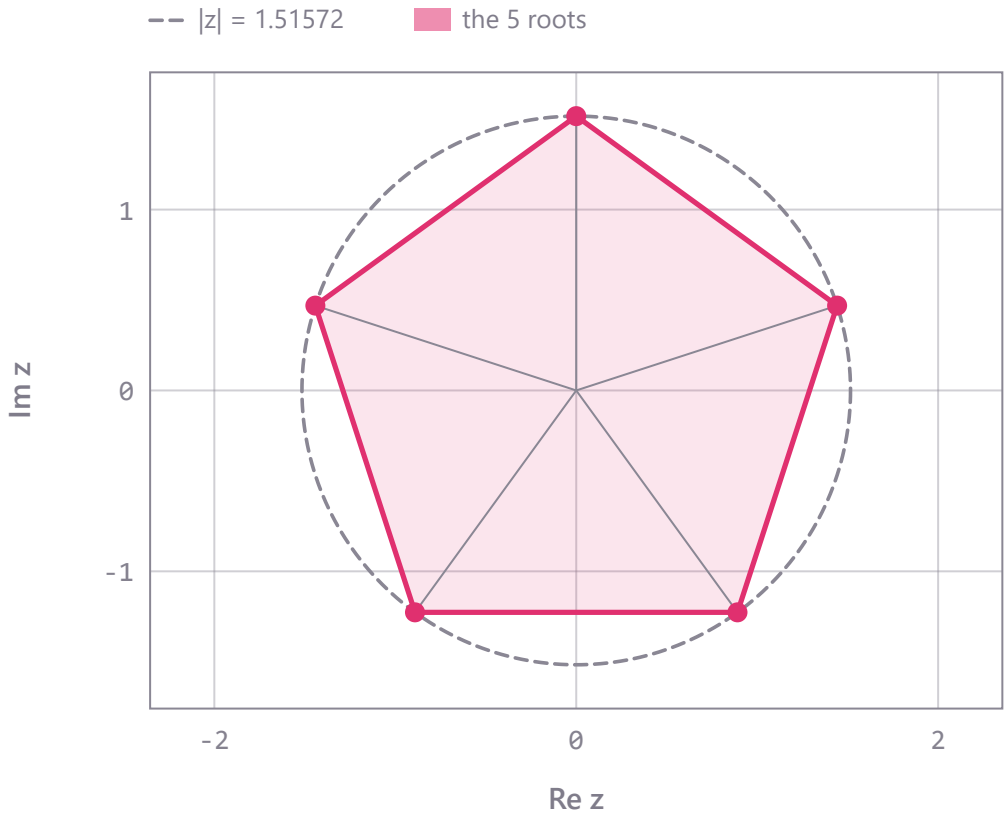
All n complex n-th roots of $z = a + bi$ (n from 2 to 12), ordered by argument from 0 to 360 degrees.

$$w_k = \sqrt[n]{r}\left(\cos\frac{\theta + 2\pi k}{n} + i\sin\frac{\theta + 2\pi k}{n}\right)$$

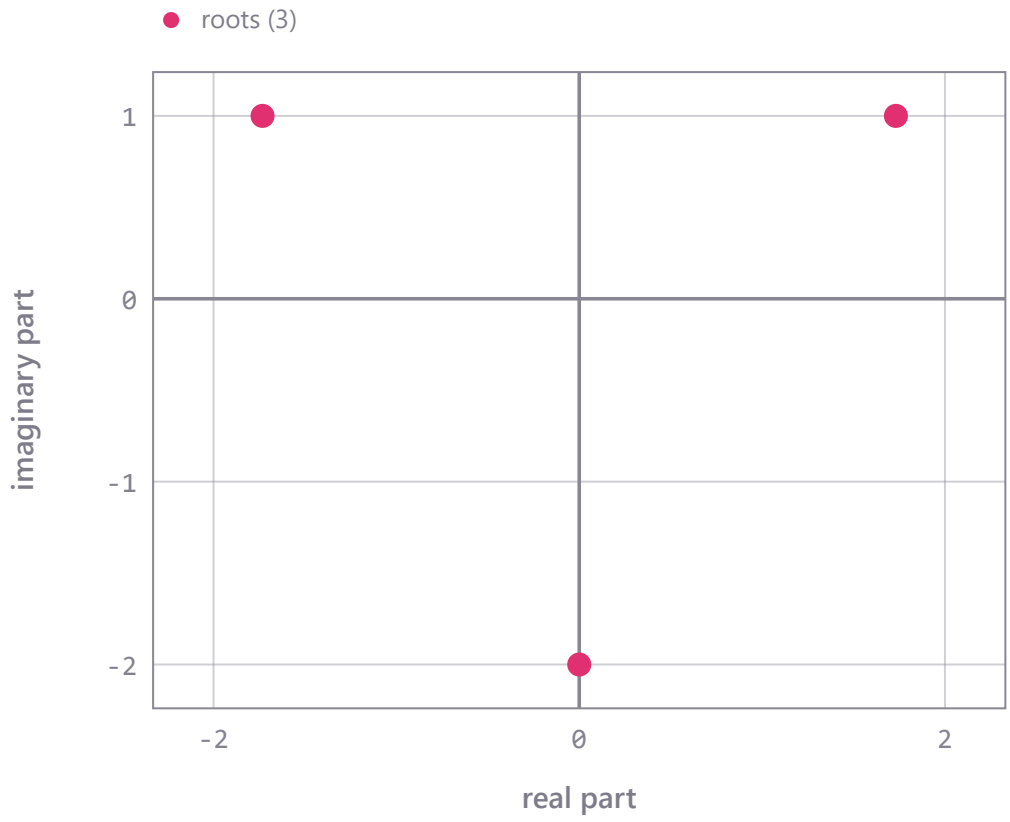
input	meaning	condition
`a`	real part of z	
`b`	imaginary part of z	
`n`	which roots (2 = square roots)	

output	meaning
`modulus`	the common modulus |z|^$(1/n)$
`count`	the number of roots, n
`roots`	the roots, by argument in [0, 360)

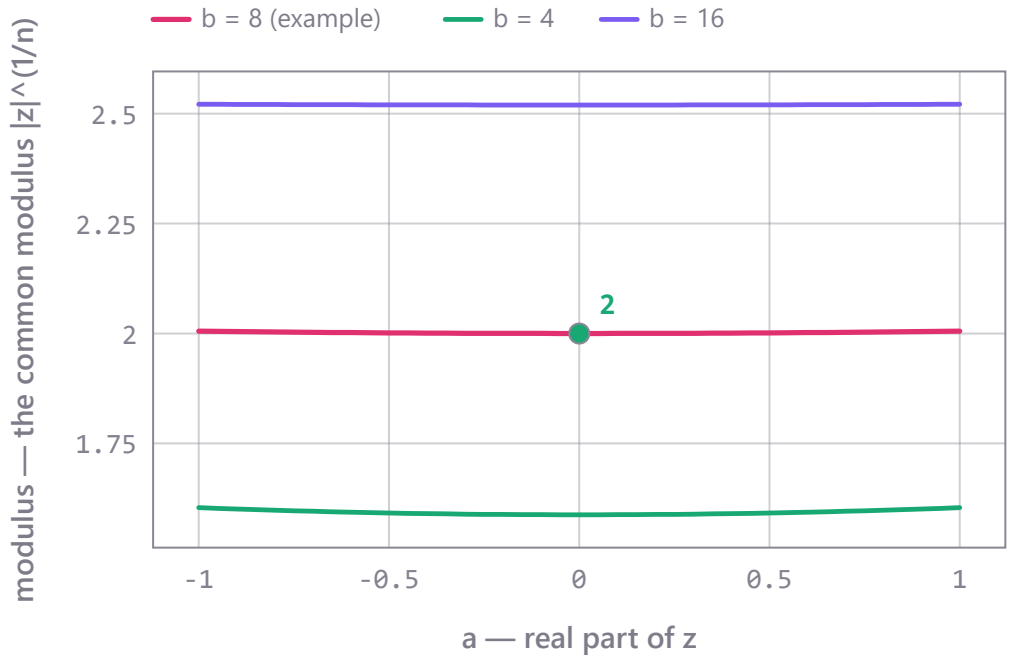
- **Method:** w_k = |z|^$(1/n)$ (cos((t + 2 pi k)/n) + i sin((t + 2 pi k)/n)), $k = 0 \,..\, n-1$
- **Assumes:** $z \neq 0$; $2 \leq n \leq 12$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The 5 complex roots of $z^5 = 0 + 8i$ returned by the package — -0.890916 − 1.22624i, 0.890916 − 1.22624i, 1.44153 + 0.468382i, 0 + 1.51572i, -1.44153 + 0.468382i — the vertices of a regular pentagon on the circle of radius 1.51572 (= 8^(1/5)).



the roots, by argument in [0, 360) (roots) returned by math-foundations-ai999x for the worked example (a = 0, b = 8, n = 3), drawn in the complex plane: 1.73205 + 1i, -1.73205 + 1i, 0 − 2i.



modulus (the common modulus |z|^(1/n)) against a (real part of z) from -1 to 1. One curve for each value of b (imaginary part of z): 8, 4, 16. Other inputs as in the worked example: n = 3. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 0, modulus = 2.

```
math_foundations_ai999x.solve('complex_nth_roots', a=0, b=8, n=3)
# -> modulus = 2, count = 3, roots = [(1.73205080757+1j), (-1.73205080757+1j), -2j]   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_nth_roots', a=-16, b=0, n=4)
# -> modulus = 2, count = 4, roots = [(1.41421356237+1.41421356237j),
(-1.41421356237+1.41421356237j), (-1.41421356237-1.41421356237j), (1.41421356237-1.41421356237j)]
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the cube roots of 8i.	SOLVED	modulus = 2, count = 3, roots = [(1.73205080757+1j), (-1.73205080757+1j), -2j]
Find all the fourth roots of -16.	SOLVED	modulus = 2, count = 4, roots = [(1.41421356237+1.41421356237j), (-1.41421356237+1.41421356237j), (-1.41421356237-1.41421356237j), (1.41421356237-1.41421356237j)]
Find the square roots of 9.	SOLVED	modulus = 3, count = 2, roots = [(3+0j), (-3+0j)]
Find the cube roots of 0.	INVALID_INPUT	The n-th roots of a complex number: 0 has only the root 0.

Solution set of |x - c| > d — `abs_value_inequality_lower_bound`

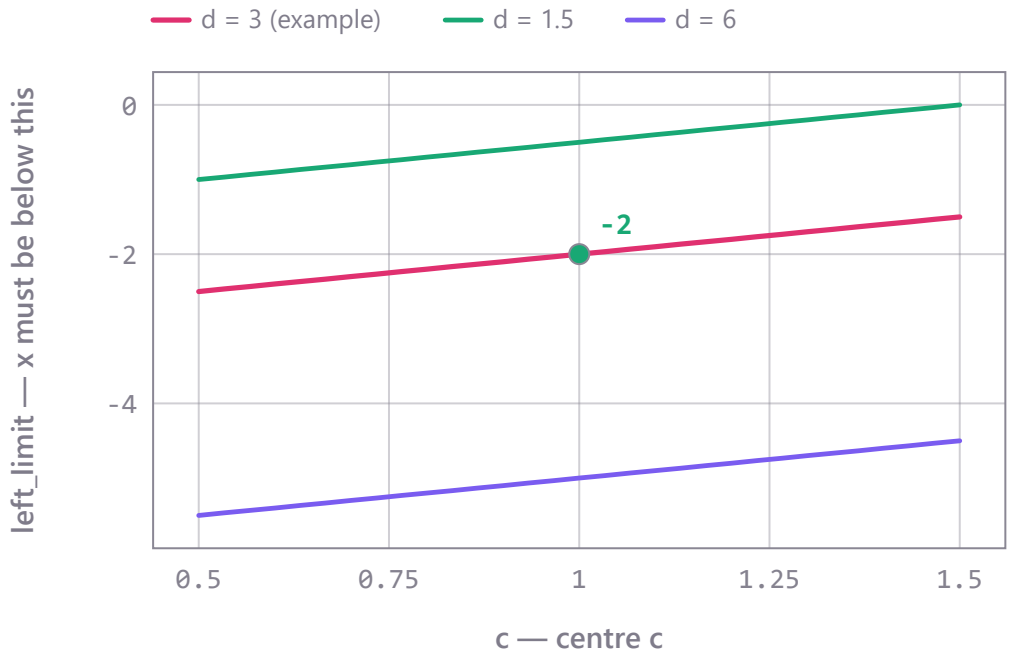
For |x - c| > d with $d \geq 0$, the solutions are x < left_limit or x > right_limit.

$$\text{right limit} - \text{left limit} = 2d, \quad \text{right limit} + \text{left limit} = 2c$$

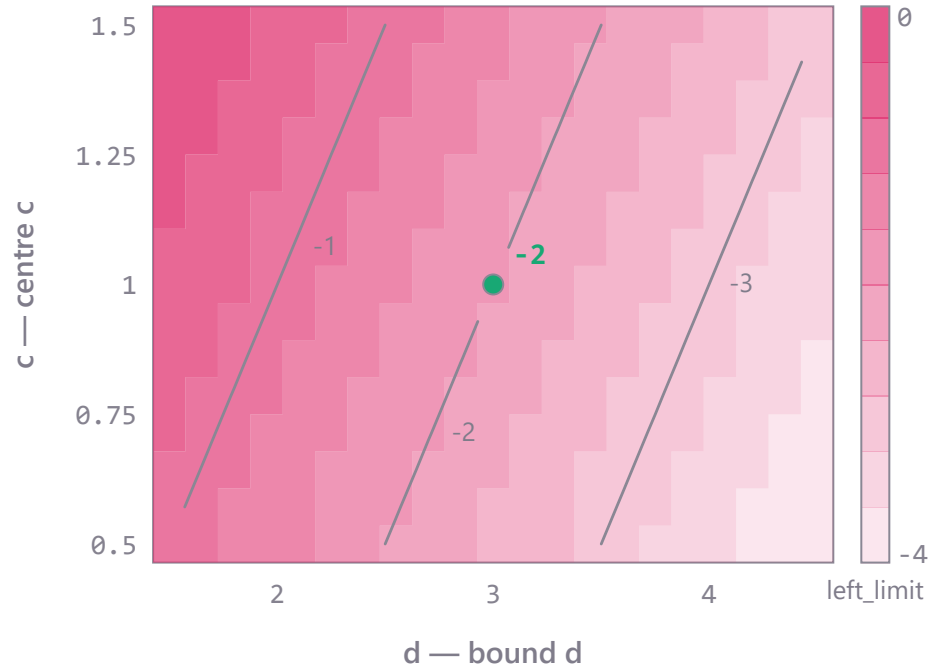
input	meaning	condition
`c`	centre c	
`d`	bound d	$d \geq 0$

output	meaning
`left_limit`	x must be below this
`right_limit`	or above this

- **Method:** |x - c| > d means $x - c < -d$ or $x - c > d$
- **Assumes:** $d \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



left_limit (x must be below this) against c (centre c) from 0.5 to 1.5. One curve for each value of d (bound d): 3, 1.5, 6. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 1, left_limit = -2.



left_limit (x must be below this) over d (bound d) from 1.5 to 4.5 and c (centre c) from 0.5 to 1.5: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light -4 to dark 0); iso-lines at left_limit = -3, -2, -1. Marked: the worked example, d = 3, c = 1, left_limit = -2.

```
math_foundations_ai999x.solve('abs_value_inequality_lower_bound', c=1.0, d=3.0)
# -> left_limit = -2, right_limit = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('abs_value_inequality_lower_bound', c=5.0, d=0.0)
# -> left_limit = 5, right_limit = 5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve |x - 1| > 3.	SOLVED	left_limit = -2, right_limit = 4
For which x is |x - 6| greater than 2?	SOLVED	left_limit = 4, right_limit = 8

Solution interval of |x - c| < d — `abs_value_inequality_upper_bound`

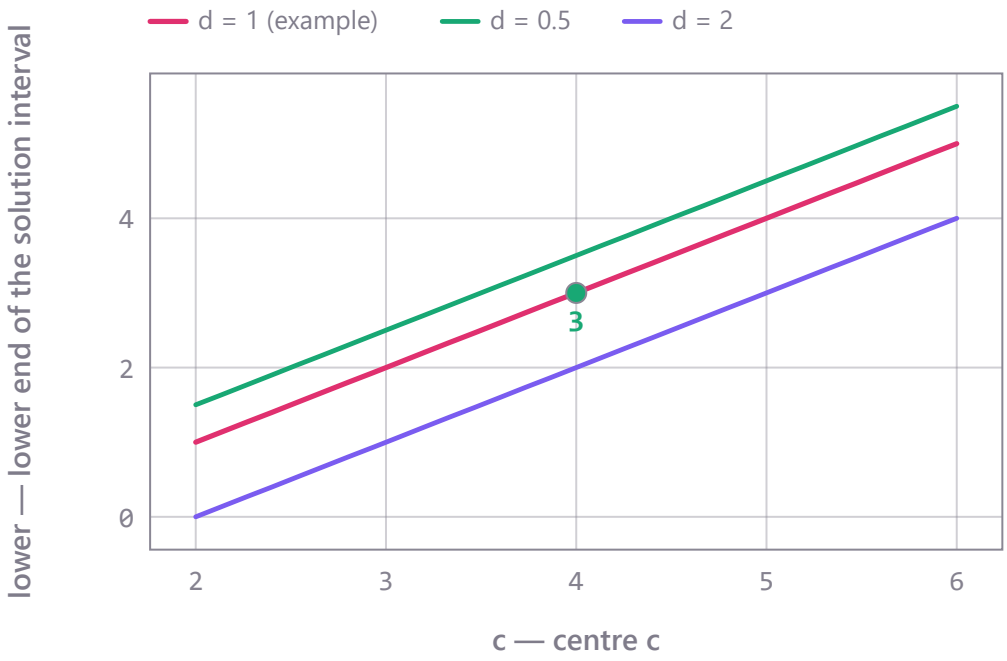
For |x - c| < d with $d > 0$, the solutions are lower < x < upper.

$$\text{upper} - \text{lower} = 2d, \quad \text{upper} + \text{lower} = 2c$$

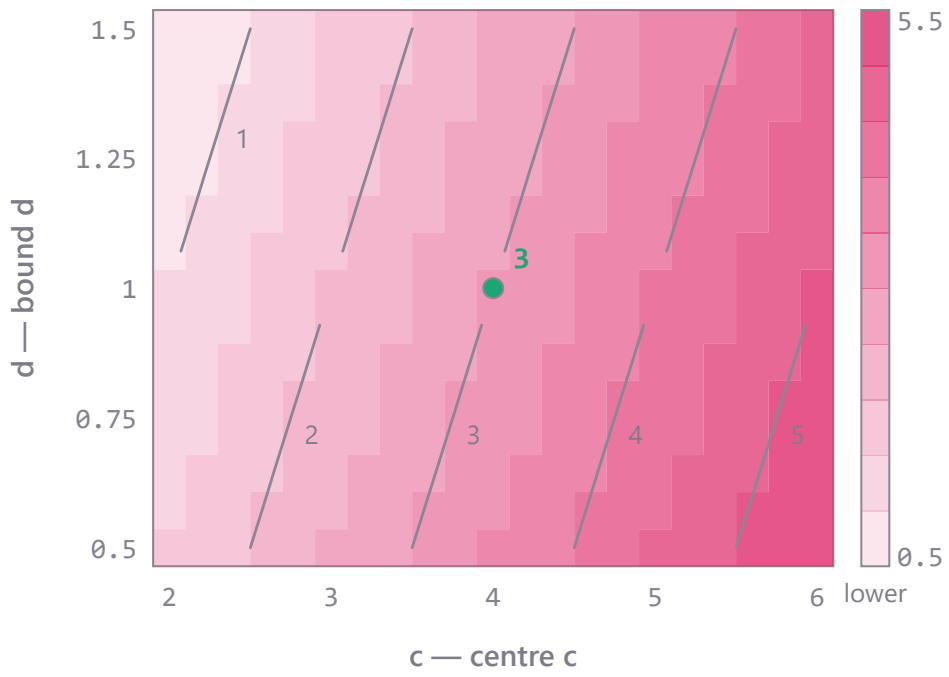
input	meaning	condition
`c`	centre c	
`d`	bound d	$d > 0$

output	meaning
`lower`	lower end of the solution interval (excluded)
`upper`	upper end of the solution interval (excluded)

- **Method:** |x - c| < d means -d < x - c < d
- **Assumes:** $d > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



lower (lower end of the solution interval (excluded)) against c (centre c) from 2 to 6. One curve for each value of d (bound d): 1, 0.5, 2. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 4, lower = 3.



lower (lower end of the solution interval (excluded)) over c (centre c) from 2 to 6 and d (bound d) from 0.5 to 1.5: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 0.5 to dark 5.5); iso-lines at lower = 1, 2, 3, 4, 5. Marked: the worked example, c = 4, d = 1, lower = 3.

```
math_foundations_ai999x.solve('abs_value_inequality_upper_bound', c=4.0, d=1.0)
# -> lower = 3, upper = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('abs_value_inequality_upper_bound', c=0.0, d=2.5)
# -> lower = -2.5, upper = 2.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve |x - 4| < 1.	SOLVED	lower = 3, upper = 5
Solve |x - 2| < 0.	INVALID_INPUT	Solution interval of |x - c| < d: the condition d > 0 does not hold for these inputs.

Solutions of an absolute value equation — `absolute_value_equation`

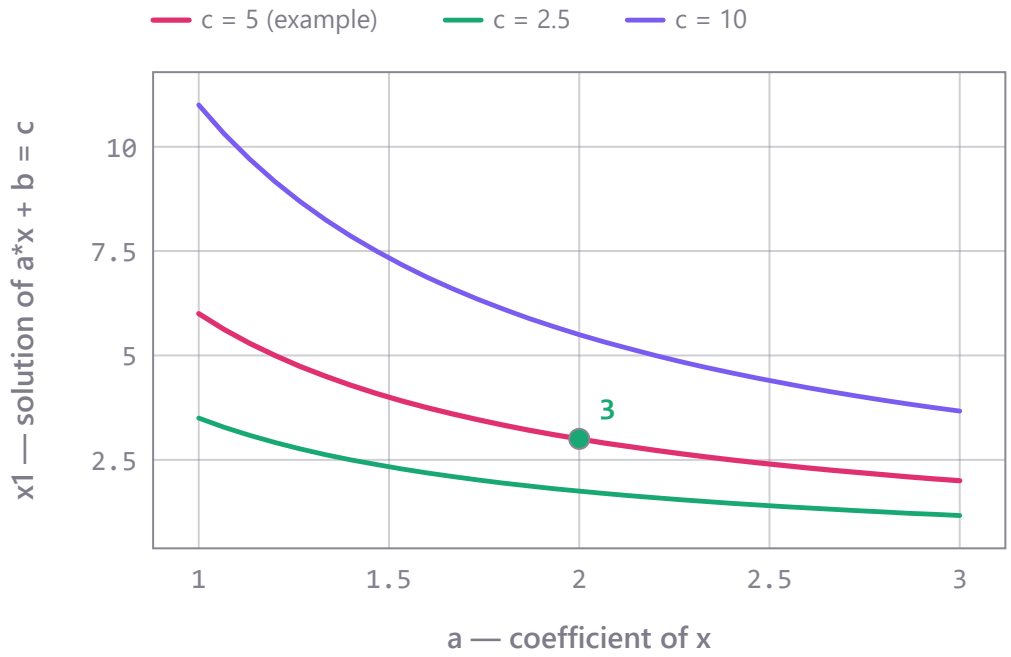
Solve $|ax + b|$ = c for x (two solutions when $c > 0$, one when c = 0).

$$ax_1 + b = c, \quad ax_2 + b = -c$$

input	meaning	condition
a	coefficient of x	$a \neq 0$
b	constant inside the absolute value	
c	right-hand side	$c \geq 0$

output	meaning
x1	solution of $ax + b = c$
x2	solution of $ax + b = -c$

- **Method:** the expression inside equals c or -c; solve each linear equation
- **Assumes:** $a \neq 0$; $c \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x1 (solution of a*x + b = c) against a (coefficient of x) from 1 to 3. One curve for each value of c (right-hand side): 5, 2.5, 10. Other inputs as in the worked example: b = -1.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, x1 = 3.

```
math_foundations_ai999x.solve('absolute_value_equation', a=2.0, b=-1.0, c=5.0)
# -> x1 = 3, x2 = -2   (SOLVED)
```

```
math_foundations_ai999x.solve('absolute_value_equation', a=1.0, b=3.0, c=0.0)
# -> x1 = -3, x2 = -3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve |2x - 1| = 5.	SOLVED	x1 = 3, x2 = -2
Solve |x + 4| = 6.	SOLVED	x1 = 2, x2 = -10

Add rational expressions with same denominator — `add_rational_expressions_same_denom`

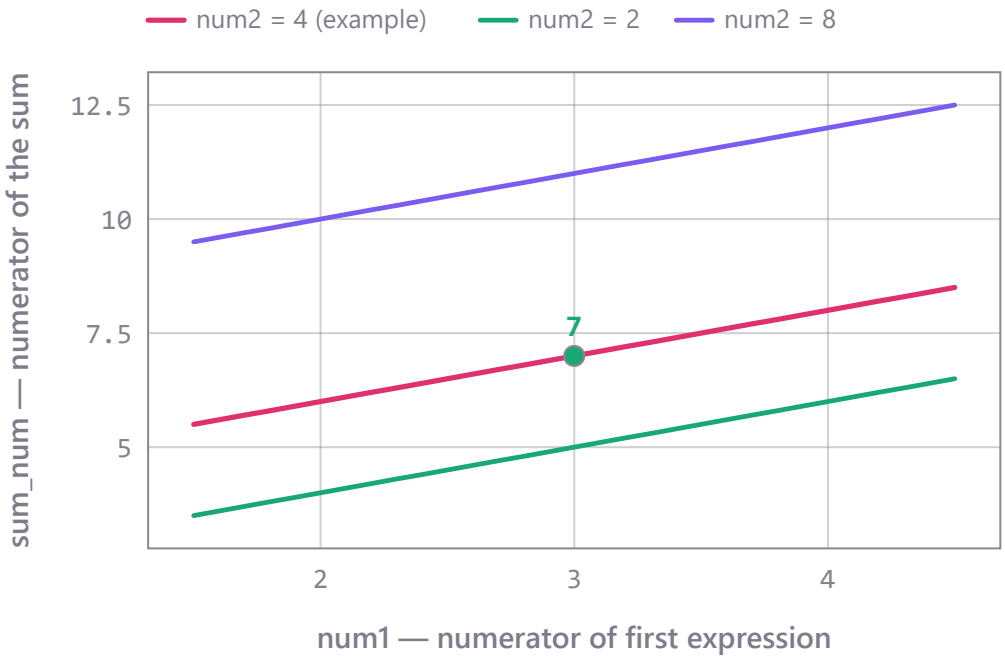
Add two rational expressions that share a common denominator.

$$\text{sum num} = \text{num}_1 + \text{num}_2, \quad \text{sum den} = \text{den}$$

input	meaning	condition
`num1`	numerator of first expression	
`num2`	numerator of second expression	
`den`	common denominator	

output	meaning
`sum_num`	numerator of the sum
`sum_den`	denominator of the sum

- **Method:** closed form verified against the defining equations
- **Assumes:** den != 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



sum_num (numerator of the sum) against num1 (numerator of first expression) from 1.5 to 4.5. One curve for each value of num2 (numerator of second expression): 4, 2, 8. Other inputs as in the worked example: den = 5.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, num1 = 3, sum_num = 7.

```
math_foundations_ai999x.solve('add_rational_expressions_same_denom', den=5.0, num1=3.0, num2=4.0)
# -> sum_num = 7, sum_den = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('add_rational_expressions_same_denom', den=4.0, num1=10.0, num2=-2.0)
# -> sum_num = 8, sum_den = 4   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Add 1/6 and 2/6.	SOLVED	sum_num = 3, sum_den = 6

Term and partial sum of an arithmetic sequence — `arithmetic_sequence`

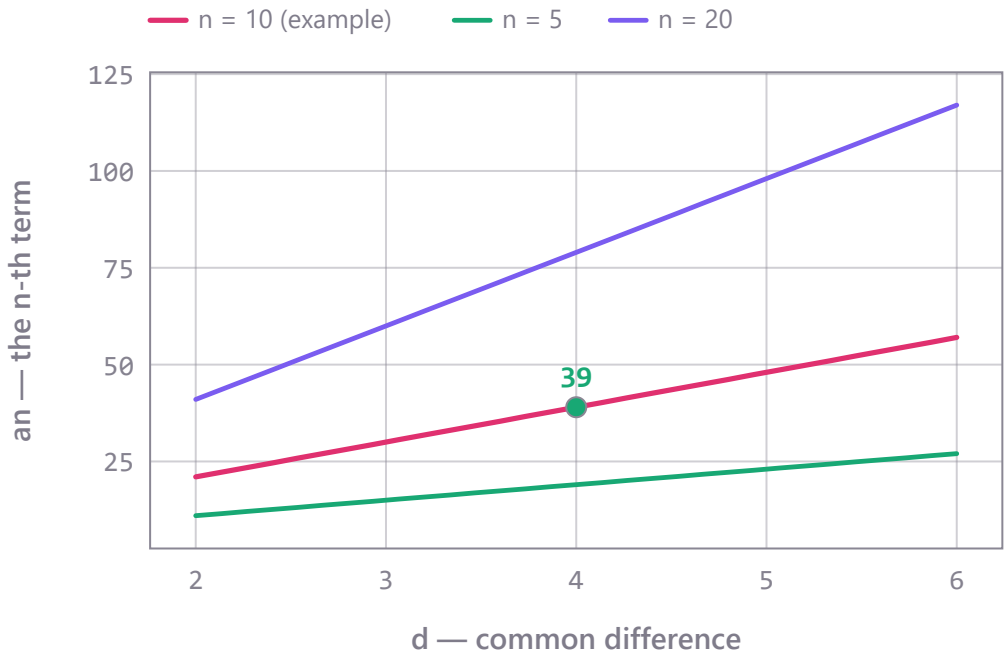
The n-th term and the sum of the first n terms of the arithmetic sequence with first term a1 and common difference d.

$$a_n = a_1 + (n-1)d, \quad 2s_n = n\,(a_1 + a_n)$$

input	meaning	condition
`a1`	first term	
`d`	common difference	
`n`	position of the term (a positive whole number)	$n \geq 1$

output	meaning
`an`	the n-th term
`sn`	the sum of the first n terms

- **Method:** the n-th term adds d to the first term $n - 1$ times; the sum pairs first and last terms
- **Assumes:** $n \geq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



an (the n-th term) against d (common difference) from 2 to 6. One curve for each value of n (position of the term (a positive whole number)): 10, 5, 20. Other inputs as in the worked example: a1 = 3.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, d = 4, an = 39.

```
math_foundations_ai999x.solve('arithmetic_sequence', a1=3.0, d=4.0, n=10.0)
# -> an = 39, sn = 210   (SOLVED)
```

```
math_foundations_ai999x.solve('arithmetic_sequence', a1=100.0, d=-7.0, n=5.0)
# -> an = 72, sn = 430   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the 10th term of the arithmetic sequence with first term 3 and common difference 4.	SOLVED	an = 39, sn = 210
Use the arithmetic sequence with a1 = 1, d = 2, n = 2.5.	INVALID_INPUT	Term and partial sum of an arithmetic sequence: n must be a whole number.

Average rate of change of a function between two inputs — `average_rate_of_change_definition`

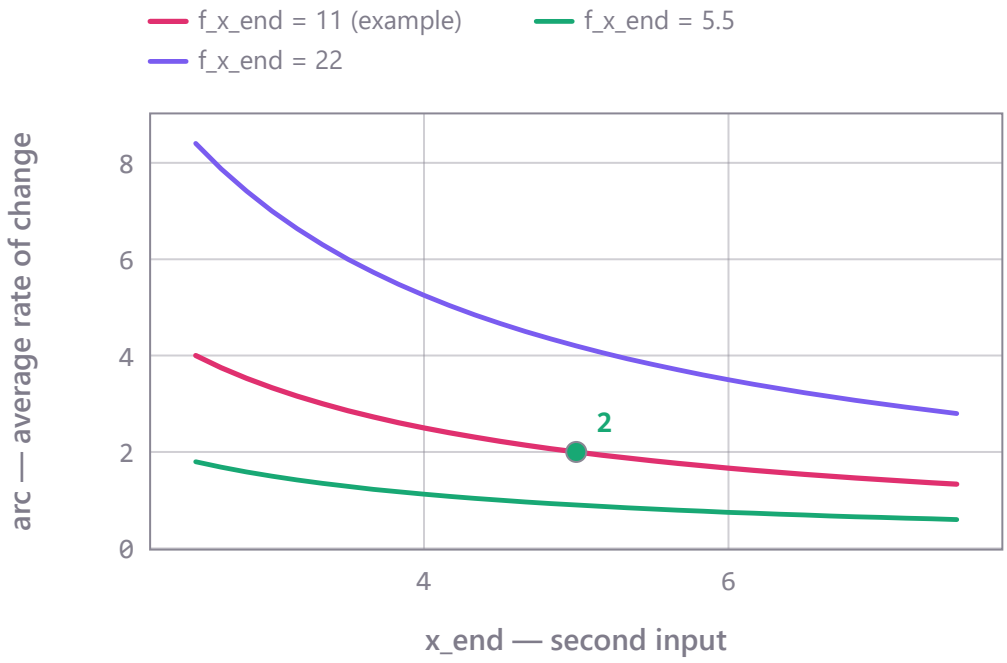
(f(x_end) - f(x_start)) / (x_end - x_start) from two values of a function.

$$\text{arc}\,(x_{\text{end}} - x_{\text{start}}) = f_{x,end} - f_{x,start}$$

input	meaning	condition
`x_start`	first input	
`f_x_start`	value of the function at the first input	
`x_end`	second input	
`f_x_end`	value of the function at the second input	

output	meaning
`arc`	average rate of change

- **Method:** change in output divided by change in input
- **Assumes:** x_end != x_start
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



arc (average rate of change) against x_end (second input) from 2.5 to 7.5. One curve for each value of f_x_end (value of the function at the second input): 11, 5.5, 22. Other inputs as in the worked example: x_start = 0.0, f_x_start = 1.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, x_end = 5, arc = 2.

```
math_foundations_ai999x.solve('average_rate_of_change_definition', x_start=0.0, f_x_start=1.0,
x_end=5.0, f_x_end=11.0)
# -> arc = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('average_rate_of_change_definition', x_start=1.0, f_x_start=4.0,
x_end=3.0, f_x_end=0.0)
# -> arc = -2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
If f(0) = 1 and f(5) = 11, what is the average rate of change of f?	SOLVED	arc = 2
If f(2) = 7 and f(2) = 9, what is the average rate of change of f?	INVALID_INPUT	Average rate of change of a function between two inputs: the condition x_end != x_start does not hold for these inputs.

Average rate of change between two points — `average_rate_of_change_linear`

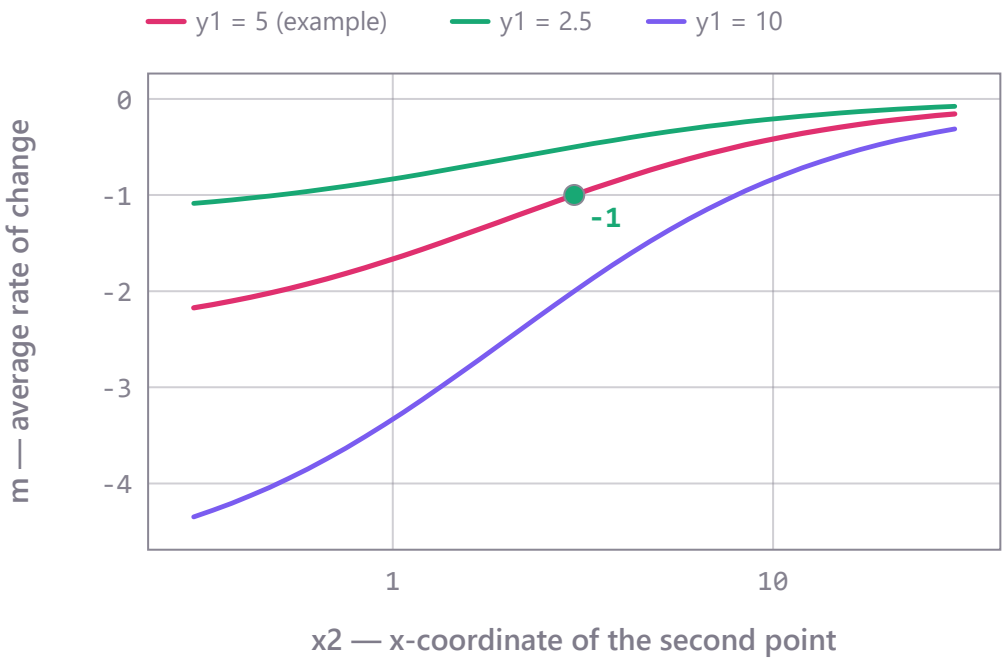
$(y_2 - y_1)/(x_2 - x_1)$ between the points (x1, y1) and (x2, y2).

$$m\,(x_2 - x_1) = y_2 - y_1$$

input	meaning	condition
`x1`	x-coordinate of the first point	
`y1`	y-coordinate of the first point	
`x2`	x-coordinate of the second point	
`y2`	y-coordinate of the second point	

output	meaning
`m`	average rate of change

- **Method:** change in y divided by change in x
- **Assumes:** $x_2 \neq x_1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



m (average rate of change) against x2 (x-coordinate of the second point) from 0.3 to 30 on a logarithmic axis. One curve for each value of y1 (y-coordinate of the first point): 5, 2.5, 10. Other inputs as in the worked example: x1 = -2.0, y2 = 0.0. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, x2 = 3, m = -1.

```
math_foundations_ai999x.solve('average_rate_of_change_linear', x1=-2.0, y1=5.0, x2=3.0, y2=0.0)
# -> m = -1   (SOLVED)
```

```
math_foundations_ai999x.solve('average_rate_of_change_linear', x1=0.0, y1=0.0, x2=4.0, y2=2.0)
# -> m = 0.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the average rate of change between the points (-2, 5) and (3, 0).	SOLVED	m = -1
What is the average rate of change from (1, 2) to (1, 8)?	INVALID_INPUT	Average rate of change between two points: the condition x2 != x1 does not hold for these inputs.

Product of two binomials — `binomial_product`

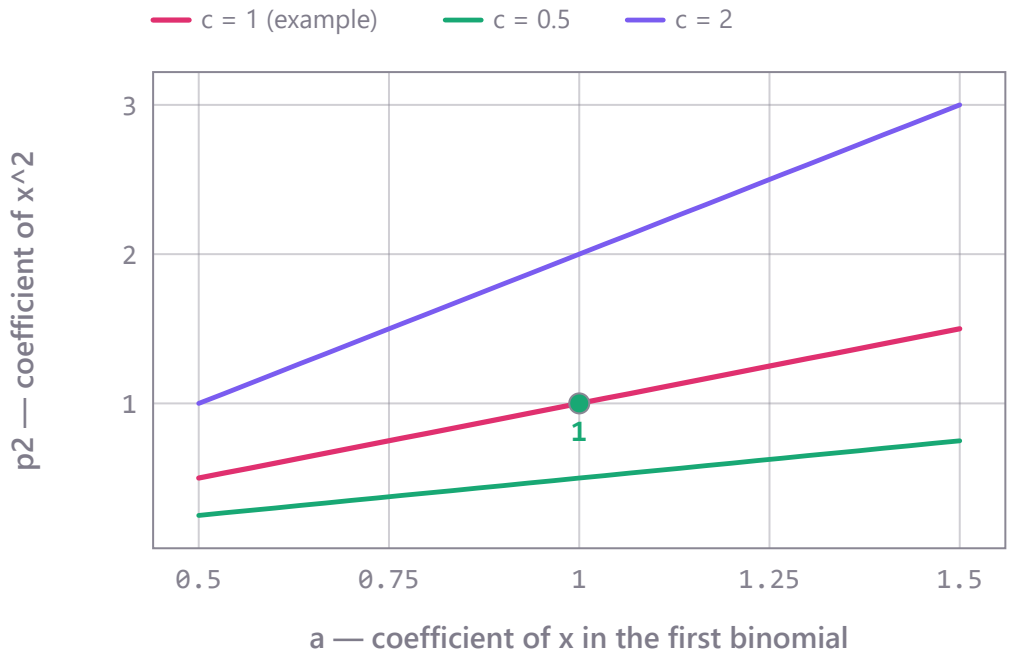
Expand $(ax + b)(cx + d)$ into $p_2x^2 + p_1x + p_0$.

$$p_2t^2 + p_1t + p_0 = (at + b)\,(ct + d)$$

input	meaning	condition
`a`	coefficient of x in the first binomial	
`b`	constant of the first binomial	
`c`	coefficient of x in the second binomial	
`d`	constant of the second binomial	

output	meaning
`p2`	coefficient of x^2
`p1`	coefficient of x
`p0`	constant term

- **Method:** distribute each term of the first binomial over the second (FOIL) and collect like terms
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



p2 (coefficient of x^2) against a (coefficient of x in the first binomial) from 0.5 to 1.5. One curve for each value of c (coefficient of x in the second binomial): 1, 0.5, 2. Other inputs as in the worked example: b = 2.0, d = 3.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, p2 = 1.

```
math_foundations_ai999x.solve('binomial_product', a=1.0, b=2.0, c=1.0, d=3.0)
# -> p2 = 1, p1 = 5, p0 = 6   (SOLVED)
```

```
math_foundations_ai999x.solve('binomial_product', a=2.0, b=-1.0, c=3.0, d=4.0)
# -> p2 = 6, p1 = 5, p0 = -4   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Expand (x + 2)(x + 3).	SOLVED	p2 = 1, p1 = 5, p0 = 6
Multiply (2x - 1) by (3x + 4).	SOLVED	p2 = 6, p1 = 5, p0 = -4

Centre and radius of a circle from its general equation — `circle_from_general_form`

For $x^2 + y^2 + Dx + Ey + F = 0$, the centre (h, k) and radius rad.

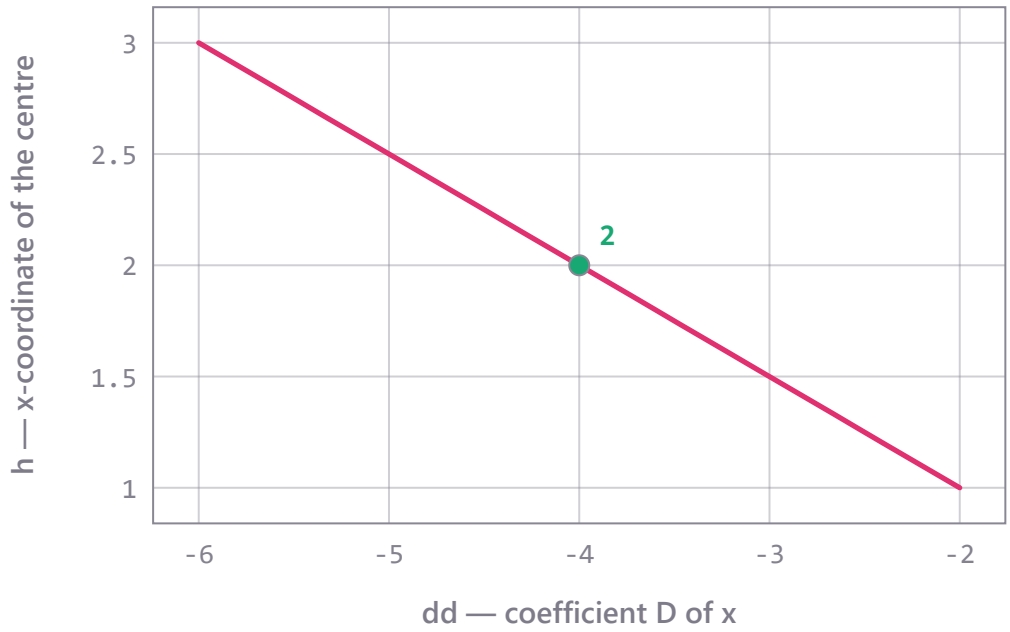
$$(t - h)^2 + (w - k)^2 - \mathrm{rad}^2 = t^2 + w^2 + d_d t + e_e w + f_f$$

input	meaning	condition
`dd`	coefficient D of x	
`ee`	coefficient E of y	
`ff`	constant F	

output	meaning
`h`	x-coordinate of the centre
`k`	y-coordinate of the centre

output	meaning
`rad`	radius

- **Method:** complete the square in x and in y
- **Assumes:** dd**$2/4 + ee$**$2/4 - ff > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



h (x-coordinate of the centre) against dd (coefficient D of x) from -6 to -2. Other inputs as in the worked example: ee = 6.0, ff = -3.0. 31 points computed by math-foundations-ai999x; marked: the worked example, dd = -4, h = 2.

```
math_foundations_ai999x.solve('circle_from_general_form', dd=-4.0, ee=6.0, ff=-3.0)
# -> h = 2, k = -3, rad = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('circle_from_general_form', dd=0.0, ee=0.0, ff=-25.0)
# -> h = 0, k = 0, rad = 5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the centre and radius of x^2 + y^2 - 4x + 6y - 3 = 0.	SOLVED	h = 2, k = -3, rad = 4
Find the centre and radius of x^2 + y^2 + 2x + 2y + 5 = 0.	INVALID_INPUT	Centre and radius of a circle from its general equation: the condition dd**2/4 + ee**2/4 - ff > 0 does not hold for thes

Product and modulus of complex numbers — `complex_product_modulus`

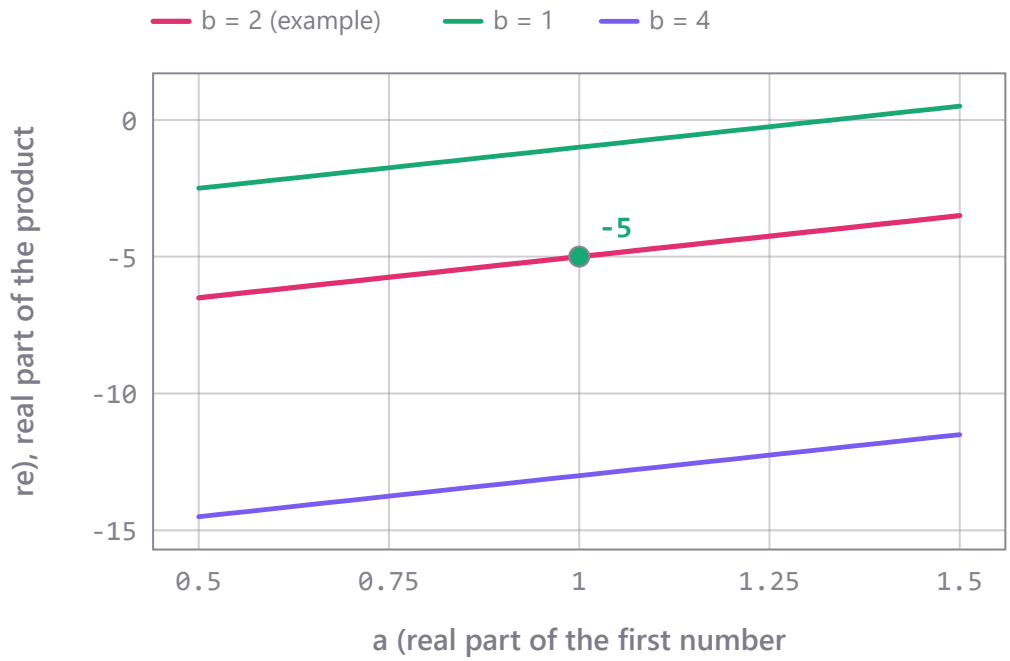
The product $(a + bi)(c + di) = re + imi$ and the modulus of the product.

$$\mathrm{Re} = ac - bd, \quad \mathrm{Im} = ad + bc, \quad \mathrm{modulus}^2 = \left(a^2 + b^2\right)\left(c^2 + d^2\right)$$

input	meaning	condition
`a`	real part of the first number	
`b`	imaginary part of the first number	
`c`	real part of the second number	
`d`	imaginary part of the second number	

output	meaning
`re`	real part of the product
`im`	imaginary part of the product
`modulus`	modulus of the product

- **Method:** distribute and use $i^2 = -1$; the modulus of a product is the product of the moduli
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



re (real part of the product) against a (real part of the first number) from 0.5 to 1.5. One curve for each value of b (imaginary part of the first number): 2, 1, 4. Other inputs as in the worked example: c = 3.0, d = 4.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, re = -5.

```
math_foundations_ai999x.solve('complex_product_modulus', a=1.0, b=2.0, c=3.0, d=4.0)
# -> re = -5, im = 10, modulus = 11.1803   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_product_modulus', a=2.0, b=-1.0, c=2.0, d=1.0)
# -> re = 5, im = 0, modulus = 5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Multiply (1 + 2i)(3 + 4i).	SOLVED	re = -5, im = 10, modulus = 11.1803

Composition of two linear functions, `compose_linear_functions`

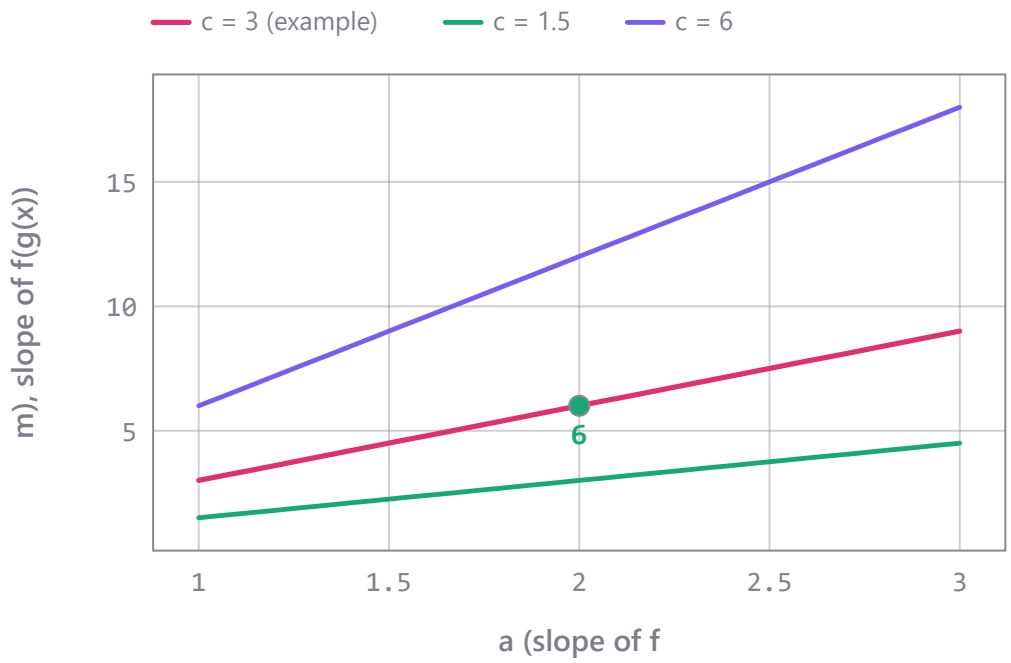
For f(x) = a*x + b and g(x) = c*x + d, the composition f(g(x)) = $mx + k$.

$$mt + k = a\,(ct + d) + b$$

input	meaning	condition
`a`	slope of f	
`b`	intercept of f	
`c`	slope of g	
`d`	intercept of g	

output	meaning
`m`	slope of f(g(x))
`k`	intercept of f(g(x))

- **Method:** substitute g(x) for x in f
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



m (slope of f(g(x))) against a (slope of f) from 1 to 3. One curve for each value of c (slope of g): 3, 1.5, 6. Other inputs as in the worked example: b = 1.0, d = -4.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, m = 6.

```
math_foundations_ai999x.solve('compose_linear_functions', a=2.0, b=1.0, c=3.0, d=-4.0)
# -> m = 6, k = -7   (SOLVED)
```

```
math_foundations_ai999x.solve('compose_linear_functions', a=-1.0, b=5.0, c=2.0, d=3.0)
# -> m = -2, k = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
If f(x) = 2x + 1 and g(x) = 3x - 4, find f(g(x)).	SOLVED	m = 6, k = -7

Compound interest, `compound_interest`

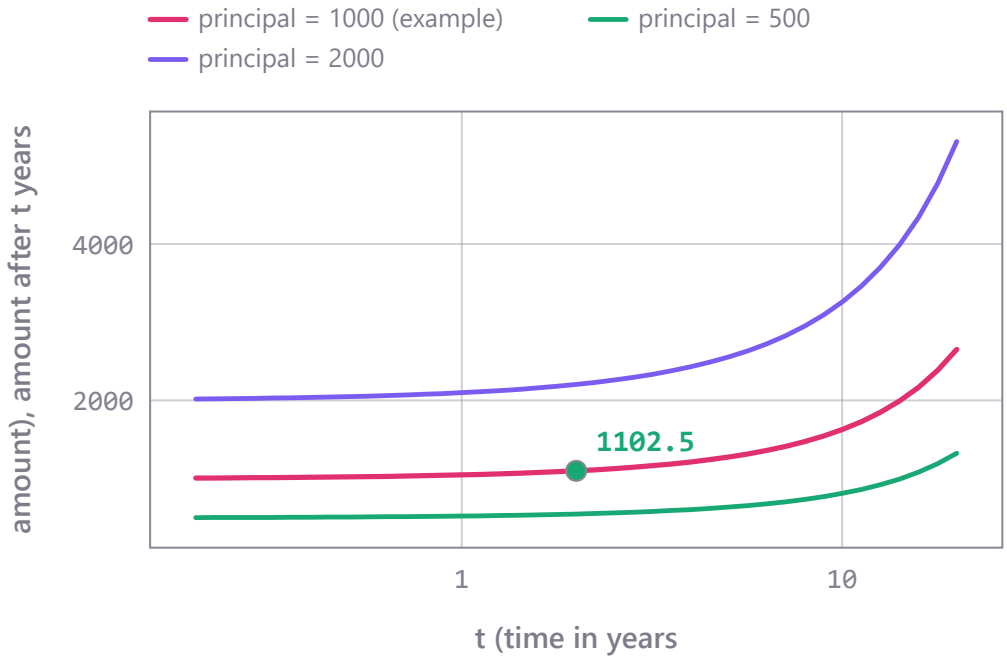
The amount after t years when a principal is compounded n times a year at an annual rate r (a decimal).

$$\ln(\text{amount}) = \ln(\text{principal}) + nt\ln\left(1 + \frac{r}{n}\right)$$

input	meaning	condition
`principal`	starting amount	principal > 0
`r`	annual interest rate as a decimal	$r > -1$
`n`	compounding periods per year	$n \geq 1$
`t`	time in years	$t \geq 0$

output	meaning
`amount`	amount after t years

- **Method:** each period multiplies the amount by $(1 + r/n)$, nt times
- **Assumes:** principal > 0; $n \geq 1$; $t \geq 0$; $1 + r/n > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



amount (amount after t years) against t (time in years) from 0.2 to 20 on a logarithmic axis. One curve for each value of principal (starting amount): 1000, 500, 2000. Other inputs as in the worked example: r = 0.05, n = 1.0. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, t = 2, amount = 1102.5.

```
math_foundations_ai999x.solve('compound_interest', principal=1000.0, r=0.05, n=1.0, t=2.0)
# -> amount = 1102.5   (SOLVED)
```

```
math_foundations_ai999x.solve('compound_interest', principal=500.0, r=0.12, n=12.0, t=1.0)
# -> amount = 563.413   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the amount after 2 years if 1000 is invested at rate 0.05 compounded 1 times a year.	SOLVED	amount = 1102.5

y-intercept of a line from its slope and a point, `convert_to_slope_intercept`

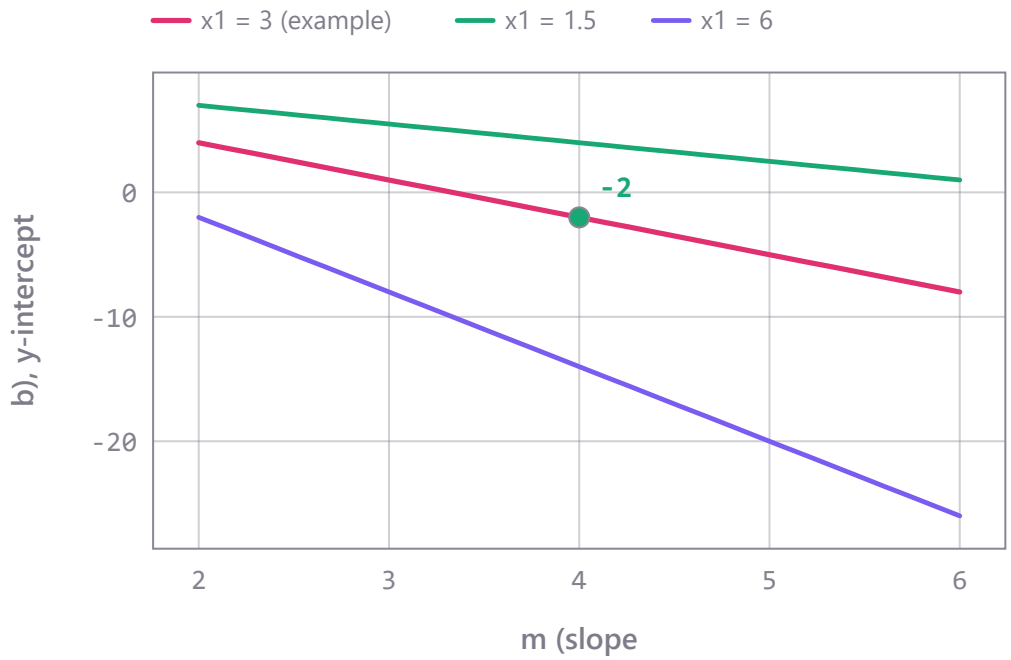
The constant b of $y = mx + b$ for the line with slope m through (x1, y1).

$$y_1 = mx_1 + b$$

input	meaning	condition
`m`	slope	
`x1`	x-coordinate of the point	
`y1`	y-coordinate of the point	

output	meaning
`b`	y-intercept

- **Method:** substitute the point into $y = mx + b$ and solve for b
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



b (y-intercept) against m (slope) from 2 to 6. One curve for each value of x1 (x-coordinate of the point): 3, 1.5, 6. Other inputs as in the worked example: y1 = 10.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, m = 4, b = -2.

```
math_foundations_ai999x.solve('convert_to_slope_intercept', m=4.0, x1=3.0, y1=10.0)
# -> b = -2   (SOLVED)
```

```
math_foundations_ai999x.solve('convert_to_slope_intercept', m=0.0, x1=5.0, y1=7.0)
# -> b = 7   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A line with slope 4 passes through (3, 10). What is its y-intercept?	SOLVED	b = -2
Find the y-intercept of the line with slope 0.5 through the point (2, 4).	SOLVED	b = 3

Determinant of 2x2 matrix, `determinant_2x2_formula`

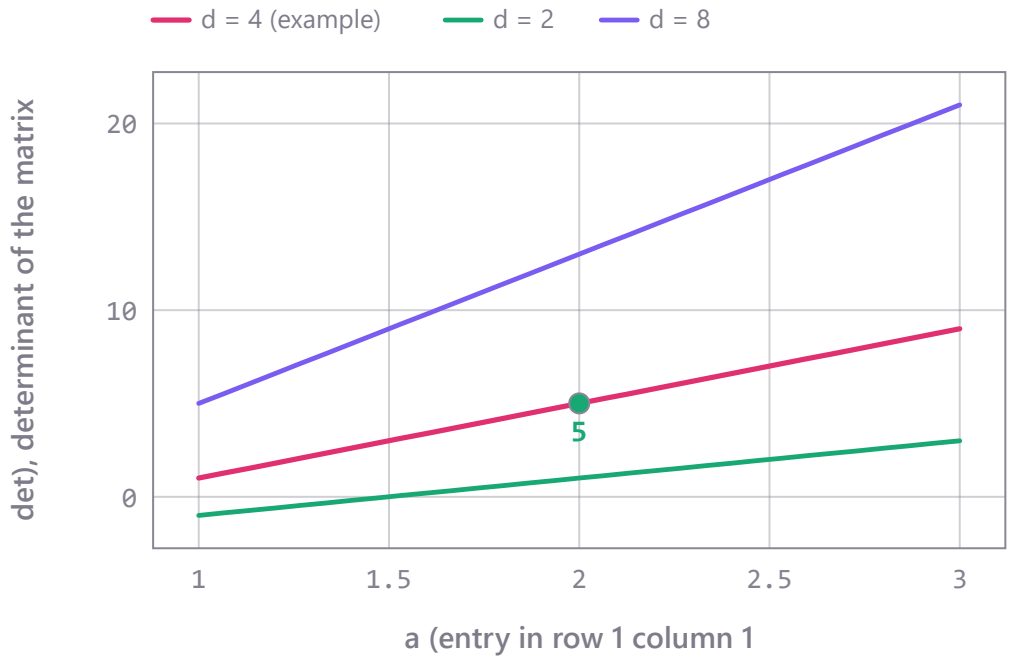
Computes the determinant of a 2x2 matrix using the standard formula.

$$\det = ad - bc$$

input	meaning	condition
`a`	entry in row 1 column 1	
`b`	entry in row 1 column 2	
`c`	entry in row 2 column 1	
`d`	entry in row 2 column 2	

output	meaning
`det`	determinant of the matrix

- **Method:** closed form verified against the defining equations
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



det (determinant of the matrix) against a (entry in row 1 column 1) from 1 to 3. One curve for each value of d (entry in row 2 column 2): 4, 2, 8. Other inputs as in the worked example: b = 3.0, c = 1.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, det = 5.

```
math_foundations_ai999x.solve('determinant_2x2_formula', a=2.0, b=3.0, c=1.0, d=4.0)
# -> det = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('determinant_2x2_formula', a=5.0, b=2.0, c=4.0, d=8.0)
# -> det = 32   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the determinant of the matrix [[3, 1], [2, 4]].	SOLVED	det = 10

Distance between two points, `distance_between_points`

The distance between (x1, y1) and (x2, y2) in the plane.

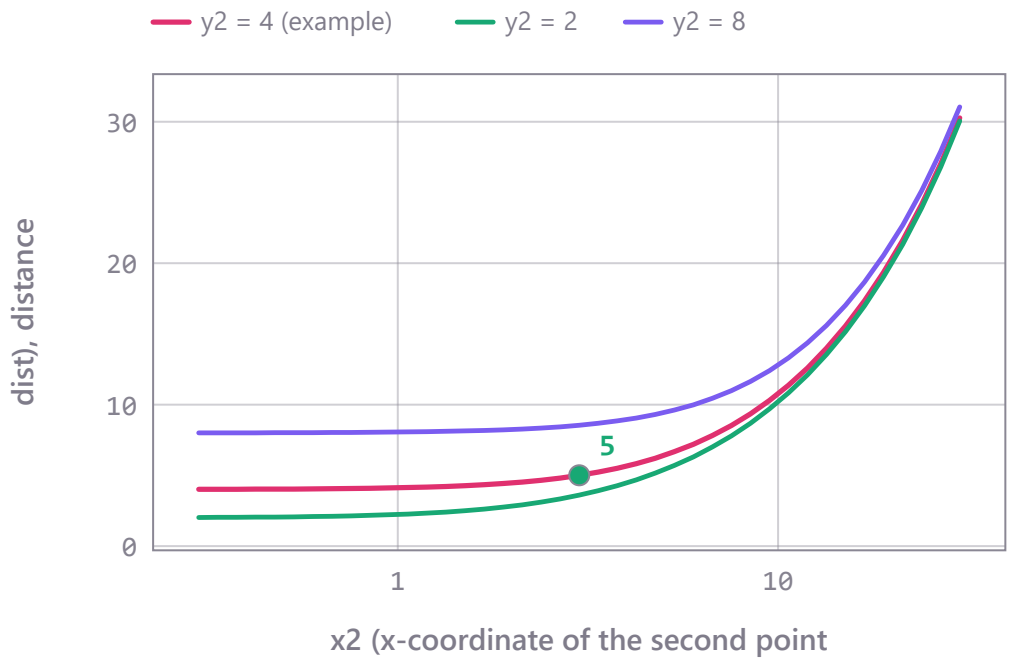
$$\text{dist}^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

input	meaning	condition
`x1`	x-coordinate of the first point	
`y1`	y-coordinate of the first point	
`x2`	x-coordinate of the second point	
`y2`	y-coordinate of the second point	

output	meaning
`dist`	distance

- **Method:** the Pythagorean theorem on the horizontal and vertical differences

- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dist (distance) against x2 (x-coordinate of the second point) from 0.3 to 30 on a logarithmic axis. One curve for each value of y2 (y-coordinate of the second point): 4, 2, 8. Other inputs as in the worked example: x1 = 0.0, y1 = 0.0. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, x2 = 3, dist = 5.

```
math_foundations_ai999x.solve('distance_between_points', x1=0.0, y1=0.0, x2=3.0, y2=4.0)
# -> dist = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('distance_between_points', x1=-1.0, y1=2.0, x2=5.0, y2=10.0)
# -> dist = 10   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the distance between (1, 1) and (4, 5).	SOLVED	dist = 5

Doubling time of exponential growth, `doubling_time`

The time for a quantity growing as $N = N_0 e^{kt}$ to double, for a growth constant $k > 0$.

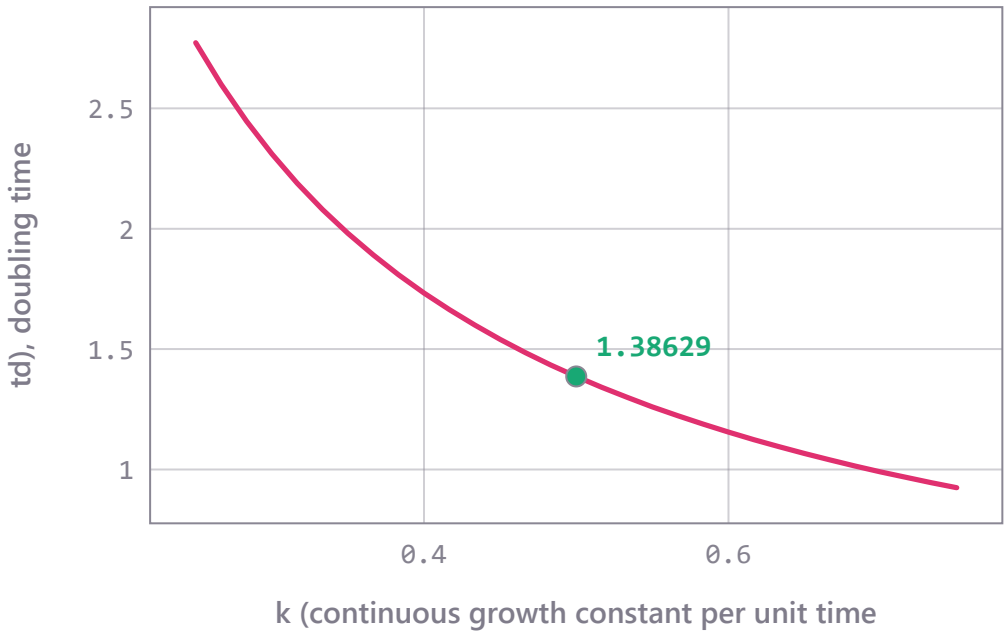
$$kt_d = \ln(2)$$

input	meaning	condition
`k`	continuous growth constant per unit time	$k > 0$

output	meaning
`td`	doubling time

- **Method:** solve $e^{kt} = 2$ for t

- **Assumes:** $k > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



td (doubling time) against k (continuous growth constant per unit time) from 0.25 to 0.75. 31 points computed by math-foundations-ai999x; marked: the worked example, k = 0.5, td = 1.38629.

```
math_foundations_ai999x.solve('doubling_time', k=0.5)
# -> td = 1.38629   (SOLVED)
```

```
math_foundations_ai999x.solve('doubling_time', k=0.1)
# -> td = 6.93147   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the doubling time for a growth constant of 0.1?	SOLVED	td = 6.93147

Solution of an exponential equation, `exponential_equation`

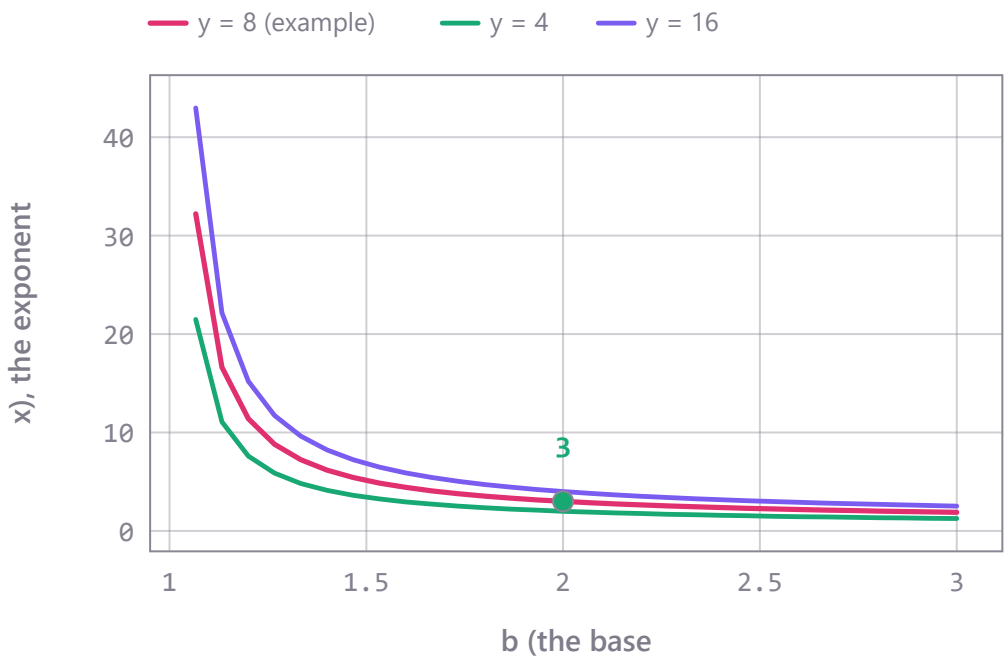
Solve $b^x = y$ for x, with $b > 0$, $b \neq 1$ and $y > 0$.

$$x \ln(b) = \ln(y)$$

input	meaning	condition
b	the base	$b > 0$
y	the value of the power	$y > 0$

output	meaning
x	the exponent

- **Method:** take logarithms of both sides: $x = \frac{\log(y)}{\log(b)}$
- **Assumes:** $b > 0$; $y > 0$; $b \neq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x (the exponent) against b (the base) from 1 to 3. One curve for each value of y (the value of the power): 8, 4, 16. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 2, x = 3.

```
math_foundations_ai999x.solve('exponential_equation', b=2.0, y=8.0)
# -> x = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('exponential_equation', b=10.0, y=0.001)
# -> x = -3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve 2^x = 32.	SOLVED	x = 5
Solve 1^x = 5.	INVALID_INPUT	Solution of an exponential equation: the condition b != 1 does not hold for these inputs.

Factoring a monic quadratic, `factor_monic_quadratic`

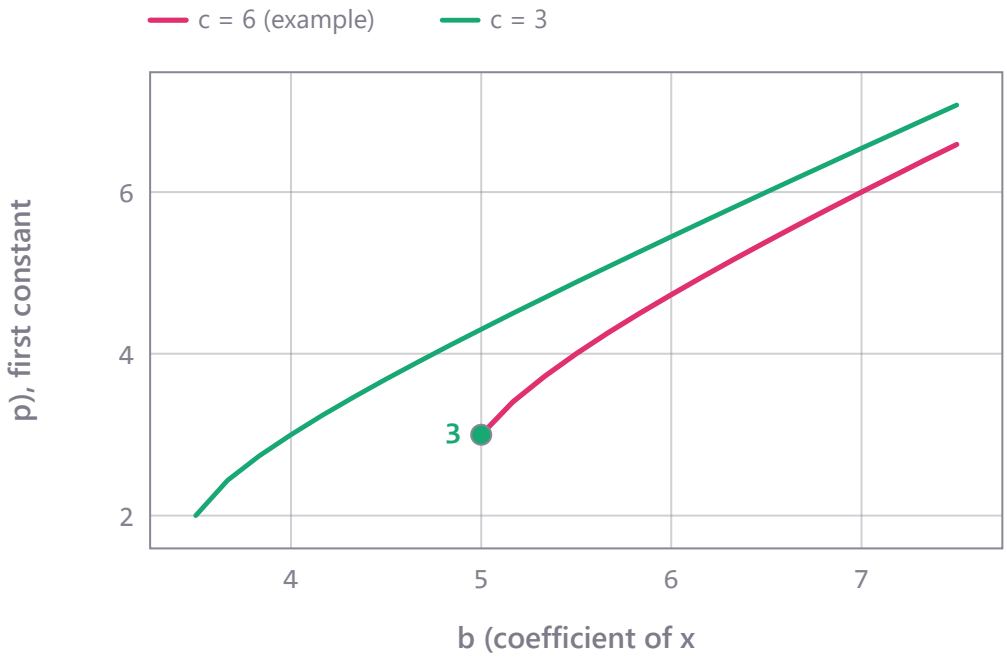
Write $x^2 + bx + c$ as $(x + p)(x + q)$ with real p and q.

$$(t + p)(t + q) = t^2 + bt + c$$

input	meaning	condition
b	coefficient of x	
c	constant term	

output	meaning
p	first constant
q	second constant

- **Method:** find p and q with $p + q = b$ and $pq = c$ (the roots of the quadratic are -p and -q)
- **Assumes:** b**$2 - 4c \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



p (first constant) against b (coefficient of x) from 2.5 to 7.5. One curve for each value of c (constant term): 6, 3. 16 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 5, p = 3.

```
math_foundations_ai999x.solve('factor_monic_quadratic', b=5.0, c=6.0)
# -> p = 3, q = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('factor_monic_quadratic', b=-1.0, c=-12.0)
# -> p = 3, q = -4   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Factor x^2 + 5x + 6.	SOLVED	p = 3, q = 2
Factor x^2 - x - 12.	SOLVED	p = 3, q = -4
Factor x^2 + 2x + 5.	INVALID_INPUT	Factoring a monic quadratic: the condition b**2 - 4*c >= 0 does not hold for these inputs.

Term and partial sum of a geometric sequence, `geometric_sequence`

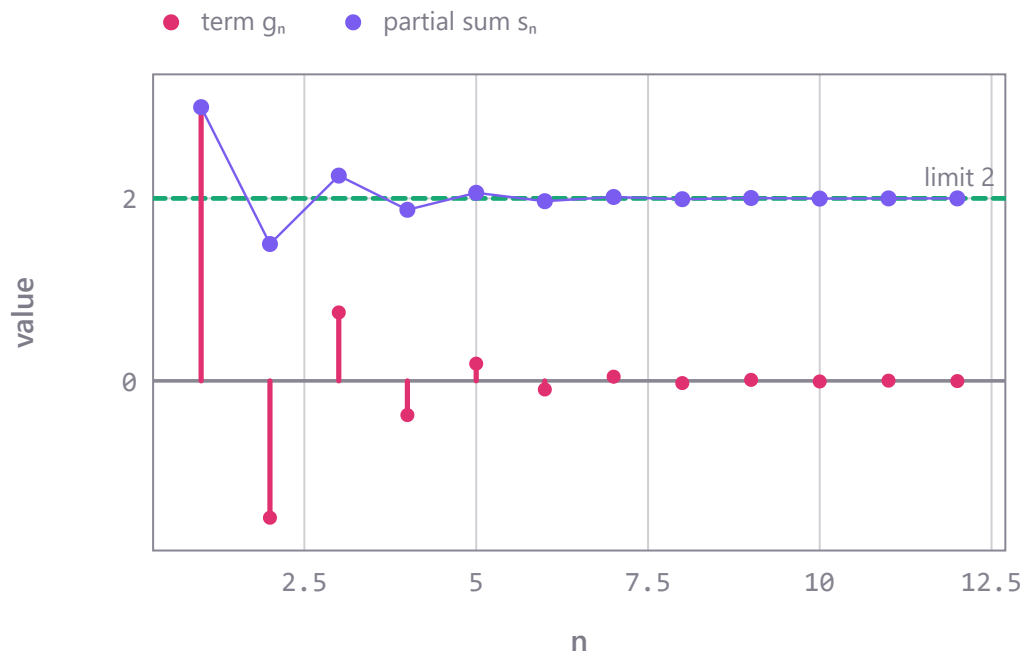
The n-th term and the sum of the first n terms of the geometric sequence with first term g1 and common ratio q $(q \neq 1)$.

$$g_n = g_1 q^{n-1}, \quad s_n\,(1-q) = g_1\,(1-q^n)$$

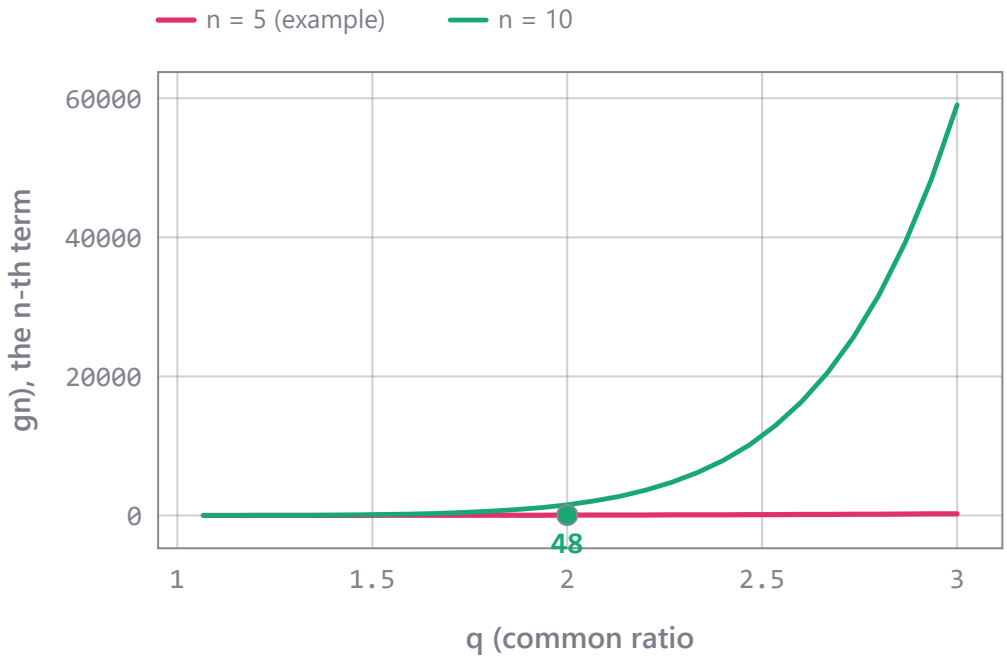
input	meaning	condition
g1	first term	
q	common ratio	$q \neq 1$
n	position of the term	$n \geq 1$

output	meaning
gn	the n-th term
sn	the sum of the first n terms

- **Method:** each term multiplies the previous one by q; the sum telescopes when multiplied by $(1-q)$
- **Assumes:** $q \neq 1$; $n \geq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The geometric sequence g_1 = 3, q = -0.5: the alternating terms (stems) and the partial sums (dots), each computed by the package, settle on the sum of the infinite series, 2; g_{12} = -0.00146484, s_{12} = 1.99951.



gn (the n-th term) against q (common ratio) from 1 to 3. One curve for each value of n (position of the term): 5, 10. Other inputs as in the worked example: g1 = 3.0. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, q = 2, gn = 48.

```
math_foundations_ai999x.solve('geometric_sequence', g1=3.0, q=2.0, n=5.0)
# -> gn = 48, sn = 93   (SOLVED)
```

```
math_foundations_ai999x.solve('geometric_sequence', g1=81.0, q=-0.3333333333333333, n=4.0)
# -> gn = -3, sn = 60   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find term 4 and the sum of the first 4 terms of the geometric sequence with first term 2 and ratio 3.	SOLVED	gn = 54, sn = 80

Line through two points, `line_through_two_points`

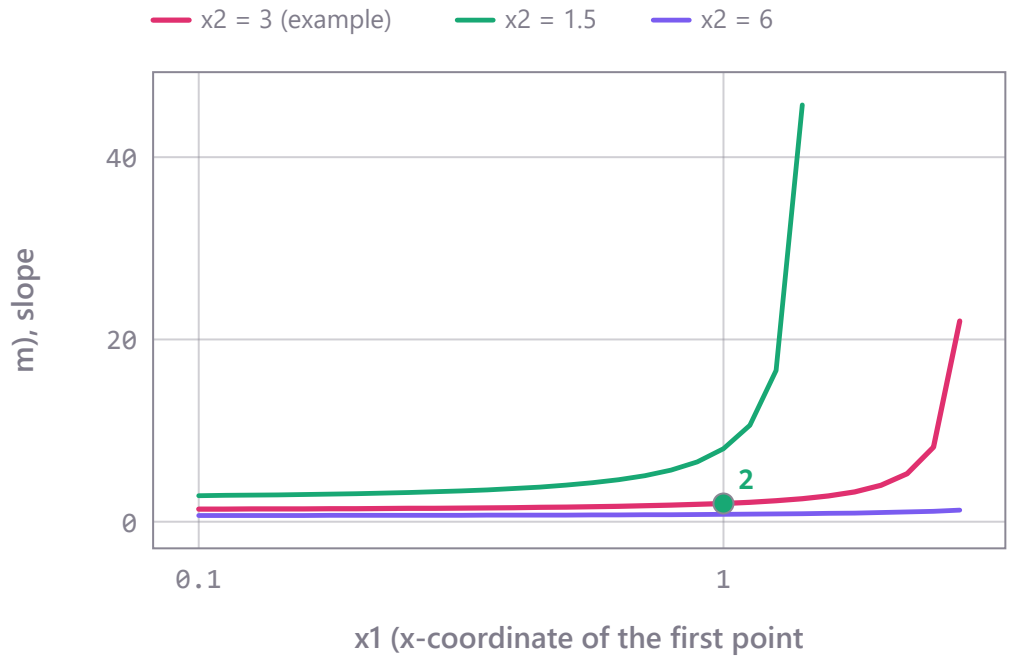
The slope m and y-intercept k of the line through (x1, y1) and (x2, y2), in the form $y = mx + k$.

$$mx_1 + k = y_1, \quad mx_2 + k = y_2$$

input	meaning	condition
`x1`	x-coordinate of the first point	
`y1`	y-coordinate of the first point	
`x2`	x-coordinate of the second point	
`y2`	y-coordinate of the second point	

output	meaning
`m`	slope
`k`	y-intercept

- **Method:** slope = rise over run; the intercept follows from one point
- **Assumes:** $x_2 - x_1 \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



m (slope) against x1 (x-coordinate of the first point) from 0.1 to 2.81838 on a logarithmic axis. One curve for each value of x2 (x-coordinate of the second point): 3, 1.5, 6. Other inputs as in the worked example: y1 = 3.0, y2 = 7.0. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, x1 = 1, m = 2.

```
math_foundations_ai999x.solve('line_through_two_points', x1=1.0, y1=3.0, x2=3.0, y2=7.0)
# -> m = 2, k = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('line_through_two_points', x1=-2.0, y1=5.0, x2=2.0, y2=-3.0)
# -> m = -2, k = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the equation of the line through (0, 1) and (2, 5).	SOLVED	m = 2, k = 1
Find the equation of the line through (3, 1) and (3, 4).	INVALID_INPUT	Line through two points: the condition x2 - x1 != 0 does not hold for these inputs.

Root of a linear equation, `linear_equation_root`

Solve $ax + b = c$ for x.

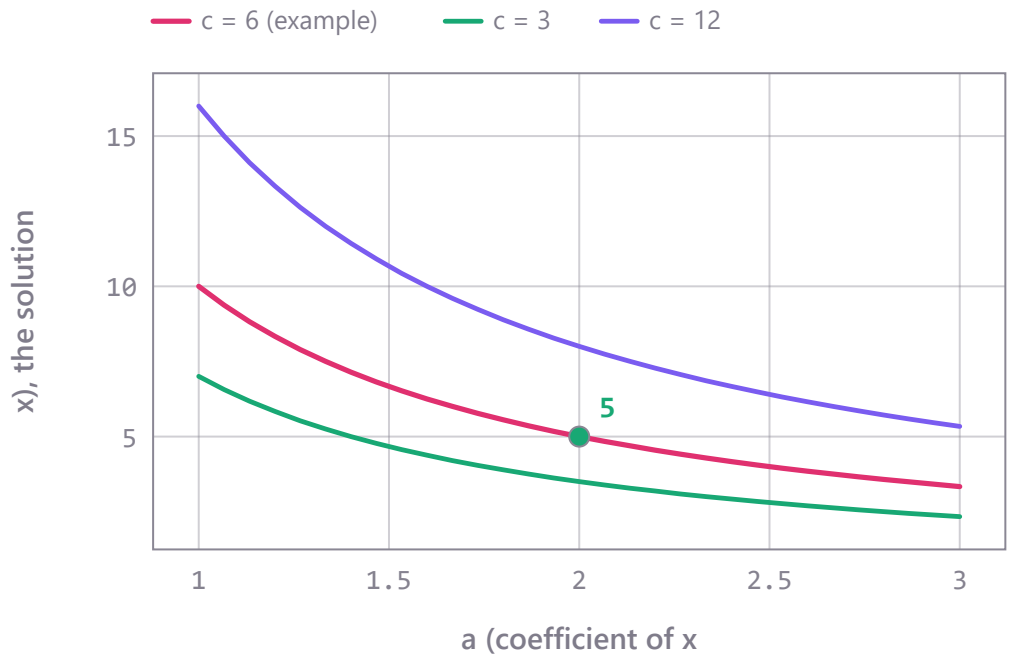
$$ax + b = c$$

input	meaning	condition
a	coefficient of x	$a \neq 0$
b	constant term on the left side	

input	meaning	condition
c	right-hand side	

output	meaning
x	the solution

- **Method:** subtract b from both sides, then divide by a
- **Assumes:** $a \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x (the solution) against a (coefficient of x) from 1 to 3. One curve for each value of c (right-hand side): 6, 3, 12. Other inputs as in the worked example: b = -4.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, x = 5.

```
math_foundations_ai999x.solve('linear_equation_root', a=2.0, b=-4.0, c=6.0)
# -> x = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('linear_equation_root', a=-3.0, b=1.0, c=1.0)
# -> x = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('linear_equation_root', a=0.5, b=0.0, c=3.0)
# -> x = 6   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve 3x + 6 = 0.	SOLVED	x = -2
Solve x - 4 = 10.	SOLVED	x = 14
Find x if 5x + 1 = 2.	SOLVED	x = 0.2
Solve the linear equation with a = 0, b = 1, c = 2.	INVALID_INPUT	Root of a linear equation: the condition a != 0 does not hold for these inputs.

question	outcome	result
Solve the linear equation with a = 4, b = 1.	NEEDS_INPUT	linear_equation_root needs c (right-hand side). Example: solve('linear_equation_root', a=2, b=-4, c=6).

Solution of a linear inequality, `linear_inequality`

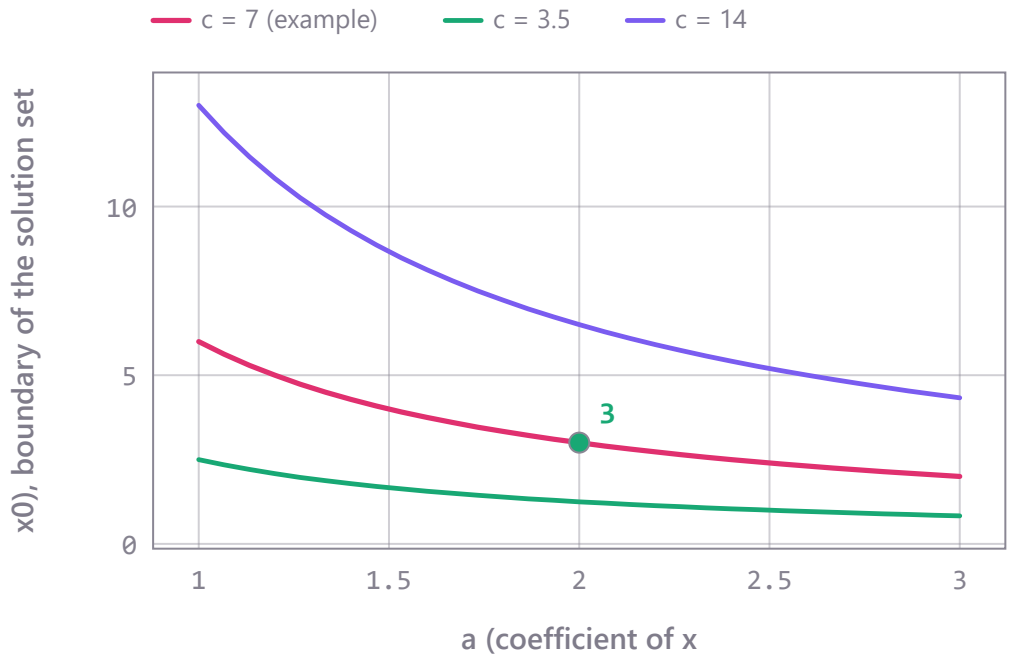
Solve $ax + b < c$ (or >): the boundary x0 and the direction s (s = 1 means the solution of the < form is $x < x_0$; $s = -1$ means it reverses to $x > x_0$ because a < 0).

$$ax_0 + b = c, \quad s\sqrt{a^2} = a$$

input	meaning	condition
`a`	coefficient of x	$a \neq 0$
`b`	constant on the left	
`c`	right-hand side	

output	meaning
`x0`	boundary of the solution set
`s`	1 if the inequality keeps its direction, -1 if it reverses

- **Method:** subtract b, divide by a; dividing by a negative number reverses the inequality
- **Assumes:** $a \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x0 (boundary of the solution set) against a (coefficient of x) from 1 to 3. One curve for each value of c (right-hand side): 7, 3.5, 14. Other inputs as in the worked example: b = 1.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, x0 = 3.

```
math_foundations_ai999x.solve('linear_inequality', a=2.0, b=1.0, c=7.0)
# -> x0 = 3, s = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('linear_inequality', a=-3.0, b=4.0, c=10.0)
# -> x0 = -2, s = -1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the inequality 2x + 1 < 7.	SOLVED	x0 = 3, s = 1
Solve the inequality -3x + 4 < 10.	SOLVED	x0 = -2, s = -1

Solution of two linear equations in two unknowns, `linear_system_2x2`

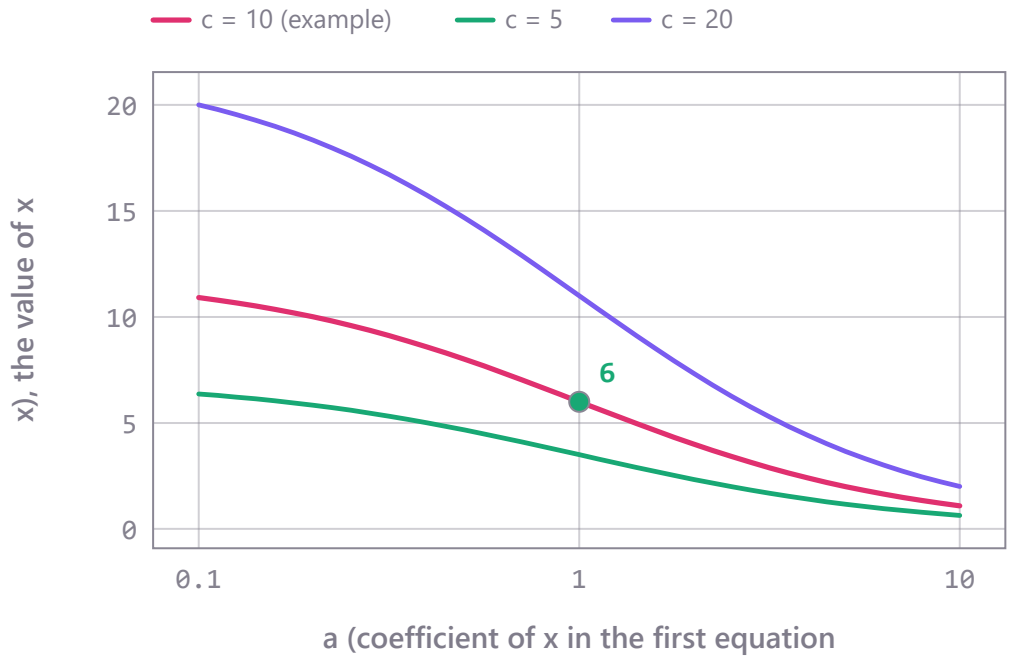
Solve $ax + by = c$ and $dx + ey = f$ for x and y.

$$ax + by = c, \quad dx + ey = f$$

input	meaning	condition
`a`	coefficient of x in the first equation	
`b`	coefficient of y in the first equation	
`c`	right-hand side of the first equation	
`d`	coefficient of x in the second equation	
`e`	coefficient of y in the second equation	
`f`	right-hand side of the second equation	

output	meaning
`x`	the value of x
`y`	the value of y

- **Method:** elimination (Cramer's rule); the determinant $ae - bd$ must not be zero
- **Assumes:** $ae - bd \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x (the value of x) against a (coefficient of x in the first equation) from 0.1 to 10 on a logarithmic axis. One curve for each value of c (right-hand side of the first equation): 10, 5, 20. Other inputs as in the worked example: b = 1.0, d = 1.0, e = -1.0, f = 2.0. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, x = 6.

```
math_foundations_ai999x.solve('linear_system_2x2', a=1.0, b=1.0, c=10.0, d=1.0, e=-1.0, f=2.0)
# -> x = 6, y = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('linear_system_2x2', a=2.0, b=3.0, c=12.0, d=4.0, e=-1.0, f=10.0)
# -> x = 3, y = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the system x + y = 10, x - y = 2.	SOLVED	x = 6, y = 4
Solve the system 2x + 3y = 12, 4x - y = 10.	SOLVED	x = 3, y = 2
Solve the system x + 2y = 3, 2x + 4y = 6.	INVALID_INPUT	Solution of two linear equations in two unknowns: the condition a*e - b*d != 0 does not hold for these inputs.

Solution of a basic logarithmic equation, `logarithmic_equation`

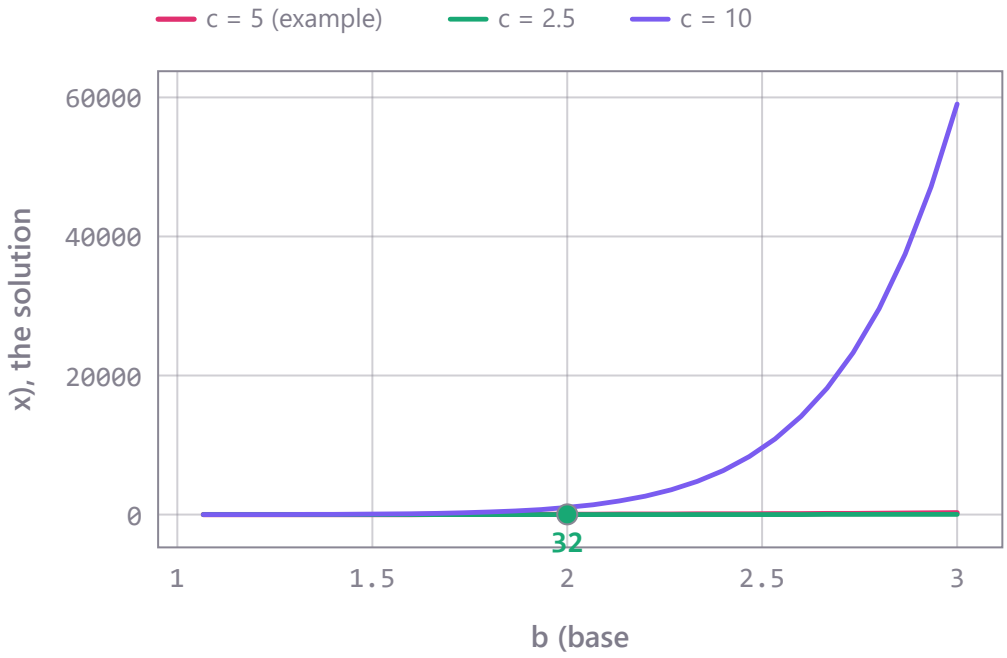
Solve log_b(x) = c for x.

$$\ln(x) = c \ln(b)$$

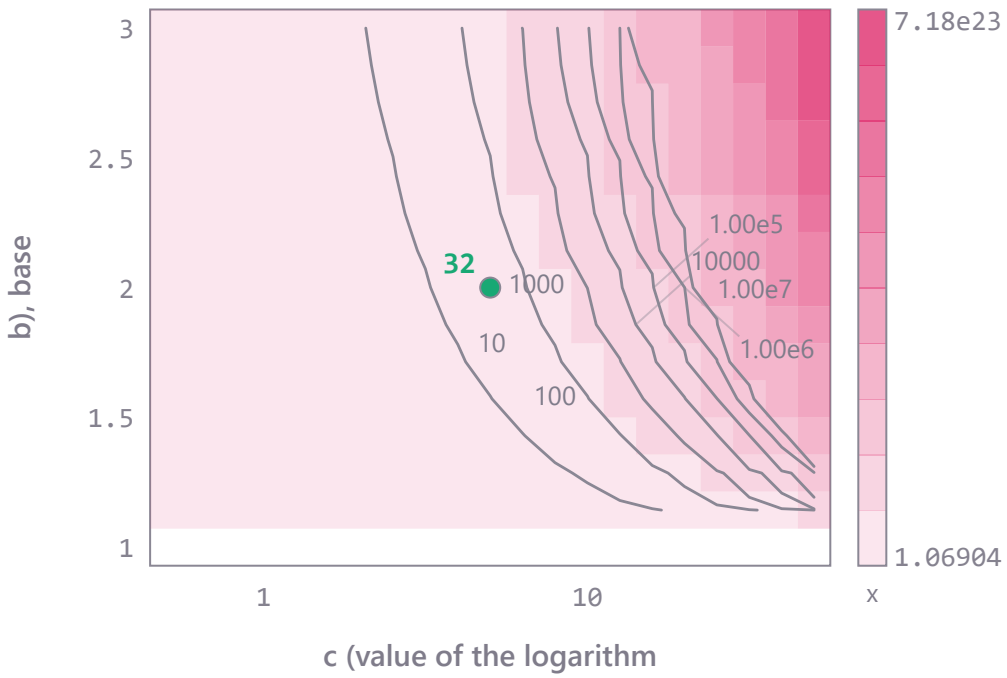
input	meaning	condition
`b`	base	$b > 0$
`c`	value of the logarithm	

output	meaning
x	the solution

- **Method:** rewrite in exponential form: $x = b^c$
- **Assumes:** $b > 0$; $b \neq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x (the solution) against b (base) from 1 to 3. One curve for each value of c (value of the logarithm): 5, 2.5, 10. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 2, x = 32.



x (the solution) over c (value of the logarithm) from 0.5 to 50 and b (base) from 1 to 3 (logarithmic horizontal axis): each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 1.06904 to dark 7.1790e+23, logarithmic shading); iso-lines at x = 10, 100, 1000, 10000, 1.00e5, 1.00e6, 1.00e7; blank cells: inputs the capability refuses. Marked: the worked example, c = 5, b = 2, x = 32.

```
math_foundations_ai999x.solve('logarithmic_equation', b=2.0, c=5.0)
# -> x = 32   (SOLVED)
```

```
math_foundations_ai999x.solve('logarithmic_equation', b=10.0, c=-2.0)
# -> x = 0.01   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve log base 3 of x = 4.	SOLVED	x = 81
Solve log base 1 of x = 2.	INVALID_INPUT	Solution of a basic logarithmic equation: the condition b != 1 does not hold for these inputs.

Determinant and inverse of a 2 by 2 matrix, `matrix_2x2_inverse`

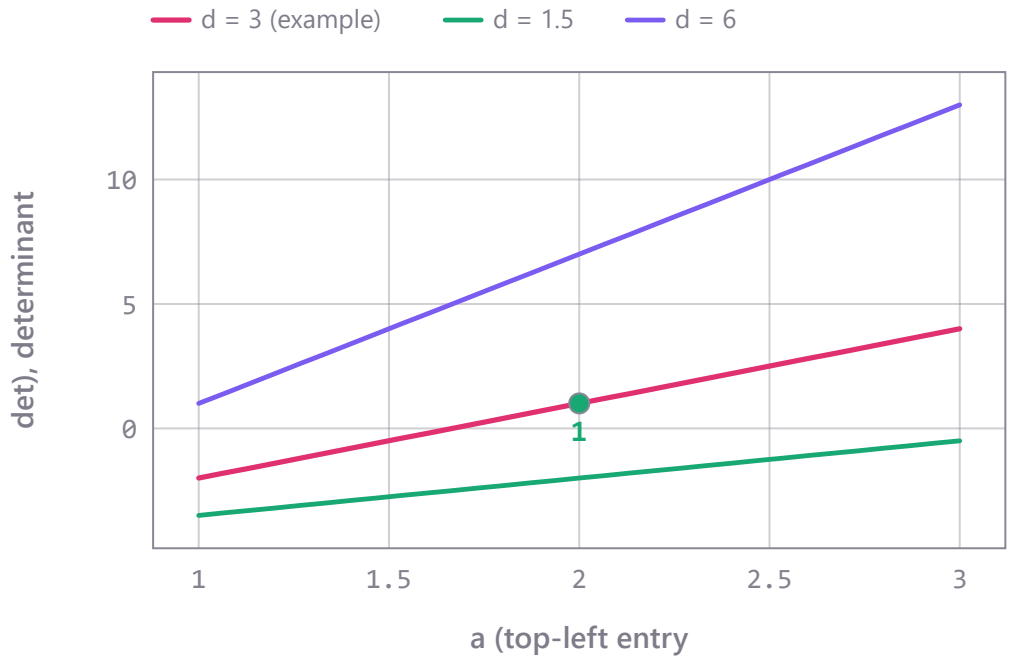
For the matrix [[a, b], [c, d]], its determinant and the entries of its inverse.

$$\det = ad - bc, \quad ai_{11} + bi_{21} = 1, \quad ai_{12} + bi_{22} = 0, \quad ci_{11} + di_{21} = 0, \quad ci_{12} + di_{22} = 1$$

input	meaning	condition
`a`	top-left entry	
`b`	top-right entry	
`c`	bottom-left entry	
`d`	bottom-right entry	

output	meaning
`det`	determinant
`i11`	inverse, top-left
`i12`	inverse, top-right
`i21`	inverse, bottom-left
`i22`	inverse, bottom-right

- **Method:** swap the diagonal, negate the off-diagonal, divide by the determinant
- **Assumes:** $ad - bc \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



det (determinant) against a (top-left entry) from 1 to 3. One curve for each value of d (bottom-right entry): 3, 1.5, 6. Other inputs as in the worked example: b = 1.0, c = 5.0. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, det = 1.

```
math_foundations_ai999x.solve('matrix_2x2_inverse', a=2.0, b=1.0, c=5.0, d=3.0)
# -> det = 1, i11 = 3, i12 = -1, i21 = -5, i22 = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('matrix_2x2_inverse', a=4.0, b=7.0, c=2.0, d=6.0)
# -> det = 10, i11 = 0.6, i12 = -0.7, i21 = -0.2, i22 = 0.4   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the inverse of the matrix [[2, 1], [5, 3]].	SOLVED	det = 1, i11 = 3, i12 = -1, i21 = -5, i22 = 2
Find the inverse of the matrix [[1, 2], [2, 4]].	INVALID_INPUT	Determinant and inverse of a 2 by 2 matrix: the condition a*d - b*c != 0 does not hold for these inputs.

Matrix scalar multiple definition, `matrix_scalar_multiple_definition`

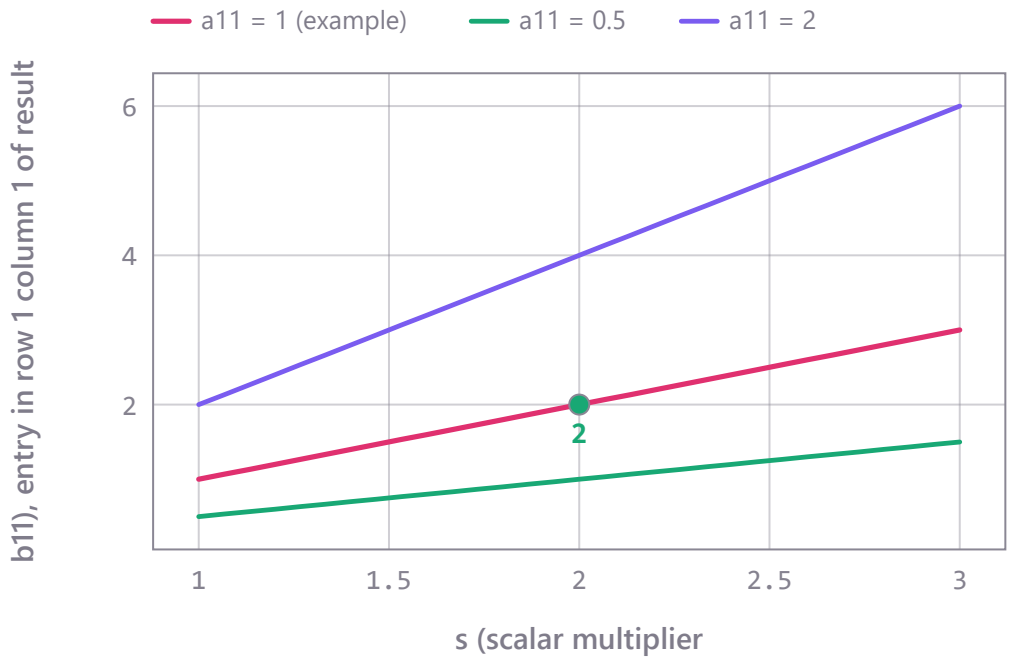
Computes the entries of a 2x2 matrix scaled by a scalar.

$$b_{11} = sa_{11}, \quad b_{12} = sa_{12}, \quad b_{21} = sa_{21}, \quad b_{22} = sa_{22}$$

input	meaning	condition
`s`	scalar multiplier	
`a11`	entry in row 1 column 1	
`a12`	entry in row 1 column 2	
`a21`	entry in row 2 column 1	
`a22`	entry in row 2 column 2	

output	meaning
`b11`	entry in row 1 column 1 of result
`b12`	entry in row 1 column 2 of result
`b21`	entry in row 2 column 1 of result
`b22`	entry in row 2 column 2 of result

- **Method:** closed form verified against the defining equations
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



b11 (entry in row 1 column 1 of result) against s (scalar multiplier) from 1 to 3. One curve for each value of a11 (entry in row 1 column 1): 1, 0.5, 2. Other inputs as in the worked example: a12 = 3.0, a21 = -4.0, a22 = 5.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, s = 2, b11 = 2.

```
math_foundations_ai999x.solve('matrix_scalar_multiple_definition', a11=1.0, a12=3.0, a21=-4.0,
a22=5.0, s=2.0)
# -> b11 = 2, b12 = 6, b21 = -8, b22 = 10   (SOLVED)
```

```
math_foundations_ai999x.solve('matrix_scalar_multiple_definition', a11=0.0, a12=7.0, a21=2.0,
a22=9.0, s=-1.0)
# -> b11 = 0, b12 = -7, b21 = -2, b22 = -9   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Calculate 3 times the matrix [[1, 2], [3, 4]].	SOLVED	b11 = 3, b12 = 6, b21 = 9, b22 = 12

Midpoint of a line segment, `midpoint_of_segment`

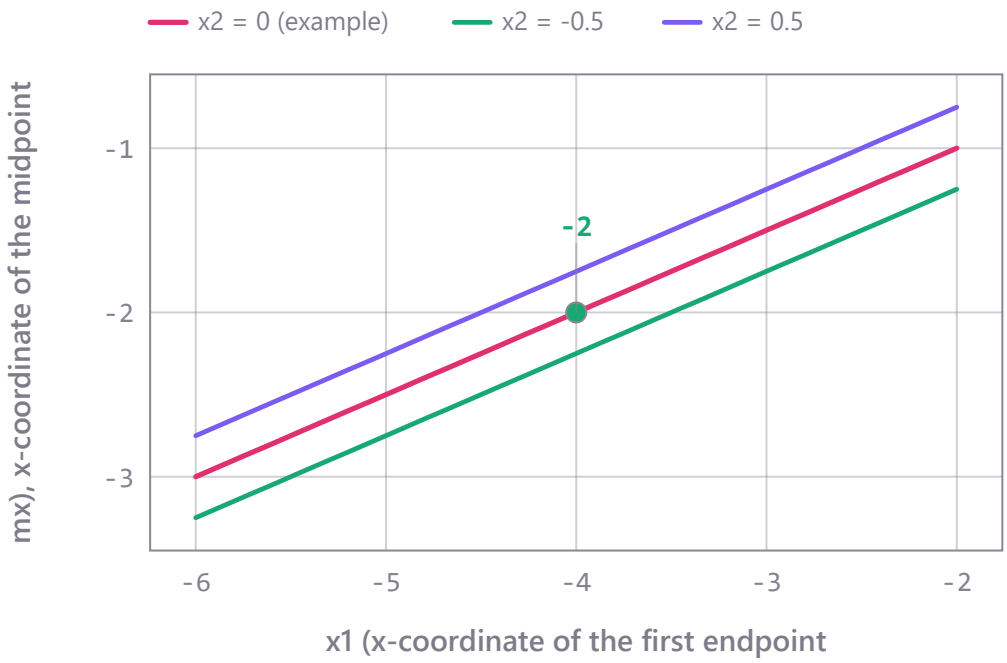
The midpoint (mx, my) of the segment from (x1, y1) to (x2, y2).

$$2m_x = x_1 + x_2, \quad 2m_y = y_1 + y_2$$

input	meaning	condition
`x1`	x-coordinate of the first endpoint	
`y1`	y-coordinate of the first endpoint	
`x2`	x-coordinate of the second endpoint	
`y2`	y-coordinate of the second endpoint	

output	meaning
`mx`	x-coordinate of the midpoint
`my`	y-coordinate of the midpoint

- **Method:** average the x-coordinates and the y-coordinates
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



mx (x-coordinate of the midpoint) against x1 (x-coordinate of the first endpoint) from -6 to -2. One curve for each value of x2 (x-coordinate of the second endpoint): 0, -0.5, 0.5. Other inputs as in the worked example: y1 = 2.0, y2 = 6.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, x1 = -4, mx = -2.

```
math_foundations_ai999x.solve('midpoint_of_segment', x1=-4.0, y1=2.0, x2=0.0, y2=6.0)
# -> mx = -2, my = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('midpoint_of_segment', x1=1.0, y1=1.0, x2=5.0, y2=3.0)
# -> mx = 3, my = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the midpoint of the segment from (-4, 2) to (0, 6).	SOLVED	mx = -2, my = 4

question	outcome	result
What is the midpoint of the line segment joining (1, 1) and (5, 3)?	SOLVED	mx = 3, my = 2

Multiply rational expressions, `multiply_rational_expressions`

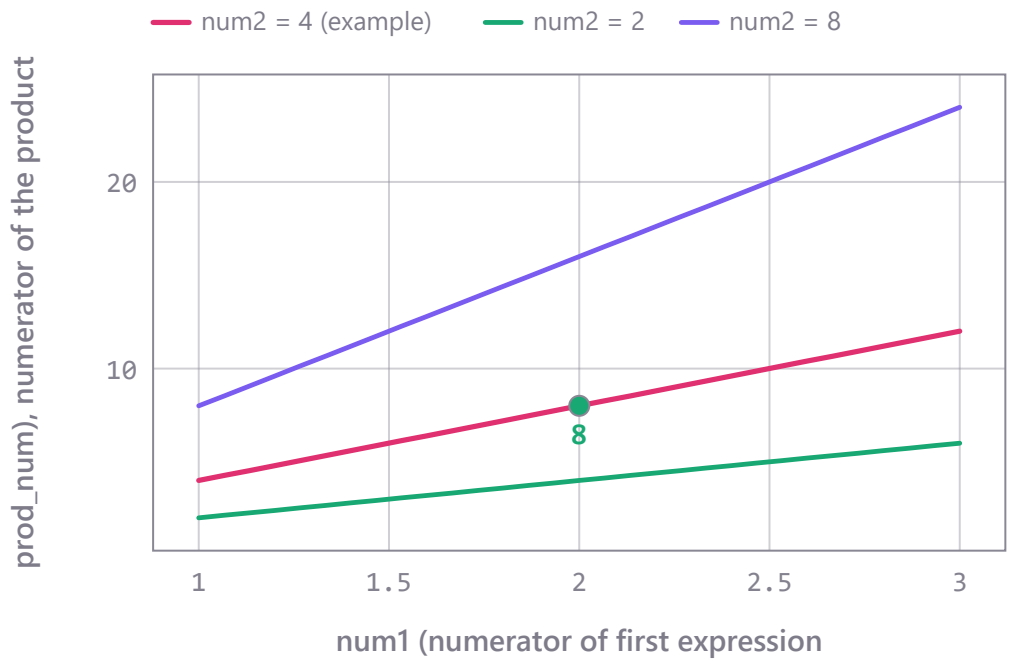
Compute the product of two rational expressions.

$$\text{prod num} = \text{num}_1\text{num}_2, \quad \text{prod den} = \text{den}_1\text{den}_2$$

input	meaning	condition
`num1`	numerator of first expression	
`den1`	denominator of first expression	
`num2`	numerator of second expression	
`den2`	denominator of second expression	

output	meaning
`prod_num`	numerator of the product
`prod_den`	denominator of the product

- **Method:** closed form verified against the defining equations
- **Assumes:** den1 != 0; den2 != 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



prod_num (numerator of the product) against num1 (numerator of first expression) from 1 to 3. One curve for each value of num2 (numerator of second expression): 4, 2, 8. Other inputs as in the worked example: den1 = 3.0, den2 = 5.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, num1 = 2, prod_num = 8.

```
math_foundations_ai999x.solve('multiply_rational_expressions', den1=3.0, den2=5.0, num1=2.0,
num2=4.0)
# -> prod_num = 8, prod_den = 15   (SOLVED)
```

```
math_foundations_ai999x.solve('multiply_rational_expressions', den1=2.0, den2=4.0, num1=1.0,
num2=3.0)
# -> prod_num = 3, prod_den = 8   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Multiply 2/3 by 5/7.	SOLVED	prod_num = 10, prod_den = 21

Perimeter of a rectangle, `perimeter_of_rectangle`

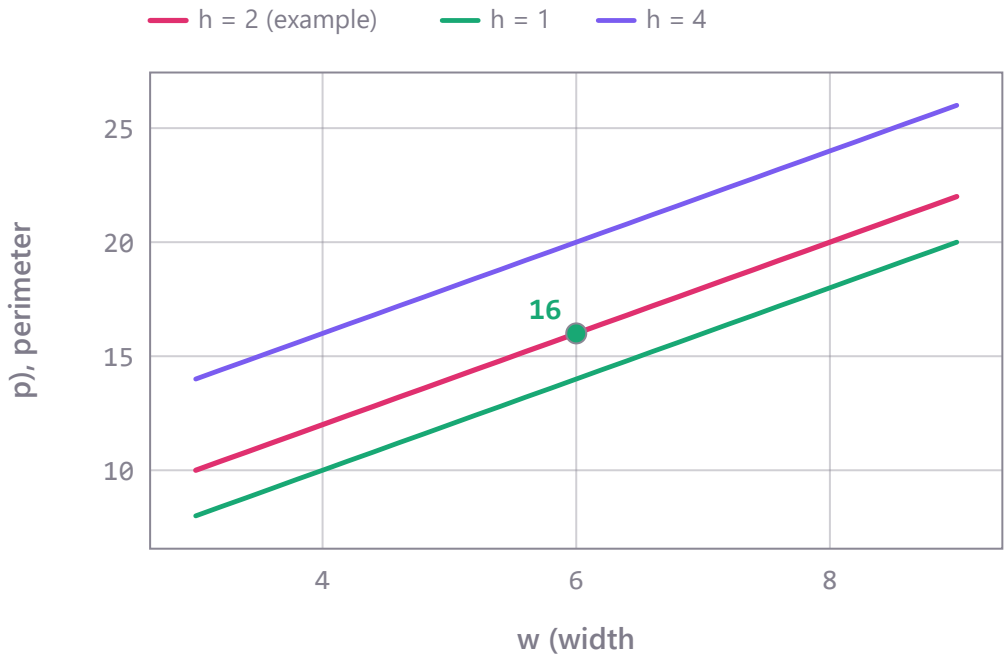
The perimeter p of a rectangle with width w and height h.

$$p - 2h = 2w$$

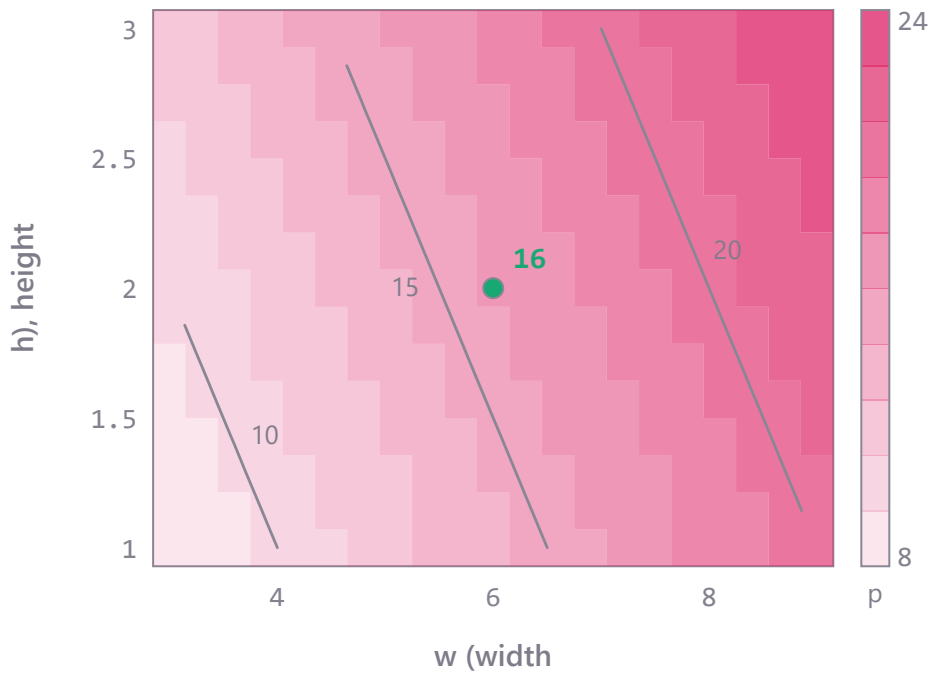
input	meaning	condition
`w`	width	$w \geq 0$
`h`	height	$h \geq 0$

output	meaning
`p`	perimeter

- **Method:** add the four sides
- **Assumes:** $w \geq 0$; $h \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



p (perimeter) against w (width) from 3 to 9. One curve for each value of h (height): 2, 1, 4. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, w = 6, p = 16.



p (perimeter) over w (width) from 3 to 9 and h (height) from 1 to 3: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 8 to dark 24); iso-lines at p = 10, 15, 20. Marked: the worked example, w = 6, h = 2, p = 16.

```
math_foundations_ai999x.solve('perimeter_of_rectangle', w=6.0, h=2.0)
# -> p = 16   (SOLVED)
```

```
math_foundations_ai999x.solve('perimeter_of_rectangle', w=2.5, h=4.0)
# -> p = 13   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the perimeter of a rectangle with width 6 and height 2?	SOLVED	p = 16
Find the perimeter of a rectangle of width 0 and height 3.	SOLVED	p = 6

Slope of a perpendicular line, `perpendicular_lines_slope_product`

The slope m_perp of a line perpendicular to a line of nonzero slope m.

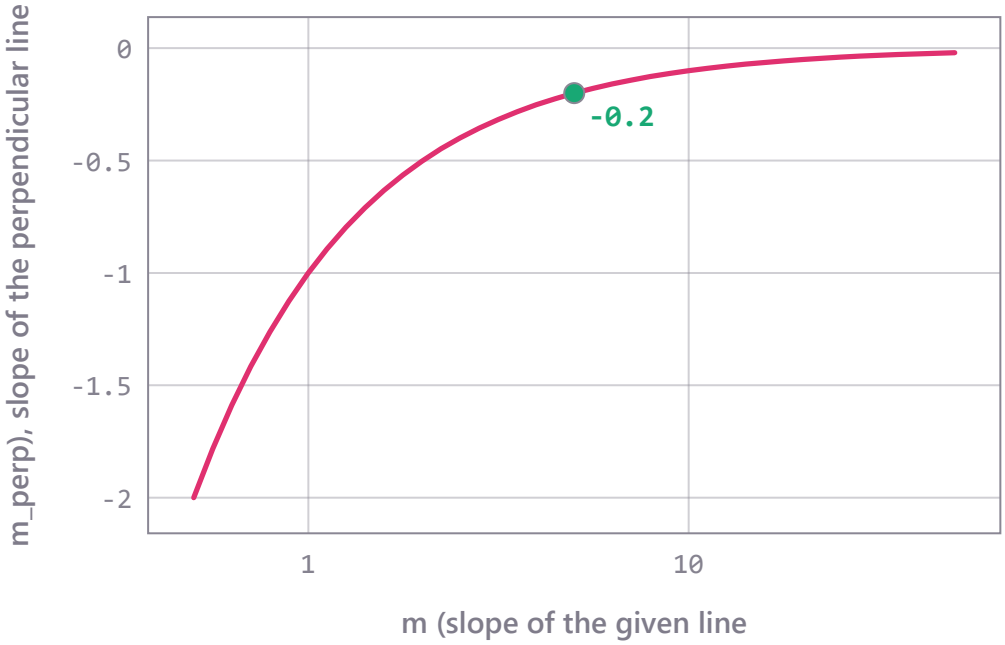
$$mm_{\text{perp}} = -1$$

input	meaning	condition
`m`	slope of the given line	$m \neq 0$

output	meaning
`m_perp`	slope of the perpendicular line

- **Method:** the product of the slopes of perpendicular lines $is -1$
- **Assumes:** $m \neq 0$

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



m_perp (slope of the perpendicular line) against m (slope of the given line) from 0.5 to 50 on a logarithmic axis. 41 points computed by math-foundations-ai999x; marked: the worked example, m = 5, m_perp = -0.2.

```
math_foundations_ai999x.solve('perpendicular_lines_slope_product', m=5.0)
# -> m_perp = -0.2   (SOLVED)
```

```
math_foundations_ai999x.solve('perpendicular_lines_slope_product', m=-0.5)
# -> m_perp = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A line has slope 5. What is the slope of a line perpendicular to it?	SOLVED	m_perp = -0.2
A line has slope 0. What is the slope of a line perpendicular to it?	INVALID_INPUT	Slope of a perpendicular line: the condition m != 0 does not hold for these inputs.

Asymptotes of a linear rational function, `rational_function_asymptotes`

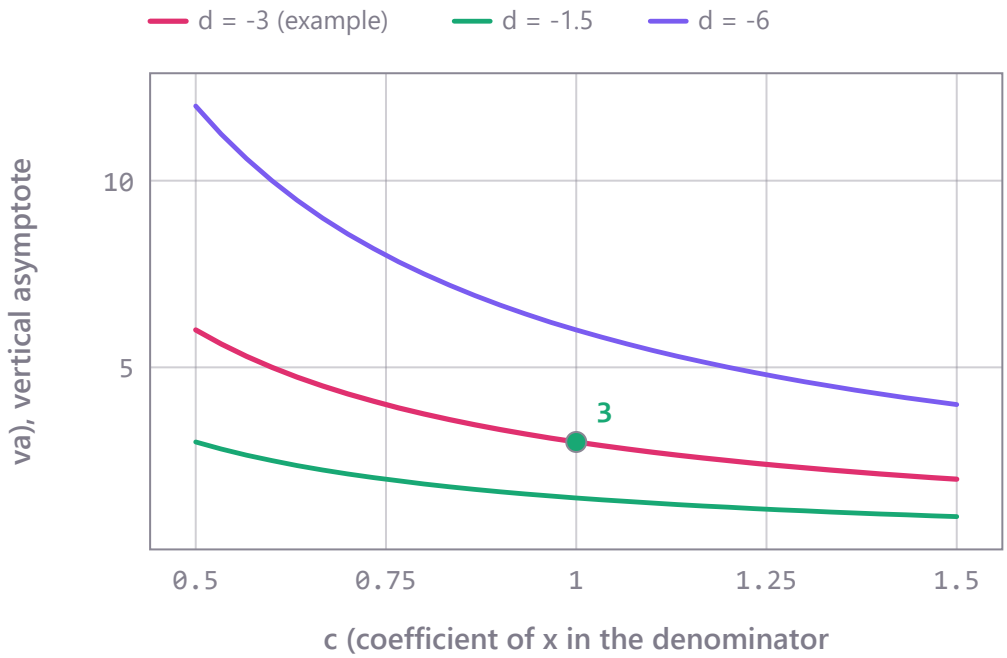
The vertical asymptote $x = va$ and horizontal asymptote $y = ha$ of f(x) = (a*x + b)/(c*x + d).

$$cv_a + d = 0, \quad ch_a = a$$

input	meaning	condition
a	coefficient of x in the numerator	
b	constant of the numerator	
c	coefficient of x in the denominator	$c \neq 0$
d	constant of the denominator	

output	meaning
`va`	vertical asymptote
`ha`	horizontal asymptote

- **Method:** the denominator vanishes at $x = -d/c$; for large x the ratio tends to a/c (if $ad = bc$ the function is constant and has no vertical asymptote)
- **Assumes:** $c \neq 0$; $ad - bc \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



va (vertical asymptote) against c (coefficient of x in the denominator) from 0.5 to 1.5. One curve for each value of d (constant of the denominator): -3, -1.5, -6. Other inputs as in the worked example: a = 2.0, b = 1.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 1, va = 3.

```
math_foundations_ai999x.solve('rational_function_asymptotes', a=2.0, b=1.0, c=1.0, d=-3.0)
# -> va = 3, ha = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('rational_function_asymptotes', a=1.0, b=0.0, c=2.0, d=4.0)
# -> va = -2, ha = 0.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the asymptotes of f(x) = (2x + 1)/(x - 3).	SOLVED	va = 3, ha = 2
Find the asymptotes of f(x) = (2x + 4)/(x + 2).	INVALID_INPUT	Asymptotes of a linear rational function: the condition a*d - b*c != 0 does not hold for these inputs.

Vertex form of a quadratic function, `vertex_form`

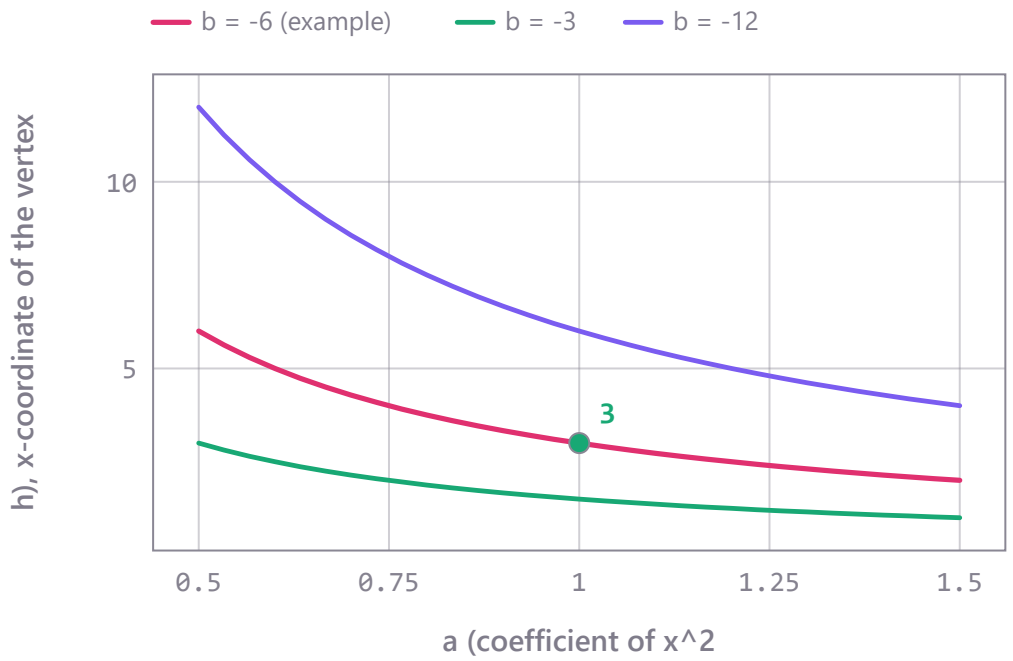
Write $ax^2 + bx + c$ as $a(x - h)^2 + k$, giving the vertex (h, k).

$$a(t-h)^2 + k = at^2 + bt + c$$

input	meaning	condition
a	coefficient of x^2	$a \neq 0$
b	coefficient of x	
c	constant term	

output	meaning
h	x-coordinate of the vertex
k	y-coordinate of the vertex

- **Method:** completing the square
- **Assumes:** $a \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



h (x-coordinate of the vertex) against a (coefficient of x^2) from 0.5 to 1.5. One curve for each value of b (coefficient of x): -6, -3, -12. Other inputs as in the worked example: c = 5.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, h = 3.

```
math_foundations_ai999x.solve('vertex_form', a=1.0, b=-6.0, c=5.0)
# -> h = 3, k = -4   (SOLVED)
```

```
math_foundations_ai999x.solve('vertex_form', a=2.0, b=8.0, c=1.0)
# -> h = -2, k = -7   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the vertex of y = x^2 - 6x + 5.	SOLVED	h = 3, k = -4
Write 2x^2 + 8x + 1 in vertex form.	SOLVED	h = -2, k = -7

Proofs and derivations

Each derivation below was certified step by step: every step follows mechanically from the ones before and from what it rests on (a certified relation, an identity, a definition or a stated hypothesis), and a wrong variant of the result is refused. Ask in one of the documented forms, or call `math_foundations_ai999x.prove("<id>")`. A statement no certified derivation proves is OUT_OF_SCOPE; a conditional result asked without its hypotheses is NEEDS_INPUT and names them.

documented question (executed)	id	steps
Prove that if A = l*w then A = l*w.	`area_of_rectangle`	1
Prove that (a + b) + c = a + (b + c).	`associative_property_of_addition`	1
Prove that if a*x + b*y = e and c*x + d*y = f then x = (e*d - b*f) / (a*d - b*c).	`cramer_rule_x_solution`	8
Prove that if a1*x + b1*y = c1 and a2*x + b2*y = c2 then x*(a1*b2 - a2*b1) = c1*b2 - c2*b1.	`determinant_cramer_two_variables`	7
Prove that a**3 - b**3 = (a - b)*(a**2 + a*b + b**2).	`difference_of_cubes_factorization`	2
Prove that (a + b) * (a - b) = a**2 - b**2.	`difference_of_squares`	6
Prove that a*(b + c) = a*b + a*c.	`distributive_property_verification`	1
Prove that if a*x + b*y = c and d*x + e*y = f then (a - k*d)*x + (b - k*e)*y = c - k*f.	`elimination_method_identity`	4
Prove that if a1*x + b1*y = c1 and a2*x + b2*y = c2 then x = (c1*b2 - c2*b1)/(a1*b2 - a2*b1).	`elimination_solution_two_variables`	6
Prove that if a*x + b = c then x = (c - b)/a.	`linear_equation_solution_formula`	2
Prove that if ((a*r1 + b)/(c*(r1 - r2)))/(x - r1) + ((a*r2 + b)/(c*(r2 - r1)))/(x - r2) = RHS_combined then (a*x + b)/(c*(x - r1)*(x - r2)) = ((a*r1 + b)/(c*(r1 - r2)))/(x - r1) + ((a*r2 + b)/(c*(r2 - r1)))/(x - r2).	`partial_fraction_decomposition_linear_factors`	8
Prove that (a + b)**2 = a**2 + 2*a*b + b**2.	`perfect_square_trinomial`	6
Prove that (a + b)*(c + d) = a*c + a*d + b*c + b*d.	`product_of_sums`	2
Prove that if h = -b/(2*a) and k = c - b**2/(4*a) then a*x**2 + b*x + c = a*(x - (-b/(2*a)))**2 + (c - b**2/(4*a)).	`quadratic_vertex_form_conversion`	4
Prove that if sqrt(x) = c then x = c**2.	`radical_equation_squaring`	2
Prove that (a - b)**2 = a**2 - 2*a*b + b**2.	`square_of_difference`	4
Prove that if a*x + b*y = c and x = d then a*d + b*y = c.	`substitution_method_identity`	2
Prove that if a1*x + b1*y = c1 and a2*x + b2*y = c2 then a1*((c1 - b1*((a1*c2 - a2*c1)/(a1*b2 - a2*b1)))/a1) + b1*((a1*c2 - a2*c1)/(a1*b2 - a2*b1)) = c1.	`substitution_solution_two_variables`	7
Prove that a**3 + b**3 = (a + b)*(a**2 - a*b + b**2).	`sum_of_cubes_factorization`	5